From Surface to Core: Opening New Windows into the Chemistry of Rocky Planets through Novel Methods in Mass Spectrometry and Machine Learning

Thesis by
Ben J. Thyer

In Partial Fulfillment of the Requirements for the
degree of
Doctor of Philosophy

CALIFORNIA INSTITUTE OF TECHNOLOGY
Pasadena, California

2026
Defended 9/28/2026

© 2026

Ben J. Thyer
ORCID: 0000-0002-5341-0316

All rights reserved

ABSTRACT

[TODO: Write abstract.]

ACKNOWLEDGEMENTS

[TODO: Write acknowledgements.]

I am deeply grateful...

The fieldwork at Draaikraal would not have been possible...

Finally, I thank family/friends for their unwavering support...

TABLE OF CONTENTS

Abstract . . . iii
Acknowledgements . . . iv
Table of Contents . . . v
Introduction . . . 1
Chapter I: A Simple Method for Helium Step Heat Experiments at High Water or Sulfur Fugacity . . . 2
1.1 Introduction . . . 3
1.2 Experimental Methods . . . 4
1.3 Theoretical Framework of Permeation Lag . . . 9
1.4 Experimental Approaches Addressing Permeation Lag . . . 14
1.5 Analytical Methods . . . 19
1.6 Results . . . 21
1.7 Discussion . . . 28
1.8 Conclusions . . . 30
1.1s A Fick's Law Formulation Describing Helium Migration Across Semi-Permeable Membranes . . . 34
1.2s Diffusive Steady State Concentration Profiles in Cylindrical Membranes . . . 38
1.3s Estimating permeation lag uncertainty for uncorrected, long heating steps . . . 60
1.4s Comparison to Literature Permeability Values . . . 64
1.5s Additional Analytical Corrections for the Empty Ampoule Calibration Experiments . . . 66
1.6s Sample Preparation and SEM Imaging of Buffered and Decomposed Sulfides . . . 69
1.7s Supplementary Data Tables . . . 70
Chapter II: Helium Step Heat Experiments on Goethite in Hydrothermal Conditions Indicate High Retentivity at Surface Temperature and Consistency with Multiple Diffusion Domain Systematics . . . 80
2.1 Introduction . . . 81
2.2 Experimental Methods . . . 84
2.3 Analytical Methods . . . 90
2.4 Results . . . 96
2.5 Discussion . . . 105
2.6 Conclusions . . . 122
2.1s Supplementary Verification: The Naïve/Informed Bracketing Method for the Encapsulated, Irradiated Win-06-01b Step Heat Experiment . . . 124

Chapter III: Sulfur-Buffered Helium Step Heat Experiments of Sulfides Reveal Remarkable Helium Retentivity and Complex Behavior: Implications for Sulfide Geochronology . . . 144
3.1 Introduction . . . 146
3.2 Experimental Methods . . . 151
3.3 Results . . . 162
3.4 Discussion . . . 179
3.5 Conclusions . . . 195
3.1s Chalcopyrite, Chalcocite, and Galena Step Heat Results and Discussion . . . 197
3.2s Complete U-Th/He Geochronology Compilation . . . 203
Chapter IV: Chemically Mapping Saturn's Outer Rings with DustMAP: an Automated Mass Calibration and Categorization Pipeline for TOF Mass Spectra . . . 209
4.1 Introduction . . . 209
4.2 Background . . . 209
4.3 The DustMAP Pipeline . . . 209
4.4 Validation . . . 209
4.5 Application: Chemical Map of Saturn's Outer Rings . . . 210
4.6 Discussion . . . 210
4.7 Conclusions . . . 210
Chapter V: New Frontiers in Thermodynamically Informed Geodynamic Modeling: a General Deep Learning Accelerated Free Energy Minimizer for Multicomponent Silicate Systems . . . 211
5.1 Introduction . . . 211
5.2 Background . . . 211
5.3 Model Architecture . . . 211
5.4 Validation . . . 212
5.5 Application to Geodynamic Modeling . . . 212
5.6 Discussion . . . 212
5.7 Conclusions . . . 212
Conclusion . . . 213
References . . . 214
Field Studies Reveal Synmagmatic Deformation and Extensive Crustal Contamination of the Upper Critical Zone, Recessive Lower Main Zone, and Floor Rocks at the Margin of the Bushveld Complex at Draaikraal, Mpumalanga, South Africa . . . 225
Introduction . . . 225
Literature Review . . . 225
Field Methods . . . 226
Field Observations . . . 226
Sample and Station Data . . . 226
Field Map . . . 226
Annotated Field Photographs . . . 226
Discussion . . . 226

Conclusions 226

INTRODUCTION

[TODO: Write short dissertation introduction, longer than abstract, detail each chapter.]

Chapter 1

A SIMPLE METHOD FOR HELIUM STEP HEAT EXPERIMENTS AT HIGH WATER OR SULFUR FUGACITY

Helium isotope studies and U-Th/He geochronology of sulfur-bearing and hydrous minerals offer direct windows into the timing and context of hydrothermal mineralization, provided the conditions governing radiogenic helium retention and loss in nature are known. However, assessing helium retentivity at elevated temperatures is currently hampered by these minerals' instability under the high-temperature, high-vacuum conditions of conventional helium step-heating experiments. This paper introduces a novel method for conducting helium step-heating experiments under hydrothermal and sulfur-saturated conditions, designed to extend our knowledge of helium diffusion kinetics in hydrous and sulfur-bearing mineral phases. In this approach, silica glass ampoules encapsulate the mineral of interest, maintaining internally sulfur- or water vapor-saturated conditions against the ultra-high vacuum of the surrounding extraction line, while the high helium permeability of fused silica permits quantification of evolved helium using standard temperature control and mass spectrometry procedures. This work presents the experimental framework and key equations necessary to deconvolve the processes of helium permeation through the silica ampoule from the thermally activated degassing of helium from the encapsulated mineral. Validation experiments measuring helium permeation through empty ampoules and ampoules containing only water, sulfur, or Durango apatite demonstrate that permeation lag is readily correctable in well-designed experiments and characterize the extent to which water and sulfur influence the helium permeability of fused silica. While sulfur-saturated experiments are straightforward to carry out with this method, several tractable complications are identified for experiments when significant water is present. This method expands standard experimental capabilities to encompass mineral phases that decompose under high vacuum, including sulfides and iron oxyhy-

droxides, opening new avenues for thermochronometric investigation of hydrothermal systems.

1.1 Introduction

Noble gas geochronology and thermochronometry ($^{40}Ar/^{39}Ar$ and U-Th/He) on hydrous and sulfur-bearing mineral phases such as hornblende, clay minerals, biotite, goethite, and pyrite have already been leveraged as a powerful tool for understanding regional thermal histories as well as the timing of iron or rare metal mineralization (McInnes et al., 2005; Heim et al., 2006; P. M. Vasconcelos et al., 2013; Nomade et al., 2004; O. Yakubovich, Podolskaya, et al., 2020; O. Yakubovich, Vikentyev, et al., 2021). The interpretation of these data, however, is often complicated by the fact that helium (or argon) within hydrous or sulfur-bearing phases often degasses differently in the laboratory and in nature (Hanson et al., 1975; Lee, 1993; Lo et al., 2000; Farley, H. B. Monteiro, et al., 2024).

Experiments aiming to characterize the migration mechanisms and rates of noble gasses through hydrous phases yield different results when conducted under hydrothermal conditions as opposed to traditional experimental methods in vacuo. Hydrothermal step heat experiments for argon in biotite and hornblende showed higher retentivity and comparatively favored diffusive loss mechanisms relative to experiments conducted in vacuo, and these results are more likely to accurately reflect loss mechanisms in nature (Harrison, 1982; Harrison et al., 1985); however, such experiments are labor-intensive and comparatively inflexible relative to modern helium step heat methods (Farley, Reiners, et al., 1999).

Recently, some workers have taken advantage of the high helium permeability and selectivity of pure silica glass when designing novel experimental methods for U-Th/He geochronology (O. Yakubovich, Podolskaya, et al., 2020; O. Yakubovich, Vikentyev, et al., 2021), but these ideas have only been applied in a limited way to characterize helium migration in sulfur-bearing or hydrous phases (e.g., monotonic ramped heating experiments: O. V. Yakubovich, Gedz, et al., 2019). This work aims to apply decades of research on helium permeability, solubility, and diffusion in fused silica glass (e.g., Jones, 1953; Norton, 1953; Rogers et al., 1954; Altemose, 1961; Swets et al., 1961; Doremus, 1966; Masaryk, 1968; Shelby, 1971; Shelby, 1972; Shelby, 1976; Shackelford

et al., 1972) to adapt existing agile step-heat methods (Farley, Reiners, et al., 1999) to investigate helium loss rates and mechanisms in sulfur-bearing and hydrous mineral phases.

Encapsulating minerals in silica glass ampoules allows experimentalists to maintain hydrothermal or sulfur-saturated conditions within the ampoule without otherwise altering the experimental design. Depending on the thermodynamics of the phase of interest, this method may stabilize the phase at elevated experimental temperatures, allowing for detailed characterization of helium migration in sulfide and hydrous minerals that hitherto has remained poorly understood (Fig. 1.1).

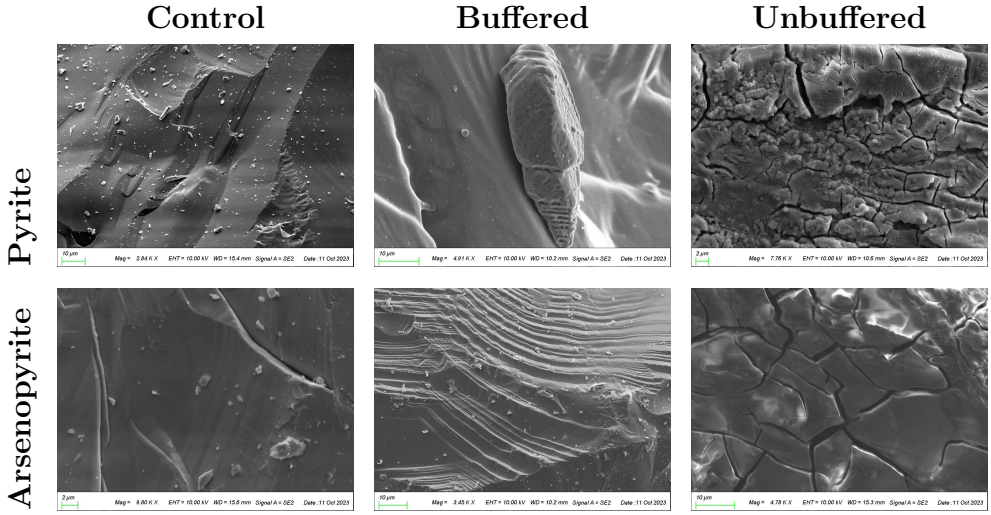


Figure 1.1: Secondary electron SEM images of pyrite (top row) and arsenopyrite (bottom row) surface textures before heat treatment (control), and after heat treatment at 400°C for at least 4 hours, both unbuffered and buffered in a silica glass ampoule with fused sulfur or mineral realgar (AsS) for pyrite and arsenopyrite respectively. The buffered samples maintain the smooth surfaces and clear cleavage faces that the controls have, showing no evidence of decomposition. The unbuffered samples show clear evidence of decomposition by devolatilization, indicated by the formation of contraction cracks and surface roughening. Using silica glass ampoules to buffer samples within controlled microenvironments can prevent phase decomposition at the elevated experimental temperatures typical of helium step heat experiments. See the supplement §1.6s for details on the experimental methods and sample preparation.

1.2 Experimental Methods

Five helium step heat experiments of sealed silica glass ampoules were undertaken at Caltech to demonstrate the applicability of equations derived here



Figure 1.2: A typical silica glass ampoule used for helium step heat experiments. The ampoule shown here contains Durango Apatite crushate sieved to 300 µm (radii ≈ 150 µm). The ampoule is roughly one cm long and one mm wide. The bars visible in the background are each separated by 1 mm.

to real helium step heat experiments of geologic materials using a projector-lamp heating apparatus, described in detail in Farley, Reiners, et al. (1999). Four of these experiments involved introducing helium into empty, water-filled, and sulfur-filled ampoules, then measuring helium permeation to assess the applicability of theory outlined in this chapter to helium step heat experiments at Caltech, and to reproduce measurements of permeability rates from the literature (Norton, 1953; Jones, 1953; Swets et al., 1961; Altemose, 1961). The fifth experiment was a standard helium step heat experiment on encapsulated Durango apatite, a widely used standard for helium diffusion experiments and U-Th/He thermochronometry (Zeitler et al., 1987; Wolf, Farley, and Silver, 1996; Farley, 2000; Shuster and Farley, 2005; Shuster, Flowers, et al., 2006; Flowers, Shuster, et al., 2007). The aim of that experiment was to be a practical test in deconvolving the process of helium permeation through silica glass ampoules from thermal helium loss of a well-studied geologic material. Table 1.1 summarizes the dimensions, geometric factors, and encapsulated materials of this work.

To prepare the silica glass ampoules, ~10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thin pipettes with a jewelry welding torch and their thin ends

Table 1.1: Silica Glass tubes that were the subject of helium step heat experiments in this work. Ampoules labeled "Quad" had evolved 4He quantified by isotope dilution using a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer, whereas ampoule labeled "MC" had both evolved 3He and 4He quantified using a Helix-MC-Plus^{10K} multi-collector mass spectrometer. Where 3He was measured, the ampoule was soaked in tank helium with an isotope ratio of ~0.15 Ra

Ampoule Contents	Mass Spec.	Inner Diameter µm	Outer Diameter µm	Geometric factor G cm^{-2}	Volume Ratio $\left(\frac{V_{glass}}{V_{gas}}\right)$	Content Mass	Helium Source
Empty	Quad	630	960	4785.3	1.32	—	Atmosphere
Empty	MC	747	1159	3263.9	1.41	—	Tank He
Sulfur	MC	529–550	835–872	5992.1	1.5	~0.2 mg	Tank He
Water	MC	487–515	734–776	7789	1.27	~0.5 mg	Tank He
Durango Apatite	Quad	850	1350	2393.5	1.52	3.18 mg	Radiogenic

were fused closed.

After the glass had cooled, degassed lab-grade sulfur, distilled water, or Durango apatite were added to some of the tubes. Two tubes were left empty. To seal, the top of each pipette was interfaced with a vacuum line and was pumped to a pressure of <0.1 torr. The encapsulated water tube was flash-frozen in liquid nitrogen and then pumped and sealed in a saltwater ice bath to selectively pump away atmospheric gasses and not water. Sublimation of sulfur and water ice during pumping was minor. The primary motivation for pumping the tube to rough vacuum was to aid the sealing process. After rough vacuum was achieved, a fine-tipped welding torch was used to seal the top of each ampoule. To determine the geometric factors for permeation of the ampoules, the inner and outer diameters of each ampoule were determined using the measurement function of the Amscope stereo microscope software, which was calibrated and tested during early development of this method.

Except for the encapsulated Durango apatite ampoule, helium gas needed to be introduced into the sealed ampoules before they were loaded into the projector lamp heating cells. About 15 nccSTP of atmospheric helium was introduced into the empty ampoule used for the quadrupole mass spectrometer experiment by heating the ampoule in a muffle furnace to >400°C in atmosphere overnight, which was sufficient to support a short permeation experiment of 4He. However, to conduct step heat permeation experiments on 3He, much larger quantities

of helium were necessary. The encapsulated sulfur and water ampoules and the other empty ampoule were soaked in 2-20 torr of high purity tank helium with an isotope ratio of about ~0.15 Ra at 300°C for about 24 hours. This procedure introduced 0.6-10 pccSTP 3He and up to 52,000 nccSTP 4He into each tube, allowing for measurements of both 3He and 4He permeation simultaneously.

The sealed ampoules were placed in copper foil packets and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley, Reiners, et al. (1999). Before each experiment began, each ampoule was washed with ethanol, air-dried, loaded in the lamp cell, pumped to high vacuum, and was held at 100°C for 30 minutes under dynamic vacuum to desorb ethanol, water, and other atmospheric species on the ampoule, copper packet, and lamp cell. Less than 0.5 ‰ of the radiogenic helium in 150µm radius Durango apatite would be expected to degas in 30 minutes at 100°C, which would amount to ~0.01 nccSTP for this experiment, using the diffusion parameters of Farley (2000) and the forward equations of Fechtig and Kalbitzer (1966). The atmospheric helium component within the encapsulated Durango Apatite ampoule constrained by permeation steps at the beginning of the experiment was much larger: 1.47 ± 0.18 nccSTP 4He.

We found that the fused silica ampoules were capable of maintaining surprisingly high internal pressures, but ampoules filled with water would often explode at temperatures above 315°C. It is recommended that experimentalists limit the temperature of water-filled ampoules to below 300°C. We have not observed any explosions of sulfur-filled ampoules at any temperature.

Heating steps and isotopic measurements for all samples were automated using custom LabView routines with the exception of the tank helium-soaked empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer after the 7th measurement. For all other ampoules, a Watlow temperature controller held the ampoule at the temperature set point for a pre-programmed duration, after which the ampoule was allowed to cool while the evolved gas was prepared for measurement for about 25 minutes. After the first 7 measurements, the empty tube experiment on the Helix-MC-Plus^{10K} was instead set to an elevated temperature manually while the evolved gas was allowed to accumulate continuously in an attempt to mitigate uncertainties related to frequently changing temperatures (supplementary Eq. (1.47); Fig. 1.19) and continued permeation at room temperature during gas cleanup and measurement. The

accumulated gas was measured every 40-120 minutes. For this experiment, the temperature was changed either during the 1 minute pump-out of the lamp cell after inletting gas into the mass spectrometer, or a period of helium accumulation for measurement. When the temperature was changed during an accumulation period, no permeability was calculated for that step, and the measured abundances of those steps were used only to track the helium contents of the tube.

Helium isotope measurements of evolved gasses from steps heats of one empty silica glass ampoule and glass-encapsulated sulfur and water ampoules equilibrated with 2-20 torr of high purity tank helium ($R \approx 0.15\ R_a$) were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from November 19th, 2023 to Feburary 15th, 2024. Helium was purified by exposing evolved gasses to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray. 4He abundances were very large during these experiments, resulting in a small but measurable 4He overspray of up to 0.68 cps at mass 3.5. These overspray signals at mass 3.5 were always less than 1% of the 3He signal at mass 3, suggesting that 4He overspray did not significantly affect 3He signals. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility. These standards show 3He was internally reproducible to within 5% (2σ), whereas 4He was internally reproducible to within 2% (2σ), where reproducibility is estimated as the standard deviation divided by the mean.

Helium-4 evolved during step heats of the empty silica glass ampoule equilibrated with atmosphere and the silica glass-encapsulated Durango apatite was

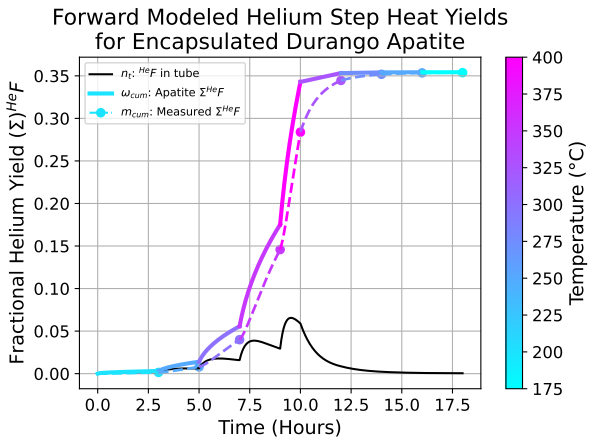


Figure 1.3: An illustrative simulation of the effects of permeation lag on step heat measurements of 150 µm radii spheres of Durango apatite encapsulated in a "standard ampoule" with an outer diameter of 1mm and inner diameter of 0.6mm, corresponding to a geometric factor G of 4350.26 cm^{-2}. Some amount of helium degassed from the apatite during an experimental step remains in the tube during measurement and permeates out in later steps. The heating schedule of this model system is given in Table 1.2. This simple model assumes new temperatures and their corresponding steady-state diffusive concentration profile within the glass ampoule are reached instantaneously. Effects of diffusive disequilibrium, if considered, would change the permeated amount for each step by less than 5%.

quantified using isotope dilution with a high purity 3He spike on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer from October 30th to November 10th, 2023. Evolved gas from the ampoule in the lamp heating cell was allowed to mix with the 3He spike and the mixture was frozen on charcoal held at 14°K within a cryostat, then helium was selectively released to the mass spectrometer by raising the temperature to 36°K.

A standard gas with a known amount of 4He is routinely measured, at least one time per experiment, to monitor drift in the relative sensitivity of 3He and 4He on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer. Relative sensitivities are determined for each standard measurement and are linearly interpolated in time for sample measurements. Blanks are approximately 5 pccSTP ^{4}He, which is very small compared to all interpreted ^{4}He measurements.

1.3 Theoretical Framework of Permeation Lag

The process of helium permeation through thin walls of an ampoule is rapid but not instantaneous on experimental timescales. Helium evolved from encapsu-

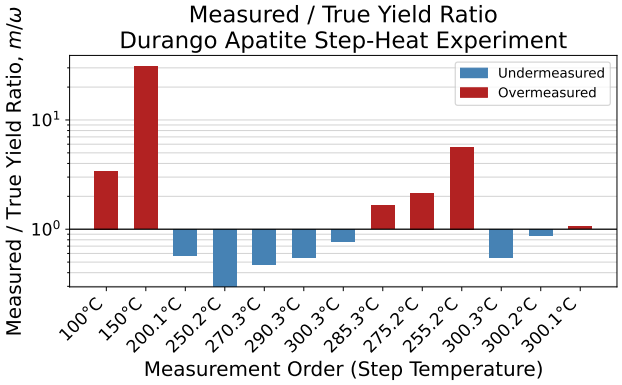


Figure 1.4: Measured-to-true yield ratio, m/ω, for the first 13 experimental steps (through the isothermal 300°C triplicate) of the real encapsulated Durango apatite step heat experiment described above (§2.2) and tabulated in full in Table 1.3, plotted in measurement order and labeled by nominal step temperature. Here m is the raw measured helium yield for a step and ω is its permeation lag-corrected yield; bars are anchored at a ratio of 1 (no anomaly) on a logarithmic axis, so a doubling and a halving of the ratio are given equal visual weight. Steps taken while ramping to higher temperatures are generally undermeasured (blue), as some of the sample's evolved helium has not yet fully permeated out of the tube at the time of measurement, while steps taken while ramping to lower temperatures are generally overmeasured (red), as helium retained in the tube from earlier, hotter steps continues to permeate out. The first two steps (100-150°C) are strongly overmeasured despite being on the heating ramp because little helium had yet evolved from the apatite, so measured yield there is dominated by pre-existing atmospheric helium draining from the tube rather than fresh sample degassing.

lated minerals is only available for measurement following permeation through the glass ampoule (Fig. 1.3), producing a predictable "permeation lag" effect in helium measurements that commonly manifests with undermeasurement of helium during prograde heating sequences and corresponding overmeasurement of helium produced during retrograde sequences (Fig. 1.5). This is not merely a theoretical concern: Fig. 1.6 shows the same artifact in real, uncorrected step heat data for the encapsulated Durango apatite experiment described above (§2.2); the correction developed below is applied to this dataset in full in §2.4 (Fig. 1.6).

This effect can be modeled (and corrected for) for a system obeying the simple equation:

$$n_t = n_0 + \omega - m \tag{1.1}$$

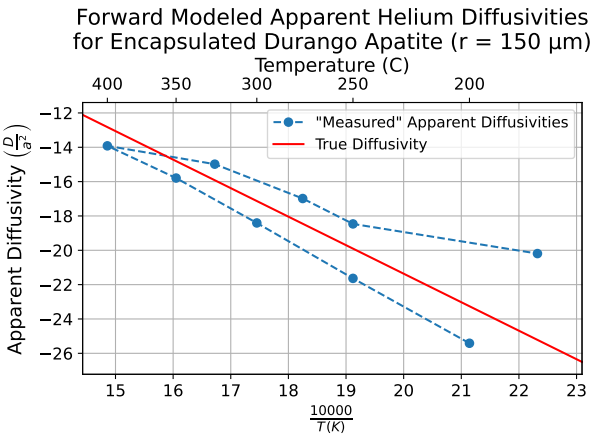


Figure 1.5: An Arrhenius plot of apparent diffusivities inverted from modeled hypothetical measurements of an encapsulated sample of Durango apatite. Permeation lag produces a characteristic effect where measurements of prograde sequences of increasing temperature produce erroneously low apparent diffusivities when inverted using the equations of Fechtig and Kalbitzer (1966), whereas retrograde (decreasing) temperature sequences exhibit the opposite effect. Diffusion parameters are taken from Farley (2000).

Table 1.2: Heating schedule for numerically modeled step heat experiment of silica glass encapsulated Durango apatite.

Temperature (°C)	Time (min)
200	180
250	120
300	120
350	120
400	60
325	120
275	120
250	120
175	120

Where n_t is the amount of helium within the ampoule after time t, n_0 is the initial amount of helium within the ampoule (which is inherited from previous heating steps), ω is the amount of helium evolved by the encapsulated sample during time t, and m is the amount of measurable helium that has permeated out of the ampoule into the high-vacuum chamber during time t.

Helium permeation through fused silica in diffusive steady state is a first order process that can be described with the simple differential equation:

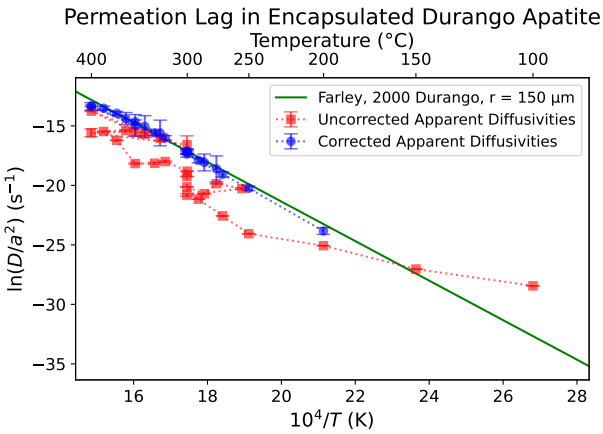


Figure 1.6: An Arrhenius plot of a helium step heat experiment of silica-glass encapsulated Durango Apatite with 2σ error bars. Without permeation lag corrections, the apparent diffusivities display highly irregular behavior that would be unrecognizable to this well-behaved sample – exactly the artifact anticipated by the simulation of Fig. 1.3 above. With permeation lag corrections, the sample agrees well with the reported diffusive parameters of Farley (2000) for the appropriate sieve size ($r = 150\mu m$). As the experiment proceeds beyond temperatures of 300°C, the activation energy of diffusion decreases to reveal a slight downward curl, a phenomenon well-documented for Durango apatite likely due to annealing of crystal defects and radiation damage (Wolf, Farley, and Silver, 1996; Farley, 2000). Apparent diffusivities are calculated with the equations of Fechtig and Kalbitzer (1966) with and without permeation lag corrections. Four measurements of the permeation lag corrected apparent diffusivities are within error of 0 at the 2σ level and thus are excluded from the plot.

$$\frac{dn}{dt} = -GKnRT \tag{1.2}$$

Where n is the number of helium atoms, t is the time, G is the geometric parameter in units of cm^{-2}, which is a function only of the geometry of the glass membrane, K is the permeation constant for helium in fused silica, which follows the Arrhenius law: $\left(K(T) = K_0 e^{-\frac{E_a}{RT}}\right)$ and has been repeatedly measured for fused silica (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), R is the gas constant, and T is the temperature in Kelvin. Equation (1.2) is identical to the permeation equation used in early studies (Norton, 1953; Altemose, 1961; Swets et al., 1961; Studt, 1962; Masaryk, 1968), except that it is formulated in terms of absolute atomic abundances instead of helium gas pressure by the ideal gas law. It is easily integrated by separation of variables

to yield a simple analytical solution (see supplement, §1.1s, for details), most intuitively expressed in terms of the fraction of the ampoule's initial helium reservoir that permeates during time t:

$$^{He}F = \frac{m}{n_0} = 1 - e^{-KGRTt} \tag{1.3}$$

Where ^{He}F is the fractional permeation yield: the fraction of the ampoule's initial helium reservoir (n_0) that permeates as m (Eq. (1.1)) during a time t of steady state permeation.

The geometric factor G for a cylindrical membrane with negligible permeation through the top or bottom is determined using the effective area of permeation given by Masaryk, 1968:

$$G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} \tag{1.4}$$

where a is the cylinder's internal radius, and b is the external radius. There is no dependence on the cylinder length when permeation through the top and bottom are negligible. This geometric constant will vary according to the experimental design and therefore must be measured by the experimentalist.

Equation (1.3) is only strictly valid with two assumptions:

1. The helium partial pressure in the "measurable" reservoir outside of the ampoule is negligible compared to the partial pressure within the ampoule. Because the ratio of typical ampoule and the diffusion cell volumes is ~1/10,000 (Farley, Reiners, et al., 1999), this assumption is essentially always reasonable.

2. The helium concentration profile within the ampoule's glass walls is in diffusive steady state between the inner and outer walls. The real steady state helium concentration profile is well-approximated by the ideal steady state profile for ampoules with a glass/gas volume ratio less than 2 (Figs. 1.11, 1.15). For these ampoules, (1.3) is accurate to within 1% after the establishment of a diffusive steady state. Additionally, the diffusive steady state assumption is ephemerally violated when the temperature is changed due to the temperature dependence of the glass/gas partitioning

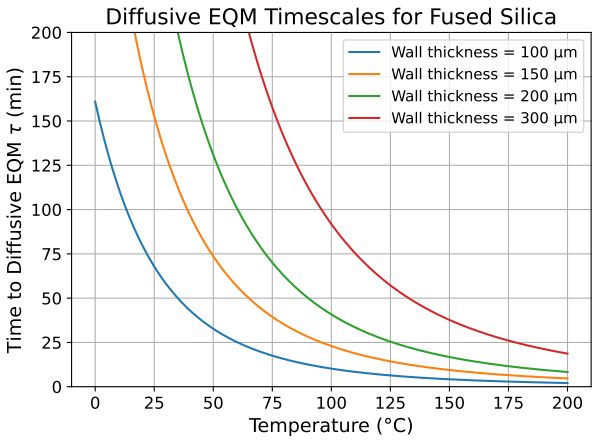


Figure 1.7: Timescales as a function of temperature to reestablish a steady state diffusive concentration profile for different glass thicknesses. These curves correspond to the timescales at which the Fourier number equals 1. 4He diffusivities are taken from Swets et al. (1961).

of helium. In this work, low-temperature heating steps are repeated to allow for the new diffusive steady state concentration profile to become established. The ampoules of this work reach diffusive equilibrium in 30-70 minutes at 80° (Fig. 1.7, supplementary figures 10 and 11). Comparisons between the abundance of helium within glass as a function of temperature limits the error introduced by thermal perturbation to less than 5% (supplementary figure 9).

1.4 Experimental Approaches Addressing Permeation Lag

A combination of analytical and experimental methods may be used to correct for permeation lag and properly account for all helium evolved from the sample. Some common practices for noble gas measurements may introduce error when using this method. For example, it is common practice to purify and collect emitted helium with the diffusion cell open to the vacuum line at room temperature for extended times, often 15-30 minutes, and it is assumed that no additional helium evolution occurs. This assumption is reasonable for commonly measured geological materials, but it can be problematic when unexamined during permeation steps of silica glass encapsulated materials because helium permeation in silica glass is significant even at room temperature. In this work, permeation during these waiting times is often mathematically corrected for using Eq. (1.3), assuming permeation continues at room temper-

ature. Alternatively, there are experimental approaches to address ongoing permeation during these waiting periods: they may be shortened to minimize their impact on the measurement, or the ampoule may be held at the warm permeation temperature during the gas purification and cryofocusing processes, so that permeation continues at the same temperature during this time without introducing complicated thermal histories.

From Eq. (1.1), it is immediately clear that when the evolved helium ω is large with respect to the amount of helium within the ampoule, n_t or n_0, then $\omega \approx m$ and permeation lag can be neglected. Unfortunately, this is not the case for most phases commonly subjected to helium step heat experiments with routine heating schedules (e.g. hour-timescale steps at 100-400°C for apatite or zircon), but it is sometimes possible to design an experiment with long time-steps where the total evolved helium is large compared to the amount of helium in-residence within the tube. Nonetheless, permeation lag can be quantified and corrected for in *any* experiment where the encapsulated material is effectively closed to helium loss at room temperature. Three approaches are described here to recover the helium evolved from an encapsulated sample (ω), from measured amounts (m): Informed Bracketing, Naïve Bracketing, and Approximation with Uncertainty.

Informed Bracketing

In typical experimental heating steps, there are two fluxes of helium controlling the ampoule's helium abundance: an influx of helium evolved from the sample and an outflux of helium due to permeation. For lower temperature steps, for which helium release from the sample is negligible ($\omega = 0$), the helium abundance within the ampoule is controlled only by the process of helium permeation, simplifying Eq. (1.1) and allowing for Eqs. (1.1) and (1.3) to form a system of equations that can be solved for n_t and n_0 given a measurement m for an isothermal heating step of known duration.

$$n_0 = \frac{m}{{}^{He}F} \tag{1.5}$$

Where ${}^{He}F$ is the fractional permeation yield defined in Eq. (1.3). Such a low temperature heating step is hereby referred to as a *permeation step* in this work. This equation requires knowledge of the glass ampoule's dimensions to precisely determine the geometric factor G using Eq. (1.4), as well as the permeability

coefficient K for helium in fused silica, which has been measured repeatedly as a function of temperature (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968). Thus, one may bracket high temperature *experimental steps* ($\omega > 0$) with low temperature *permeation steps* ($\omega = 0$) which are used to calculate the amount of helium residing in the ampoule between steps (n_t, n_0) to solve for the helium evolved from the natural sample, ω.

Naïve Bracketing

The permeability coefficient K of helium through nominally anhydrous silica glass is well known at relevant experimental temperatures for water-free glass (Norton, 1953; Altemose, 1961; Swets et al., 1961; Rogers et al., 1954; Masaryk, 1968). However, because water diffuses into glass and changes its helium permeability (Shelby, 1972; Tsuru et al., 2011), the helium permeability constant K must be regularly measured in hydrothermal experiments to monitor drift. Fortunately, even if K and/or G are not known for any reason, the process of helium permeation through glass remains a first order process that obeys Eq. (1.3), which is simplified into the recognizable decay equation by substituting $\lambda = KGRT$:

$$n_t = n_0 e^{-\lambda t} \tag{1.6}$$

λ can be experimentally determined alongside n_0 by using a non-linear least squares optimization to fit a consecutive series of at least three isothermal permeation step measurements to the following equation, expressed in terms of cumulative measured helium over cumulative time:

$$m_{cum} = n_0 \left(1 - e^{-\lambda t_{cum}}\right) \tag{1.7}$$

This equation can instead be stated in terms of individual measurements, which is a superior approach in that it avoids introducing a bias into the fit by accumulating errors and uncertainties from earlier measurements to the cumulative yields of later measurements.

$$m_{i+1} = n_0 \left(1 - e^{-\lambda t_{i+1}}\right) - n_0 \left(1 - e^{-\lambda t_i}\right) \tag{1.8}$$

This non-linear least-squares optimization can be accomplished, for example, using the SciPy implementation of the Levenberg-Marquardt algorithm (Virtanen et al., 2020), or the "Solve" function in Microsoft Excel.

It is important to note that this naïve method for determining n_0 requires that enough cumulative time has passed to reveal the curvature of cumulative helium yield against time, or else the problem will be poorly conditioned. This can be assured in practice by waiting until the rate of helium permeation has slowed markedly after three or more isothermal permeation measurements before advancing to the next high-temperature experimental step.

One strength of this method is that, in addition to the initial helium contents of the ampoule n_0, it simultaneously determines the product of the permeability and geometric factor for the silica glass ampoule. Once $\lambda = KGRT$ has been satisfactorily constrained over a range of temperatures, the activation energy E_a of permeation and the product of the geometric factor and permeability at infinite temperature, K_0G, can be fit using linear regression. Determining these values allows for extrapolation of the permeability to the high temperatures of the experimental steps, which can be useful to an experimentalist who wishes to design a heating schedule wherein the permeation lag is negligible for some experimental steps but does not know the permeability coefficient K and/or the geometric factor G.

Approximation with Uncertainty

Experimentalists may opt to treat permeation lag as uncertainty without applying corrections for convenience. Permeation lag is a significant effect for typical heating schedules operating on timescales of a few hours (e.g., Farley, 2000; Shuster, Farley, et al., 2004; Figs. 1.3 and 1.5), but its relative significance decreases with step duration and increased temperature (Fig. 1.8). Consider an arbitrarily long experimental step: after some time, a steady state $\left(\frac{dn}{dt} = 0\right)$ will be reached where the helium permeation rate out of the ampoule is balanced by the production rate of helium within the ampoule from the degassing of the natural sample. The upper bound for this steady state concentration as a function of measured helium abundance (Eq. (1.9)) is derived in the supplement, §1.3s (Eq. (1.57)).

$$n^{ss} \leq \frac{m}{\lambda t - 1} \tag{1.9}$$

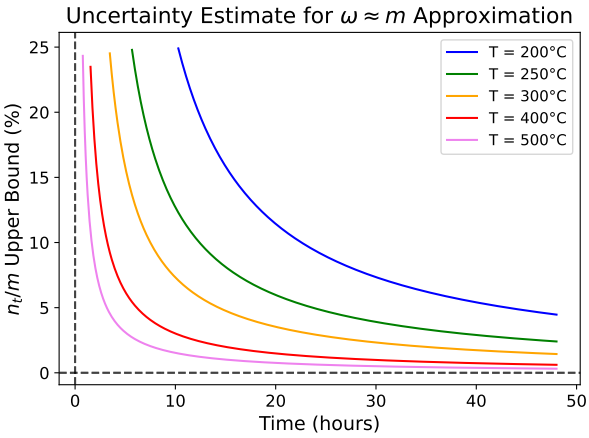


Figure 1.8: For long steps and high temperatures, the abundance of helium within the ampoule will reach a steady state and will be constant as the measurable helium outside the ampoule in the lamp cell increases. As the step gets longer, this amount becomes small relative to the amount of measurable helium, and the measurable helium can be used to approximate the total amount of degassed helium with only minor inaccuracy. Eq. (1.9) is plotted here to show the upper bound relative amount of helium trapped within the ampoule normalized to the measurable amount. The geometric factor for these calculations is 4350 cm^{-2}, corresponding to an ampoule with an inner and outer diameter of 0.6 and 1 mm respectively. Permeability values (K) are for 4He in silica glass and are taken from Swets et al. (1961).

Equation (1.9) only applies when step heats are long enough (roughly 5-24 hours depending on temperature and ampoule geometry, Fig. 1.22) to assume steady state permeation ($\lambda t >\sim 5$), which requires that G and K are known or can be conservatively estimated. It also assumes that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample is constant over time at a given temperature. Typically, the helium production rate Ω decreases over time in most experiments, which would produce a smaller steady state He concentration than what is predicted by Eq. (1.9). Ω decreases over time for helium loss governed by volume diffusion (Wolf, Farley, and Silver, 1996; Reiners and Farley, 1999; Farley and McKeon, 2015; Farley, 2018; Farley, H. B. Monteiro, et al., 2024), or first order processes (O. V. Yakubovich, Y. A. Shukolyukov, et al., 2010; Wilson and Bisson, 1973), except for the case of initially inhomogeneous distributions in at low helium loss fractions (Goldsmith et al., 2020; Ketcham et al., 2011; Shuster, Farley, et al., 2004; Farley, Wolf, et al., 1996) or radiation damage annealing (Cherniak, 2019; Guenthner et al., 2013; Flowers, Shuster, et al., 2007; Flowers, Ketcham, et al., 2009; Shuster,

Flowers, et al., 2006). For experiments where these factors are of interest, the bracketing method to explicitly handle permeation lag is superior.

Using the steady state concentrations calculated with Eq. (1.9) from a long duration step (1) and the previous step (0) as upper bound estimates in Eq. (1.1) yields the inequality that can be used to bound the amount of helium evolved from the sample during step 1 without bracketing steps:

$$m_1 - \frac{m_0}{(\lambda t)_0 - 1} \leq \omega_1 \leq m_1 \left(1 + \frac{1}{(\lambda t)_1 - 1}\right) \tag{1.10}$$

The relative uncertainty $\left(\frac{m}{n^{ss}} = \frac{1}{\lambda t - 1}\right)$ is plotted in Fig. 1.8 as a function of time and temperature for a standard ampoule with geometric factor G of 4350.26 cm^{-2}. To achieve relative uncertainties below ~ 5%, steps longer than 24 hours are typically required at temperatures below 300°C.

1.5 Analytical Methods

Permeabilities were calculated for the empty ampoule experiments by solving Eq. (1.3) for permeability K.

Besides the encapsulated Durango experiment, which contained a helium source, the helium contents (n_0, n_t) of the other ampoules changed only by decreasing due to helium permeation through silica glass throughout the experiments. These experiments were concluded when measured helium permeate of a high temperature step was indistinguishable from blank. The values of n_0 and n_t for each step were directly quantified after the experiments concluded using the total yield. Assuming that the helium permeated during the short pump-out time of the lamp cell is negligible, the total yield is simply the sum of all the measured abundances of helium for each step. The atmosphere-equilibrated empty tube experiment on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer was pumped for 15 minutes between steps, the encapsulated apatite experiment was pumped for 5 minutes between steps, and the Helix-MC-Plus^{10K} experiments on silica glass ampoules equilibrated in 2-20 torr tank helium were pumped for only 1 minute between steps. This pump-out time was short enough to neglect for all but the atmosphere-equilibrated empty tube, whose longer 15-minute pump-out and short (45-minute) heated steps required small explicit pump-out and gas-preparation corrections using Eq. (1.3); see the supplement, §1.5s, for the full derivation.

One high temperature measurement – the last step of the empty ampoule experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer – showed unusual, non-linear signal evolution and required a bespoke recalculation to recover its 4He and 3He yields; see the supplement, §1.5s, for details.

The encapsulated apatite experiment included high temperature steps ranging from 100-400°C, bracketed by a sequence of two or three permeation steps. Equation (1.5) was applied to solve for the helium contents of the ampoule (n_0) between steps for permeability corrections. Only the last of each sequence of room temperature measurements (shown in **bold** in Table 1.3) was used to solve for n_0 using Eq. (1.5), because the temperature records showed that the encapsulated apatite had not yet cooled to room temperature by the time the first permeation step at room temperature began, although each pair of permeation steps always yielded values of n_0 within uncertainty (2σ) of one another. Measured helium abundances of the other permeation steps taken before reestablishing an equilibrium steady state concentration profile (shown in *italics* in Table 1.3) were added or subtracted to the n_0 value to propagate the determined helium contents of the ampoule back or forward in time to the high temperature experimental steps for permeation lag corrections using Eq. (1.1). Measurement uncertainties are estimated using standard deviations of sensitivities and blanks determined by standards. These measurement uncertainties are propagated through Eqs. (1.3) and (1.1) in quadrature, in addition to a general 10% uncertainty for the use of Eq. (1.3) following the recommendations of this work (see discussion). 4 of the 25 high temperature experimental steps of the encapsulated Durango apatite produced corrected yields within error (2σ) of 0 and were excluded from Fig. 1.6 (see figure caption). Uncertainty for a step's fractional cumulative yield was estimated as the sum of the uncertainty of its fractional yield plus the uncertainties of the measured fractional yields of all previous measurements, added in quadrature. The total yield was estimated based on the precisely-measured mass of the encapsulated Durango apatite (3.18 mg) and its helium concentration of 194.2 nccSTP/mg (Farley, 2000), a value independently verified in-lab (supplement, §1.5s). No uncertainty in the total yield estimation is propagated through to the diffusivities reported here.

Numerical Diffusive Permeation Model for Arbitrary Thermal History and Diffusive Disequilibrium

Most of the permeation measurements performed on the Helix-MC-Plus^{10K} mass spectrometer were complicated by additional permeation occurring at variable temperature during long cooling times following high temperature experimental steps. Such a complication can be effectively avoided by limiting the duration of time for which the diffusion cell is opened to the rest of the vacuum line, or by keeping the sample at-temperature during this equilibration time. To address this, a numerical diffusive permeation model is used to calculate the ampoule's expected permeation fraction (n_t/n_0) for each step with a complex thermal history (see supplement, §1.2s, for details).

1.6 Results

Predicted and measured fractional yields for all permeation steps agree well (within 1σ; Fig. 1.9, supplementary Figs. 1.23–1.24) with the predictive equations presented in this work and with the permeabilities reported by Swets et al. (1961) and Jones (1953) for the empty and sulfur-filled ampoules across all studied temperatures (24-300°C) and step fractional yields (up to ~80%), but not for the water-filled ampoule.

Conversely, helium permeation for the ampoule containing water was inconsistent with the permeabilities reported by Swets et al. (1961) and Jones (1953). For these hydrothermal step heat experiments, measured helium partial yields are systematically high by about 25% relative to the expected values (supplementary Figs. 1.9, 1.24), meaning that permeation was systematically faster than expected for the ampoule filled with water. An analysis of this discrepancy and a hypothesis for its origin is presented in the discussion.

Permeation ratios for 3He and 4He measured in the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer are plotted in supplementary Fig. 1.25. These data present significant scatter, but generally agree with the reported results of Jones (1953), except for measurements at room temperature, which cluster tightly at a permeation ratio of 1:1 for the two isotopes. Additionally, fields corresponding to the ratios of apparent permeabilities for the water-filled and sulfur-filled experiments are plotted for the temperature ranges of those experiments. Results for those experiments are plotted as fields because each step integrated permeation behavior over a

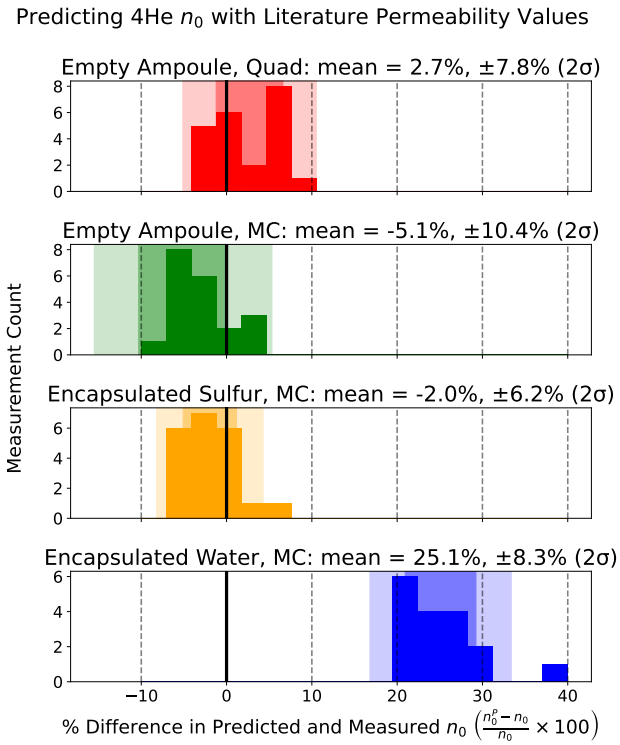


Figure 1.9: Histograms showing the accuracy of 4He n_0 predictions using permeabilities of Swets et al. (1961) from either Eq. (1.21) or a numerical diffusion model from the measured value, with shaded regions behind the histograms corresponding to 1 and 2 sigma (σ) from the average relative difference between measured and predicted values. The measured n_0 values of the empty and sulfur-filled ampoules are within 1σ of the predicted values, suggesting that the permeation model and literature values apply well to those systems, and such models can be used to reliably recover helium contents of permeating ampoules for use in permeation lag corrections. On the other hand, measured and predicted 4He n_0 for water-filled ampoules are well resolved beyond 2σ, suggesting that their helium permeability differs strongly from the literature values. Internal reproducibilities (1σ) of a gas standard repeatedly analyzed during these experiments were better than 1% for 4He, requiring that most of the scatter in these data be related to factors outside of analytical precision, such as long/inconsistent ramp times during heating or uncertainties or rounding of heating times.

complex (non-isothermal) cooling history. These experiments also agree with the results of Jones (1953) and the empty ampoule experiment. The qualitative result of Jones (1953) that silica glass is only slightly more permeable for 3He than 4He is supported by these data, which are in marked contrast to the reported extreme differences in 3He and 4He diffusivity in silica glass of McMurtry et al. (2019), who report instantaneous isotopic fractionations up to

a factor of 1000.

The permeation lag-corrected helium diffusivities for the encapsulated Durango apatite experiment are within uncertainty of the Arrhenius relationship for 150µm radii Durango apatite reported by Farley (2000), except for the first two plotted points and the 400°C step (Fig. 1.6). Farley (2000) also observed some low apparent diffusivities for helium in Durango apatite in early steps, although rounding of the concentration profile by diffusive loss or alpha ejection is not expected for interior fragments of large gemmy apatites. The high temperature downward curvature of the apparent diffusivities in Fig. 1.6, however, is well-understood. Wolf, Farley, and Silver (1996) and Farley (2000) discuss similar downward curvature of the diffusion array of Durango apatite at high temperatures. The annealing of radiation damage 'traps' and crystalline defects is the favored explanation for this irreversible change in diffusivity at high temperatures for Durango apatite (Farley, 2000; Shuster, Flowers, et al., 2006). Overall, the permeation lag-corrected apparent diffusivities for encapsulated Durango Apatite reproduce the same results of unencapsulated Durango apatite (Fig. 1.6), suggesting that permeation lag can be reasonably deconvolved from thermal helium emission from minerals or other solid materials of interest when performing step heat experiments to characterize helium retentivity and loss mechanisms in solids.

Table 1.3: Step heat data for the silica glass encapsulated Durango apatite experiment. ${}^{He}F_{Tube}$ is the fractional permeation yield (Eq. (1.3)) of the bracketing permeation step; ${}^{He}F_{Apatite}$ is the permeation lag-corrected cumulative fractional yield of helium released from the apatite. n_0 values (initial ^{4}He content of the ampoule, Eq. (1.5)) are shown in **bold** for permeation steps used directly to solve for n_0, and in *italics* for all other steps, where n_0 is instead propagated forward or backward in time from the nearest bolded permeation step using Eq. (1.1). Blank "Type" entries denote room-temperature measurements preceding the permeation step used to solve for n_0, taken before the diffusive steady state concentration profile was reestablished (see text). Uncertainties are 2σ. Corrected $\ln(D/a^2)$ values shown as "–" correspond to steps where the corrected apatite yield (ω) was within 2σ of 0 and were excluded from the Arrhenius plot (Fig. 1.6).

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	${}^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	${}^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	24.3	0.00704 ± 0.00046	4.76‰	**1.48 ± 0.18**				
Experimental	240	100.0	0.1700 ± 0.0011		*1.47 ± 0.18*	0.05 ± 0.24	0.00803%	–	-28.39
	15	28.0	0.00721 ± 0.00039		*1.35 ± 0.16*				
	15	24.0	0.00639 ± 0.00039		*1.34 ± 0.16*				
Permeation	15	24.0	0.00630 ± 0.00039	4.71‰	**1.34 ± 0.16**				
Experimental	240	150.0	0.3090 ± 0.0023		*1.33 ± 0.16*	0.01 ± 0.21	0.00983%	–	-26.98
	15	28.1	0.00542 ± 0.00039		*1.03 ± 0.13*				
	15	24.3	0.00484 ± 0.00039		*1.03 ± 0.13*				
Permeation	15	24.2	0.00485 ± 0.00039	4.73‰	**1.02 ± 0.13**				
Experimental	240	200.1	0.9416 ± 0.0085		*1.02 ± 0.13*	1.64 ± 0.23	0.276%	-23.80 ± 0.28	-25.02
	15	29.9	0.00947 ± 0.00041		*1.72 ± 0.19*				
	15	23.8	0.00786 ± 0.00040		*1.71 ± 0.19*				
Permeation	15	24.0	0.00804 ± 0.00040	4.72‰	**1.70 ± 0.19**				
Experimental	60	250.2	1.097 ± 0.011		*1.69 ± 0.19*	3.70 ± 0.48	0.874%	-20.21 ± 0.20	-24.01
	15	33.0	0.02653 ± 0.00050		*4.30 ± 0.44*				

continued on next page

(continued)

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
	15	23.5	0.02016 ± 0.00046		*4.27 ± 0.44*				
Permeation	15	24.0	0.02003 ± 0.00047	4.72‰	**4.25 ± 0.44**				
Experimental	60	270.3	2.508 ± 0.027		*4.23 ± 0.44*	5.26 ± 0.83	1.73%	-19.04 ± 0.22	-22.51
	15	33.9	0.04414 ± 0.00067		*6.98 ± 0.70*				
Permeation	15	23.6	0.03246 ± 0.00057	4.68‰	**6.93 ± 0.70**				
Experimental	60	290.3	4.979 ± 0.058		*6.90 ± 0.70*	9.0 ± 1.3	3.18%	-17.87 ± 0.20	-21.10
	15	33.6	0.06816 ± 0.00095		*10.9 ± 1.1*				
Permeation	15	23.6	0.05050 ± 0.00078	4.66‰	**10.8 ± 1.1**				
Experimental	60	300.3	7.243 ± 0.091		*10.8 ± 1.1*	9.5 ± 1.7	4.72%	-17.34 ± 0.25	-20.07
	15	34.2	0.0843 ± 0.0012		*13.1 ± 1.3*				
	15	23.5	0.06160 ± 0.00094		*13.0 ± 1.3*				
Permeation	15	24.1	0.06097 ± 0.00095	4.72‰	**12.9 ± 1.3**				
Experimental	60	285.3	5.919 ± 0.080		*12.9 ± 1.3*	3.6 ± 1.7	5.30%	-18.04 ± 0.71	-20.65
	15	36.8	0.0729 ± 0.0011		*10.5 ± 1.1*				
Permeation	15	23.6	0.04846 ± 0.00084	4.64‰	**10.4 ± 1.1**				
Experimental	80	275.2	5.382 ± 0.077		*10.4 ± 1.1*	2.5 ± 1.3	5.71%	-18.6 ± 1.1	-19.80
	15	37.0	0.05243 ± 0.00092		*7.53 ± 0.76*				
Permeation	15	23.6	0.03487 ± 0.00070	4.66‰	**7.48 ± 0.76**				
Experimental	100	255.2	3.382 ± 0.052		*7.44 ± 0.76*	0.60 ± 0.90	5.81%	–	-20.22
	15	37.1	0.03206 ± 0.00069		*4.66 ± 0.48*				
Permeation	15	23.6	0.02147 ± 0.00056	4.64‰	**4.63 ± 0.48**				
Experimental	60	300.3	3.684 ± 0.059		*4.61 ± 0.48*	6.68 ± 0.91	6.89%	-17.17 ± 0.20	-20.76
	15	36.3	0.0518 ± 0.0010		*7.60 ± 0.77*				
Permeation	15	23.6	0.03513 ± 0.00077	4.65‰	**7.55 ± 0.77**				
Experimental	120	300.2	9.78 ± 0.17		*7.51 ± 0.77*	11.1 ± 1.2	8.69%	-17.13 ± 0.22	-19.21
	15	35.7	0.0595 ± 0.0012		*8.85 ± 0.90*				

continued on next page

(continued)

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	23.6	0.04087 ± 0.00089	4.65‰	**8.79 ± 0.90**				
Experimental	180	300.1	14.58 ± 0.27		*8.75 ± 0.90*	13.8 ± 1.2	10.9%	-17.08 ± 0.18	-18.75
	15	35.9	0.0547 ± 0.0012		*7.96 ± 0.81*				
Permeation	15	23.6	0.03782 ± 0.00089	4.78‰	**7.91 ± 0.81**				
Experimental	60	320.3	7.49 ± 0.14		*7.87 ± 0.81*	11.4 ± 1.5	12.8%	-15.96 ± 0.18	-17.94
	15	36.9	0.0813 ± 0.0017		*11.7 ± 1.2*				
Permeation	15	23.5	0.0541 ± 0.0012	4.64‰	**11.7 ± 1.2**				
Experimental	60	330.4	11.18 ± 0.22		*11.6 ± 1.2*	14.5 ± 1.9	15.1%	-15.54 ± 0.21	-18.08
	15	37.4	0.1076 ± 0.0023		*14.9 ± 1.5*				
Permeation	15	23.7	0.0691 ± 0.0016	4.67‰	**14.8 ± 1.5**				
Experimental	60	350.4	19.78 ± 0.41		*14.7 ± 1.5*	28.3 ± 2.8	19.7%	-14.61 ± 0.15	-18.11
	15	37.0	0.1615 ± 0.0035		*23.3 ± 2.4*				
Permeation	15	23.5	0.1071 ± 0.0024	4.64‰	**23.1 ± 2.4**				
Experimental	60	370.4	32.97 ± 0.71		*23.0 ± 2.4*	41.3 ± 4.0	26.4%	-13.90 ± 0.15	-16.15
	15	37.3	0.2207 ± 0.0049		*31.4 ± 3.2*				
Permeation	15	23.5	0.1447 ± 0.0033	4.65‰	**31.1 ± 3.2**				
Experimental	60	385.4	42.32 ± 0.94		*31.0 ± 3.2*	44.8 ± 4.8	33.6%	-13.48 ± 0.17	-15.41
	15	37.8	0.2393 ± 0.0055		*33.5 ± 3.4*				
Permeation	15	23.5	0.1548 ± 0.0037	4.66‰	**33.2 ± 3.4**				
Experimental	60	360.4	27.53 ± 0.64		*33.1 ± 3.4*	15.3 ± 4.1	36.1%	-14.35 ± 0.51	-15.33
	15	39.6	0.1620 ± 0.0039		*20.9 ± 2.1*				
Permeation	15	23.6	0.0963 ± 0.0024	4.65‰	**20.7 ± 2.1**				
Experimental	60	340.4	14.78 ± 0.35		*20.6 ± 2.1*	7.4 ± 2.6	37.3%	-15.00 ± 0.90	-15.63
	15	40.0	0.1029 ± 0.0026		*13.2 ± 1.4*				
Permeation	15	23.5	0.0610 ± 0.0017	4.64‰	**13.1 ± 1.4**				
Experimental	100	325.2	12.37 ± 0.31		*13.1 ± 1.4*	6.8 ± 1.6	38.4%	-15.6 ± 1.1	-13.69

continued on next page

(continued)

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
	15	40.1	0.0589 ± 0.0017		*7.50 ± 0.78*				
Permeation	15	24.0	0.0351 ± 0.0011	4.72‰	**7.44 ± 0.78**				
Experimental	240	400.1	101.1 ± 2.7		*7.40 ± 0.78*	110.0 ± 3.2	56.2%	-13.281 ± 0.076	-13.70
	15	34.9	0.1096 ± 0.0031		*16.3 ± 1.7*				
Permeation	15	23.8	0.0757 ± 0.0022	4.67‰	**16.2 ± 1.7**				
Experimental	90	350.3	14.88 ± 0.41		*16.1 ± 1.7*	5.8 ± 1.9	57.2%	-14.92 ± 0.92	-15.42
	15	39.5	0.0567 ± 0.0017		*7.06 ± 0.74*				
Permeation	15	24.0	0.0330 ± 0.0011	4.71‰	**7.00 ± 0.74**				
Experimental	60	400.4	14.21 ± 0.40		*6.97 ± 0.74*	18.1 ± 1.4	60.1%	-13.31 ± 0.31	-15.50
	15	40.4	0.0866 ± 0.0026		*10.9 ± 1.1*				
Permeation	15	24.0	0.0508 ± 0.0016	4.71‰	**10.8 ± 1.1**				
Experimental	120	300.2	7.65 ± 0.22		*10.7 ± 1.1*	0.7 ± 1.2	60.2%	–	-16.50
	15	39.4	0.0291 ± 0.0011		*3.79 ± 0.41*				
Permeation	15	23.7	0.01758 ± 0.00077	4.68‰	**3.76 ± 0.41**				

1.7 Discussion

Except for ampoules with encapsulated water, the experimental results of this work are predicted within 10% when assuming the literature values for permeabilities for 4He from Swets et al. (1961) and the 3He/4He permeability ratio of Jones (1953). This suggests that the complication of permeation lag is a tractable complexity, enabling the use of fused silica ampoules to control the sulfur and water fugacity of helium step heat experiment, but that literature values for helium permeability do not apply to hydrothermal experiments and require the "naïve bracketing approach" described in the text.

Resolving the Paradoxically High Apparent Helium Permeability in Water Filled Tubes

The high measured permeability of helium for water-filled silica glass ampoules (supplementary Figs. 1.9, 1.24) is initially surprising and warrants further explanation, as the dissolved water and/or hydroxyl ion content of fused silica glass is only known to decrease helium permeability (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968). It is hypothesized here to result from an increase in the partial pressure of helium within the gas phase of the ampoule due to volume displacement of the gas phase by liquid water. This results from the low solubility and high diffusivity of helium in water, which allow for helium to be concentrated within the gas phase while remaining in equilibrium with the glass throughout the entire ampoule, even where obstructed by water.

The diffusivity of helium in water is about 7×10^{-5} cm^2/s at room temperature (Verhallen et al., 1984), whereas the diffusivity of helium in glass is three orders of magnitude slower: about 2×10^{-8} cm^2/s (Swets et al., 1961). Helium diffusion in water can be considered instantaneous relative to the timescales of helium diffusion through glass, so the helium concentration in the gas phase is expected to be in equilibrium with the glass throughout the entire tube even where mediated by liquid water. Therefore, adding water to an ampoule changes the volume available for the gas phase while not affecting the effective area of permeation.

Helium partial pressures in the gas phase can be increased by volume displacement assuming that helium strongly partitions into the gas phase over glass and water. Helium solubilities in water are similar to but slightly lower than glass. At room temperature, the solubility of helium in water is ~

$4 \times 10^{-6} \frac{moles}{m^3 * Pa}$ (Sander, 2015) or $\sim 2 \times 10^{17} \frac{atoms}{cm^3 * atm}$. The solubility of helium in silica glass is $\sim 6 \times 10^{17} \frac{atoms}{cm^3 * atm}$ (Swets et al., 1961), a factor of 3 higher. At room temperature, the concentration of helium atoms in glass and in water is about 3% and 1% respectively of the concentration of helium atoms in the gas phase (supplementary Eq. (1.40)). The net result of this differential partitioning is that the effective permeability of helium roughly scales with the inverse volume fraction of the gas phase, following the equation:

$$K_{eff} = K_{dry} \frac{V_{water} + V_{gas}}{V_{gas}} \tag{1.11}$$

Where K_{eff} is the effective helium permeability, K_{dry} is the nominal helium permeability at anhydrous conditions, and V_{water} and V_{gas} are the relative volumes of water and gas respectively. This effect is potentially larger than the decrease in helium permeability of fused silica glass due to elevated water contents; therefore an ampoule that is filled with significant water may paradoxically permeate helium faster than the same ampoule without water, despite the lower helium permeability of the glass itself.

Evolving Helium Permeability in Water Filled Tubes

Helium permeability measurements of water-filled ampoules show decreasing fractional yields for identical steps (Fig. 1.10). This observation is consistent with previous studies that show decreased helium permeability in silica glass following hydrothermal treatment (Tsuru et al., 2011; Masaryk, 1968). Other studies have shown that diffusion of water in silica glass is complex, but it is always significantly slower than diffusion of helium in silica glass. At 250°C, the effective diffusivity of both hydroxyl groups and molecular water in silica glass is initially ~10^{-11} cm^2/s and further slows by a factor of 4 over the first 500 hours of observation (Oehler and Tomozawa, 2004). Helium, on the other hand, has a diffusivity of ~10^{-6} cm^2/s at 250°C (Swets et al., 1961). For a 200 µm thick silica ampoule, the diffusive timescale ($\tau = \frac{L^2}{D}$) of water is about 76 years, whereas the diffusive timescale of helium in the same ampoule is less than 7 minutes. Even so, we observe a gradual decrease in helium permeability of about 7% over the course of a few days at 100°C in this work (Fig. 1.10). The implication of this observation is that the helium permeability in water-filled tubes must be continuously monitored using naïve bracketing (Eqs. (1.7), (1.8)) during a hydrothermal step heat experiment to track (or exclude) significant temporal evolution of helium permeability throughout the experiment.

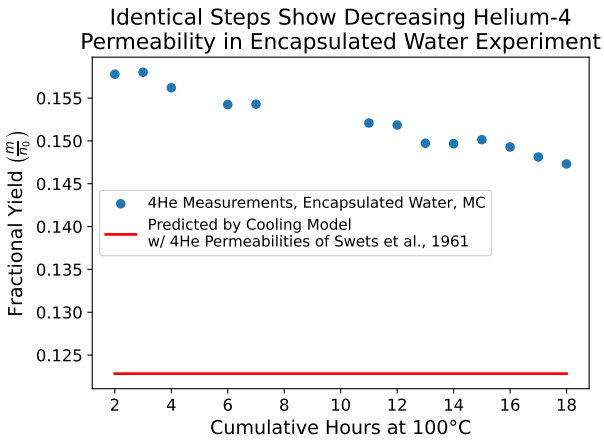


Figure 1.10: Identical permeation steps at 100°C with identical cooling times of the encapsulated water tube show higher 4He fractional yields than predicted with a numerical diffusion model and literature values of permeability from Swets et al. (1961). These identical steps also show a striking systematic decrease in fractional yields as the ampoule is held at 100°C, suggesting that the permeability of the ampoule is slowing during the experiment, possibly due to diffusion of water and hydroxyl groups into the silica glass (Tsuru et al., 2011; Oehler and Tomozawa, 2004). This trend is not visible in the 3He data due to higher scatter in the data.

Previous studies have shown that the diffusion of water-derived species (OH^-, H_2O, H^+) into silica glass slows significantly as the water concentration increases within the glass (Oehler and Tomozawa, 2004; Davis and Tomozawa, 1995). Thus, it may be possible that a hydrothermal pretreatment (or early high-temperature heating steps) could sufficiently increase the trace water content of the silica glass for it to enter a regime of slower solubilization of water, thereby stabilizing the trace water content and helium permeability of silica glass for hydrothermal experiments. This hypothesis would be a fruitful direction for future work related to hydrothermal helium step heat experiments in silica glass ampoules.

1.8 Conclusions

Encapsulating minerals in pure silica glass allows helium step heat experiments to be conducted within controlled microenvironments of high water or sulfur partial pressure. Helium permeation through fused silica glass is a rapid but non-instantaneous first-order process that complicates the interpretation of step heat data; however, its effects can be reliably deconvolved from the sample's true helium release using the equations and procedures described

above. Although there are many ways to address permeation lag, a simple general workflow is suggested below:

1. Pull a fused silica tube (e.g., 3 mm inner diameter, 4 mm outer diameter stock) into a pipette with the desired inner diameter and wall thickness for the experiment.

2. Load the mineral sample and the desired buffering phase (e.g., water or sulfur) into the ampoule and seal it, taking care not to heat the sample itself during sealing.

3. If possible, measure the ampoule's inner and outer diameters with a calibrated microscope. This is not strictly necessary, but is helpful for planning the experiment: these measurements give the geometric factor G (Eq. (1.4)), which, combined with literature permeability values, allows the rate constant of permeation λ to be estimated in advance.

4. Program the step heat schedule so that, except during the elevated-temperature experimental steps themselves, the ampoule is held at all times at a low-but-warm permeation temperature at which helium is expected to be immobile in the sample (e.g., ~40–100°C). Minimize pump-out times between steps: 1-5 minutes is ideal.

5. After every heating step, program an automated resting period, before measurement, in which the ampoule is held at the permeation temperature for at least the diffusive timescale $\tau = \frac{L^2}{D}$, where L is the ampoule's wall thickness and D is the diffusivity of helium in fused silica glass (e.g., from Swets et al., 1961). Doubling this estimated timescale is a reasonable safety margin that should be conservative enough even for hydrothermal experiments. This resting period allows a steady state helium concentration profile to establish within the glass immediately after each heating step, which is required for accurate permeation corrections (Eqs. (1.1), (1.5)). Because the resting period alone is already long enough to permeate a significant portion of the ampoule's residual helium, a permeation step can simply be programmed as a step with no additional heating time.

6. At several points throughout the experiment, run a series of at least three consecutive permeation steps to determine the permeation rate constant

λ (Eq. (1.8)). This value can drift over the course of hydrothermal experiments and should therefore be checked repeatedly.

7. Bracket every high-temperature experimental step with at least one permeation step, using either the interpolated rate constant for experiments where this value resolvably drifts or the average measured rate constant for experiments where there is no resolvable drift, to calculate the ampoule's helium contents between steps (Eqs. (1.1), (1.5)).

For high temperature experiments of retentive phases experimentalists may opt to treat permeation lag as uncertainty, provided the rate constant of permeation is known at the experimental temperature. This can only be done for steps where $\lambda t > 5$. No permeation steps are necessary for this approach.

Supplementary Information

for Chapter 1: *A Simple Method for Helium Step Heat Experiments at High Water or Sulfur Fugacity*

1.1s A Fick's Law Formulation Describing Helium Migration Across Semi-Permeable Membranes

We derive the relevant equations describing helium permeation through glass by beginning with Fick's Law, similarly to previous works (Rogers et al., 1954; Altemose, 1961).

$$J = -D\frac{d\varphi}{dx} \tag{1.12}$$

Where J is the area-normalized Fickian diffusive flux in units of $\frac{atoms}{s \times cm^2}$, D is the diffusivity (of helium in glass) in units of $\frac{cm^2}{s}$, and $\frac{d\varphi}{dx}$ is the concentration gradient, in units of $\frac{\left(\frac{atoms}{cm^3}\right)}{cm}$ or $\frac{atoms}{cm^4}$.

For steady state permeation across a planar membrane, the concentration gradient is constant. If we take l to be the glass thickness and we rewrite the Fickian flux J as $\frac{F}{A}$, where F is the absolute atom flux in units of $\frac{atoms}{s}$, and A is simply the membrane area, we can rewrite Fick's law as:

$$F = -DA\frac{\Delta\varphi}{l} \tag{1.13}$$

Finally, we assume that the concentration of helium within the glass membrane is Henrian, i.e. proportional to the gas pressure in contact with the membrane. Experimental work has shown that helium permeation through silica glass is Henrian at pressures of several hundred bars and lower (Shelby, 1976), well above the experimental helium partial pressures for this work. We define this proportionality factor as the solubility, S, in units of $\frac{atoms}{atm \times cm^3}$. Now we may write the expression in terms of the gas pressure gradient, or the difference of helium partial pressures on each side of the membrane.

$$F = -DA\frac{S\Delta P}{l} \tag{1.14}$$

Diffusivity and solubility are material properties that are functions of temperature following the Arrhenius equation, and therefore their mathematical product (permeability, K) also follows an Arrhenius relation with temperature (Rogers et al., 1954; Swets et al., 1961). Now, recognizing that K = DS, we may express the equation in terms of permeability $\frac{atoms}{s \times cm \times atm}$.

$$F = -\frac{KA\Delta P}{l} \tag{1.15}$$

Studt (1962) and Masaryk (1968) also use this equation to describe helium permeation through silica glass, taken from Darcy's Law for gas flow through semi-permeable membranes (Carman, 1956). This ordinary, first-order differential equation is well suited for describing permeation through a silica membrane when the permeate pressures on each side of the membrane are constant (Norton, 1953; Altemose, 1961; Swets et al., 1961). Cases where permeate partial pressure changes during the experiment (such as this work) require solving the first order ODE above to relate the abundance of an analyzed permeate to the abundance of unanalyzed permeate remaining in the depleting reservoir.

Recalling that F is an atomic flux in units of $\frac{atoms}{s}$ that could be notated as $\frac{dn}{dt}$:

$$\frac{dn}{dt} = -\frac{KA\Delta P}{l} \tag{1.16}$$

We assume that the helium partial pressure in the ultra-high vacuum (UHV) heating cell surrounding the ampoule is 0, so that the pressure gradient equals the partial pressure within the ampoule ($\Delta P = P$). This is a valid assumption for this work because the heating cell is 4-5 orders of magnitude more voluminous than the glass ampoule. The assumption therefore is only appreciably violated for the trivial case in which effectively all the helium present within the tube is permeated out and the helium partial pressures within and outside of the ampoule are equal. In this case, net helium flux goes to zero even though the ampoule is not entirely evacuated. Relative to the amount of helium within the ampoule at time 0 (n_0), the amount of helium remaining in the ampoule after infinite time *(n_{eqm})* will be set by the volume ratio of the ampoule and the lamp cell.

$$\frac{n_{eqm}}{n_0 - n_{eqm}} \approx \frac{n_{eqm}}{n_0} = \frac{V_{ampoule}}{V_{lamp\ cell}} \approx 10^{-4}$$

For this experimental setup, only 1 part in 10000 helium atoms will remain in the ampoule after infinite time in the closed heating cell.

Before integrating Eq. (1.16), it is useful to experimentalists working with mass spectrometry to express the pressure term P in terms of atoms n, the atomic abundance of helium within the ampoule. Helium is well-approximated as an ideal gas at experimental conditions, so the ideal gas law (PV = nRT) relates the pressure, P, to the atomic abundance, n, yielding Eq. (1.17):

$$\frac{dn}{dt} = -\frac{AKnRT}{lV} \tag{1.17}$$

Where R is the gas constant $\left(1.36 \times 10^{-22} \frac{cm^3 atm}{K\ atoms}\right)$, T is the temperature in Kelvin, and V is the volume in cm^3. It is now obvious that the helium atoms in the gas phase (n) decrease over time. Because Eq. (1.17) assumes a steady state helium concentration profile in the glass walls of the ampoule in equilibrium with the helium partial pressure inside the ampoule, one must make another essential assumption to continue: that the permeation timescale be small with respect to the diffusive timescale so that the helium concentration profile of the glass effectively remains in an equilibrium with the helium atoms in the gas phase (n). If the inner gas volume is small relative to the volume of the ampoule glass, then $\frac{dn}{dt}$ will be relatively large and the helium atoms in the gas phase (n) will decrease faster than diffusion's ability to maintain an equilibrium concentration profile in the glass, violating the assumption of a steady state concentration profile in equilibrium with the inner gas phase. Then, Eq. (1.17) and all equations derived from it will be invalid. In other words, the timescale of diffusion must be much smaller than the timescale of permeation ($\tau_D \ll \tau_P$) for Eq. (1.17) and all derived equations to apply. A more thorough analysis of this assumption is presented in *Diffusive Steady State Assumption for Permeation.*

Eq. (1.17) can be simplified by defining a geometric factor (G) in units of cm^{-2} that incorporates the spatial parameters for area, volume, and length of the experimental glass ampoule:

$$\frac{dn}{dt} = -GKnRT \tag{1.18}$$

Eq. (1.18) can be integrated using a straightforward separation of variables to get a useful general equation describing helium reservoirs depleting via permeation.

$$n_t = n_0 e^{-KGRTt} \tag{1.19}$$

Where n_t is the amount of helium atoms within the reservoir after time t in seconds, and n_0 is the amount of helium atoms within the reservoir at time t = 0.

Eq. (1.19) can equivalently be expressed in terms of the fraction of the initial helium reservoir that has permeated during time t, which is often a more intuitive quantity when relating this equation to a measured amount of helium. Defining the fractional permeation yield ${}^{He}F \equiv \frac{m}{n_0}$, where $m \equiv n_0 - n_t$ is the amount of helium permeated and available to be measured during time t, Eq. (1.19) can be recast as:

$$ {}^{He}F = \frac{m}{n_0} = 1 - e^{-KGRTt} \tag{1.20}$$

Eq. (1.19) can be solved for permeability (K) for use in experimentally determining permeability for experiments in which the permeating helium source is significantly depleted during the experiment, yielding Eq. (1.21):

$$ K = -\frac{\ln(n_t) - \ln(n_0)}{GRTt} \tag{1.21}$$

In typical experimental measurements of helium permeability through glass, permeation rates are measured in real time, and the pressure gradient is constant throughout the measurement, so an equation like Eq. (1.15) is used to solve for permeability instead (Norton, 1953; Swets et al., 1961; Altemose, 1961).

Permeation Through Cylindrical Membranes

For non-planar membranes, the effective area of permeation, A, is not straightforward to evaluate because the surface area on each side of the membrane differs. Masaryk (1968) gives the effective area of permeation for a cylinder (such as for this experiment) as:

$$ A = \frac{2\pi hl}{\ln\left(\frac{b}{a}\right)} \tag{1.22}$$

Where a is the cylinder's internal radius, b is the cylinder's external radius, and h is its length. This effective area of permeation (Eq. (1.22)) can be used with the equation for the internal volume of the cylinder: $V = h\pi a^2$ to evaluate the geometric factor G (Eq. (1.18)) for a cylinder in terms of the ampoule geometry. Reducing yields Eq. (1.23):

$$G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} \tag{1.23}$$

Interestingly, h cancels so that the permeation rate has no dependence on the ampoule length when permeation through the top and bottom are negligible. Eq. (1.23) displays the intuitive behavior that thinner walls and narrower tubes lead to more efficient permeation. Where G is used elsewhere in this work, it refers to the cylindrical geometric factor given in Eq. (1.23). Most silica ampoules sealed at Caltech for helium step heat experiments were 0.6-1.2 mm wide with 150-300 µm thick glass, translating to values of G between 2000-7000 cm^{-2}. In this work, if no value of G is specified, then a typical 200 µm thick ampoule with an outer diameter of 1 mm is assumed, corresponding to $a = 0.03$ cm, $b = 0.05$ cm and $G = 4350.26$ cm^{-2}.

1.2s Diffusive Steady State Concentration Profiles in Cylindrical Membranes

To establish a steady state concentration profile, the diffusive timescale must be much faster than the permeation timescale, otherwise the helium concentration profile within the glass will respond too slowly to maintain equilibrium with the rapidly depleting internal gas phase (Fig. 1.11). The ratio of the permeation timescale and diffusive timescale is a strong function of the ratio of the ampoule's glass and internal gas volumes with a very minor dependence on temperature (Fig. 1.12 below). An ampoule with a glass/gas volume ratio of 2 has a diffusive timescale that is ~5.6% of the permeation timescale (Eq. (1.27) below). Numerical diffusion-permeation simulations show that Eq. (1.21) accrues less than 1% error due to deviation from the idealized helium concentration profile (Figs. 1.13 and 1.14 below). Smaller glass/gas volume ratios are better: the ampoules in the main text have glass/gas volume ratios ~1.25-1.5. Numerical simulations show that this error is determined nearly entirely by the volume ratio of the ampoule glass and gas and is otherwise insensitive to ampoule geometry (Figs. 1.15 and 1.16 below).

The derivation for Eq. (1.21) always assumes an equilibrium steady state helium concentration profile across the ampoule. The most important corollary of this assumption is that the timescale of permeation must be much larger than the diffusive timescale ($\tau_D \ll \tau_P$) for Eq. (1.21) and all derived equations to apply.

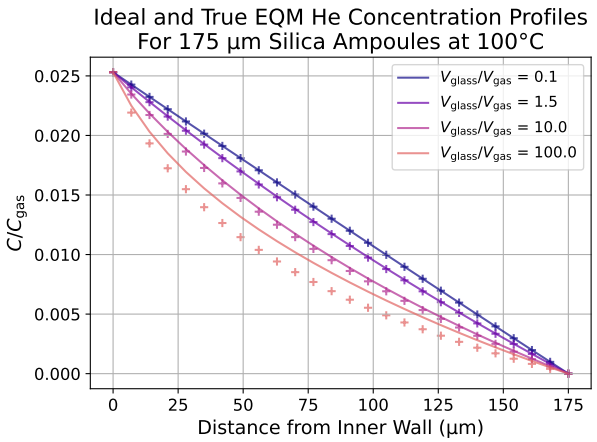


Figure 1.11: A visualization of ideal diffusive steady state concentration profiles (demarcated by the '+' symbols) are contrasted with the numerically calculated "effective" steady state concentration profile, represented by the solid lines. These effective concentration profiles are not strictly in steady state, but they are effectively in equilibrium with a rapidly decreasing internal helium reservoir. These profiles are retrieved by numerically simulating a model ampoule permeating helium for at least one diffusive timescale. At low glass/gas volume ratios, the idealized and true concentration profiles are indistinguishable. Ideal steady state concentration profiles for cylindrical membranes are strongly curved at high glass/gas volume ratios and become linear as the glass/gas volume ratio approaches 0. For closed ampoules where internal pressure is not constant, but instead decreases due to permeation, the internal pressure decreases faster than the glass concentration profile can respond to the new internal helium pressure, so the concentrations are higher (and less curved) than the ideal profile, because of memory of higher internal helium pressures. 4He diffusivities and solubilities are taken from Swets et al. (1961).

An analysis is presented here exploring the ampoule geometries that permit the use of the analytical solutions for helium permeation through silica glass.

The diffusive timescale is the timescale at which diffusion is generally expected to establish a steady state concentration profile between two (effectively) constant reservoirs. For a diffusive domain size L, the timescale is given as:

$$\tau_D = \frac{L^2}{D} \tag{1.24}$$

The permeation timescale is analogous to an "e-folding time" or "mean life" in radioactive decay, the time after which point the original abundance of something decaying by a first order process following the form $\frac{dn}{dt} = -\lambda n$ would have decreased by a factor of Euler's number $(1/e)$. After one permeation

timescale has passed, more than half of the original helium has permeated through the ampoule walls, leading to the intuition that the dimensionless timescale ratio $\frac{\tau_D}{\tau_P} \ll 1$ in order for Eq. (1.21) and Eq. (1.19) to apply, because only vanishingly small changes in the total helium are allowed within one diffusive timescale in order for an equilibrium helium concentration profile to be stabilized and maintained.

$$\tau_P = \frac{1}{\lambda} = \frac{1}{GRTK} \tag{1.25}$$

The ratio of these two timescales is a weak function of temperature and a strong function of the ratio of ampoule glass and internal gas volumes $\left(\frac{V_{glass}}{V_{gas}}\right)$ (Fig. 1.12). By the equation for the volume of cylinders:

$$V_{\frac{glass}{gas}} = \frac{h\pi\left(b^2 - a^2\right)}{h\pi a^2} = \frac{\left(b^2 - a^2\right)}{a^2} \tag{1.26}$$

$\frac{\tau_D}{\tau_P}$ can be expanded and recast to be a function of only temperature and $V_{\frac{glass}{gas}}$. Recalling that L = b-a and K = SD, and the identity of G for cylinders from Eq. (1.23).

$$\frac{\tau_D}{\tau_P} = \frac{L^2}{D}GRTK = \frac{(b-a)^2}{D}\frac{2}{a^2 \ln\left(\frac{b}{a}\right)}RTSD$$

Factoring $(b^2 - a^2) = (b-a)(b+a)$, substituting, and simplifying:

$$\frac{\tau_D}{\tau_P} = V_{\frac{glass}{gas}} \bullet \frac{(b-a)}{(b+a)}\frac{2}{\ln\left(\frac{b}{a}\right)}RTS$$

Now using Eq. (1.26) to relate the terms containing a and b to $V_{glass/gas}$, all spatial terms can be expressed in terms of this ratio.

$$\left(\frac{b}{a}\right) = \sqrt{V_{\frac{glass}{gas}} + 1}$$

For example:

$$b = a\sqrt{V_{\frac{glass}{gas}} + 1}$$

$$(b-a) = a\left(-1+\sqrt{V_{\frac{glass}{gas}}+1}\right)$$

$$(b+a) = a\left(1+\sqrt{V_{\frac{glass}{gas}}+1}\right)$$

$$\frac{\tau_D}{\tau_P} = S(T)\bullet\, RT\, \bullet V_{\frac{glass}{gas}} \bullet \frac{-1+\sqrt{V_{\frac{glass}{gas}}+1}}{1+\sqrt{V_{\frac{glass}{gas}}+1}} \bullet \frac{2}{\ln\left(\sqrt{V_{\frac{glass}{gas}}+1}\right)} \quad (1.27)$$

Eq. (1.27) is plotted for three temperatures as a function of glass/gas volume ratio in Fig. 1.12. $\frac{\tau_D}{\tau_P}$ increases monotonically and nearly linearly in log-log space with the glass/gas volume ratio of ampoules, whereas temperature plays only a minor role. Ampoules with high glass/gas volume ratios and $\frac{\tau_D}{\tau_P}$ ratios can be considered "glass dominated". For these glass-dominated ampoules, the relative difference in surface area between the inner and outer surfaces is large; resulting in a more profoundly curved ideal concentration profile (Fig. 1.11 above, Eq. (1.32)). For glass dominated ampoules, this ideal curved concentration profile is only realized if the internal pressure is held constant, otherwise, this internal helium partial pressure decays rapidly due to permeation relative to the rate of diffusion within the glass, thereby stabilizing a helium concentration profile in the glass that will be less curved and higher than the idealized steady state of Eq. (1.32) (Fig. 1.11 above). When normalized to the concentration, quantity, or pressure of helium within the gas phase of the ampoule, this quasi-steady state concentration profile is reached within one diffusive timescale and remains constant as diffusion and permeation proceeds, regardless of the ratio of glass and gas volumes (this property was tested by varying the permeation time between one and ten diffusive timescales, which had no effect of the concentration profile normalized to the concentration, abundance, or partial pressure of helium in the gas phase). Indeed, all model ampoule calculations from Figs. 1.13–1.15 were initialized to have this effective steady state concentration profile, which is done by allowing the model ampoule to permeate and diffuse for one diffusive timescale, after which the concentration profile is multiplied by the simulated $\frac{n_0}{n_t}$ to renormalize it to the original helium contents of the ampoule. The helium content of the gas is then recalculated using the relation $n_{gas} = n_0 - n_{glass}$, and the permeated helium outside of the

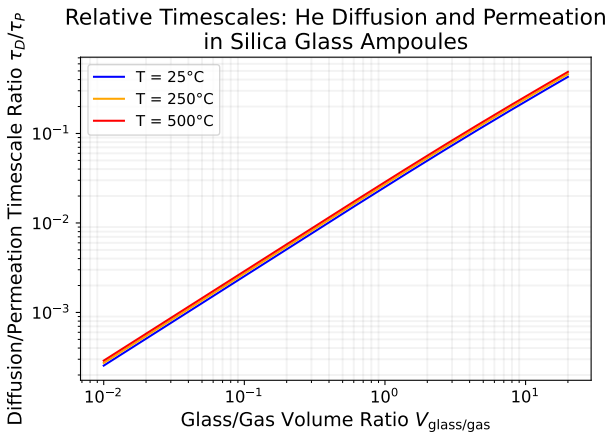


Figure 1.12: The accuracy of Eq. (1.19) and equations derived from it is dependent on a steady state concentration profile in equilibrium with the helium gas concentration inside the ampoule. If the timescale of permeation is comparable to or lower than the timescale of diffusion for the ampoule, then the gas concentration inside the ampoule will change faster than the response of the concentration profile within the glass, and these equations will become inaccurate. In other words, these equations are only accurate when $\tau_D/\tau_P \ll 1$. For helium permeation in silica glass ampoules, the ratio of the diffusive and permeation timescales is a strong function of the ratio of glass and gas volume, and a weak function of temperature. τ_D/τ_P is nearly linear with V_{glass}/V_{gas} in log-log space, forming subtly-offset parallel arrays at different temperatures. 4He solubilities are taken from Swets et al. (1961).

ampoule is reset to 0. Numerical simulations of model ampoules with glass/gas volume ratios above ~ 0.1 that were initialized to the ideal steady state of Eq. (1.32) recorded an early spike in relative error with Eq. (1.21) that was maximized before one diffusive timescale because of an incorrect initial steady state assumption. This artifact vanished completely when the true steady state profile was used.

Numerical simulations of model ampoules of varying dimensions, thicknesses, and geometric factors show that the only geometric parameter that directly affects the accuracy of the analytical solutions, e.g. Eq. (1.21), is the glass/gas volume ratio of the ampoules (Figs. 1.13–1.15). This geometric dependence is much more important than subtle effects of temperature (Figs. 1.12; 1.15). Ampoules from this work have glass/gas volume ratios between 1.2 and 1.6; for which Eq. (1.21) is accurate to within 1% for temperatures less than 500°C (Fig. 1.14). For more precise handling, odd thermal schedules, or for ampoules with higher glass/gas volume ratios, permeation fractions would need

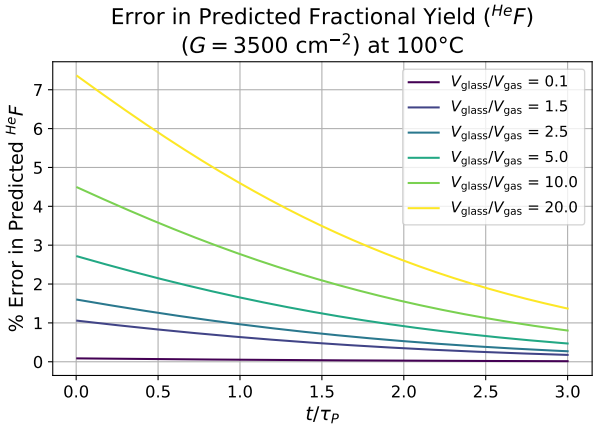


Figure 1.13: Numerical simulations confirm that Eq. (1.21) is accurate for ampoules with low glass/gas volume ratios. For ampoules with higher glass/gas volume ratios, the relative error is not constant but depends on the duration of the step. The relative error reaches a maximum at a step shorter than the permeation timescale (τ_P) for ampoules with glass/gas volume ratios higher than 1, then decreases as steps continue and most of the original gas within the ampoule has been permeated. 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

to be numerically solved to achieve sufficient accuracy to use a permeation measurement to constrain the helium contents of an ampoule.

To understand the steady state helium concentration profile in the glass, it is necessary to model the radial diffusion through a symmetrical, hollow cylinder. Carslaw and Jaeger (1959) give the equation for the change in concentration by diffusion in a cylinder (eqn. 5, page 17, second edition), expressed here in terms of concentration and diffusivity.

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} + \frac{1}{r^2}\frac{\partial^2 C}{\partial \theta^2} + \frac{\partial^2 C}{\partial z^2}\right] \tag{1.28}$$

For typical ampoules used in this work, the surface area of the top and bottom together is about 5-10% of the surface area of the walls, but they are neglected here for simplicity. Assuming that the concentration profiles are radially symmetrical and uniform with height (z), Eq. (1.28) is reduced to:

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right] \tag{1.29}$$

At steady state, $\frac{dC}{dt} = 0$, so

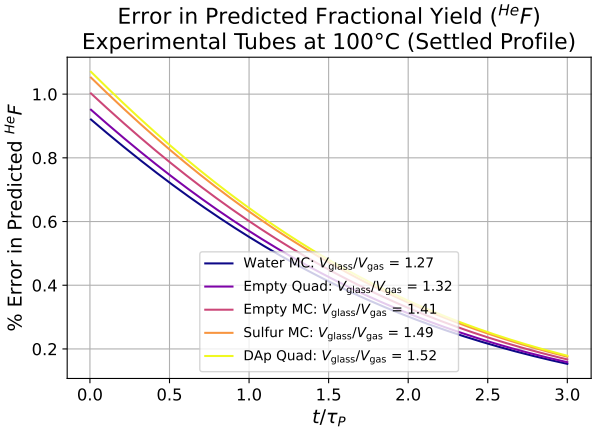


Figure 1.14: Numerical experiments of this work show that these equations will be accurate to within 1% for ampoule glass/gas volume ratios less than 2 for temperatures less than 500°C, which corresponds to $\tau_D/\tau_P < 0.056$. Experimental ampoules at Caltech have glass/gas volume ratios generally between 1.25 and 1.5, and thus Eq. (1.21) is accurate to within 1% for ampoules with concentration profiles in steady state. 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

$$\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} = 0$$

Which, by the product rule, can be expressed as:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C}{\partial r}\right) = 0$$

Now, the $\frac{1}{r}$ term can be dropped, and both sides can be integrated with respect to r to yield:

$$r\frac{\partial C}{\partial r} = A$$

Where A is a constant of integration. Now both sides are integrated again with respect to r, leaving the equation describing the geometry of the diffusive steady state concentration profile of a cylinder (Eq. (1.30)):

$$\frac{\partial C}{\partial r} = \frac{A}{r}$$

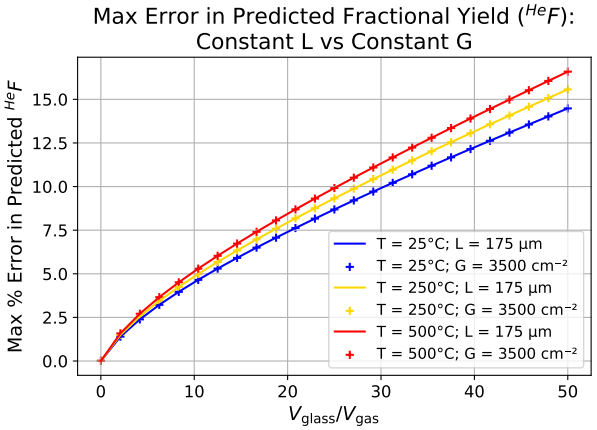


Figure 1.15: Numerical simulations of two suites of model ampoules with varying glass/gas volume ratios and a constant thickness (L) or permeation geometric factor (G) show that the only factors that determine the applicability of Eq. (1.21) for helium permeation of closed ampoules are glass/gas volume ratio (primarily) and temperature (secondarily). These two suites have different geometries and permeation rates, yet the deviation of the numerical model from Eq. (1.21) is the same for different ampoules with the same glass/gas volume ratio at the same temperature. This is clear evidence that the ratio of diffusive and permeation timescales is the primary control on the accuracy of Eq. (1.21). 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

$$C(r) = \int \frac{A}{r} dr$$

$$C(r) = Aln(r) + B \tag{1.30}$$

Where B is another constant of integration. Constants A and B can be solved for once the boundary conditions are defined, that is, the helium concentrations at the inner and outer walls of the glass ampoule. The concentration of helium at the outer wall is effectively 0 because the high vacuum heating cell is essentially an infinite volume compared to the glass ampoule. At the inner wall, the concentration is governed by the solubility and helium partial pressure of the encapsulated volume. Letting a be the radius of the ampoule inner wall, b be the radius of the outer wall, and P_{He}^{gas} be the partial pressure of helium in the encapsulated gas of the ampoule, the following boundary conditions are enforced:

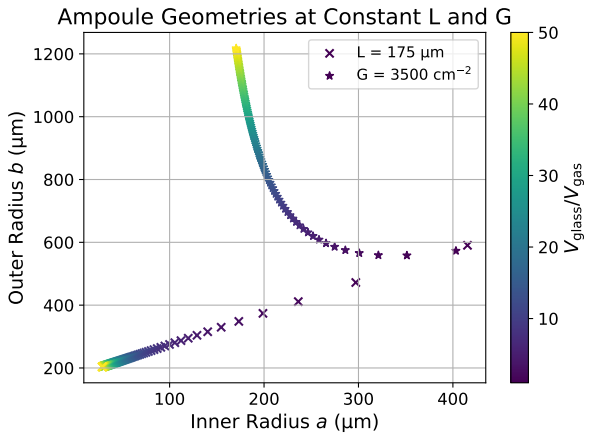


Figure 1.16: Model ampoules with uniformly distributed glass/gas volume ratios between 0.01 and 50 and one other constraint (either constant thickness or constant geometric permeation factor G, Eq. (1.23)) have defined inner and outer radii. These are the same suites of ampoules for which helium permeation was numerically simulated in Fig. 1.15.

$$C(r = b) = 0$$

$$C(r = a) = S(T) \bullet P_{He}^{gas}$$

Now a system of equations can be formulated so that A can be solved first, then B.

$$0 = Aln(b) + B$$

$$S(T) \bullet P_{He}^{gas} = Aln(a) + B$$

$$S(T) \bullet P_{He}^{gas} = Aln(a) - Aln(b)$$

$$S(T) \bullet P_{He}^{gas} = A\ ln\left(\frac{a}{b}\right)$$

$$A = \frac{S(T) \bullet P_{He}^{gas}}{ln\left(\frac{a}{b}\right)}$$

$$B = -\frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} ln(b)$$

Substituting A and B back into Eq. (1.30):

$$C(r) = \frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} \ln(r) - \frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} \ln(b)$$

$$C(r) = S(T) \bullet P_{He}^{gas} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)} \tag{1.31}$$

Using the ideal gas law and the volume for the cylinder, the partial pressure of helium within the ampoule can be related to the geometry of the ampoule and the abundance of helium atoms within the ampoule:

$$P_{He}^{gas} = \frac{n_{He}^{gas} RT}{V_{gas}}$$

$$V_{gas} = \pi a^2 h$$

$$P_{He}^{gas} = \frac{n_{He}^{gas} RT}{\pi a^2 h}$$

$$C(r) = S(T) \bullet \frac{n_{He}^{gas} RT}{\pi a^2 h} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)} \tag{1.32}$$

Where h is the length of the ampoule.

Because the concentration profile of the ampoule is radially symmetrical, Eq. (1.32) can be expressed as a radial density of helium atoms instead to eliminate the dependence on h. The concentration ($C(r)\frac{atoms}{cm^3}$) can be converted into a radial density ($\epsilon(r)\frac{atoms}{cm}$) by multiplying by the circumference of a circle with radius r and the cylinder height h.

$$\epsilon(r) = C(r) * 2\pi r h \tag{1.33}$$

Substituting in $C(r)$ from Eq. (1.32) and the geometric constant of permeation G from Eq. (1.23) $\left(G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} = -\frac{2}{a^2 \ln\left(\frac{a}{b}\right)}\right)$:

$$\epsilon(r) = \ S(T) \bullet \frac{2n_{He}^{gas} rRT}{a^2} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)}$$

$$\epsilon(r) = \ -S(T) \bullet n_{He}^{gas} GRTr \ln\left(\frac{r}{b}\right) \tag{1.34}$$

To find the proportion of helium within the glass at steady state, the radial density is integrated from $r = a$ to $r = b$.

$$n_{He}^{glass} = \int_a^b \epsilon(r)dr = \int_a^b -S(T) \bullet n_{He}^{gas} GRTr \ln\left(\frac{r}{b}\right)dr$$

$$n_{He}^{glass} = -S(T) \bullet n_{He}^{gas} GRT \bullet \int_a^b \left[rln(r) - rln(b)\right] dr$$

Applying the sum rule, then integration by parts to one term yields the indefinite integral:

$$\int \left[rln(r) - rln(b)\right] dr = \frac{1}{2}r^2 \ln(r) - \frac{r^2}{4} - \frac{1}{2}r^2 \ln(b) + C$$

Evaluating the definite integral from the bounds of $r = a$ to $r = b$ yields:

$$\int_a^b \left[rln(r) - rln(b)\right] dr = \frac{1}{2}b^2 \ln(b) - \frac{b^2}{4} - \frac{1}{2}b^2 \ln(b) - \frac{1}{2}a^2 \ln(a) + \frac{a^2}{4} + \frac{1}{2}a^2 \ln(b)$$

Reducing and simplifying:

$$\int_a^b \left[rln(r) - rln(b)\right] dr = \frac{1}{2}a^2 \left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}$$

$$n_{He}^{glass} = -S(T) \bullet n_{He}^{gas} GRT \bullet \left(\frac{1}{2}a^2 \left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right) \tag{1.35}$$

Eq. (1.35) can be easily expressed in terms of total helium in the ampoule by substituting for the relationship $n_{He}^{gas} = n_0 - n_{He}^{glass}$

$$n_{He}^{glass} = -S(T) \bullet \left(n_0 - n_{He}^{glass}\right) GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)$$

$$\frac{\left(n_0 - n_{He}^{glass}\right)}{n_{He}^{glass}} = \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1}$$

$$\frac{n_0}{n_{He}^{glass}} = 1 + \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1} \tag{1.36}$$

A Numerical Model for Diffusive Permeation of Helium Through Silica Glass Ampoules

A numerical model to estimate diffusive permeation is invaluable, as it provides a flexible way to estimate the permeation fraction $\left(\frac{m}{n_0}\right)$ even when the required assumptions for the analytical solutions are not met. For example, a numerical approach can calculate the expected permeation fraction for a non-isothermal time-temperature history, and it needn't obey the assumption that the helium concentration profile in the glass is in diffusive equilibrium with the partial pressure of helium in the encapsulated gas phase. As will be investigated below, the equilibrium concentration profile assumption is ephemerally broken when temperatures are changed, and for ampoules with high glass/gas ratios it is never true.

We numerically solve the radial diffusion equation for cylinders (Eq. (1.29)) using the Forward-Time Central-Space (FTCS) Finite Difference Method (FDM) along a radial transect of a model ampoule of arbitrary geometry for small, discrete time intervals. The system is spatially discretized into 100 cells along a radial transect, and numerical stability was insured by solving for the time interval of a single step, Δt, such that $\frac{D\Delta t}{\Delta x^2} = 0.5$, satisfying the Courant-Friedrichs-Lewy (CFL) condition.

Helium concentrations along the transect (except for the boundaries, where r = a and r = b) are updated according to the following equation, an FTCS discretization of the radially symmetric diffusion equation for a cylinder (Eq. (1.29)):

$$\frac{dC_r}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right]$$

$$\Delta C_r \approx \Delta t \ \bullet D\left[\frac{C_{r-\Delta r} - 2C_r + C_{r+\Delta r}}{\Delta r^2} + \frac{1}{r}\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r}\right] \tag{1.37}$$

Dirichlet boundary conditions were enforced at the inner and outer boundaries. The concentration at the model ampoule's outer boundary (where r = b) was held at 0, while the concentration at the inner boundary (where r = a) was calculated at each discrete step by multiplying the solubility (in units of $\frac{atoms}{atm \times cm^3}$) by the helium partial pressure within the ampoule, according to Eq. (1.38) or (1.39).

$$C_{He}^{glass} = S(T) \bullet P_{He}^{gas} \tag{1.38}$$

Eq. (1.38) can be related to the atomic abundance of helium in the gas phase using the ideal gas law and the formula for the volume of a cylinder:

$$P_{He}^{gas} = \frac{n_{gas} RT}{V_{gas}}$$

$$V_{gas} = \pi a^2 h$$

$$C_{He}^{glass} = S(T) \bullet \frac{n_{gas} RT}{\pi a^2 h} \tag{1.39}$$

For simulations where the internal pressure is not constant, n_{gas} changes after every step and Eq. (1.38) is reapplied to the inner boundary cell where r = a. While the helium concentrations in the glass and in the encapsulated gas phase are both dependent on the length of the ampoule (h), they compensate for one another such that the absolute atom flux into and out of the ampoule is not a function of h. Within the numerical model, h was varied by orders of magnitude and the results of the simulations did not change, confirming this property.

It is sometimes useful to normalize the helium concentration of the glass to the concentration of helium in the gas. To do this, the internal helium partial pressure is related to an atomic or molar concentration using the ideal gas law:

$$C_{He}^{gas} = \frac{n}{V} = \frac{P_{He}^{gas}}{RT}$$

The pressure term vanishes in the ratio of the two concentrations and depends only on temperature:

$$C_{He}^{glass/gas} = \frac{C_{He}^{glass}}{C_{He}^{gas}} = S(T) \bullet RT \tag{1.40}$$

Eq. (1.40) is plotted in Fig. 1.17. It shows that this partitioning ratio strongly favors the gas and is only a weak function of temperature, varying between 2.5 and 2.8% between 0 and 300°C. This implies that departures from the equilibrium steady state assumption due to temperature changes will be small and perhaps negligible, which is confirmed in Fig. 1.19.

In order to simulate permeation, the atomic abundances of helium within the gas phases both outside the ampoule (at high vacuum) and inside the ampoule must be tracked, as do the fluxes between the glass and these two gas phases. These fluxes were calculated simply as the diffusive flux in $\frac{atoms}{s \bullet cm^2}$ as is given by Fick's law (Eq. (1.12)). For spatially discrete cells throughout the ampoule at any radius except for the boundaries, the diffusive flux can be approximated with the finite difference:

$$J_r = -D\frac{\partial C_r}{\partial r} \approx -D\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r} \tag{1.41}$$

However, Eq. (1.41) cannot be applied at the ampoule's inner wall when r = a or outer wall when r = b because there is no remaining glass on one side of the cell. In that case, second order terms from Taylor expansions centered at $a + \Delta r$ and $b - \Delta r$ can be used to better approximate the $\frac{\partial C_a}{\partial r}$ and $\frac{\partial C_b}{\partial r}$ terms. Letting $f(r) = \frac{\partial C_r}{\partial r}$ and $f'(r) = \frac{\partial^2 C_r}{\partial r^2}$:

$$J_a = -D\frac{\partial C_a}{\partial r} \approx -D\left(\frac{\partial C_{a+\Delta r}}{\partial r} + \frac{\partial^2 C_{a+\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_{a+2\Delta r} - C_a}{2\Delta r} + \frac{C_a - 2C_{a+\Delta r} + C_{a+2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{1.42}$$

$$J_b = -D\frac{\partial C_b}{\partial r} \approx -D\left(\frac{\partial C_{b-\Delta r}}{\partial r} - \frac{\partial^2 C_{b-\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_b - C_{b-2\Delta r}}{2\Delta r} - \frac{C_b - 2C_{b-\Delta r} + C_{b-2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{1.43}$$

The helium contents in the measurable (external, high vacuum) reservoir and the inner encapsulated gas phase are updated every step using fluxes determined from Eq. (1.43) and (1.42) respectively. To convert these area-normalized fluxes into absolute atomic abundances, these fluxes are multiplied

by the time increment (Δt) and the surface area of the cylinder at the outer and inner walls respectively.

Modeling permeation for ampoules that experienced complex thermal histories

In order to compare the experimental results for ampoules that experienced complex cooling histories with the expectations of the first order permeation model developed in this work, the numerical model developed above was used. The model predicts the relative helium permeation for the three experiments on the Helix-MC-Plus^{10K} multi-collector mass spectrometer for both 4He and 3He by simulating helium permeation of a model ampoule with the same dimensions and thermal history of the real experiment. For ampoules with multiple measurements of the geometric dimensions, the smaller set of measurements is used for the numerical model.

To simulate permeation during cooling of an ampoule, the temperature is decreased at every step following a parabolic rate law determined using the Watlow temperature records of the cooling ampoule. A parabolic function is fit to cooling rates evaluated as a function of temperature. Cooling rates were algorithmically calculated for discrete time intervals using the Watlow temperature files of the relevant ampoule in Python. Temperatures were taken to be the mid-point between the maximum and minimum temperatures of the interval over which the temperature change was evaluated. The quadratic functions fit to these data were forced to have a single root at T = 24°C so that the simulated ampoules would cool to a final temperature of 24°C, a common room temperature of the Noble Gas Laboratory at Caltech.

The model ampoules were initially assigned an arbitrary and unitless amount of helium, and the permeation predictions were output as step fractional yields $\left(\frac{n_t}{n_0}\right)$, where n_0 was the amount of helium in the ampoule at the beginning of the step. Before the first heating step was simulated, the ampoule was assumed to be in diffusive steady state at 24°C. Afterwards, the ampoule was not assumed to be in diffusive equilibrium except for isothermal steps at times longer than the diffusive timescale at that temperature, so that the concentration profile of the model ampoule was tracked throughout the entire heating history. For steps above room temperature, the ampoule was simulated to immediately transition to the target temperature, then the helium concentration profile,

influx, and outflux of the glass was tracked for the nominal (programmed) heating times for times smaller than the diffusive timescale at that temperature. After a heated step, an equilibrium concentration profile was reinitialized at that temperature only if the equilibrium equation (Eq. (1.21)) was used for permeation timescales longer than the diffusive timescale, then the temperature of the ampoule was cooled continuously according to the quadratic rate law determined from the experimental ampoule's thermal history while helium diffusion and permeation was continuously simulated.

Errors related to violation of the steady state concentration profile assumption due to temperature change

The equilibrium concentration profile is always a function of temperature because the partitioning of helium between the gas phase and the glass at the ampoule's inner boundary is temperature dependent (Shackelford et al., 1972). A new steady-state helium concentration profile must be established every time the temperature is changed. Because of this behavior, it is advantageous to characterize the timescales necessary to achieve a steady state diffusive profile at different temperatures, as well as the magnitude of the uncertainties that result from applying Eq. (1.21) immediately after a temperature change, when the diffusive steady state concentration profile is violated.

A new steady state concentration profile must first be established when the temperature is changed for Eq. (1.21) to strictly apply. It is possible to design experiments where permeation measurements are taken only after a diffusive steady state concentration profile is established within the ampoule, which is further discussed below.

It may unduly inconvenience the experimentalist to experimentally enforce the diffusive steady state condition when the uncertainties introduced by violating the assumption are small. In Fig. 1.17, Eq. (1.40) is plotted for relevant experimental temperatures using the solubilities reported by Swets et al. (1961). Fig. 1.17 shows that the helium volume density of glass at the ampoule's inner boundary will be between 2.5 and 3% of the volume density of equilibrated gaseous helium. The overall variation in helium partitioning is small ($\leq$12%) across temperatures relevant to step heat experiments on encapsulated goethite (0-300°C). There will be excess helium within the glass when changing from a higher temperature experimental step ($\geq$175°C) to a lower temperature

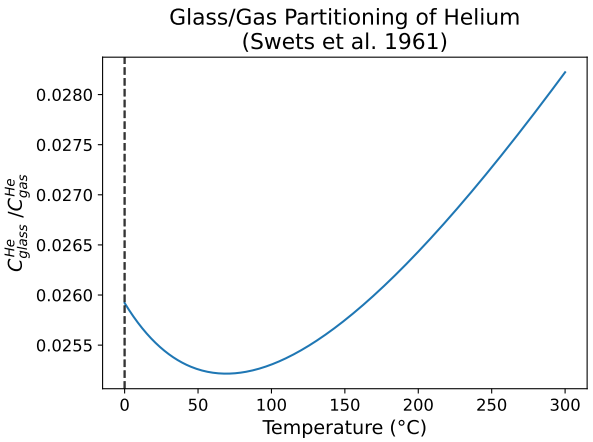


Figure 1.17: Helium preferentially partitions into the gas phase over silica glass. Eq. (1.40) is plotted here: $\frac{C^{glass}_{He}}{C^{gas}_{He}} = S \cdot RT$. It is the ratio of concentrations (in units of atoms/mL) for helium in silica glass over gas at equilibrium using solubility values of 4He from Swets et al. (1961). This partitioning is only a weak function of temperature, so the concentration ratio is between 2.5% and 2.9% for temperatures between 0 and 300°C.

permeation step (≤120°C) that would result in a slightly higher measured value than what would be predicted by Eq. (1.19). Other temperature shifts could also result in a deficit of dissolved helium instead of an excess (e.g. when shifting temperatures from 70°C to 25°C: Figs. 1.18, 1.19).

$$Disequilibrium\ Excess = \ n^{glass}_{He}\ (T_1) - n^{glass}_{He}\,(T_2) \tag{1.44}$$

The disequilibrium excess of Eq. (1.44) can be made into a relative error by dividing by the ideal amount of permeated helium measured in a permeation step from Eq. (1.19).

$$Relative\ Disequilibrium\ Error\ = \frac{n^{glass}_{He}\ (T_1) - n^{glass}_{He}\,(T_2)}{n_0(1 - e^{-\lambda(T_2)t})} \tag{1.45}$$

To evaluate Eq. (1.44) and (1.45), n^{glass}_{He} must be calculated. An upper bound can be done by assuming a steady state diffusion profile and an ideal open tube geometry (with small tops and bottoms neglected for simplicity). Assuming that the helium partial pressure outside of the ampoule is 0 and defining the abundance of helium within the gas phase as n^{gas}_{He} . The radial concentration profile at diffusive steady state as a function of radius and temperature is

given in Eq. (1.34). Radially integrated, the total amount of helium dissolved within the glass is given in Eq. (1.36), which can be solved for n_{He}^{glass} , yielding Eq. (1.46).

$$n_{He}^{glass} = \frac{n_0}{1 + \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1}} \tag{1.46}$$

Where a is the radius of the ampoule's inner wall and b is the radius of the ampoule's outer wall. For legibility, the terms that are not a function of temperature are grouped as the parameter H.

$$H = -GR \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)$$

Now Eq. (1.46) can be expressed as:

$$n_{He}^{glass} = \frac{n_0}{\left(\frac{1}{HTS} + 1\right)}$$

$\frac{n_{He}^{glass}}{n_0}$, the fraction of helium atoms within the ampoule dissolved within glass, is a function only of temperature for a given ampoule. This quantity is evaluated using Eq. (1.46) for several tubes of varying glass/gas volume ratios that share a geometric factor for permeation of $G = 4350$ cm^{-2} and plotted in Fig. 1.18. Although these model tubes would all permeate helium at the same rate in diffusive steady state, they have different sizes, thicknesses, and relative internal (gas) and glass volumes, which affect how long they would need to achieve a steady state diffusive concentration profile after thermal perturbation and how severely diffusive disequilibrium would cause the amount of permeated helium to vary from the predicted amount from Eq. (1.18). Fig. 1.18 gives the intuitive result that ampoules containing more glass relative to gas will produce larger helium excesses/deficits when permeating immediately after thermal perturbation.

Now that the amount of helium atoms dissolved in the glass is defined as a function of temperature, Eq. (1.46) evaluated at the old and new temperatures (T_1 and T_2 respectively) can be substituted into Eq. (1.45) to estimate the relative measured helium excess caused by thermal perturbation as a function of time. Considering that the predicted value of n_0 is proportional to the

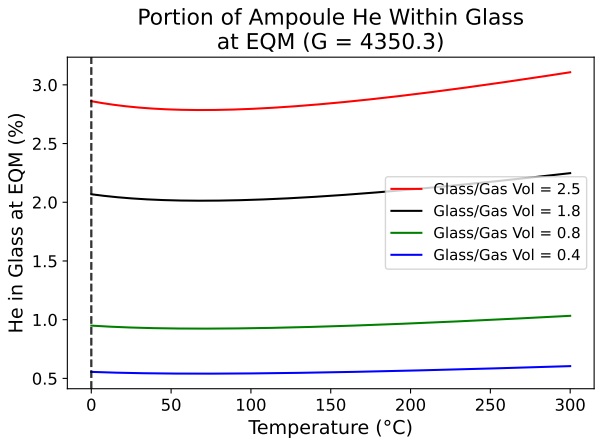


Figure 1.18: Most helium trapped within the ampoules of this work resides in the gas phase. Permeation from ampoules with more glass relative to gas will deviate more strongly from the expected permeation at diffusive steady state (Eq. (1.19)) after thermal perturbation. At diffusive equilibrium, the concentration of helium at the ampoule's inner wall is defined by the equilibrium concentration ratio with the gas phase plotted in Fig. 1.17. The concentration at the outer wall is effectively 0. The concentration profile at any point along a transect through the tube is defined by Eq. (1.34). Eq. (1.35) integrates this concentration profile along the radius to calculate the total amount of helium within the glass, and this value is divided by the total amount of helium within the gas to show the relative distribution of helium between glass and gas for ampoules as a function of inner and outer radii and temperature. It is solved here using solubilities of 4He from Swets et al. (1961) for several tubes with varying glass/gas volume ratios that share the same geometric constant of permeation G. Red, black, green, and blue lines correspond to inner radii a of 270, 300, 400, and 500 microns and outer radii of 507, 500, 533, and 601 microns respectively. The black line follows the "standard ampoule" of this work.

measured amount of helium in a permeation step according to Eq. (1.18), the *Relative Disequilibrium Excess* defined below applies to both the measured value of helium atoms m and to the calculated value of helium within the ampoule at the beginning of the permeation step n_0.

$$Relative\ Disequilibrium\ Excess = \frac{\left(\frac{n_0}{\left(\frac{1}{HT_1S_1}+1\right)} - \frac{n_0}{\left(\frac{1}{HT_2S_2}+1\right)}\right)}{n_0(1-e^{-KGRT_2t})}$$

Upon factoring and simplifying, n_0 cancels out, removing the dependence of this error estimation on knowledge of the true initial helium content of the

ampoule.

$$\textit{Relative Disequilibrium Excess} = \frac{\left(\frac{1}{\left(\frac{1}{HT_1S_1}+1\right)} - \frac{1}{\left(\frac{1}{HT_2S_2}+1\right)}\right)}{1-e^{-KGRT_2t}} \tag{1.47}$$

The relative excess helium that would be measured in a permeation step following a high-temperature experimental step is plotted in percent in Fig. 1.19 as a function of permeation time according to Eq. (1.47) for several initial experimental temperatures and permeation temperatures. This relative error decreases as the new equilibrium concentration profile is established, approaching the permeation behavior expected from Eq. (1.18). These excesses assume an instantaneous transition from experimental step to permeation step and therefore are upper bounds. These excesses are dependent also on the ampoule dimension; in Fig. 1.19, a standard ampoule ($a = 0.03$ cm, $b = 0.05$ cm) is assumed. Thicker ampoules will produce higher magnitudes of excesses (or deficits) for the same heating schedules. Within the temperature range 0-300°C the maximum excess helium that could be introduced into a permeation step measurement due to diffusive disequilibrium would be for an 80°C permeation step following an experimental step at 300°C. Even for those circumstances, the maximum relative error would be $\leq$2% for a 3-hour permeation step (Fig. 1.19). It is important to recognize, however, that this error estimation does not account for non-instantaneous cooling, although fast thermal response for similar experiments and the exponential temperature dependence of diffusion and permeation typically permit experimentalists to neglect fast but non-instantaneous heating and cooling (Farley, Reiners, et al., 1999). These calculations show that, at least for ampoules of similar dimensions to those sealed at Caltech, the error in Eq. (1.19) resulting from violating the diffusive steady state assumption of Eq. (1.21) is small.

For cases where more precise determinations of n_0 and n_t from the application of Eq. (1.19) on a permeation step are desired, it is possible to experimentally ensure compliance with the diffusive steady state concentration assumption. For example, the ampoule could be programmed to be held at the target permeation temperature (for example, 80°C) immediately after a high temperature experimental step instead of completely cooling (to ~25°C) while evolved gasses are processed and helium is purified from the high temperature

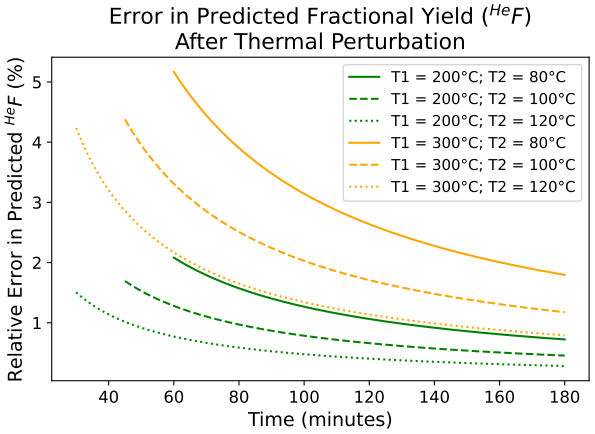


Figure 1.19: When changing temperatures, the inner boundary condition and diffusive steady state concentration profile of helium within a silica glass ampoule change, so the amount permeated will be different from the expected permeate amount of Eq. (1.19) that assumes a diffusive steady state profile has already been established at the new temperature. This difference is calculated here as the difference of helium dissolved within the tube at the starting and final temperature (neglecting the change in helium abundance in the gas phase) and is normalized to the expected permeation amount at the final temperature as a function of time for several temperature jumps, using Eq. (1.35) and solubilities of 4He from Swets et al. (1961) for a standard ampoule (with inner and outer radii of $a = 300$ microns and $b = 500$ microns respectively). Generally, larger temperature jumps produce larger relative errors. For a standard ampoule, the error of Eq. (1.19) will be less than 2% for any permeation step 3 hours or longer following a 300°C experimental step.

experimental step. To do this, one must be aware of the timescale necessary to reach diffusive equilibrium at a given temperature. At Caltech, helium purification after heating steps typically takes 30 minutes, which is only enough time to establish steady state concentration profiles for ampoules <175 µm thick at 100°C (main text Fig. 1.7). For larger ampoules or cooler permeation step temperatures, an additional delay may need to be programmed in order to experimentally enforce the diffusive steady state assumption. Alternatively, expected permeation fractions for any arbitrary thermal schedule can be numerically solved, provided that the solubility and diffusivity are each known for the experimental temperature range.

The Fourier number can be used to determine the time necessary to establish diffusive equilibrium of the helium concentration profile within the ampoule. The Fourier number is a dimensionless number that describes the extent of

diffusive equilibration or homogenization of a system by heat or mass transfer given a characteristic length scale and time (Ostrogorsky and Glicksman, 2015). It is the time divided by the diffusive timescale:

$$Fo = \frac{Dt}{L^2} = \frac{t}{\tau} \tag{1.48}$$

Where t is time and L is the diffusive length scale. For the purposes of helium permeation through silica glass ampoules, L is the glass thickness of the ampoule. When $Fo > 1$, a system is generally said to be effectively homogenized or at equilibrium with respect to diffusion, but it is somewhat arbitrary to define a specific Fourier number at which one can confidently say that diffusion has effectively re-established an equilibrium concentration profile (Ostrogorsky and Glicksman, 2015). To test the value of $Fo = 1$ as the relevant minimum value, we use the numerical diffusive permeation model described above for a characteristic ampoule where a = 300 µm and b = 500 µm. Dirichlet boundary conditions are imposed at the ampoule's inner and outer boundaries, where r = a and r = b, holding $C^{Glass}_{r=b} = 0$ and $C^{Glass}_{r=a} = C^{Gas}_{He} \bullet \left(\frac{C^{Glass}_{He}}{C^{Gas}_{He}}\right)_{EQM}$ as calculated in Eq. (1.40) and plotted in Fig. 1.17. For the analysis of the Fourier number during the approach to the diffusive equilibrium concentration profile in Figs. 1.20 and 1.21, C^{Gas}_{He} is arbitrary and held constant, so plotted concentrations are normalized to C^{Gas}_{He}. A steady state concentration profile (Eq. (1.32)) is taken as the initial condition. The absolute concentration is technically dependent on the ampoule height, but this dependence on h vanishes when normalizing the glass concentration to the gas concentration. The results of these calculations are plotted in Figs. 1.20 and 1.21, where Fig. 1.20 shows the transition from an equilibrium concentration profile at 300°C to 100°C, and Fig. 1.21 shows the more dramatic establishment of an equilibrium concentration profile at 100°C from an initial helium concentration of 0 throughout, such as could be expected after suddenly introducing helium into an ampoule that previously contained no helium. Both figures show that an equilibrium concentration profile is well-established for Fourier number $Fo = 1$, and Fig. 1.20 in particular shows that the difference in concentration profiles is small and nearly reestablished even when the Fourier number $Fo = 0.1$. We conclude that the timescale associated with $Fo = 1$ is the relevant time needed to hold the ampoule to ensure that Eq. (1.19) strictly applies without errors due to thermally-forced disequilibrium in the helium concentration profile.

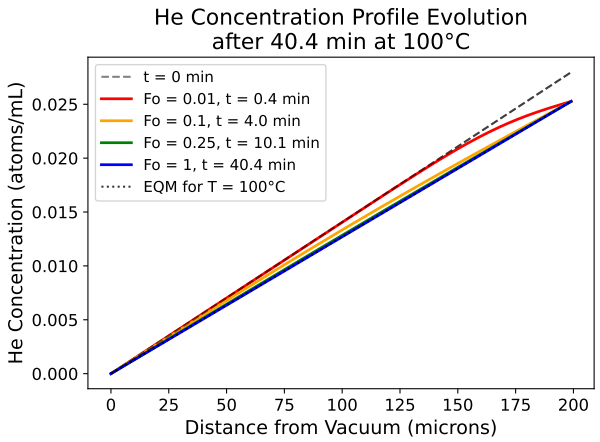


Figure 1.20: A numerical simulation of the evolution of 4He concentration profiles in silica glass ampoules from steady state at 300°C to steady state at 100°C. The steady state concentration profile is only subtly different for different temperatures. The diffusive timescale $\tau = L^2/D$ corresponding to a Fourier number of 1 is effectively sufficient to reestablish a steady state concentration profile at the new temperature. The concentration is arbitrary and is normalized to the concentration of helium in the gas phase of the ampoule, which is held constant for this simulation (therefore the ideal steady state of Eq. (1.34) applies regardless of the glass/gas volume ratio). The inner boundary condition is determined as a function of temperature using the values of Eq. (1.40), plotted in Fig. 1.17. The ampoule simulated here is a standard ampoule with an outer radius of 500 microns and inner radius of 300 microns.

The diffusive timescales associated with $Fo = 1$ for permeability values taken from Swets et al. (1961) are plotted in main text Fig. 1.7 for a few different thickness values (L) as a function of temperature for typical permeation step temperatures used in this study.

1.3s Estimating permeation lag uncertainty for uncorrected, long heating steps

The uncertainty of the helium evolution of the encapsulated natural material during a heating step (ω) due to permeation lag correction can be estimated by calculating the maximum possible helium residing in the tube between steps, i.e. for the values for $n_{0,}$ and n_t: n_0^{UB} and n_t^{UB}. From the permeation lag equation (main text Eq. (1.1)), upper bound estimates for $n_{0,}$ and n_t bound the encapsulated material contribution ω within:

$$m - n_0^{UB} \leq \omega \leq m + n_t^{UB} \tag{1.49}$$

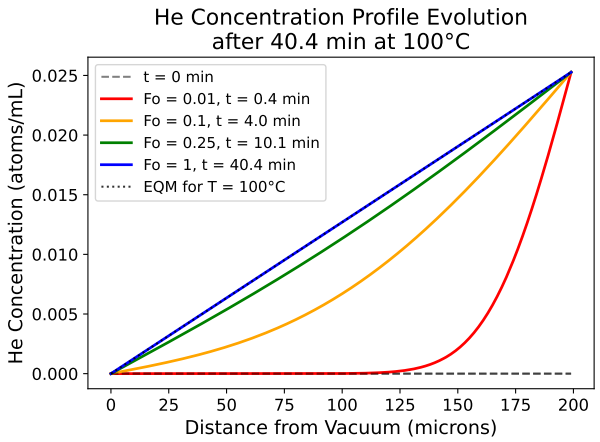


Figure 1.21: Similar to Fig. 1.20, but starting from a helium concentration of 0 throughout the ampoule. This is most useful to visualize the incomplete establishment of a steady state concentration profile for timescales corresponding to Fourier numbers up to 0.25. When the Fourier number is 1, the concentration profile is in diffusive steady state.

Upper bounds for n_t for a given high temperature experimental step can be calculated if step heats are long enough to assume steady state permeation, assuming G and K are known or can be conservatively estimated. To proceed, one must make the generally conservative assumption that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample ($\Omega = \frac{\omega}{t}$) is constant over time at a given temperature. In reality, Ω will decrease over time for helium loss governed by first order processes (O. V. Yakubovich, Y. A. Shukolyukov, et al., 2010; Wilson and Bisson, 1973) or diffusion (except for the case of initially inhomogeneous distributions, the memory of which is often erased early in any diffusion experiment: Shuster, Farley, et al., 2004). Eq. (1.51), given below, shows that the amount of helium trapped in the tube (n_t) is proportional to the rate of helium outgassing from the natural encapsulated sample (Ω), so the approximation that this rate is constant will overestimate the amount of unanalyzed helium remaining in the ampoule after a high temperature experimental step. The rate of helium degassing over time will decrease for degassing controlled by second-order volume diffusion *except* for the minor case of the alpha-ejection effect at low He-release fractions (Farley, Wolf, et al., 1996; Ketcham et al., 2011) or for the more serious effects of radiation damage annealing (Goldsmith et al., 2020) and mineral decomposition (Farley, H. B. Monteiro, et al., 2024). In any case, sequential isothermal steps immediately reveal when $\frac{\omega}{t}$ increases over time.

A new first order differential equation like Eq. (1.18) is formulated, now incorporating a constant source term (Ω) to conservatively approximate the helium influx into the ampoule from the degassing natural sample.

$$\frac{dn}{dt} = \Omega - \lambda n \tag{1.50}$$

Where $\Omega = \frac{\omega}{t}$ and $\lambda = KGRT$. Immediately, the steady state quantity of helium atoms (n_{ss}) present in the ampoule can be found by solving Eq. (1.50) for the steady state condition where $\frac{dn}{dt} = 0$.

$$n_{ss} = \frac{\Omega}{\lambda} = \frac{\omega}{\lambda t} \tag{1.51}$$

Additionally, the helium concentration within the ampoule during the approach to the steady state concentration can be solved for by integrating Eq. (1.50) with respect to t. First, the integrating factor for Eq. (1.50) is identified: $I = e^{\lambda t}$, which is multiplied to both sides to yield Eq. (1.52).

$$e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n = e^{\lambda t}\Omega \tag{1.52}$$

Then both sides are integrated with respect to *t:*

$$\int \left(e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n \right) dt = \int e^{\lambda t}\Omega dt$$

Because $\frac{d}{dt}(ne^{\lambda t}) = e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n$ by the product rule Eq. (1.52) reduces to:

$$n_t e^{\lambda t} = \frac{\Omega}{\lambda}e^{\lambda t} + C \tag{1.53}$$

Where C is an unknown constant introduced by indefinite integration. This expression becomes more intuitive when both sides are divided by the integration factor and by substituting the steady state concentration from Eq. (1.51).

$$n_t = n_{ss} + Ce^{-\lambda t} \tag{1.54}$$

The integration constant C can be solved by enforcing the condition that $n_t(t = 0) = n_0$, leading to:

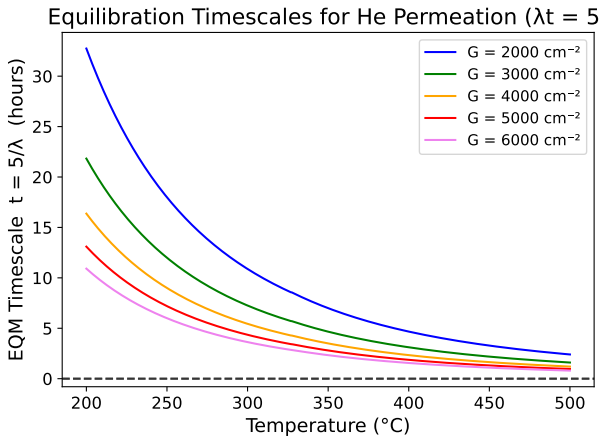


Figure 1.22: The upper limit of helium contents within the ampoule between high temperature experimental steps can be reasonably estimated without further measurement, provided permeative equilibrium has been reached. For typical ampoules with geometric factors between 3000 and 4000 cm^{-2}, this criterion is generally met for steps longer than 24 hours, or within 5 hours at 350°C or higher.

$$n_0 = n_{ss} + C \tag{1.55}$$

Solving Eq. (1.55) for C and substituting into Eq. (1.54) yields Eq. (1.56). Now an inequality is assigned to the equation to reflect the initial assumption that the isothermal helium outgassing rate (Ω) is constant *or decreasing*.

$$n_t \leq \ n_{ss} + (n_0 - n_{ss})\, e^{-\lambda t} \tag{1.56}$$

For long timescales (i.e. when $e^{-\lambda t} \approx 0$), Eq. (1.56) reduces to $n_t \leq \ n_{ss}$. This is effectively true for steps long enough such that $t \geq \frac{5}{\lambda}$, for which $e^{-\lambda t} \leq 0.007$. It is advisable to only consider the Approximation with Uncertainty method for steps of sufficiently long duration such that $t \geq \frac{5}{\lambda}$ so that the estimation for n_t can be reduced to simply the steady state value and not be dependent upon the estimation for n_0 or its uncertainties. The timescales required to reach permeation equilibrium are plotted as a function of temperature in Fig. 1.22 below for some select values of G, the geometric factor.

To estimate the upper bound on the permeation lag (n_t^{UB}), one must consider the circumstances that would produce the highest possible value for n_t. In addition, the constraint that $t \geq \frac{5}{\lambda}$ is enforced so that previous heating steps

are not significant, simplifying the problem significantly. Because n_t scales with ω by Eq. (1.51), the upper bound estimate n_t^{UB} must assume the upper bound estimate for ω, which is given in Eq. (1.49) as $m + n_t^{UB}$. Substituting $\omega^{UB} = m + n_t^{UB}$ into Eq. (1.51) to express n_t^{UB} in terms of the measured value m:

$$n_t^{UB} = n_{ss}(\omega^{UB} = m + n_t^{UB}) = \frac{m + n_t^{UB}}{\lambda t}$$

Solving for n_t^{UB} yields Eq. (1.57):

$$n_t^{UB} = \frac{m}{\lambda t - 1} \tag{1.57}$$

The upper bound amount of helium-4 within the ampoule normalized to the measured value in percent at steady state permeation is plotted in main text Fig. 1.8 using Eq. (1.57) $\left(\frac{n_t^{UB}}{m} = \frac{1}{\lambda t - 1}\right)$, assuming a standard tube where G = 4350.26 cm^{-2} and the permeability values given in Swets et al. (1961). Main text Fig. 1.8 shows that, in order to push this uncertainty to be less than 5%, even very hot steps require long step durations. For a 5% uncertainty using this method, a 10-hour step duration is required even at 400°C. Much longer durations are required for the same precision at lower temperatures: at 250°C, a step duration of 35 hours is required to push the uncertainty beneath 5%.

1.4s Comparison to Literature Permeability Values

Predicted and measured fractional yields for all permeation steps in ampoules besides the encapsulated water experiment agree well (within 1σ: Figs. 1.9, 1.24) with the predictive equations presented in this work and with the permeabilities reported by Swets et al. (1961) and Jones (1953) for all studied temperatures (24-300°C; Fig. 1.23) and step fractional yields (up to ~80%). One outlier measurement on the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer was excluded from Figs. 1.9 and 1.24.

Permeation ratios for 3He and 4He measured in the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer are plotted in Fig. 1.25. These data present significant scatter, but generally agree with the reported results of Jones (1953), except for measurements at room temperature, which cluster tightly at a permeation ratio of 1:1 for the two isotopes. Additionally, fields corresponding to the ratios of apparent permeabilities for the water-filled

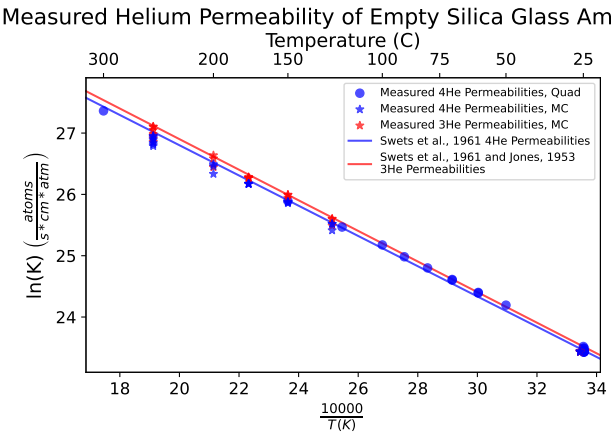


Figure 1.23: Helium permeabilities measured in silica glass ampoules at Caltech on a quadrupole mass spectrometer (4He only) and on a Helix-MC-Plus^{10K} (both 3He and 4He) multi-collector mass spectrometer are compared to literature values. Permeabilities of 3He and 4He are very similar, but 3He permeates slightly faster. The final measurement on the Helix-MC-Plus^{10K} mass spectrometer is an outlier and is excluded from the plot, as it is most sensitive to the value of the total yield remaining in the ampoule. These experiments on empty tubes at Caltech show excellent agreement with the literature values with allowance for some scatter. This result shows that existing understanding of helium permeability in glass can be reasonably leveraged for use in helium step heats of silica glass encapsulated geological samples. Literature 4He permeabilities are taken from Swets et al. (1961); whereas 3He permeabilities from the literature are calculated by multiplying the 3He/4He permeability ratios of Jones (1953) with the 4He permeabilities of Swets et al. (1961).

and sulfur-filled experiments are plotted for the temperature ranges of those experiments. Results for those experiments are plotted as fields because each step integrated permeation behavior over a complex (non-isothermal) cooling history. These experiments also agree with the results of Jones (1953) and the empty ampoule experiment. The qualitative result of Jones (1953) that silica glass is only slightly more permeable for 3He than 4He is supported by these data, which are in marked contrast to the reported extreme differences in 3He and 4He diffusivity in silica glass of McMurtry et al. (2019), who report instantaneous isotopic fractionations up to a factor of 1000.

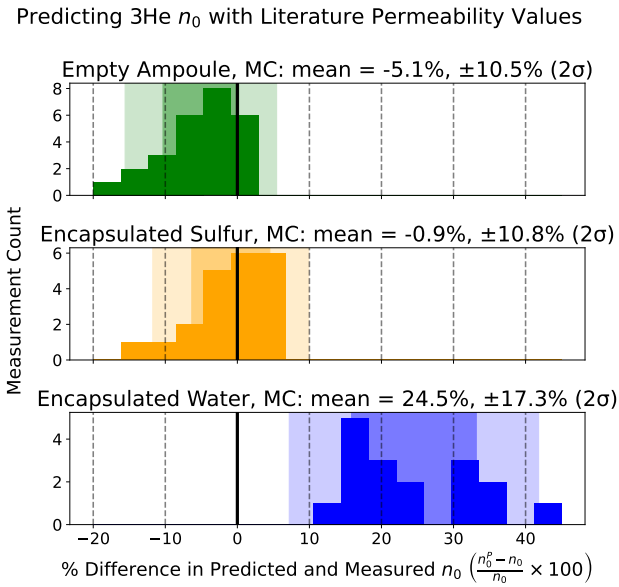


Figure 1.24: Similar to Fig. 1.9, but with 3He permeabilities taken from Jones (1953) and Swets et al. (1961). Shaded regions behind the histograms correspond to 1 and 2 sigma (σ) from the average relative difference between the predicted and measured values. The 3He permeation model and literature permeability values apply well to empty and sulfur-filled ampoules, and such models can be used to reliably recover 3He contents of permeating ampoules for use in permeation lag corrections. On the other hand, measured and predicted values of 3He n_0 for water-filled ampoules are well resolved beyond 2σ, suggesting that their permeability differs strongly from the literature values. Internal reproducibilities (1σ) of a gas standard repeatedly analyzed during these experiments were better than 3% for 3He, requiring that much of the scatter in these data be related to factors outside of analytical precision such as long/inconsistent ramp times during heating or uncertainties or rounding of heating times.

1.5s Additional Analytical Corrections for the Empty Ampoule Calibration Experiments

Pump-Out and Gas-Preparation Corrections for the Atmosphere-Equilibrated Empty Ampoule

Using the helium permeability values of Swets et al. (1961) for silica glass, an ampoule with a geometric factor (G) of 6000 cm^{-2} would permeate 0.2‰ of its helium in 1 minute at 26°C (Eq. (1.3)), which is negligible for the scope of this work. The atmosphere-equilibrated empty ampoule with a geometric factor (G) of ~4785 cm^{-2} would permeate 2.8‰ of its helium in its 15-minute pumping time between steps. For that experiment only, helium loss during pumping is accounted for by applying the permeation relationship of Eq. (1.3)

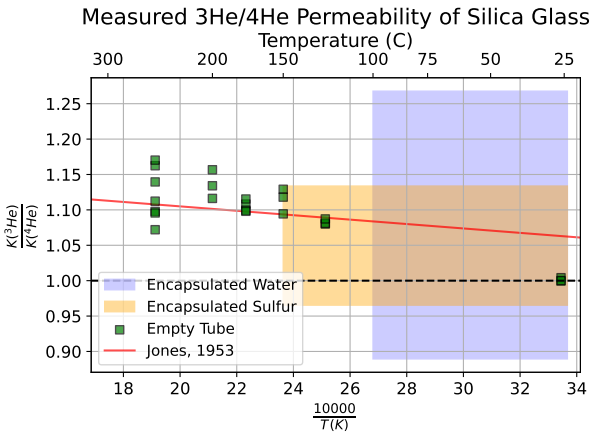


Figure 1.25: Ratios of calculated permeabilities from step heat experiments on silica glass ampoules on the Helix-MC-Plus^{10K} mass spectrometer at Caltech. The last measurement on the empty tube experiment is an outlier and was excluded from the plot. Experiments on encapsulated water and encapsulated sulfur were allowed to cool before isotopic measurements. Because of slow cooling rates, simple corrections could not be applied to these two experiments, which effectively queried permeabilities at a range of temperatures within a single step. Therefore, the ratios of their apparent permeabilities are plotted as a field with 2σ standard deviation across the experimental temperature range. These three experiments show general, albeit imprecise, agreement to the results of Jones (1953), except for the measurements at room temperature. Jones (1953) was likely able to measure the relative permeabilities of 4He and 3He with much higher precision because his experimental vessel was charged with 3He at a partial pressure of around 0.5 atmospheres, over one billion times higher than the 3He partial pressure within the fused silica ampoule used in this work. Interestingly, three room-temperature measurements of helium permeation of an empty tube at Caltech yielded essentially identical permeability of 3He and 4He.

($n_0 = n_t/(1 - {}^{He}F)$) to solve for n_0, the amount of helium in the ampoule before pump-out, from n_t, the amount remaining after pumping with T = 25°C and t = 15 minutes. A permeability coefficient (K) for helium in silica glass had to be assumed in order to apply this correction, so the value of 1.54×10^{10} $\frac{atoms}{s \times cm \times atm}$ at 25°C given by the Arrhenius law reported by Swets et al. (1961) was used. The validity of this choice is supported by the data; the average calculated permeability at room temperature for this experiment after correcting for helium lost during the 15-minute pump out is also 1.54×10^{10} $\frac{atoms}{s \times cm \times atm}$.

For that experiment only, a correction was also applied to the high temperature

steps because they were heated for short durations (45 minutes for all but one heated step), so that the permeation of the ampoule at room temperature during gas cryofocusing represented a significant portion of the step's yield. This was similarly corrected by calculating the expected fractional yield at room temperature during the gas preparation time, which is the step's total time minus the pump out time and the heated time, (29-31 minutes for all heated steps). For that correction, the permeation relationship of Eq. (1.3) ($n_0 = n_t/(1 - {}^{He}F)$) is applied again to solve for the unknown n_0, the amount of helium within the ampoule immediately after the completion of the heated step, from n_t, the amount of helium within the ampoule when the gas is inlet into the mass spectrometer about 30 minutes later after spiking and helium purification. This cooling and gas preparation correction is not applied to the room temperature measurements for which the temperature remained constant during gas preparation.

Anomalous Signal Recalculation for the Tank Helium-Soaked Empty Ampoule

The last high temperature step of the empty ampoule experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer had an unusual and non-linear signal evolution at the beginning of the measurement for both 3He and 4He. For this measurement, 4He was recalculated using a linear regression on the last three of six measurement cycles with an R^2 of 0.985 to regress to the 4He content at time 0. Typically, the signal evolution for a measurement like this would have a negative slope because the helium in the mass spectrometer is consumed as it is ionized and measured, but the slope for this measurement was positive. 3He from this measurement was calculated based on the recalculated 4He and an assumed isotope ratio for the measurement of 0.07 Ra. Three more measurements were taken over the next few days to ensure complete degassing of the ampoule. Two of these measurements yielded helium above blank, but were small compared to the final high temperature step. The yields of these steps were all considered in the calculation of the experiment's total yield.

Independent Verification of the Encapsulated Durango Apatite Mass

The total yield for the encapsulated Durango apatite experiment (main text, §2.3) is estimated from the precisely-measured mass of the encapsulated apatite (3.18 mg) and a helium concentration of 194.2 nccSTP/mg (Farley, 2000). This

concentration is independently supported by the measurement of 8 different >0.5 mg aliquots of the same Durango apatite sample in-lab, which yield a concentration of 194 ± 9 nccSTP/mg.

1.6s Sample Preparation and SEM Imaging of Buffered and Decomposed Sulfides

Samples of pyrite, arsenopyrite, and marcasite were crushed under ethanol, and grains larger than 100 µm were dried for each mineral and subdivided into four populations: one control, and three for heat treatment in atmosphere, ultra-high vacuum, or encapsulated within a glass ampoule along with an excess of fused sulfur (or realgar: As3S3 for arsenopyrite). The encapsulated populations were loaded into 3 mm inner diameter, 4 mm outer diameter silica glass tubes with fused bottoms and open tops, along with sufficient fused sulfur or crystalline realgar to guarantee vapor saturation at 400°C. These tubes were interfaced with a vacuum line and were pumped to 60 millitorr before the tops of the tubes were sealed.

The open-air populations for each mineral were carefully placed on glass microscope slides within borosilicate petri dishes and the encapsulated populations were submerged in pure quartz sand within capped, deep petri dishes. These petri dishes containing the encapsulated and open-air populations of sulfides were placed on a temperature controlling hot plate under a fume hood and were held at 400°C for 8-11 hours, at which point the hot plate was turned off and the samples were left to cool.

The ultra-high vacuum population for pyrite and marcasite were loaded into copper packets and mounted to a type K thermocouple within a high vacuum line. A silver wool based sulfur trap was installed between the samples and the turbo pump. The high-vacuum line was closed and pumped to <10-5 torr. A Watlow temperature controller held the samples at 400°C for 4.5 hours with a projector lamp illuminating the sample through a sapphire window. This procedure was repeated for each mineral sample of pyrite and marcasite. The arsenopyrite sample was not heated in vacuum to avoid contaminating the line with arsenic. To simulate thermal decomposition in an anoxic environment, a population of arsenopyrites was thermally treated in a large glass capsule similarly to the encapsulated populations, but without a realgar "buffer".

The control and thermally treated samples were mounted to carbon tape and

carbon coated, then were loaded in the Zeiss 1550 Scanning Electron Microscope (SEM) at Caltech. Images of the surface topography of each population were taken using the secondary electron detector and an Electron High Tension (EHT) of 10 kV at a working distance of 10-15 mm.

1.7s Supplementary Data Tables

The following tables present the reduced step-by-step helium permeation data underlying the empty-tube, encapsulated-water, and encapsulated-sulfur calibration experiments summarized in the ampoule dimensions table of the main text, Table 1.1 and the permeability/isotope-ratio comparison figures above. m is the measured helium abundance of the step; n_0 and n_t are the helium contents of the ampoule at the beginning and end of the step (Eq. (1.19)); n_t/n_0 predicted and measured are the fractional yields expected from the permeability of Swets et al. (1961) (and, for 3He, the isotope ratios of Jones (1953)) and as measured, respectively.

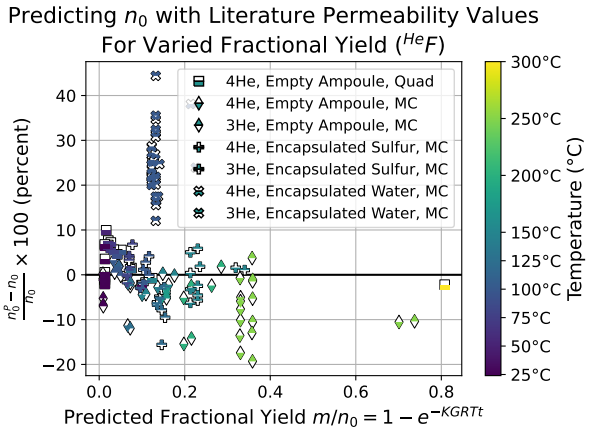


Figure 1.26: Helium measurements (m) of a permeating ampoule are related to the initial helium contents of the ampoule (n_0) by Eq. (1.21), using permeability values taken from Swets et al. (1961) and Jones (1953) (for 3He only). The n_0^P variable is the predicted value of n_0, the true value. For the empty ampoules, except for 3 measurements on the Helix-MC-Plus^{10K}, predicted values (n_0^P) are calculated using Eq. (1.19) (with some corrections applied to the quadrupole experiment: see main text for details). For the encapsulated sulfur and water ampoules, as well as 3 measurements of the blank ampoule on the Helix-MC-Plus^{10K}, predicted values (n_0^P) were determined by forward modeling diffusion and permeation on ampoules as they were held at high temperature and subsequently allowed to cool (see main text for details). This plot shows that the predicted values are correct to within ~10% for empty and sulfur-filled ampoules regardless of temperature or extent of permeation, whereas n_0^P values for water-filled ampoules are systematically high. The final Helix-MC-Plus^{10K} measurement on the empty ampoule yields an outlier n_0^P that was ~80% high — this measurement is not plotted here.

Table 1.4: Distilled helium permeation data for the empty silica glass ampoule equilibrated with atmosphere, measured by isotope dilution on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer ("Empty Tube, Pfeiffer Quad" in the ampoule dimensions table of the main text, Table 1.1). n_0 and n_t are the initial and final helium contents of the ampoule for each step (Eq. (1.19)); n_t "before cooling" is only populated for high-temperature experimental steps, for which the room-temperature pump-out/cool-down correction discussed in the main text is applied to obtain the "after cooling" value used to calculate permeability. $\ln(K)$ values are the natural log of the permeability coefficient in $\frac{atoms}{s \times cm \times atm}$.

Temp (°C)	Time (min)	m 4He (nccSTP)	n_0 4He (nccSTP)	n_t 4He (before cooling) (nccSTP)	n_t 4He (after cooling) (nccSTP)	n_t/n_0 measured	$10000/T$ (K^{-1})	$\ln(K)$ measured
25	77	0.1877	12.87		12.68	0.9854	33.56	23.52
25	77	0.1793	12.64		12.47	0.9858	33.56	23.49
50	45	0.2902	12.43	12.21	12.14	0.9819	30.96	24.19
24.8	74	0.1593	12.11		11.95	0.9868	33.58	23.46
100	45	0.7129	11.92	11.26	11.2	0.9452	26.81	25.18
24.7	74	0.145	11.17		11.03	0.9870	33.59	23.44
60	45	0.3094	11	10.75	10.69	0.9771	30.03	24.40
70	45	0.3619	10.66	10.35	10.3	0.9712	29.15	24.60
80	45	0.4225	10.27	9.902	9.849	0.9640	28.33	24.80
24.7	74	0.1272	9.823		9.696	0.9870	33.59	23.44
24.8	74	0.1252	9.67		9.545	0.9871	33.58	23.44
24.8	75	0.1238	9.519		9.395	0.9870	33.58	23.43
70	45	0.3209	9.37	9.098	9.049	0.9709	29.15	24.61
60	45	0.2534	9.025	8.82	8.771	0.9773	30.03	24.39
24.8	74	0.113	8.748		8.635	0.9871	33.58	23.44
90	45	0.4231	8.611	8.231	8.188	0.9558	27.55	24.98
119.8	45	0.6625	8.166	7.543	7.504	0.9236	25.46	25.47
24.8	74	0.09578	7.484		7.388	0.9872	33.58	23.43
150	45	0.9371	7.368	6.465	6.431	0.8775	23.64	25.89
25	581	0.6179	6.414		5.796	0.9037	33.56	23.43
300	90	4.549	5.78	1.238	1.231	0.2141	17.45	27.36
24.7	74	0.01673	1.228		1.211	0.9864	33.59	23.49
400	90	1.178	1.208		0.03073			
24.7	74	0.00117	0.03065					

Table 1.5: Distilled 3He permeation data for the empty silica glass ampoule equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Empty Tube, Helix MC" in the ampoule dimensions table of the main text, Table 1.1). Rows marked '–' in the predicted and measured n_t/n_0 and permeability columns are cooling-only intervals for which no isothermal permeability was calculated (see main text).

Temp (°C)	Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He K measured ($\frac{atoms}{s\ cm\ atm}$)
26	73.6	0.0516	6.16×10^{-4}	5.787	5.736	0.9902	0.9911	1.53×10^{10}
–	113.3	0.269	3.89×10^{-3}	5.736	5.467			
26	73.7	0.048	8.51×10^{-4}	5.467	5.419	0.9902	0.9912	1.50×10^{10}
26	75.3	0.0485	9.92×10^{-4}	5.419	5.37	0.9900	0.9910	1.50×10^{10}
–	112.9	0.271	5.67×10^{-3}	5.37	5.099			
–	126.6	0.418	8.80×10^{-3}	5.099	4.682			
125	52.9	0.299	6.31×10^{-3}	4.682	4.382	0.9275	0.9361	1.18×10^{11}
125	53.1	0.311	6.57×10^{-3}	4.382	4.071	0.9273	0.9290	1.31×10^{11}
125	53.1	0.291	6.15×10^{-3}	4.071	3.78	0.9273	0.9285	1.32×10^{11}
–	73.6	0.422	8.67×10^{-3}	3.78	3.358			
150	73.6	0.502	8.56×10^{-3}	3.358	2.856	0.8512	0.8506	1.95×10^{11}
150	52.9	0.299	4.47×10^{-3}	2.856	2.557	0.8907	0.8952	1.86×10^{11}
150	88.4	0.45	6.28×10^{-3}	2.557	2.107	0.8240	0.8240	1.94×10^{11}
175	53.1	0.318	4.55×10^{-3}	2.107	1.788	0.8427	0.8489	2.59×10^{11}
175	53.1	0.272	4.00×10^{-3}	1.788	1.516	0.8427	0.8478	2.61×10^{11}
175	53.1	0.228	3.54×10^{-3}	1.516	1.288	0.8427	0.8498	2.57×10^{11}
175	53.1	0.193	3.15×10^{-3}	1.288	1.095	0.8427	0.8504	2.56×10^{11}
–	53.1	0.164	2.82×10^{-3}	1.095	0.9317			
200	53.2	0.174	3.23×10^{-3}	0.9317	0.7582	0.7843	0.8139	3.07×10^{11}
200	53.1	0.16	3.16×10^{-3}	0.7582	0.5978	0.7847	0.7887	3.55×10^{11}
200	73.6	0.175	3.80×10^{-3}	0.5978	0.4224	0.7146	0.7070	3.75×10^{11}
250	53.2	0.152	3.93×10^{-3}	0.4224	0.2704	0.6410	0.6406	6.02×10^{11}
250	53.1	0.0977	2.95×10^{-3}	0.2704	0.1727	0.6413	0.6396	6.06×10^{11}
250	53.1	0.0607	2.12×10^{-3}	0.1727	0.1119	0.6413	0.6499	5.85×10^{11}
250	53.1	0.0379	1.56×10^{-3}	0.1119	0.07407	0.6414	0.6639	5.58×10^{11}
250	52.9	0.0249	1.13×10^{-3}	0.07407	0.04914	0.6427	0.6634	5.57×10^{11}
250	53.1	0.016	7.70×10^{-4}	0.04914	0.03315	0.6415	0.6746	5.32×10^{11}
250	159.9	0.026	1.31×10^{-3}	0.03315	0.007191	0.2625	0.2169	6.86×10^{11}
250	52.9	0.00494	2.52×10^{-4}	0.007191	0.00225	0.6424	0.3129	1.58×10^{12}
High T	52.2	0.00225		0.00225	0			

Table 1.6: Distilled 4He permeation data for the same empty silica glass ampoule experiment as Table 1.5. $K(3/4)$ is the ratio of the measured 4He to 3He permeability coefficients for each step; 3/4 He measured (Ra) is the measured 3He/4He ratio of the step, normalized to the atmospheric ratio.

Temp (°C)	Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	4He K measured ($\frac{atoms}{s\,cm\,atm}$)	K_3/K_4	3He/4He measured (Ra)
26	73.6	263.7	1.33	29670	29410	0.9908	0.9911	1.52×10^{10}	1.0031	0.1408
–	113.3	1282	7.99	29410	28130					0.1509
26	73.7	247.4	1.95	28130	27880	0.9908	0.9912	1.51×10^{10}	0.9984	0.1396
26	75.3	250	2.34	27880	27630	0.9905	0.9910	1.50×10^{10}	0.9990	0.1397
–	112.9	1322	13.1	27630	26310					0.1475
–	126.6	1971	20	26310	24340					0.1524
125	52.9	1446	14.9	24340	22890	0.9332	0.9406	1.09×10^{11}	1.0786	0.1489
125	53.1	1509	15.6	22890	21380	0.9330	0.9341	1.21×10^{11}	1.0795	0.1483
125	53.1	1413	14.6	21380	19970	0.9330	0.9339	1.21×10^{11}	1.0856	0.1483
–	73.6	2032	20.8	19970	17940					0.1495
150	73.6	2399	22.2	17940	15540	0.8630	0.8663	1.73×10^{11}	1.1271	0.1505
150	52.9	1498	12.7	15540	14040	0.8996	0.9036	1.70×10^{11}	1.0922	0.1438
150	88.4	2238	16.7	14040	11800	0.8377	0.8406	1.74×10^{11}	1.1155	0.1447
175	53.1	1626	10.8	11800	10170	0.8556	0.8622	2.34×10^{11}	1.1053	0.1409
175	53.1	1404	8.46	10170	8771	0.8556	0.8620	2.34×10^{11}	1.1121	0.1395
175	53.1	1211	6.28	8771	7560	0.8556	0.8619	2.34×10^{11}	1.0959	0.1353
175	53.1	1042	4.8	7560	6518	0.8556	0.8622	2.34×10^{11}	1.0939	0.1332
–	53.1	895.4	3.67	6518	5623					0.1314
200	53.2	952.2	3.41	5623	4670	0.8020	0.8306	2.77×10^{11}	1.1107	0.1311
200	53.1	887.5	2.99	4670	3783	0.8023	0.8100	3.15×10^{11}	1.1275	0.1300
200	73.6	987.1	3.27	3783	2796	0.7370	0.7391	3.26×10^{11}	1.1485	0.1278

continued on next page

(continued)

Temp (°C)	Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	4He K measured $(\frac{atoms}{s\,cm\,atm})$	K_3/K_4	3He/4He measured (Ra)
250	53.2	898	3.01	2796	1898	0.6693	0.6788	5.23×10^{11}	1.1520	0.1218
250	53.1	609.8	2.07	1898	1288	0.6696	0.6787	5.24×10^{11}	1.1569	0.1153
250	53.1	412.2	1.42	1288	875.7	0.6696	0.6799	5.21×10^{11}	1.1231	0.1059
250	53.1	278.8	0.995	875.7	596.9	0.6697	0.6817	5.18×10^{11}	1.0778	0.0978
250	52.9	187.7	0.689	596.9	409.2	0.6709	0.6855	5.13×10^{11}	1.0868	0.0955
250	53.1	128.8	0.486	409.2	280.4	0.6697	0.6852	5.11×10^{11}	1.0411	0.0893
250	159.9	215.4	0.953	280.4	64.98	0.2990	0.2318	6.56×10^{11}	1.0452	0.0867
250	52.9	41.86	0.205	64.98	23.12	0.6706	0.3558	1.40×10^{12}	1.1246	0.0849
High T	52.2	23.12		23.12	0					0.0700

Table 1.7: Distilled 3He isothermal permeation data for the silica glass ampoule containing distilled water, equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Encapsulated Water, Helix MC" in the ampoule dimensions table of the main text, Table 1.1). Predicted n_t/n_0 values marked '–' correspond to steps for which the ampoule had not yet reached a diffusive steady state concentration profile (see main text).

Temp (°C)	Hot Time (min)	Cold Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured
100	60	26	0.116	1.22×10^{-3}	0.6447	0.5289	–	0.8204
100	60	25.1	0.0942	1.00×10^{-3}	0.5289	0.4347	0.8677	0.8219
100	60	25	0.0757	8.06×10^{-4}	0.4347	0.359	0.8678	0.8259
100	60	25.1	0.0578	6.18×10^{-4}	0.359	0.3012	0.8678	0.8389
100	60	53.5	0.053	5.72×10^{-4}	0.3012	0.2482	0.8589	0.8241
100	60	25.2	0.0382	4.34×10^{-4}	0.2482	0.21	0.8678	0.8459
100	60	25.1	0.0337	4.06×10^{-4}	0.21	0.1763	0.8678	0.8396
100	60	795.6	0.0493	2.52×10^{-4}	0.1763	0.127	0.7741	0.7201
100	60	28.1	0.0197	9.42×10^{-5}	0.127	0.1072	0.8666	0.8446
100	60	43.7	0.0173	8.39×10^{-5}	0.1072	0.08988	0.8622	0.8382
100	60	24.9	0.0138	6.99×10^{-5}	0.08988	0.0761	0.8673	0.8467
100	60	24.9	0.0132	7.60×10^{-5}	0.0761	0.06295	0.8678	0.8272
100	60	25	0.00977	6.32×10^{-5}	0.06295	0.05318	0.8678	0.8448
100	60	25.1	0.00952	6.36×10^{-5}	0.05318	0.04366	0.8679	0.8210
100	60	25.1	0.00764	5.16×10^{-5}	0.04366	0.03602	0.8678	0.8250
100	60	25	0.00572	3.86×10^{-5}	0.03602	0.0303	0.8678	0.8413
100	60	25	0.00578	3.90×10^{-5}	0.0303	0.02451	0.8678	0.8091
100	60	25.1	0.00363	2.45×10^{-5}	0.02451	0.02089	0.8679	0.8520
REST			0.0209	1.39×10^{-4}	0.02089	0		

Table 1.8: Distilled 4He isothermal permeation data for the same encapsulated-water ampoule experiment as Table 1.7.

Temp (°C)	Hot Time (min)	Cold Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He/4He measured (Ra)
100	60	26	604	1.77	3871	3267	–	0.8440	0.1379
100	60	25.1	515.5	1.52	3267	2751	0.8772	0.8422	0.1315
100	60	25	434.8	1.29	2751	2317	0.8772	0.8420	0.1252
100	60	25.1	361.9	1.07	2317	1955	0.8772	0.8438	0.1150
100	60	53.5	325.4	0.965	1955	1629	0.8688	0.8335	0.1171
100	60	25.2	251.3	0.745	1629	1378	0.8772	0.8458	0.1095
100	60	25.1	212.6	0.63	1378	1165	0.8772	0.8457	0.1140
100	60	795.6	344.8	0.802	1165	820.7	0.7858	0.7042	0.1029
100	60	28.1	127.2	0.276	820.7	693.5	0.8761	0.8450	0.1116
100	60	43.7	111.3	0.222	693.5	582.2	0.8719	0.8395	0.1122
100	60	24.9	88.54	0.166	582.2	493.7	0.8768	0.8479	0.1120
100	60	24.9	74.96	0.121	493.7	418.7	0.8773	0.8481	0.1262
100	60	25	62.69	8.84×10^{-2}	418.7	356	0.8772	0.8503	0.1121
100	60	25.1	53.28	7.18×10^{-2}	356	302.7	0.8773	0.8503	0.1285
100	60	25.1	45.45	6.11×10^{-2}	302.7	257.3	0.8772	0.8499	0.1209
100	60	25	38.41	5.30×10^{-2}	257.3	218.9	0.8773	0.8507	0.1071
100	60	25	32.42	4.81×10^{-2}	218.9	186.4	0.8772	0.8519	0.1284
100	60	25.1	27.46	4.35×10^{-2}	186.4	159	0.8773	0.8527	0.0950
REST			159	0.258	159	0			0.0945

Table 1.9: Distilled 3He isothermal permeation data for the silica glass ampoule containing fused sulfur, equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Encapsulated Sulfur, Pfeiffer Quad" in the ampoule dimensions table of the main text, Table 1.1; note this isothermal calibration series was run on the Helix-MC-Plus^{10K}).

Temp (°C)	Hot Time (min)	Cold Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured
80	60	25.6	0.699	2.78×10^{-2}	9.569	8.87	0.9299	0.9270
80	60	61.3	0.736	2.89×10^{-2}	8.87	8.134	0.9222	0.9170
80	60	25.1	0.591	2.17×10^{-2}	8.134	7.543	0.9300	0.9274
80	60	25	0.544	1.83×10^{-2}	7.543	6.999	0.9301	0.9279
80	60	25	0.494	1.60×10^{-2}	6.999	6.505	0.9301	0.9294
100	60	91.8	0.759	2.42×10^{-2}	6.505	5.746	0.8831	0.8834
100	60	25.9	0.601	2.01×10^{-2}	5.746	5.145	0.8966	0.8955
100	60	52.0	0.575	2.06×10^{-2}	5.145	4.57	0.8912	0.8882
100	60	26	0.463	1.72×10^{-2}	4.57	4.108	0.8966	0.8988
100	60	89.9	0.498	2.04×10^{-2}	4.108	3.609	0.8834	0.8787
120	60	24.8	0.51	2.27×10^{-2}	3.609	3.099	0.8537	0.8586
120	60	52.8	0.457	2.18×10^{-2}	3.099	2.641	0.8481	0.8524
120	60	24.9	0.356	1.76×10^{-2}	2.641	2.286	0.8537	0.8653
120	60	61.3	0.318	1.56×10^{-2}	2.286	1.967	0.8460	0.8607
120	60	25.1	0.243	1.17×10^{-2}	1.967	1.724	0.8537	0.8766
150	60	25.0	0.42	2.00×10^{-2}	1.724	1.304	0.7695	0.7562
150	60	25.0	0.283	1.32×10^{-2}	1.304	1.021	0.7696	0.7828
150	60	690.8	0.351	1.03×10^{-2}	1.021	0.6699	0.6598	0.6563
150	60	25	0.147	3.98×10^{-3}	0.6699	0.5226	0.7696	0.7802
150	60	53.3	0.125	3.33×10^{-3}	0.5226	0.3972	0.7643	0.7601
150	60	25.0	0.0893	2.40×10^{-3}	0.3972	0.308	0.7695	0.7752
remainder			0.308	8.48×10^{-3}				

Table 1.10: Distilled 4He isothermal permeation data for the same encapsulated-sulfur ampoule experiment as Table 1.9.

Temp (°C)	Hot Time (min)	Cold Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He/4He measured (Ra)
80	60	25.6	3354	10.8	52330	48980	0.9348	0.9359	0.1499
80	60	61.3	3617	12	48980	45360	0.9276	0.9262	0.1464
80	60	25.1	2949	10	45360	42410	0.9349	0.9350	0.1441
80	60	25	2760	9.6	42410	39650	0.9350	0.9349	0.1419
80	60	25	2587	9.13	39650	37060	0.9350	0.9348	0.1373
100	60	91.8	3980	15.3	37060	33080	0.8913	0.8926	0.1371
100	60	25.9	3088	13.5	33080	30000	0.9041	0.9066	0.1399
100	60	52.0	2978	14.7	30000	27020	0.8990	0.9007	0.1389
100	60	26	2517	13.1	27020	24500	0.9041	0.9068	0.1322
100	60	89.9	2687	14.4	24500	21810	0.8916	0.8903	0.1334
120	60	24.8	2769	14.6	21810	19040	0.8646	0.8731	0.1327
120	60	52.8	2551	13	19040	16490	0.8592	0.8661	0.1290
120	60	24.9	2097	10.4	16490	14400	0.8646	0.8728	0.1221
120	60	61.3	1948	9.38	14400	12450	0.8572	0.8647	0.1176
120	60	25.1	1573	7.36	12450	10870	0.8645	0.8736	0.1110
150	60	25.0	2435	11.3	10870	8439	0.7868	0.7761	0.1242
150	60	25.0	1806	8.2	8439	6633	0.7868	0.7859	0.1128
150	60	690.8	2151	8.5	6633	4481	0.6807	0.6756	0.1173
150	60	25	924.6	3.42	4481	3557	0.7868	0.7937	0.1146
150	60	53.3	753.9	2.75	3557	2803	0.7818	0.7880	0.1197
150	60	25.0	560.4	2.08	2803	2242	0.7868	0.8001	0.1146
remainder			2242	8.72					0.0988

Chapter 2

HELIUM STEP HEAT EXPERIMENTS ON GOETHITE IN HYDROTHERMAL CONDITIONS INDICATE HIGH RETENTIVITY AT SURFACE TEMPERATURE AND CONSISTENCY WITH MULTIPLE DIFFUSION DOMAIN SYSTEMATICS

U-Th/He geochronology of Goethite (FeOOH) is a powerful method to quantitatively date weathering and erosion processes on Earth's surface; however, precise interpretation of goethite U-Th/He dates is challenging because uncertainty persists regarding the mineralogical and environmental conditions under which radiogenic helium is retained and how it may be lost over time. Standard vacuum step heating experiments have been successfully used on a variety of mineral phases to characterize helium migration, but the results of such experiments on goethite have been dominated by dehydration-induced helium loss, raising questions about their relevance to helium retention and loss in nature (Farley, H. B. Monteiro, et al., 2024). Observations that thermal decomposition of goethite is greatly slowed in the presence of high water pressure led us to develop a new method to conduct helium step heat experiments under hydrothermal conditions (up to 86 bars and 300°C) within standard experimental setups. Using this method, we report results from the first helium step heat experiment of proton-irradiated goethite under hydrothermal conditions. Irradiated goethite sample Win-06-01b of the Flinders Ranges, Southern Australia and distilled water were encapsulated in a pure silica glass ampoule and were subjected to multiple prograde and retrograde thermal cycles over a temperature range of 170°C to 250°C under water vapor saturated conditions. Step durations varied from 2 hours to over a week, with low-temperature (<120°C) permeation steps bracketing higher-temperature steps to account for residual helium within the ampoule. The results show apparent helium diffusivities approximately seven log units lower than those observed in vacuum, with apparent diffusivities following a consistent linear

array on an Arrhenius plot for cumulative fractional 3He loss exceeding 7%. Delta plots generated using this reference array with a calculated activation energy of 157.1± 5.3kJ/mol are consistent with a volume-diffusion helium loss mechanism with MDD (Multiple Diffusion Domain) systematics. These results show high helium retentivity in natural goethite under hydrothermal conditions that extrapolate to robust helium retention at surface conditions even over billion-year timescales. The method presented in this work can be systematically applied to experimentally investigate effects of crystallite size, chemical composition, and radiation damage to refine these results and to assess the applicability of existing vacuum step heat data to helium retention and loss in goethite in nature.

2.1 Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024). It is uniquely suited to date weathering and ferruginization processes on Earth's surface, including laterite/duricrust formation and recrystallization (H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Allard et al., 2018), and ferruginization of organic matter or paleosols (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann, Reichenbacher, et al., 2017).

As with any radiochronometer, including U-Th/He geochronology and thermochronometry, knowledge of the conditions under which the system is open to loss or gain of parent or daughter nuclides from the environment is essential to make meaningful geological interpretations based on measured ages. For goethite U-Th/He geochronology, step heats of irradiated samples are often used to reveal the degree of partial helium loss using the 4He/3He method (Shuster, P. M. Vasconcelos, et al., 2005), with reported losses always non-zero but relatively low, ranging from 2-30% (Shuster, P. M. Vasconcelos, et al., 2005; Heim et al., 2006; Hofmann, Reichenbacher, et al., 2017; Deng et al., 2017). Although one study attempts to explain the variable loss principally as a result of variable radiation damage (Bassal et al., 2022), the effects of temperature, crystallite size and chemical composition on helium retentivity in goethite have not been experimentally determined and remain elusive (Farley, H. B. Monteiro, et al., 2024).

Laboratory experiments of helium migration under controlled conditions have proven extremely useful in understanding the mechanisms and rates of helium migration in geological materials and have produced sensible results that are predominantly explainable by thermally activated volume diffusion (Reiners and Farley, 1997; Farley, 2000; Farley, 2007; Cherniak and Watson, 2011; Farley, 2018), albeit variably complicated by the effects of radiation damage and annealing (Shuster, Flowers, et al., 2006; Guenthner et al., 2013). This approach has also been leveraged for goethite; however, some complications arise when applying this approach to goethite. Above surface temperatures, goethite decomposes to hematite and water, often along crystallite elongation directions (Goss, 1987). Farley et al. (2024) showed experimentally that helium and water release are highly correlated even at temperatures less than 200°C, suggesting that goethite decomposition into hematite is frequently the dominant process governing helium migration during vacuum step-heating experiments. The results of in-vacuum experiments therefore need not reflect helium migration in goethite in nature.

Still, existing helium step heat measurements of goethite in vacuum likely remain informative to an extent, as goethite dehydration kinetics are highly influenced by some of the same factors that could be responsible for variable helium retention in natural goethite, like variable grain size (Navrotsky et al., 2008; Guo and Barnard, 2011; Goss, 1987), crystallinity (Schwertmann et al., 1985) or substitution of aluminum or other cations within the goethite structure (Majzlan et al., 2003; Bassal et al., 2022). For example, Shuster et al. (2005) initially conducted vacuum step-heating experiments on two vitreous, polycrystalline goethite samples, in which uniformly distributed 3He was introduced by proton irradiation for comparison with radiogenic 4He that was partially lost in nature. The results of Shuster, P. M. Vasconcelos, et al. (2005) could be well-fit with a model of two distinct diffusion domains or migration mechanisms with significant differences in helium retention under surface conditions, e.g. crystallite interiors and intergranular space. Although these results may have been impacted by dehydration in vacuum, certain aspects may remain applicable to helium loss in goethite in nature if helium loss in nature and the laboratory advance through the samples with similar geometries. One important point is that if laboratory experiments degas domains in order of their helium retentivity in nature, even if the rates of loss are affected by goethite dehydration, then a $^4He/^3He$ experiment in vacuum can still be used to

measure helium loss (Shuster, P. M. Vasconcelos, et al., 2005; P. M. Vasconcelos et al., 2013).

An additional complexity is that natural goethite almost always occurs as aggregates of minute crystallites with cross-sections smaller than 1 µm in nature. Neglecting to account for multiple diffusion domains can lead to inaccurate determinations of activation energy (E_a) and frequency factor (D_o/a^2) for helium diffusion (Farley, 2018). Detailed step-heating with temperature cycling, as has been done for hematite (Farley and Flowers, 2012; Farley, 2018), is needed to characterize domain size and activation energy for polycrystalline samples.

Molecular simulations indicate that helium diffusion in perfectly crystalline goethite should be highly anisotropic, essentially reducing to 1-dimensional diffusion along the b-axis (typically the long axis of the polycrystalline aggregate) that would amount to near-instantaneous helium escape from goethite (Bassal et al., 2022), which is clearly disputed by observed (U–Th)/He dates as old as ~275 Ma (Waltenberg, 2012) and dates that are concordant with $^{40}Ar/^{39}Ar$ ages of other phases (P. M. Vasconcelos et al., 2013). However, the models presented by Bassal et al. (2022) suggest that crystallographic defects caused by radiation damage could block these channels, thereby reducing He loss. While these simulations are informative, they alone are insufficient for drawing conclusions about the mineralogical and environmental conditions needed for accurately dating the crystallization of goethite with the (U–Th)/He method and require experimental verification.

Until now, no studies have reported helium step-heating results for goethite under hydrothermal (high pH_2O) conditions, which could help stabilize goethite or slow its decomposition to illuminate the processes relevant to helium loss in natural settings. If experimental conditions can be controlled to slow the decomposition of goethite such that the rate of natural helium loss in goethite greatly exceeds the rate of goethite decomposition into hematite and water, it would open up an experimental approach to systematically assess how factors like crystallite size, temperature, radiation damage, and chemical composition could affect helium mobility and retentivity in nature, enabling estimates of helium retention in previously untested natural samples.

2.2 Experimental Methods

This study builds upon previous helium step heat studies of the dense colloform goethite sample Win-06-01b (occasionally referred to as Win0601b) collected within float at the Flinders Ranges in Southern Australia (Waltenberg, 2012; H. Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages: 180-135 Ma (Waltenberg, 2012). To produce a homogeneous distribution of 3He, Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster and Farley, 2005) and subsequently was subjected to helium step heat measurements in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). In their step heat experiment of Win-06-01b, Farley, H. B. Monteiro, et al. (2024) additionally measured evolved water using capacitance manometry to show that helium and water release were tightly correlated for goethite step heats in vacuum. Although they used subtly different heating schedules, the results of Farley, H. B. Monteiro, et al. (2024) clearly agree well with H. Monteiro (2017).

In this study, an aliquot of the same irradiated Win-06-01b goethite used by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) was encapsulated with MilliQ water within a silica glass tube. Silica glass was chosen over other laboratory glasses such as borosilicate because it is the most permeable glass type to helium (Norton, 1953; Altemose, 1961). To prepare the silica glass tube, 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen glass-welding torch and the pipette ends were fused closed. After the glass had cooled, 1.18 mg of irradiated Win-06-01b goethite blades less than 2mm in length were carefully transferred to the bottom of the tube in the presence of an ionizing fan. Less than a microliter of distilled water was transferred to the sealed pipette using a thin glass capillary, submerging the goethite, after which the tube was gently tapped to liberate the bubbles and ensure that the water and goethite were efficiently packed at the bottom of the pipette. To seal the pipette, the contents of the ampoule were flash-frozen in liquid nitrogen and transferred to a saltwater-ice bath at a temperature of -10°C. While the pipette end of the tube remained frozen in the saltwater-ice bath, the top was interfaced with a vacuum line and the tube was pumped

to a pressure of <0.1 torr. The advantage of this approach was that water would remain solid within the tube, only slowly sublimating to vacuum while atmospheric gasses would be readily pumped away. While it was desirable to remove atmospheric gasses to reduce experimental complexity by removing potentially reactive gasses like oxygen, the primary motivation for pumping the tube to rough vacuum was to aid the sealing process. When the tube contents were under vacuum, fused glass would deform inwards and self-seal, greatly improving the ease and quality of the seal while requiring far less heat input from the torch. After an acceptably low pressure was attained, the tube was closed to the vacuum line and the saltwater-ice bath was quickly replaced with a liquid nitrogen bath, ensuring that no ice was melted. The liquid nitrogen bath was preferable during sealing as it more effectively protected the goethite and ice from the high temperatures of the glass-welding torch. A fine-flamed welding torch was used to seal the top of the ampoule, which dropped into the liquid nitrogen bath and was retrieved by passing the liquid nitrogen through a sieve. Fig. 2.1 shows the sealed ampoule after the completion of the step heats with all of the original water remaining, a testament to the quality of the ampoule's seal, which had sustained a massive pressure gradient of over 11 orders of magnitude across the ampoule walls, with an inner vapor pressure of about $3x10^4$ torr (~40 bars: Haar et al. (1984)) against the ~10^{-7} torr ultra-high vacuum of the lamp cell while the ampoule was held at 250°C for over a week.



Figure 2.1: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. After the experiment, a slight reddish tint was visible on the thin edges of some goethite grains, which is likely a small amount of hematite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

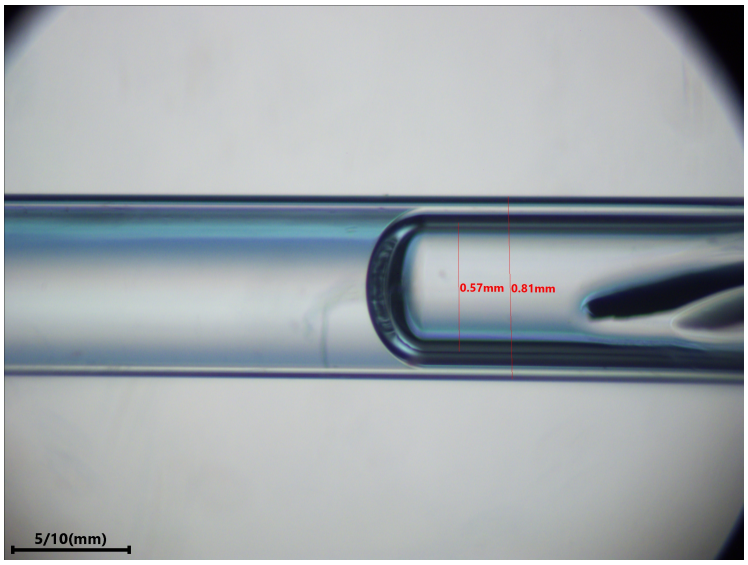


Figure 2.2: Measurements of the inner and outer diameter of the ampoule used in the hydrothermal step heats of Win-06-01b. The inner walls of the ampoule are only clearly visible within air bubbles.

The sealed ampoule, now referred to as HT-Win-06-01b, was placed in a copper foil packet and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley, Reiners, et al. (1999). Before the experiment began, the ampoule was loaded in the lamp cell, pumped to high vacuum, and was held at 70°C for 4 days to degas atmospheric helium entrained within the ampoule.

HT-Win-06-01b was subjected to multiple prograde-retrograde heating cycles to investigate possible MDD behavior. Step durations varied from 2 hours to about a week; the total experiment was conducted over about 40 days and included experimental steps ranging from 170-250°C. We elected not to heat the sample hotter because high temperatures generated very high vapor pressures. In early trials, we found that similar ampoules most often withstood the ~86 bar vapor pressures associated with liquid water saturation at 300°C (Haar et al., 1984), but they would explode when further heated to temperatures beyond 315°C.

Experimental temperature records show ~1°C amplitude oscillations around the target temperature. While we cannot exclude the possibility that sustained temperature gradients were present throughout the sample due to the use of the comparatively large ~1 mm thick ampoule, we are confident that the conclusions of this work are robust. The ampoule was in direct contact with the thermocouple junction and was in front of thermocouple junction with respect to the projector lamp heating source, suggesting that if there were a temperature discrepancy, the sample would've been hotter than the recorded temperature, which wouldn't affect the conclusions of this work. Additionally, during early trials of this method for helium step heats, silica-glass encapsulated Durango apatite had been subjected to step heats at Caltech and gave indistinguishable corrected 4He yields to unencapsulated Durango apatite, suggesting that the temperature recorded by the thermocouple for our experimental setup is sufficiently accurate.

Helium isotope measurements of evolved gasses from hydrothermal steps heats of glass-encapsulated irradiated HT-Win-06-01b goethite were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Helium was purified by exposing evolved gasses to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding

into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray, which was never significant during the experiment. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility.

A cold blank was measured at the beginning of the experiment, showing a negligible 3He blank essentially at 0 cps (technically this measurement yielded a slightly negative intercept to time 0) and a 4He blank signal corresponding to about 1.25 pccSTP, which was similar to 4He values attained during low temperature permeation step measurements. Permeation steps were thus assumed to measure shifting 4He blanks during the experiment, and the entire blank correction was added to all 4He uncertainties.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight in a 50°C oven, then completely degassed by heating to ~1100 °C in a resistance furnace. The final step measurement was conducted on a Thermo Fisher Helix SFT mass spectrometer. A hot furnace blank subjected to the same volume splits as the final HT-Win-06-01b measurement gave a 3He blank of 6.8 fccSTP and a 4He blank of 0.31 nccSTP. This blank correction was less than 1% of the measured 4He and negligible for 3He. A re-extract of HT-Win-06-01b immediately following the measurement yielded above-blank 3He and 4He signals, which were added to the total measurement, although both constituted less than 1% of the final step yield. Sensitivity was calculated by measuring a 3He-enriched air standard with known 3He and 4He contents immediately before the measurement.

Permeation Lag Correction Because HT-Win-06-01b was encapsulated in an imperfectly helium-transmissive silica glass ampoule, permeation lag was an

expected complication in these measurements. To address this, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile at experimental timescales, following Thyer et al., 2026. For 3He, measured signals at these low temperature permeation steps were entirely controlled by the process of permeation through silica glass, and therefore could be used to estimate the amount of residual 3He within the ampoule between high-temperature experimental steps. However, for 4He, the measured signals at low temperature permeation steps were most often indistinguishable from the blank, complicating the application of the Naive Bracketing approach as discussed in Thyer et al., 2026 to correct for permeation lag in this dataset.

As a result, the permeation lag is instead addressed using the "Treatment as Uncertainty" approach described by Thyer et al., 2026. In this approach, an upper bound constraint for the helium content of the ampoule at the end of each high-temperature step was calculated using Eq. (1.9) (Chapter 1).

The application of this equation requires knowledge of the lower-bound permeability of the system. The permeability of ampoules used in hydrothermal step heat experiments cannot reasonably be assumed to match literature values for anhydrous silica glass, and therefore the 3He permeability of the ampoule was determined using sequential isothermal permeation steps as described in the naive method outlined by Thyer et al., 2026.

10 series of 3-4 consecutive permeation steps held at a single temperature between 90-110°C for 120-180 minutes each with 26 minutes of cooling time before being inlet into the Helix-MC-Plus10k were identified as suitable for this exercise. The 3He measurements in these series were fit to Eq. (1.8) using a non-linear least squares optimization (Fig. 2.3). This is accomplished with the Levenberg-Marquardt algorithm implemented in the curve_fit function in the SciPy package of python (Virtanen et al., 2020), weighted by each measurement's own uncertainty. One additional series that otherwise qualified was not well fit by this model and was not used further for clarity, though its fit value of λ was within uncertainty of the other series' fit values for λ.

An initial guess of $\lambda \approx 6\text{x}10^{-6}$ sec^{-1} and $n_0 \approx \frac{m_i}{1-e^{-\lambda t_i}}$ (evaluated at the first measurement of a series) pccSTP was passed to the optimizer for each series. Uncertainties from the model fits were calculated using the covariance matrices generated for each of the model fits. Although the initial helium amount n_0

was a free parameter in the optimizations, the rate constant of permeation (λ) was the only parameter of interest for this exercise:

All measured apparent 3He permeabilities were equal to or in excess of the literature values for anhydrous silica glass (Fig. 2.4), similar to the observations of Thyer et al., 2026. 4He permeability itself could not be independently measured in this experiment due to blank-level signals during the low temperature permeation steps and is assumed to similarly exceed the literature values for anhydrous silica glass. To evaluate the uncertainty due to the permeation lag, we use the literature values as a conservative (lower bound) estimate of the true permeability. λt (where $\lambda = GRTK$) is estimated here using the 4He permeability K reported by Swets et al., 1961, while the 3He permeability is taken to be the product of the 4He permeability and the relative 3He/4He permeability reported by Jones, 1953. The geometric constant G is calculated using the measured inner and outer diameter of the ampoule (Fig. 2.2).

The first three experimental steps had values of λt lower than 4, and therefore the upper bound of the helium content of these steps could not be determined. These measurements are omitted from further analysis. Later experimental steps with values of λt lower than 4 could be determined using Eq. (1.56) from the supplement to Chapter 1 because upper bounds on previous steps' residual helium content could be evaluated.

2.3 Analytical Methods

Finite Difference Diffusion Model of Cooling Tube One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder containing an arbitrary amount of helium gas within a helium-free chamber using the procedure described in (Thyer et al., 2026). The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step and a steady state concentration profile was assumed for the new arbitrary charge at T = 24°. Permeability

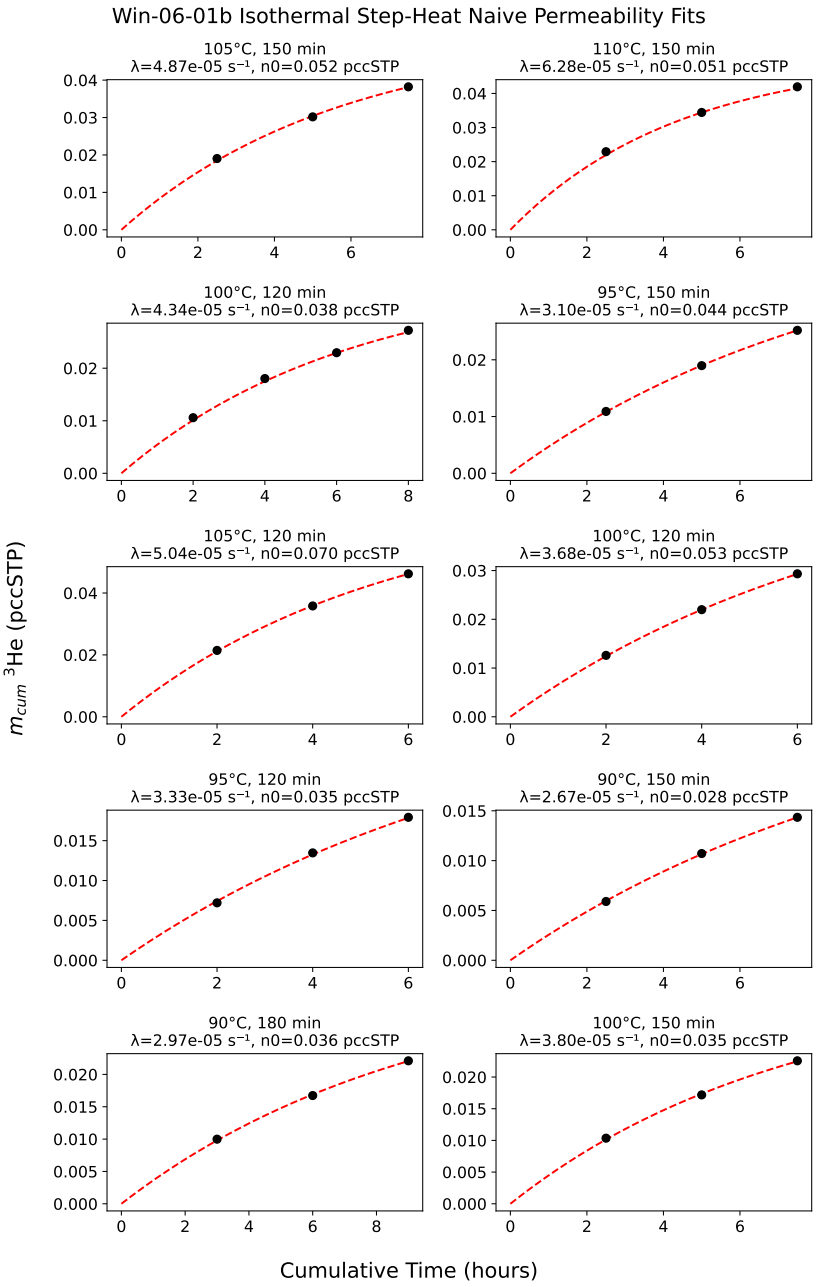


Figure 2.3: Cumulative yield against cumulative time for helium-3 yields for isothermal permeation steps. A non-linear least squares optimization fits λ and n_0 of Eq. (1.8) (Chapter 1) to data from a series of permeation steps like these. The permeability K is calculated according to the relation $K = \lambda/GRT$ and is plotted as a function of temperature in Fig. 2.4.

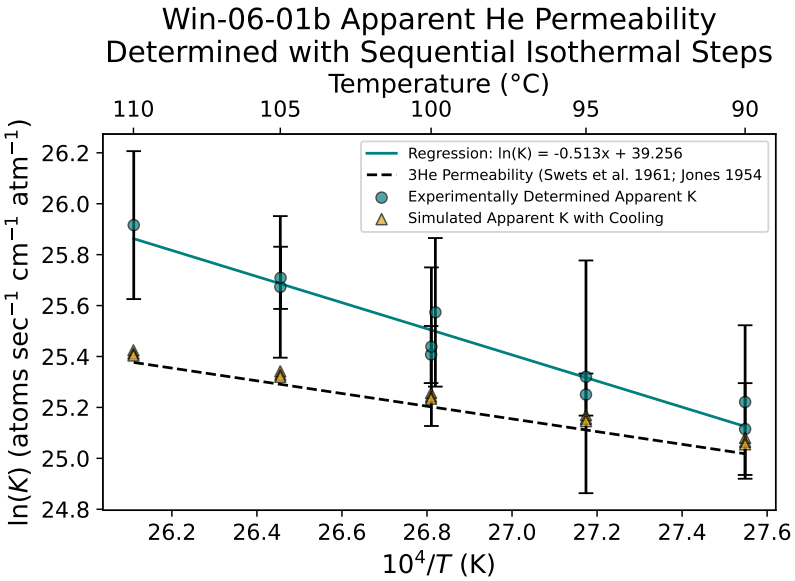


Figure 2.4: Arrhenius plot of apparent permeation coefficients as a function of inverse temperature for HT-Win-06-01b. The activation energy is higher than anhydrous silica glass, as expected (Masaryk, 1968; Shelby, 1972). It is unexpected that the glass ampoule of HT-Win-06-01b has higher apparent 3He permeability than the literature values from Swets et al., 1961 and Jones, 1953, but this can be explained by an increase in the partial pressure of the gas phase within the tube due to displacement by high He diffusivity, low He solubility water. Yellow points labeled "Diffusion Model with cooling, Anhydrous" were produced to evaluate the expected excess apparent permeability due to the 25 minute cooldown period before measurement. This is accomplished by simulating the measured apparent permeability for a dry silica glass ampoule that continued to permeate helium while cooling for an additional 26 minutes before measurement after being held at 90–110°C for 120–180 minutes, similarly to the measured values. Variability in these points is due to differences in the length of time the ampoule was heated but is very small. They show a near constant offset from the expected values of Swets et al., 1961 and Jones, 1953 that is much smaller than the uncertainty in the regression of the fit values determined experimentally in teal.

values for 3He were assumed as the product of the 4He permeability values given by Swets et al. (1961) and the 3He/4He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

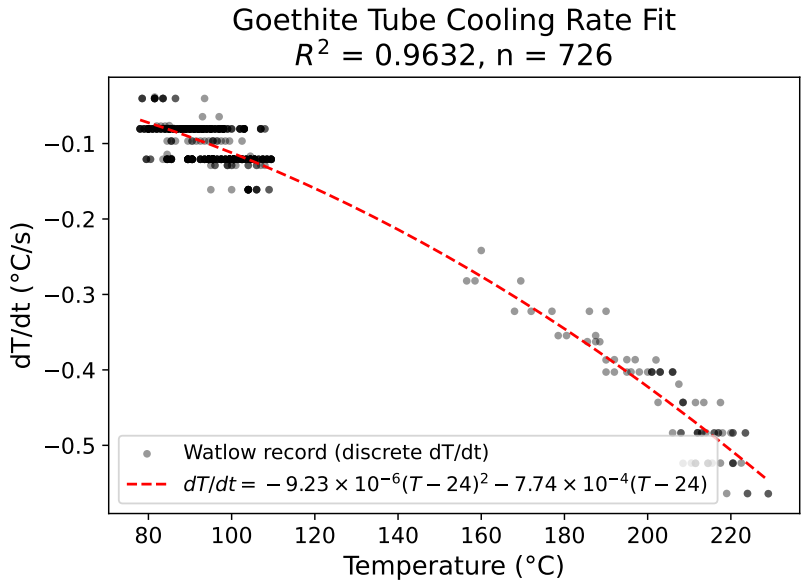


Figure 2.5: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

Helium-in-Goethite Diffusivity Inversions Apparent values of D/a^2 for 3He and 4He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of

the ampoules before and after each step, evaluated using (1.9). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag (${}^{i}_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\Big(\Sigma^{i-1}F_m,\ \Sigma^i F_m + {}^{i}_{SS}n_t\Big) \\ {}^{LB}D_\omega &= D\Big(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^i F_m\Big) \end{aligned} \tag{2.1}$$

For the 3He diffusivities, a linear regression was fit to the steps with cumulative 3He fractional yield ≥7% (12 steps, Fig. 2.7), yielding an R^2 statistic of 0.989. The fit's slope and intercept are used as a reference line to calculate "Delta" values for the diffusivities of 3He and 4He (Fig. 2.8). No uncertainty on the total amounts of 3He and 4He is propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the linear regression is significantly larger than the uncertainty related to the total yield, so it is neglected.

Helium Isotope Ratio Uncertainty for Step Heats The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotopes systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The 4He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal 3He permeation measurements show to underestimate the ampoule's true permeability (Fig. 2.4); 4He permeability itself could not be independently measured in this experiment (supplement, Helium-4 Blank Corrections). For the purposes of evaluating the measured isotope ratios, the relative 3He/4He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a water-filled ampoule in Chapter 1 (Thyer et al., 2026): 1.10 ± 0.19 (2σ). 1.29 is in great excess of realistic values of the relative 3He/4He permeability, which is expected to be between 1.06-1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (1.9) (Chapter 1): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $$\frac{\omega_4}{\omega_3} = \frac{m_4}{m_3}$$

2. $$\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1}}$$

3. $$\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1} - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$$

4. $$\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to 4He, whereas $\delta\lambda$ applies

to 3He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

SEM Study of Thermally Treated Polycrystalline Goethite In early trials of the "glass tube method" for helium step heat experiments with controlled microenvironments presented here, sub-samples of goethite morphologically similar to Win-06-01b from Igarapé Bahia (Shuster, P. M. Vasconcelos, et al., 2005; H. Monteiro, 2017) were heated in vacuum using the projector lamp heating apparatus of Farley, Reiners, et al. (1999) and in sealed glass ampoules with gypsum or water additionally added to provide moisture to slow/prevent dehydration of goethite. All samples were held at 300°C for 4-11 hours, and the glass ampoules were broken and the goethites were allowed to air dry before all samples were placed on carbon tape and carbon coated. These samples were imaged at Caltech on Secondary Electron mode by the ZEISS 1550VP Field Emission SEM.

The resistance-furnace degassing step is detailed further in the Supplementary Information, in its Totals section. That same supplement also records, as a secondary and independent verification of the results and conclusions presented here, the naïve/informed bracketing method applied to this same dataset: an alternative treatment that explicitly tracks the ampoule's helium content step by step, using the isothermal-permeability fits established above, rather than bounding it as above.

2.4 Results

The complete corrected step-heat dataset underlying the diffusivities, delta plots, and Arrhenius regressions discussed below is distilled in Tables 2.1 and 2.2, reporting the lower-bound, central, and upper-bound values from the "treatment as uncertainty" permeation-lag correction for ^{3}He and ^{4}He respectively.

Table 2.1: ^{3}He step-heat dataset for HT-Win-06-01b under the "treatment as uncertainty" method: measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), the lower- and upper-bound true step production (ω_{LB}, ω_{UB}), and the corresponding lower-bound, central, and upper-bound apparent diffusivities. "–" denotes early steps with $\lambda t < 4$, for which bounds could not be determined (see 2.2). Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots (Fig. 2.18 and others derived from this dataset).

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
170	300	0.9630	0.0000	1.1660	–	–	–	–	–
180	463	0.5442	0.9630	0.2413	–	–	–	–	–
190	550	0.5292	1.6241	0.1538	–	–	–	–	–
200	500	0.5190	2.2112	0.1430	0.3651	0.6619	-22.158	-21.775	-21.503
210	475	0.5770	2.7646	0.1440	0.4341	0.7210	-21.715	-21.406	-21.160
220	450	0.6333	3.3799	0.1447	0.4894	0.7781	-21.345	-21.068	-20.843
230	425	0.7899	4.0551	0.1685	0.6452	0.9584	-20.823	-20.604	-20.392
225	908	0.7513	4.8885	0.0716	0.5827	0.8229	-21.516	-21.246	-21.148
215	450	0.1415	5.6842	0.0349	0.0699	0.1764	-22.836	-22.124	-21.900
205	520	0.0763	5.8529	0.0184*	0.0413*	0.0947*	-23.479*	-22.863*	-22.646*
195	1106	0.0670	5.9491	0.0080*	0.0487*	0.0750*	-24.053*	-23.731*	-23.618*
185	740	0.0227	6.0246	0.0049*	0.0147*	0.0276*	-24.839*	-24.404*	-24.207*
175	920	0.0180	6.0610	0.0036*	0.0131*	0.0216*	-25.168*	-24.847*	-24.665*
200	260	0.0224	6.0814	0.0163*	0.0189*	0.0388*	-23.534*	-23.360*	-22.812*
210	520	0.0812	6.1081	0.0181*	0.0649*	0.0993*	-22.984*	-22.758*	-22.554*
220	1167	0.5766	6.1914	0.0447	0.5585	0.6213	-21.587	-21.553	-21.475
230	370	0.3577	6.7796	0.0903	0.3130	0.4480	-20.948	-20.811	-20.579
240	320	0.6767	7.1563	0.1776	0.5864	0.8543	-20.062	-19.913	-19.667
245	250	0.5549	7.8582	0.1888	0.3773	0.7437	-20.176	-19.778	-19.474
250	210	0.6934	8.4743	0.2786	0.5047	0.9721	-19.626	-19.296	-18.942

continued on next page

(continued)

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
245	270	0.4659	9.2086	0.1413	0.1872	0.6071	-20.801	-19.874	-19.601
240	320	0.2857	9.7206	0.0749	0.1445	0.3606	-21.176	-20.487	-20.250
235	370	0.2199	10.0357	0.0517	0.1450	0.2716	-21.284	-20.864	-20.650
230	420	0.1400	10.2736	0.0303	0.0883	0.1703	-21.885	-21.421	-21.224
225	652	0.1494	10.4279	0.0206	0.1191	0.1700	-22.009	-21.781	-21.650
230	500	0.1550	10.5908	0.0271	0.1344	0.1820	-21.605	-21.462	-21.299
235	450	0.1951	10.7582	0.0361	0.1681	0.2313	-21.258	-21.107	-20.936
240	400	0.2425	10.9638	0.0483	0.2063	0.2908	-20.913	-20.750	-20.566
245	350	0.3150	11.2328	0.0690	0.2667	0.3840	-20.495	-20.326	-20.125
250	300	0.3716	11.5689	0.0939	0.3026	0.4656	-20.181	-19.973	-19.743
250	300	0.3640	11.9627	0.0899	0.2700	0.4539	-20.261	-19.959	-19.734
250	300	0.3452	12.3490	0.0852	0.2553	0.4304	-20.285	-19.979	-19.755
250	10250	8.4829	12.7167	0.0494	8.3977	8.5324	-19.989	-19.976	-19.969
240	785	0.2261	21.2187	0.0209	0.1767	0.2470	-21.026	-20.778	-20.689
Total			146.1483						

Table 2.2: ^{4}He step-heat dataset and absolute ^{4}He/^{3}He ratios for HT-Win-06-01b under the "treatment as uncertainty" method, structured as in Table 2.1. "–" denotes steps for which bounds could not be determined. The Total row reports the bulk (total-yield) ^{4}He/^{3}He ratio, i.e. the normalizing value used to bulk-normalize these ratios for the plot in Fig. 2.18. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots, as in Table 2.1.

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{UB}}$
170	300	0.0192	0.0000	0.0280	–	–	–	–	–	–	–	–
180	463	0.0305	0.0192	0.0152	–	–	–	–	–	–	–	–
190	550	0.0558	0.0660	0.0179	–	–	–	–	–	–	–	–
200	500	0.1024	0.1375	0.0313	0.0845	0.1337	-26.207	-25.965	-25.619	–	197.3	–
210	475	0.1460	0.2500	0.0404	0.1147	0.1864	-25.314	-25.022	-24.717	248.2	253.0	268.8
220	450	0.2091	0.4055	0.0530	0.1686	0.2620	-24.407	-24.152	-23.876	326.3	330.1	349.3
230	425	0.3243	0.6319	0.0760	0.2714	0.4004	-23.425	-23.213	-22.956	403.7	410.6	432.3
225	908	0.3483	0.9799	0.0363	0.2722	0.3845	-23.807	-23.527	-23.413	442.6	463.6	474.4
215	450	0.0716	1.3487	0.0196	0.0353	0.0912	-24.945	-24.225	-23.976	429.1	505.9	537.1
205	520	0.0385	1.4359	0.0103*	0.0189*	0.0487*	-25.661*	-24.942*	-24.701*	392.4*	504.2*	534.6*
195	1106	0.0336	1.4912	0.0043*	0.0233*	0.0379*	-26.165*	-25.796*	-25.675*	446.7*	500.9*	515.3*
185	740	0.0079	1.5467	0.0019*	0.0037*	0.0098*	-27.586*	-26.808*	-26.592*	224.7*	350.1*	369.1*
175	920	0.0152	1.6020	0.0033*	0.0133*	0.0185*	-26.475*	-26.340*	-26.141*	843.2*	843.2*	982.0*
200	260	0.0107	1.6351	0.0091*	0.0073*	0.0198*	-25.787*	-25.411*	-24.791*	375.6*	475.5*	554.7*
210	520	0.2166	1.6547	0.0535*	0.2075*	0.2701*	-23.067*	-23.021*	-22.786*	2667.1*	2667.1*	3103.8*
220	1167	0.3187	1.8714	0.0269	0.2652	0.3456	-23.499	-23.302	-23.214	471.7	552.7	563.2
230	370	0.2113	2.6766	0.0590	0.1844	0.2703	-22.391	-22.249	-21.993	575.1	590.8	628.0
240	320	0.4286	3.1616	0.1243	0.3697	0.5529	-21.360	-21.203	-20.931	612.1	633.4	674.8
245	250	0.3988	3.6430	0.1521	0.2746	0.5510	-21.289	-20.899	-20.556	665.9	718.7	778.4
250	210	0.4440	4.2359	0.2020	0.2918	0.6460	-20.906	-20.469	-20.072	541.7	640.2	702.8

continued on next page

(continued)

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{UB}}$
245	270	0.3123	4.7263	0.1053	0.1103	0.4176	-22.043	-20.981	-20.680	469.1	670.4	721.4
240	320	0.1927	5.0846	0.0559	0.0874	0.2485	-22.376	-21.575	-21.315	505.5	674.3	718.4
235	370	0.1517	5.3031	0.0393	0.0958	0.1910	-22.386	-21.922	-21.688	599.8	689.6	729.9
230	420	0.0991	5.4714	0.0236	0.0598	0.1227	-22.956	-22.448	-22.232	608.3	707.9	745.9
225	652	0.0995	5.5835	0.0151	0.0760	0.1146	-23.136	-22.863	-22.721	608.2	666.5	689.2
230	500	0.1051	5.6969	0.0202	0.0900	0.1253	-22.679	-22.523	-22.345	652.1	678.1	707.5
235	450	0.1369	5.8251	0.0279	0.1167	0.1648	-22.290	-22.128	-21.940	674.6	701.7	734.0
240	400	0.1781	5.9762	0.0390	0.1502	0.2171	-21.892	-21.719	-21.518	705.1	734.6	770.9
245	350	0.2305	6.1807	0.0556	0.1915	0.2861	-21.479	-21.290	-21.070	694.4	731.9	771.7
250	300	0.2777	6.4385	0.0758	0.2221	0.3535	-21.134	-20.906	-20.659	703.3	747.3	793.3
250	300	0.2654	6.7374	0.0725	0.1896	0.3379	-21.250	-20.908	-20.661	653.7	729.2	774.1
250	300	0.2489	7.0231	0.0680	0.1764	0.3169	-21.282	-20.932	-20.647	646.8	721.0	765.4
250	10250	6.8042	7.2901	0.0430	6.7362	6.8472	-20.704	-20.691	-20.682	800.6	802.1	803.3
240	785	0.1853	14.1156	0.0187	0.1423	0.2040	-21.688	-21.422	-21.325	770.5	819.4	838.0
Total			145.8145								997.7	

The diffusivities determined from the corrected step heat 3He and 4He yields for HT-Win-06-01b are always significantly lower than the diffusivities determined from step heat experiments on the same sample in vacuum, and evolutions that are distinct from those determined from the vacuum experiments. The Arrhenius plot of the 3He diffusivities initially follow a declining concave-up trend until ~3.5% cumulative 3He yield, then slowly decreasing in sub-parallel arrays during prograde and retrograde heating cycles from 3.5-7% cumulative 3He yields (Fig. 2.7, Fig. 2.8). The Arrhenius plot of the 4He diffusivities differs in that it is initially concave-down until a cumulative 4He yield of ~0.1%, after which it maintains a relatively constant linear array about one log unit lower than the 3He array at the end of the experiment (Fig. 2.7). The 3He diffusivities settled into a linear array which was suitable for linear regression after ~7% of the cumulative 3He yield was lost (Fig. 2.6). All remaining diffusivity measurements of the experiment were included in the linear regression, so that diffusivities from 3He loss fractions of about 7-15% were included. The linear regression yields an R^2 of 0.989, an activation energy of 157.1± 5.3kJ/mol, and a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering only the uncertainty in the regression intercept (D_0/a^2).

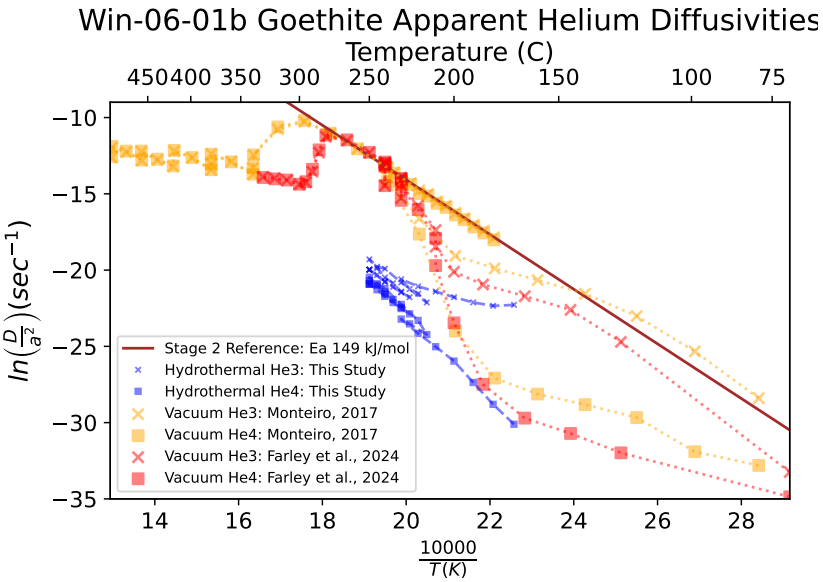


Figure 2.6: Arrhenius plot of 3He and 4He diffusivities from HT-Win-06-01b with other step heat experiments of Win-06-01b from H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). The "Stage 2 reference line" is fit to helium step heats from H. Monteiro (2017). Error bars on the data from this study are not plotted for clarity. A diffusive domain corresponding to the Stage 2 reference line would have a closure temperature of about 45°C assuming a cooling rate of 10°C/Ma.

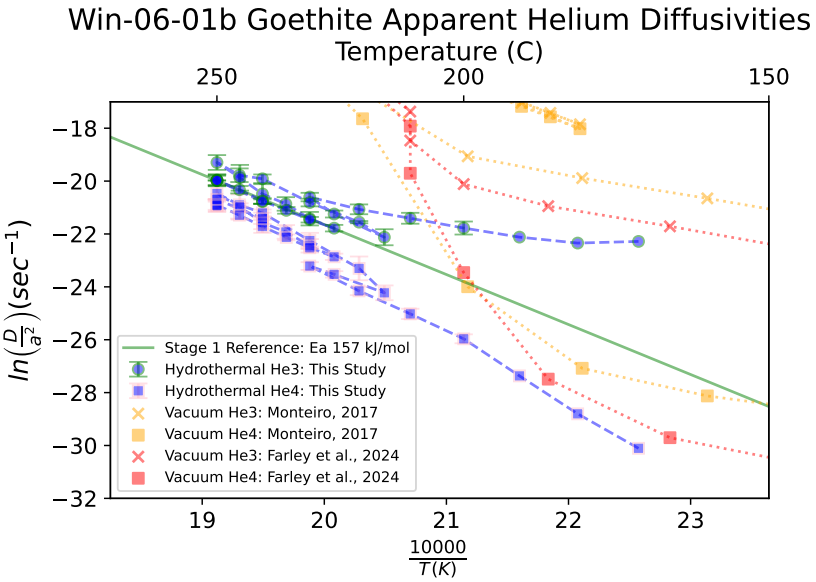


Figure 2.7: Linear regression of the last 11 measurements of 3He diffusivities in HT-Win-06-01b defining the "Stage 1" reference line. Apparent diffusivities under hydrothermal conditions are nearly 8 log units lower than in vacuum. A diffusion domain corresponding to the Stage 1 reference line has a closure temperature of 98.1±8.6°C assuming a cooling rate of 10°C/Ma. Apparent diffusivities calculated from 4He step heats are sometimes within error of those calculated for 3He, however they are occasionally resolved at the 2σ level and are about a single log unit lower than apparent diffusivities calculated for 3He after a cumulative fractional yield of 14% 3He.

A delta plot was generated using the intercept and slope parameters from the 3He regression as a reference line, similarly to previous work (Farley, 2018). The 3He apparent diffusivity delta values monotonically decrease until a cumulative 3He loss of 7%, after which the delta values are within error of Delta = 0: the reference line (Fig. 2.8). A series of low-yield measurements on the first retrograde cycle produce high-error diffusivities which appear as scattered, uncertain delta values just after a cumulative 3He loss of 4% for both 3He and 4He.

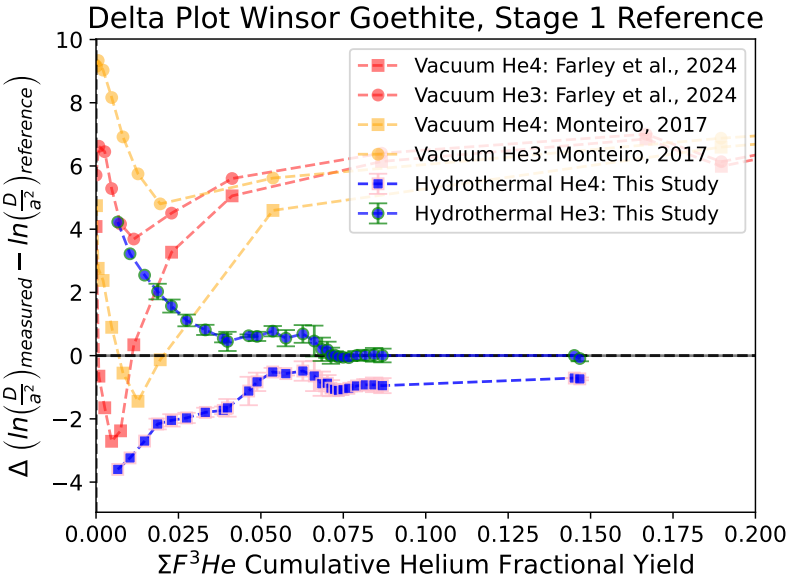


Figure 2.8: Delta plot similar to those presented in (Farley, 2018), or r/r0 plots in Lovera et al. (1991) and Farley and Flowers (2012). Delta plots are the difference between apparent log diffusivities and a reference line; they are designed to remove the temperature dependence of the dominant process of helium migration to reveal how apparent diffusivities evolve as an experiment proceeds. 3He monotonically decreases until settling on a temporary plateau from 4–6% 3He fractional yield, then settling on the reference line from 7–15% 3He fractional yield, which is consistent with MDD systematics. 4He diffusivities and delta values are generally highly uncertain but initially show an increase in delta values that settles to a slow increase or plateau about one log unit below the 3He reference line by a 3He fractional yield of 5%, which is consistent with diffusive loss of radiogenic 4He in the least retentive domains interrogated in HT-Win-06-01b. Apparent diffusivities for helium in Win-06-01b in hydrothermal conditions are up to 7 log units lower than the same sample in vacuum.

SEM imagery taken of thermally treated natural goethites in vacuum and

open air showed the development of contraction cracks on the outside and throughout the interior of goethite grains heat treated in vacuum and in open air (Fig. 2.10 and Fig. 2.14), while goethite heat-treated when encapsulated with water or gypsum did not form cracks (Fig. 2.11 and Fig. 2.12). Goethite heat treated in open air turned red, whereas goethite heat treated in vacuum remained black.

2.5 Discussion

Measurement and Numerical Simulation of Apparent Permeability of Experimental Ampoule The apparent permeability values used in the numerical simulation (§2.3) are taken from the literature (Swets et al., 1961; Jones, 1953) and explicitly apply to an ampoule without water. However, the purpose of these simulations was simply to quantify the excess apparent permeabilities that would be observed for ampoules with a cooling history similar to that of the hydrothermal experiments. Because the predicted offset is universally small and a weak function of temperature or permeability, these simulations demonstrate that the much larger experimentally observed permeability is not an artifact of the cooling history of the hydrothermal experiments, reaffirming the conclusion that the hydrothermal experiments are indeed more permeable than the vacuum experiments.

The difference in log permeability between the actual and apparent permeabilities of the anhydrous simulation were significantly smaller than nearly all the 2σ uncertainties of the apparent permeabilities determined above for the hydrothermal Win-06-01b experiment, which ranged from 0.08 to 0.46 log units. Thus, no trends with step duration could be identified in the effective permeability values.

Therefore, the literature values underestimate the helium permeabilities of the ampoule used in this experiment, which is permissible because these values are used to calculate an upper bound constraint of the helium content of the ampoule, and lower permeabilities produce more conservative (higher) upper bound constraints.

The elevated measured permeability runs counter to expectations for a hydrothermal experiment: hydration of silica glass is generally reported to slow helium permeation at all temperatures (Tsuru et al., 2011), under the common paradigm that ions such as $(OH)^-$ clog interstitial space within the amorphous

glassy network to slow permeation and decrease helium solubility (Norton, 1953; Shackelford, 1999). A similarly elevated permeability was independently observed at 100°C in a separate silica glass ampoule at Caltech partially filled with MilliQ water and equilibrated with ~0.1 torr of tank helium, as reported by Thyer et al. (2026), who posit that the elevated permeability is due to the low solubility and high diffusivity of helium in water, effectively shrinking the volume of the gas phase while maintaining equilibrium with the ampoule's interior throughout, leading to higher partial pressures of helium within the ampoule and therefore higher apparent permeabilities.

A linear regression of $\ln(K)$ against inverse temperature (Fig. 2.4) fit to these isothermal 3He permeation measurements over 90-110°C yields an R^2 of 0.936 and an activation energy of 10192±1882 (2 s.e.) cal/mol, a factor of two higher than expected for pure silica glass with or without water (Masaryk, 1968; Shelby, 1972), but similar to glasses with a high abundance of network-modifying cations such as Na, K, Pb, or Li (Norton, 1953; Altemose, 1961). Because 4He signals during these isothermal steps were blank-level (supplement, Helium-4 Blank Corrections), only the ampoule's 3He permeability could be measured directly this way, using the fitting procedure described above.

Consistency with MDD Systematics Previous work on helium step heats of goethite in vacuum have demonstrated degassing behavior in the laboratory that is inconsistent with thermally activated volume diffusion through goethite. For example, apparent diffusivities calculated from helium step heat experiments of goethites with uniform distributions of 3He produced by proton irradiation following the methods of Shuster, Farley, et al. (2004) will increase in an isothermal experiment (Farley, H. B. Monteiro, et al., 2024), which is inconsistent with volume diffusion through an unchanging crystal.

Farley, H. B. Monteiro, et al. (2024) reported that helium and water evolution in goethite vacuum step heating experiments are highly correlated and concluded that the crystal was decomposing into water and hematite during the experiment.

The goal of this work was to attempt to slow this decomposition process below the rate of thermal helium loss in unchanging goethite. SEM imagery shows that goethite that is thermally treated when encapsulated with water (or water for gypsum) does not extensively dehydrate like goethite heat-treated

in vacuum or open air (Fig. 2.9–Fig. 2.14). While it is difficult to prove that decomposition or recrystallization is not playing a role in the results presented in this work, the apparent diffusivities observed for the Win-06-01b goethite under hydrothermal conditions are much lower than those observed in previous step heats of Win-06-01b in vacuum (Fig. 2.6, Fig. 2.7, Fig. 2.8, Fig. 2.15), implying that processes other than dehydration may be responsible for the observed apparent diffusivities.

All helium step heat data on goethites are from naturally occurring polycrystalline aggregates with unclear and potentially widely variable grain sizes (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024), which are expected to follow MDD systematics. If volume diffusion is responsible for the data reported here, they too would be expected to follow MDD systematics. In MDD systems, apparent bulk diffusivities of a homogeneously distributed diffusant at a single temperature will decrease as an experiment proceeds, as would delta or r/r0 values at any temperature (where $\Delta = ln\left(\frac{D}{a^2}\right)_{measured} - ln\left(\frac{D}{a^2}\right)_{reference}$) (Lovera et al., 1991; Farley, 2018), assuming that the activation energy of the reference line used in such plots matches that of the process controlling diffusion. For the activation energy of 160 kJ/mol retrieved by regression in this work, the homogeneously distributed 3He step heat data follow the expected MDD systematics and monotonically decrease within a delta plot (Fig. 2.8), which is consistent with volume diffusion in a MDD system.

The strong plateau in the delta plot and statistically robust regression over a wide range of fractional cumulative 3He yield (from 7-15% ΣF_{3He} with an R^2 of 0.989) suggests that the frequency factor obtained by the regression may be dominated by a particular diffusion domain, rather than a mix of diffusion domains as is most often observed in the apparent diffusivities for polycrystalline hematite (Farley, 2018). The 4He apparent diffusivities are generally lower than the 3He apparent diffusivities and increased at the beginning of the experiment, which is consistent with rounding of 4He concentration profiles due to loss of radiogenic helium in the least retentive domains (Shuster, Farley, et al., 2004; Shuster, P. M. Vasconcelos, et al., 2005).

Many vacuum experiments in goethite also record declining delta values for early fractional yields of 3He (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024) that are generally not represented in the 4He data. This effect may be from helium migrating out of fast pathways along grain

boundaries that would retain little helium in nature (Shuster, P. M. Vasconcelos, et al., 2005). Goethite aggregates are known to have interstices that are filled by chemisorbed non-stoichiometric water that represents the earliest water to be lost in vacuum (Naono et al., 1987). Non-stoichiometric hydrogen is known to persist in goethite even after heating to 140°C (Yapp and Poths, 1995). It is therefore reasonable to posit that artificial 3He observed in less-retentive domains may be trapped by non-stoichiometric water or hydroxyl groups within interstices of goethite aggregates. Dehydration of non-stoichiometric interstitial water may be associated with early fractions of degassing water and 3He from goethite samples during step heats, and could potential represent the LRD (low retention domain) of Shuster, P. M. Vasconcelos, et al. (2005), and the 3He offgassed during "Stage 1" in vacuum step heats (Farley, H. B. Monteiro, et al., 2024). The high variability of the portion of 3He hosted in early-degassing "Stage 1" fractions within vacuum step heats is sensible considering observations that goethite formed at different temperatures is known to have highly variable crystallinity, crystallite ordering, surface area, and non-stoichiometric water contents (Schwertmann et al., 1985). This population of 3He would likely also be the first to be lost under hydrothermal conditions.

While the results presented here are strong experimental evidence for high helium retentivity in goethite (Fig. 2.15, Fig. 2.16), they do not yet independently provide any evidence to support or refute hypotheses that helium diffusion in crystalline goethite is anisotropic or affected by radiation damage (Bassal et al., 2022).

Goethite Dehydration and/or Recrystallization During Helium Step Heat Experiments The aim of helium step heat experiments is to characterize the mechanisms and rates helium loss in nature to aid in the interpretation of geochronologic and helium geochemical data. While the results of this experiment are encouraging in that they differ remarkably from experiments on the same and similar samples in vacuum that are primarily controlled by dehydration processes (Farley, H. B. Monteiro, et al., 2024), this experiment may also be affected to some degree by processes aside from volume diffusion that may potentially be unrelated to helium loss in goethite in nature.

Aside from nanoparticles (Guo and Barnard, 2011; Navrotsky et al., 2008), bulk goethite is unstable with respect to hematite and water at the pressures

and temperatures of this experiment (Majzlan et al., 2003). The SEM study of goethite heat-treated in vacuum and open air shows that clearly visible volume contraction cracks form throughout the interiors and surfaces of highly-dehydrated goethite grains, while goethite heat-treated in water-saturated conditions does not (Figs. 2.10, 2.11, 2.12, 2.13, 2.14). Clearly, dehydration is occurring at a much smaller scale, if at all, in the hydrothermal experiments. While this evidence excludes the possibility of pervasive dehydration in the hydrothermal experiments, it does not exclude the possibility of some dehydration occurring at a scale too small to be observed in the SEM images. Future work on textural evolution of natural goethite aggregates during low fraction dehydration would be useful to resolve the extent of "hidden" dehydration that may occur before the formation of SEM-visible cracks.

Dehydration processes may be responsible for some or all of the helium loss of Win-06-01b-HT, although if so they are markedly slowed relative to step heat experiments in vacuum (Fig. 2.16–Fig. 2.17). Even if dehydration completely controls helium loss in the step heat experiment of Win-06-01b-HT, these new data presented here would represent upper bounds for the apparent diffusivities of helium at the temperatures and fractional cumulative yields (ΣF_{He}).

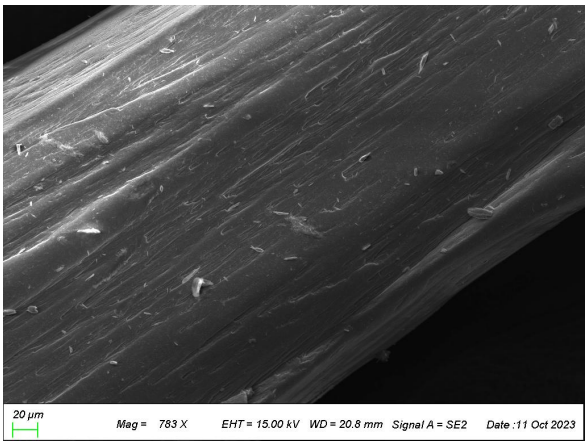


Figure 2.9: Secondary electron image of Bahia goethite aggregate before treatment.

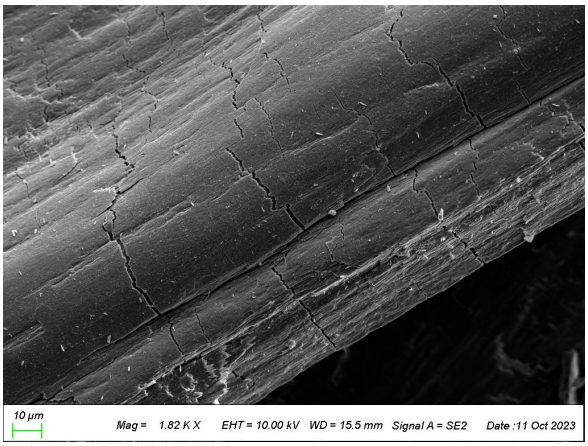


Figure 2.10: Secondary electron image of Bahia goethite aggregate heated at 300°C for 11 hours in vacuum. Cracks developed both parallel and perpendicular to the long axis of the polycrystalline aggregate, representing volume contraction due to dehydration/decomposition of goethite into hematite.

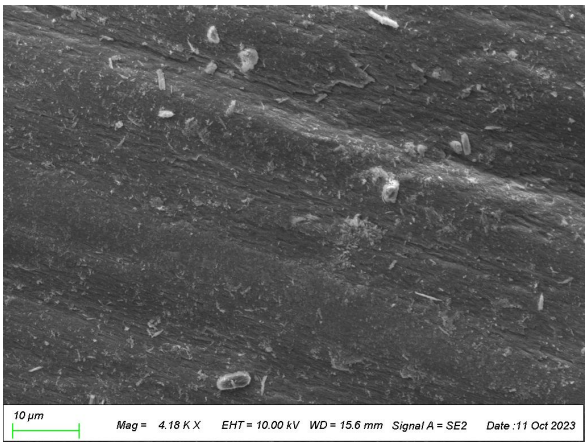


Figure 2.11: Secondary electron image of Bahia goethite aggregate heated at 300°C in a glass tube with gypsum. Despite treatment at high heat, no cracks indicative of dehydration were visible on the goethite surface, indicating that the gypsum dehydrated first and produced enough water pressure to stop/slow dehydration and prevent the development of microscopic cracks. At higher temperatures, goethite and gypsum reacted, forming a surface layer of other minerals.

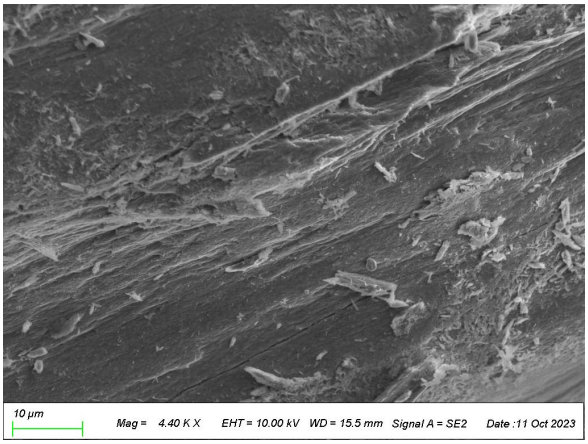


Figure 2.12: Secondary electron image of Bahia goethite heated at 300°C for 4 hours in a glass tube with water. No cracks were observed when investigating the goethite at similar scales to Figs. 2.9–2.10. Most of the aggregate's surface resembled Fig. 2.11. The area pictured had the roughest texture on the grain. A very small crack parallel to the long axis may indicate some amount of dehydration had occurred, or it could have been inherited from before hydrothermal treatment, since even the control untreated grain had some rough surfaces and rare long-axis-parallel cracks.

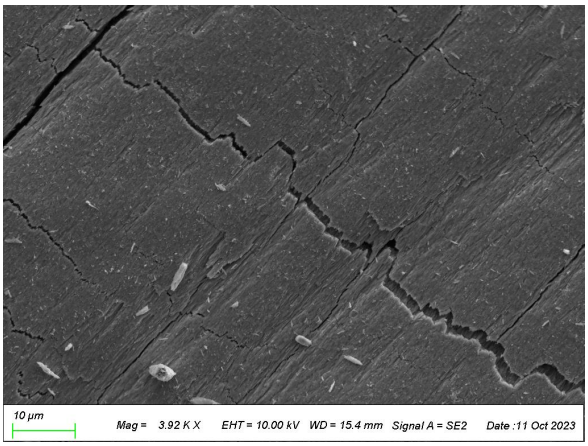


Figure 2.13: Secondary electron image of Bahia goethite heated at 300°C for 11 hours in vacuum, at the same scale as Fig. 2.12 and Fig. 2.11. This image shows rough cracks at various scales both parallel and orthogonal to the long axis of the goethite aggregate. These are typical decomposition features that would be expected in Fig. 2.12 and Fig. 2.11 if significant decomposition had occurred.

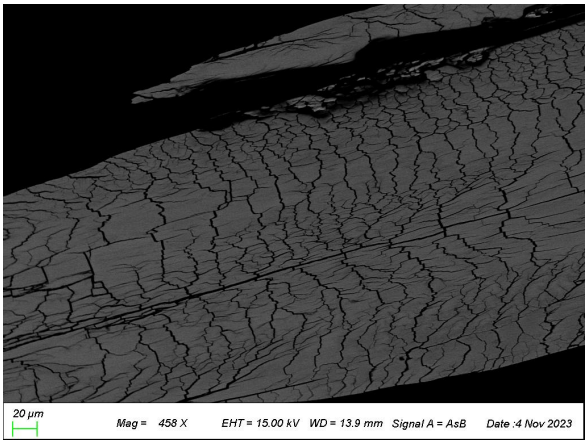


Figure 2.14: Backscatter electron image of a polished cross section of epoxy-mounted, vacuum-heated Bahia goethite (from the same subsample as Fig. 2.10) exposing the grain's interior. This image clearly shows that the cracks visible on the exterior of dehydrated goethite are pervasive throughout the interior as well.

The process of recrystallization is another potential complication in interpreting helium step heats of goethite in hydrothermal conditions. Poorly crystalline synthetic goethite held in hydrothermal conditions is known to recrystallize into more orderly, crystalline goethite aggregates, even at temperatures as low as 125°C in as little as 16 hours (Schwertmann et al., 1985). In this study, natural goethite was held in contact with liquid water and water vapor at hydrothermal conditions as harsh as ~40 bars and 250°C for as long as a week. These conditions appear to have visibly dissolved and reprecipitated silica from the ampoule's inner walls (Fig. 2.1), qualitatively testifying to the ability of the hydrothermal fluid to dissolve and recrystallize rock-forming oxides. It is unquestionable that recrystallization of some goethite could have occurred under these conditions; however, it is unclear if conditions were favorable to recrystallize this specimen of goethite, Win-06-01b, which is amongst the most helium-retentive and crystalline dated goethite samples (Waltenberg, 2012; Farley, H. B. Monteiro, et al., 2024). Less crystalline goethites in nature are commonly recrystallized, which appears to result in helium loss, resetting the U-Th/He age (H. S. Monteiro, P. M. P. Vasconcelos, et al., 2018; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Heim et al., 2006). Intuitively, it is difficult to imagine that dissolution-precipitation mediated recrystallization would not result in complete loss of helium within recrystallized regions. Therefore, assuming an initially uniform distribution of 3He throughout the crystal from proton irradiation, the upper bound extent of recrystallization would be equivalent to the 3He cumulative fractional yield, which is only 14.7% by the end of the hydrothermal experiment. Remaining helium is unlikely to pass through these now helium-free recrystallized areas if grain boundaries in goethite serve as fast pathways like they do in hematite (Farley, 2018). Recrystallization could be the dominant process controlling helium loss in this experiment, but only if it is faster than diffusive loss, in which case the data presented here would represent new upper bounds for apparent diffusivities for helium in Win-06-01b goethite. In that case, it stands to reason that processes such as dehydration and recrystallization may be faster than volume diffusion of helium in crystalline goethite in nature and may be the primary drivers of helium loss.

High Helium Retentivity of Win-06-01b-HT Apparent diffusivities calculated from helium step heats on the hydrothermal experiment on Win-06-

01b goethite are clearly resolved and much lower than the apparent diffusivities calculated for the same sample in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) for all steps after the first few (Fig. 2.6, Fig. 2.7, Fig. 2.8). Diffusivities calculated from the hydrothermal experiment are ~8 log units lower than the earlier experiments in vacuum on the same sample. This high helium retentivity is explored through three lenses: closure temperature of cooling samples (Dodson, 1973), isothermal diffusion (Wolf, Farley, and Kass, 1998), and through comparison with step heats of other polycrystalline samples. Strictly speaking, both closure temperatures and retention estimations under isothermal diffusion only apply to helium loss occurring exclusively via thermally activated volume diffusion with known domain size. While the results presented here on a polycrystalline sample are consistent with a thermally activated volume diffusive process for helium loss in goethite under hydrothermal conditions, they do not exclude other processes (such as goethite dehydration/decomposition/recrystallization) or potential complexities related to radiation damage and crystal defects on bulk diffusion (Bassal et al., 2022), nor do they permit estimation of the diffusivity at infinite temperature (D0).

If the slope and intercept on the regression of 3He apparent diffusivities from this experiment are taken to represent a real diffusion domain for goethite, then a closure temperature can be calculated for this domain. Considering uncertainty only in the intercept (D_0/a^2) term and a cooling rate of 10°C/Ma, the closure temperature for that domain would be 98.1± 8.6°C. This would correspond to the closure temperature of a ~0.2µm radius hematite crystal, or a ~650µm radius apatite crystal.

Goethite is formed near Earth's surface and thus U-Th/He geochronology is most typically applied to date surface processes (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Hofmann, Reichenbacher, et al., 2017; Allard et al., 2018). For crystals of goethite grown near the surface that are found near the surface, the concept of a closure temperature is poorly applicable, as it necessarily assumes a cooling rate (Dodson, 1973). In such cases, a better measure of retentivity is an analysis of isothermal retentivity over time (Wolf, Farley, and Kass, 1998), which is commonly expressed in terms of t'/t, where t' is the helium age and t is the isothermal holding time (most often the 'real age'). For an isothermal holding time of 100 Ma at 40°C,

a diffusive domain with the parameters obtained from the regression of 3He apparent diffusivities of Win-06-01b-HT has a t'/t = 0.968, where t'/t = 1 corresponds to perfect retentivity. Fig. 2.15 shows isothermal holding time against helium age for this same domain for 4 different temperatures. At 25°C, this domain is nearly quantitatively retentive of helium even at billion-year time scales. The results presented here qualitatively support that goethite is highly retentive of helium at surface conditions, even at 100-million-year timescales.

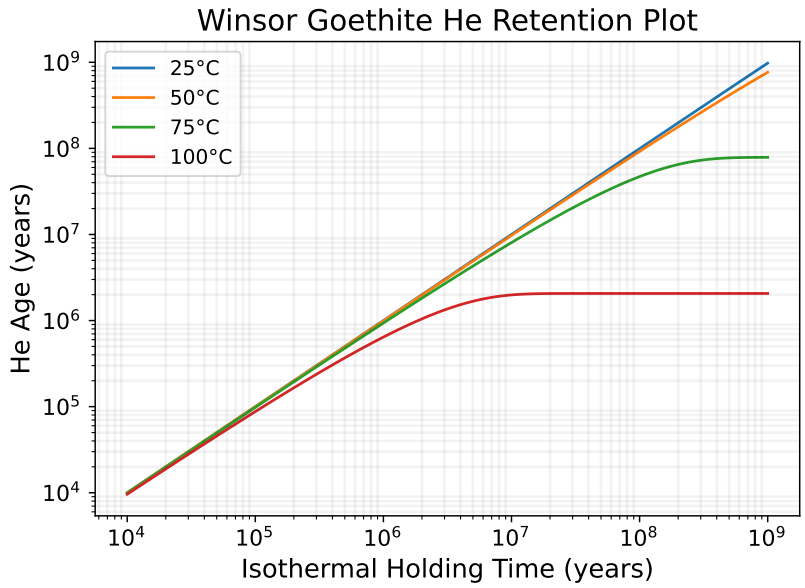


Figure 2.15: Isothermal retention plot (Wolf, Farley, and Kass, 1998) for helium in goethite corresponding to the "Stage 1" reference line regression on the hydrothermal experiment. Helium hosted in a diffusion domain with these diffusion parameters would be nearly perfectly retentive at billion-year timescales at 25°C. At 40°C, the equilibrium age for this domain (the helium age at which radiogenic helium production and diffusive loss are equal) is 33.2 billion years, attained after 50.4 billion years.

It is difficult to use Arrhenius plots to compare many samples' helium retentivities. Farley, H. B. Monteiro, et al. (2024) report that linear regressions on apparent diffusivities calculated from helium step heat data on goethite in vacuum can be fit to yield widely variable diffusive parameters even for a single sample (e.g. activation energies ranging from 107 to 176 kJ/mol within a single experiment on YAN-0201 AC). Similarly, regressions on helium step heat data of polycrystalline materials can yield erroneous activation energies when not applied carefully (Farley, 2018). Because of the complexity of comparing Arrhenius plots of calculated apparent diffusivities for polycrystalline samples with differing heating schedules, it is advantageous to convert heating schedules to some parameter that can be compared easily between experiments. One effective way to reduce the complexity of step heat experiments to visualize helium retentivity is to "flatten" a thermal history to a dimensionless diffusive timescale: the Fourier number ($Fo = Dt/a^2$). This approach offers similar benefits to Delta plots (Farley, 2018) or r/r0 plots (Lovera et al., 1991), but is optimized to qualitatively compare the helium retentivities of polycrystalline samples for a given thermal history (expressed as diffusive time).

For a given diffusive domain, a Fourier number of $Fo \approx 1$ corresponds to the complete degassing of that domain. Complex heating schedules can be converted to Fourier numbers by assuming a reference line similarly to a Delta or r/r0 plot (Lovera et al., 1991; Farley, 2018). This procedure is especially useful for MDD (Multiple Diffusion Domain) systems, where the apparent diffusivities decrease over the course of a step heat experiment as the least retentive domains are degassed. For a single spherical diffusion domain, the fraction of helium released is a function of the Fourier number alone (Crank, 1979; Fechtig and Kalbitzer, 1966). If the activation energy of the reference Fourier number is the same as the activation energy of diffusion, then the amount of helium released in that domain then is an exact function of the reference Fourier number multiplied by a constant. Thus, a plot with reference Fourier numbers against cumulative fractional helium yields for different samples will yield the same geometries and interrelationships regardless of the intercept used to calculate the reference Fourier numbers; the values of the reference Fourier numbers will be arbitrary and only meaningful as relative diffusive time. Another implication is that different aliquots of the same sample should appear identical in this space regardless of heating schedule (see Win-06-01b in Fig. 2.17, or overlapping hematite samples in Fig. 2.16). Many such plots were generated with a variety

of activation energies between 149 kJ/mol to 171 kJ/mol, but the relative positions of the degassing curves for all samples are preserved, illustrating the utility of this approach even for comparing helium retentivity between different minerals.

Fig. 2.16 compares the step heat results of Win-06-01b-HT with 11 step heat experiments of 3 samples of polycrystalline hematite and 9 step heat experiments of 2 samples of goethite in Vacuum, including Win-06-01b. In a thermal-history normalized space, it is obvious that goethite in vacuum and hematite define well-separated populations and goethite in vacuum is far less retentive of helium than hematite. Most interestingly, the most retentive sample on these plots is clearly the Win-06-01b-HT goethite under hydrothermal conditions, at least for the 14.7% of the total 3He hosted in the least retentive domains. This implies that the helium retentivity of stable goethite may even be higher than for hematite of similar size, although step heat experiments on irradiated single-crystal goethite of controlled size under hydrothermal conditions would be needed to fully assess this possibility. This plot also makes it clear that step heat results for different, unrelated goethite samples in vacuum are more similar to one another than step heat results of the same sample at different water activities.

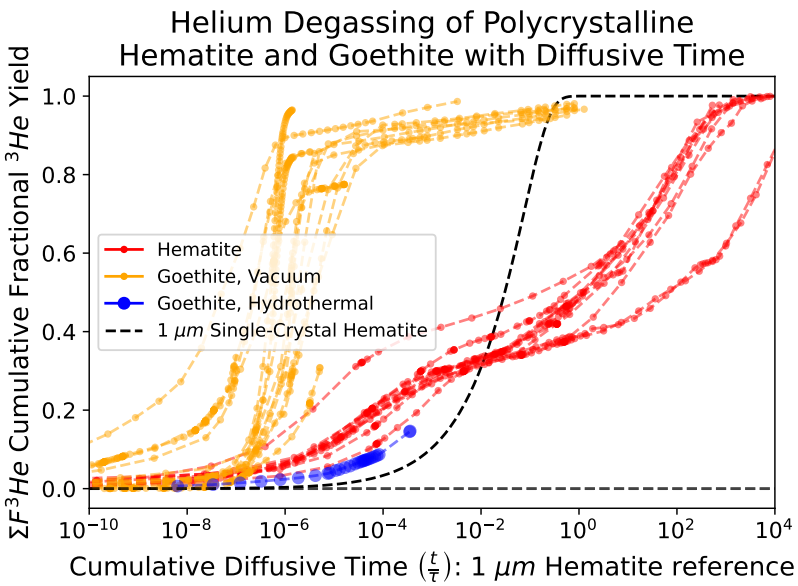


Figure 2.16: Heating schedules for polycrystalline goethite and hematite samples are flattened to reference Fourier numbers, dimensionless numbers that can be thought of as integrated thermal history, allowing for easy comparison of entire helium step heat experiments. The 3He cumulative fractional yield is plotted on the y axis as a function of Fourier number in log scale on the x axis. Curves on the left of the plot degas earlier within a given thermal history than the curves on the right. Hematite and goethite step heats in vacuum each cluster closely together, forming distinct populations. Most strikingly, the hydrothermal Win-06-01b-HT experiment is the most helium-retentive polycrystalline sample on the plot, at least until the experiment's end at a cumulative fractional yield ∼15% F3He. In vacuum, all hematite samples are significantly more retentive of helium than all goethite samples, with some limited exceptions for the least retentive domains holding <5% of the total helium. Win-06-01b step heat experiments in vacuum are indistinguishable from the Capao samples in vacuum (the two leftmost goethites are Capao samples; the rest of the Capao samples closely cluster with Win-06-01b in vacuum). The differences in the curves here reflect differences in grain size distributions for hematite, and some combination of grain size and dehydration rate (due to variable water pressure in the experimental vessel and crystal aggregate size) for goethite. Vacuum step heats of 11 aliquots of 4 samples of polycrystalline hematite from the Grand Canyon, Arizona (Farley and Flowers, 2012) and the Gogebic Range, Michigan (Farley and McKeon, 2015), and 9 aliquots of Win-06-01b and Capao goethite from Minas Gerais, Brazil (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024) are plotted here. The reference D/a^2 values are calculated for 1µm domains of hematite ($\ln(D_0) = 0.66$; $E_a = 171$ kJ/mol: (Farley, 2018)), and the reference curve for the degassing of 1µm spherical domains of hematite with uniformly distributed 3He is calculated using the forward equations of Fechtig and Kalbitzer (1966).

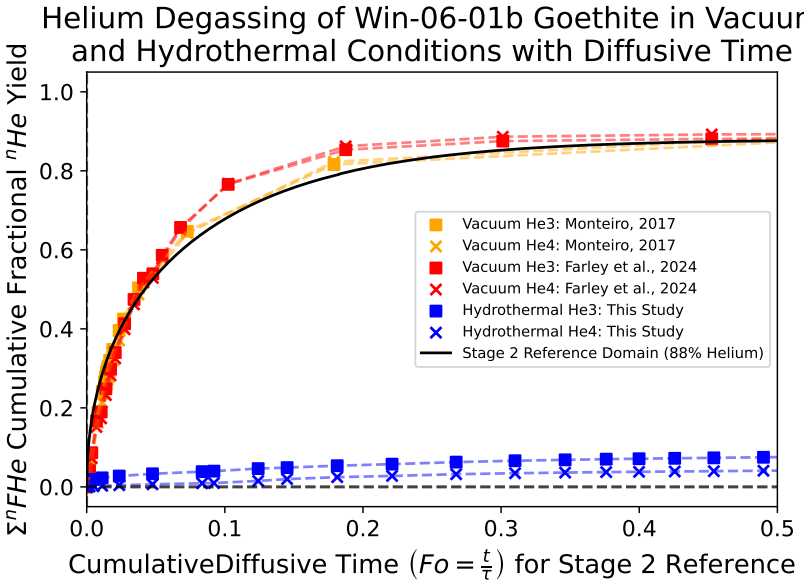


Figure 2.17: Win-06-01b step heats flattened to the Fourier number calculated from the "Stage 2" reference domain from the Win-06-01b step heat of H. Monteiro (2017). In this space, the helium retentivity contrast between the same goethite sample in vacuum and under hydrothermal conditions is striking. This plot shows that nearly 88% of helium loss for Win-06-01b in vacuum can be well-explained by a single thermally activated process following an Arrhenius law, corresponding to the "Stage 2" reference line from the experiment of H. Monteiro (2017), unlike polycrystalline hematite samples, which show a complex helium release pattern against a reference Fourier number, as expected for polycrystalline samples with complex grain size distributions. Whether or not the frequency factor D_0/a^2 for the "Stage 2" process Arrhenius law corresponds to grain size, aggregate size, or another parameter relevant to helium loss in goethite in nature is unclear. This process seems to be slightly faster for the experiment of Farley, H. B. Monteiro, et al. (2024), potentially due to lower water pressures from a liquid nitrogen trap. Both experiments of H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) show that helium diffusivity (and the 4/3 ratio) suddenly decreases after a cumulative fractional yield of 3He around 88%, signifying the beginning of "Stage 3" (Farley, H. B. Monteiro, et al., 2024), despite different heating schedules.

Profiling Radiogenic Helium Loss with $\frac{^4He}{^3He}$ Ratios of Win-06-01b-HT
Bulk-normalized 4He/3He ratios in step heats of irradiated samples measure rounding concentration profiles by natural helium loss, if the loss mechanism in nature and in the laboratory share the same geometry (Shuster, Farley, et al., 2004; Farley, H. B. Monteiro, et al., 2024). Aside from the first experimental step, bulk-normalized 4/3 ratios for all experiments begin at vanishingly small values and steeply increase at first, with the rate of increase slowing as the experiments proceed (Fig. 2.18). This behavior is consistent with complete then partial loss of radiogenic helium in nature in the least retentive domains and is observed in all step heat experiments on irradiated goethite (Farley, H. B. Monteiro, et al., 2024). The hydrothermal experiment, however, records lower bulk-normalized 4/3 ratios than the vacuum experiments at any given $\Sigma F_{3\mathrm{He}}$ for the duration of the experiment. Similarly, the approach to the 4/3 plateau varies between the three experiments: a high bulk normalized 4/3 He plateau is most decisively represented in the Farley, H. B. Monteiro, et al. (2024) experiment, whereas the 4/3 ratios in the step heats of H. Monteiro (2017) transition more gradually before reaching a plateau later in the experiment, while the increase in 4/3 ratios is slowest for the hydrothermal experiment (Fig. 2.18). This potentially could be explained by the relative dominance of dehydration processes or their mode of spatial propagation as a function of water pressure, where the smoothest 4/3 spectra are generated by higher water pressures. The experiment in Farley, H. B. Monteiro, et al. (2024) was open to a liquid nitrogen trap during heating, which would maintain lower water pressures than H. Monteiro (2017), who in turn held Win-06-01b at much lower water pressures than the hydrothermal experiment presented in this work. While the hydrothermal experiment did not reach a plateau, it may have if the experiment was allowed to proceed further.

Interestingly, both vacuum experiments show a maximum bulk normalized 4/3 He ratio of ~ 1.1 at or before $\Sigma F_{3\mathrm{He}} = 0.88$, after which the bulk normalized 4/3 He decreases to ~0.9, suggesting that these values are independent of heating schedule and may reflect a material property of Win-06-01b.

Both vacuum experiments record a somewhat high 4/3 ratio for the first measurement that is not observed in the hydrothermal experiment (Fig. 2.18). This feature represents a very small amount of the total helium and could be related to dehydration of poorly retentive domains in the vacuum, or it may have

been lost in the hydrothermal experiment while the ampoule was held at 70°C to degas entrained atmosphere before the initiation of the experiment. This small abundance of poorly retained 4He observed in the vacuum experiments could be related to physisorption of atmospheric helium (Protin et al., 2016). In any case, it represents a negligible fraction of the 4He contained in Win-06-01b.

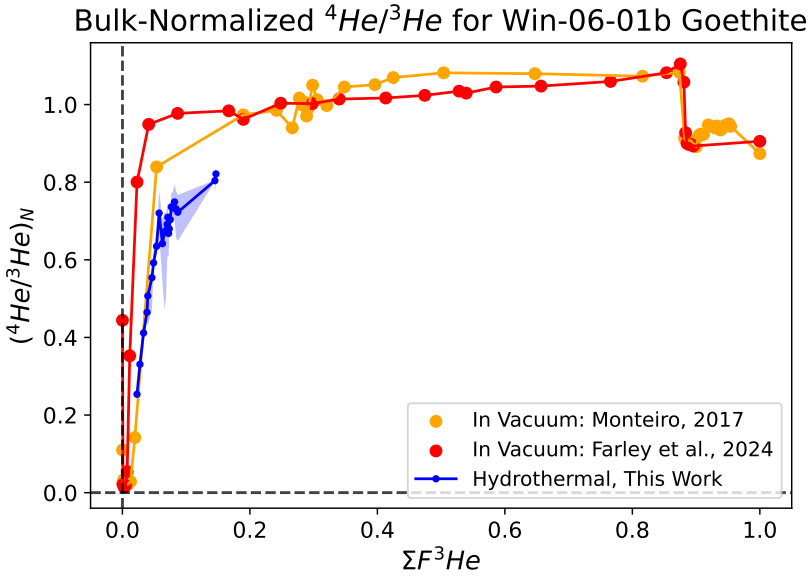


Figure 2.18: Bulk-normalized 4/3 ratios for Win-06-01b goethite step heats against cumulative fractional 3He yield. The hydrothermal experiment shows lower bulk-normalized 4/3He ratios at a given ΣF_{3He} for the entire duration of the experiment, besides the very first step. All experiments' 4/3He ratios begin at very small values and increase steeply at first, which is consistent with complete then partial loss of radiogenic helium in nature for the least retentive helium domains. The rate of increase slows between ΣF_{3He} of 0.05 to 0.2 for all experiments. The hydrothermal experiment is plotted as a shaded envelope spanning the "treatment as uncertainty" lower- and upper-bound 4/3He ratios; 10 of 34 analyses are excluded, either because the upper and lower bounds of ω cannot be computed for the early steps before λt was large enough to assume a steady-state permeation condition, or because the measured 3He signal for that step is below 0.1 pccSTP and too low-signal to give a reliable ratio.

2.6 Conclusions

Predicting the degree of helium retention and loss to guide interpretation of U-Th/He dates of natural goethite based on factors such as grain size, crystallinity, radiation damage, defect density, impurities, and aluminum substitution remains challenging. Previous efforts to understand Helium loss in natural goethite with step heating experiments in vacuum have instead queried the kinetics of dehydration (Farley, H. B. Monteiro, et al., 2024). It remains unclear how informative existing vacuum helium step heat data are with respect to helium retention and loss in natural samples. Because goethite dehydration is slowed or stopped at high water pressure, helium step heats under hydrothermal conditions are more likely to interrogate the processes controlling radiogenic helium loss in natural goethite. The method presented in this study leverages the high helium permeability of silica glass to easily conduct helium step heat experiments under hydrothermal conditions, achieving internal water pressures up to 86 bars at 300°C using standard ultra-high vacuum experimental setups (Farley, Reiners, et al., 1999).

The experiment conducted on irradiated Win-06-01b represents the first hydrothermal helium step heat experiment for goethite. Results demonstrate that apparent diffusivities under hydrothermal conditions are approximately seven natural log units lower than those observed in vacuum, suggesting that helium retention is more robust in natural environments than suggested by vacuum step heat experiments. The observed diffusivities align with a volume-diffusion loss mechanism consistent with MDD systematics, yielding an activation energy of 157.1± 5.3kJ/mol.

Further studies, including hydrothermal experiments on variable size fractions of single-crystal goethite, are necessary to confirm the mechanisms of helium migration in goethite. This methodological advancement also provides a foundation for systematically exploring the effects of factors such as grain size, crystallinity, chemical composition, and radiation damage on helium mobility in goethite.

Supplementary Information

for Chapter 2: *Helium Step Heat Experiments on Goethite in Hydrothermal Conditions Indicate High Retentivity at Surface Temperature and Consistency with Multiple Diffusion Domain Systematics*

2.1s Supplementary Verification: The Naïve/Informed Bracketing Method for the Encapsulated, Irradiated Win-06-01b Step Heat Experiment

This supplement documents the naïve/informed bracketing method: an independent, more bespoke treatment of the HT-Win-06-01b dataset that explicitly tracks the helium content of the ampoule step by step, rather than bounding it as in the main text's "treatment as uncertainty" approach (§2.2, §2.3). It is retained here solely as a secondary, independent verification of the main text's results and conclusions, not as the primary basis for them: the results of the two methods' distilled datasets and derived quantities agree, corroborating the main text's simpler treatment. This supplement covers only the naïve/informed bracketing dataset and the bespoke, step-by-step corrections applied to it (the Raw Data and Corrections Columns through Totals sections below); the isothermal-permeation-fit methodology and cooling-tube numerical model that establish the ampoule's elevated permeability – the evidentiary basis for the main text's conservative literature-permeability assumption, and a prerequisite for the corrections applied below – are described in full in the main text (§2.2).

Summary

Helium isotope measurements of evolved gasses from hydrothermal steps heats of glass-encapsulated irradiated HT-Win-06-01b goethite of the Flinders Ranges (H. Monteiro, 2017), were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Raw data from these measurements are found on the first tab "Raw Data and Corrections" of the "IrrWin0601bTube012624FullDataBJT.xlsx" file. Because the sample was encapsulated in an imperfectly helium-transmissive silica glass ampoule, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile for experimental timescales. For 3He, the resulting low-temperature measurements were above blank for the majority of the experiment and were entirely controlled by the process of permeation through silica glass, and therefore were used to estimate the amount of residual helium within the ampoule between high-temperature experimental steps. These corrections are carried out in the "Raw Data and Corrections" tab using permeability values derived internally within the experiment from the "Naïve Apparent Lambdas" tab. Diffusivities calculated using the equations of

Fechtig and Kalbitzer (1966) for 3He and 4He can be found on the "Diffusivities" tab. During the step heat experiment, 116 measurements of high 3He MM gas standard were analyzed to track instrumental drift of 3He and 4He sensitivities and to quantify internal uncertainties using reproducibility. For the final step, the glass ampoule was broken and the remainder of the helium was degassed in a resistance furnace at 1100°C. Data from this final step can be found on the "Totals" tab of "IrrWin0601bTube012624FullDataBJT1.xlsx".

Raw Data and Corrections Columns

Raw Data

"Nominal (Heated) Time" refers to the time programmed for the step heat. These times are used for both calculating apparent diffusivities with the equations of Fechtig and Kalbitzer (1966) and for permeation lag corrections.

The "Watlow Time" column gives the last timestamp recorded in the Watlow temperature log at the target temperature. It is always slightly higher than the nominal time because the recorded time includes the initial time required to heat the sample up to the programmed temperature: typically around 30 seconds to two minutes. This column was generated by a python script that read these values from the Watlow text file temperature records. It was used in early stages of data analysis to identify when Watlow errors within Labview had held the sample for longer than the programmed step duration. When these discrepancies were identified, the "Nominal Time" was updated to reflect the time the sample was held at the programmed temperature. These times are not directly used for any of the data reduction.

The "Total Time" column is the difference between the analysis timestamp and the previous analysis timestamp. These timestamps are recorded during the inlet of evolved, purified Helium into the Helix-MC-Plus^{10K}, which immediately precedes the pump out of the lamp cell chamber to the turbo pump. Therefore, the "Total Time" represents the total amount of time that helium was allowed to permeate through the silica glass ampoule into the UHV lamp cell, including both the time the sample spent "at temperature" and the time that the sample was cooling to or at room temperature (24-26°C). Since the pump out times were only 1 minute and always occurred when the sample was coolest (and permeation was slowest), permeation from the ampoule during pump out is neglected in the data reduction. Less than one thousandth of the helium

remaining in the ampoule would permeate out within one minute at room temperature (Eq. (1.6), Chapter 1, with log permeability (ln(K)) at 26°C of 23.88 ln(atoms/cm*atm*s)). Due to the formatting of the time stamps, these values have a 1-minute resolution.

The "Cooling Time" is the difference between the "Total Time" and "Nominal Time". It is greater than or equal to the time to purify evolved gas in preparation for analysis: ~26 minutes. This value is used for permeation corrections when low-temperature permeation steps with cooling times greater than 26 minutes are used.

Abundances, 4He Blank Corrections, Measured 4He/3He Sensitivity and Uncertainty Calculations

3He and 4He sensitivities were calculated for each analysis of the MM standard gas, assuming that each standard aliquot contained 157.50±0.41(2σ) nccSTP 4He and 3.567±0.040(2σ) pccSTP 3He. These values were obtained from sensitivities from a known amount of a standard gas with similar helium abundances and with the signals from 10 aliquots of the new MM standard tank in June 2023.

1 sigma uncertainties related to internal reproducibility were estimated for each sensitivity calculation as the rolling standard deviation of sensitivities calculated from 5 MM standards. In other words, the internal reproducibility of the sensitivity was estimated as the standard deviation of each calculated sensitivity and the 2 sensitivities immediately before and after. Uncertainties related to external reproducibility from the amount of 3He and 4He in each standard aliquot were added to the total uncertainty as well.

Sensitivities and their uncertainties were interpolated to times between standard analyses using a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) from the SciPy package in python. These interpolated sensitivities and uncertainties were used to convert the 3He and 4He signals (cps and fA respectively) to absolute atomic abundances and uncertainties in units of pccSTP and nccSTP respectively. The sensitivities and interpolations are graphically displayed in the "STDs" tab in a plot generated using the Matplotlib package in python.

Helium-4 Blank Corrections

Throughout the experiment, low temperature permeation steps yielded unpredictable 4He signals that did not exponentially decay like the 3He signals did. This effect is most obvious when considering analyses where cooling times were constant (~1560 seconds). In these cases, 3He signals always decreased exponentially whereas the 4He signals often increased (or decreased) erratically. A blank was measured at the beginning of the experiment and showed negligible blank for 3He (regressing to below 0 cps at the inlet time) and a very low value for 4He ($4.84\text{x}10^{-2}$ fA). Low temperature permeation steps sometimes yielded much higher signals despite failing to conform to the expectation of exponential decay that is observed in the same steps' 3He measurements. For instance, the 4He signal in a cool permeation step at 105°C from reference file He30509.txt was $7.93\text{x}10^{0}$ fA, even when the high temperature experimental step preceding it (He30502.txt) yielded a similar value: $8.28\text{x}10^{0}$ fA. For the same pair of analyses, the 3He signal matched expectations and varied by two orders of magnitude. 4He from permeation steps never fit the equation for cumulative permeation, $m_{cum} = n_0 \left(1 - e^{-\lambda t}\right)$ (Eq. (1.7), Chapter 1), and thus could not be used to infer the permeability for 4He or for the residual amount of 4He remaining in the ampoule (n_0). For comparison, the same equation always fit the low temperature permeation steps well for 3He when the cooling time of each analysis was constant.

The erratic behavior in the 4He data is best explained by a shifting blank during the experiment, which was not measured or quantified directly. It was largest in the middle of the experiment and returned to low values by the end of the experiment that were similar to the blank measured before the experiment. Because the blank for 4He was not well controlled and appeared to have affected many of the values observed even in the high temperature experimental steps, a blank correction was deemed necessary. The 4He data of the permeation steps are interpreted here to be dominated by blanks, so high temperature experimental steps were coarsely blank corrected by subtracting the average 4He signal of the subsequent permeation steps. The entirety of this correction was conservatively added to the 2σ uncertainty of the experimental data for 4He. One blank corrected measurement was within error of zero after permeation corrections and was not plotted. The permeation lag was estimated for the 4He data using the 3He data and assuming that the 4He

permeability was 0.91357±0.14603 (2σ) that of the 3He permeability. These relative permeability values for 3He and 4He were measured at Caltech for a similar silica glass ampoule half-filled with water and equilibrated with ~ 0.1 torr of tank helium. This process is described in greater detail in the Permeation Lag Corrections section.

Some 3He permeation steps that followed especially low-yield experimental steps were within error of baseline 3He cps values (similarly to the measured blank at the beginning of the experiment). These low measurements always preceded by measurements where 3He was quantified above blank, so permeation lag corrections were calculated using those values instead. In these cases, uncertainty from the permeation lag corrections dominate the uncertainty of the diffusivities, but do not affect the interpretation of the dataset.

Bracketing Method Ratios

4He/3He ratios are presented with reference to the irradiated Winsor Goethite bulk ratio (~1000 or ~1000 Ra) and as standalone values. Uncertainties are propagated in quadrature from the measured 3He and 4He abundances and from the bulk ratio, when applicable. The blank and permeation corrected ratios have the same general shape as the uncorrected ratios. The ratios plateau around a cumulative 3He fraction of 0.06 at a bulk normalized 4He/3He ratio of 0.75. The early plateau in 4He/3He ratios with 3He fraction occurs similarly in the vacuum step heat experiments of the same sample, in agreement with previous studies (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024), suggesting a relatively low loss of radiogenic helium for the Winsor Goethite. The vacuum and hydrothermal experiments differ in the bulk-normalized 4He/3He ratio at the plateau, with the hydrothermal experiment plateauing at a lower 4He/3He ratio of 0.75 times the bulk ratio, whereas the vacuum experiments plateau at the bulk ratio. This could be explained by a 4He-retentive inclusion in the hydrothermal experiment of this work similar to those discussed by P. M. Vasconcelos et al. (2013), or may reflect trapping of 3He in newly formed hematite during the early steps of the vacuum experiment.

Permeation Lag Corrections for 3He

Helium evolved from the irradiated goethite must first permeate through the silica glass ampoule into the high vacuum of the lamp cell to be analyzed, a

process which is non-instantaneous. Measurements can therefore underestimate the helium production of the sample during a step, as some of the produced helium will remain within the ampoule after a high temperature step, and measurements can also overestimate the amount of produced helium if there is some amount of helium remaining from previous step heats within the ampoule at the beginning of a new experimental step. To address this effect, we quantify the amount of helium remaining within the ampoule between high temperature experimental steps, and apply corrections using the rearranged form of Eq. (1.1) (Chapter 1), solved for m:

$$m = n_0 + \omega - n_t$$

Where ω is the true helium production of the sample in a high temperature experimental step, n_0 is the amount of helium within the ampoule at the beginning of the step, n_t is the amount of helium within the ampoule at the end of the experiment, and m is the amount measured. Values of ω and m are most often within 10% of one another but differ by up to 47.2% for 3He within this dataset. n_0 and n_t are tracked in the "3He n_0 (pccSTP)" column, where n_t for any measurement is the n_0 value of the next row.

At low temperatures (<120°C), when no new helium is evolved from the sample, the measured value is simply the difference between the helium contents of the ampoule before the measurement and after (Eq. (1.1) with $\omega = 0$):

$$m = n_0 - n_t$$

In this case, all the helium measured is a result of helium permeation through the silica glass ampoule. Helium permeation is a first order process that follows Eq. (1.6) (Chapter 1), with $\lambda = KGRT$:

$$n_t = n_0 e^{-KGRTt}$$

Where K is the temperature-dependent permeability coefficient, G is a geometric factor which can be determined through measurements of the ampoule's inner and outer diameter, R is the gas constant, T is the temperature in Kelvin, and t is the time in seconds. For a cylindrical ampoule, the value of G can

be calculated with measurements of the inner and outer radii using Eq. (1.4) (Chapter 1). Inner and outer radii were measured using the amscope program at 11x zoom setting. These radii were both measured on opposing sides of the ampoule where there were air bubbles, allowing for the accurate sighting of the inner walls of ampoule. The inner diameter, *(2a)* was measured at 505 and 540 microns; the outer diameter *(2b)* was measured at 787 and 832 microns (Fig. 2.2). The geometric factor G was calculated for both pairs of measurements and averaged to give a value of G of 6708.6 cm^{-1}.

For permeation steps, Eqs. (1.1) (with $\omega = 0$) and (1.6) apply because the temperature is too low to evolve any helium from the sample, and the $\omega = 0$ form of Eq. (1.1) can be substituted into Eq. (1.6) to relate a measurement to n_0. This equation is calculated in the column entitled "3He n0 calc (pccSTP) (m/(1-e^-λt))". Errors in the measured value m and the permeability K are propagated in quadrature. This is the informed-bracketing relation of Eq. (1.5) (Chapter 1), expressed with Eq. (1.3)'s exponential form for ${}^{He}F$:

$$n_0 = \frac{m}{(1-e^{-KGRT})}$$

The "3He n_0 (pccSTP)" column tracks the accepted value of n_0 (and $n_{t,;}$ which is n_0 in the subsequent column) for ultimate use in permeation corrections using Eq. (1.5). Permeation steps colored in pink and magenta are "Informed Bracket" measurements. They set the accepted value of n_0 at the time of their measurement using Eq. (1.5). In other words, the "3He n_0 (pccSTP)" column inherits the value calculated by Eq. (1.5) in the "3He n0 calc (pccSTP) (m/(1-e^-λt))" column for permeation steps colored pink, whereas the permeation steps colored darker magenta set the value of n_0 with an altered version of Eq. (1.5) because they cooled to room temperature and sat permeating for longer than 1560 seconds before measurement. Their respective n0 values are calculated using an altered form of Eq. (1.5) (Eq. (2.2)) in a column entitled "3He n0 calc long cool":

$$n_0 = \frac{m}{\left(1 - (e^{-K_{hot}GRT_{hot}t_{hot}} * e^{-K_{cold}GRT_{cold}t_{cold}})\right)} \tag{2.2}$$

T_{cold} is 26°C, K_{cold} is measured internally in a process described in more detail below, and t_{cold} is equal to the value in the "Cooling time" column minus 1560

seconds, because the effective permeabilities for the warmer temperatures used here were calculated from measurements that cooled for 1560 seconds.

Most experimental steps within this study are separated by multiple permeation steps that each decrease the value of n_0. Values in the "3He n_0 (pccSTP)" column are color coded according to how they are calculated. Uncolored cells denote the default method of tracking the decreasing values of n0, which is handled with Eq. (1.6). The calculated n_0 value for uncolored step i in the ith row is $n_0^i = n_0^{i-1} e^{-RG(KTt)_{i-1}}$. This default behavior is followed when cooling times for the previous step were 26 minutes, to avoid applying a value for the effective permeation constant 'K' that doesn't capture the additional effects of long cooling. It is advantageous to use this approach because it allows for the n0 value to be set by earlier measurements with larger signals, avoiding potential errors related to near-blank measurements. Errors associated with K and n_0^{i-1} are propagated through Eq. (1.6) in quadrature.

Values in the "3He n_0 (pccSTP)" that are colored brown or tan are propagated by adding or subtracting measured amounts of helium from the previous or subsequent step's determined value of "3He n_0 (pccSTP)", and simply adding the uncertainty from the previous or subsequent n_0 and the corresponding measured value of helium m. Tan cells are calculated by adding that cell's measured value m to the subsequent row's value of n_0, while the brown cells are calculated by subtracting the previous row's measured value m from its value of n_0.

Permeation steps/series in rows colored grey at the end of the experiment were arbitrarily long and were considered to effectively empty the tube of any helium, so that the value of n_0 for the subsequent cell was 0.

Permeation steps that are colored blue were taken at room temperature and were used to calculate the tube's permeability at room temperature *(K_{cold})* for use in Eq. (2.2).

Calculating the Permeability of the Ampoule at Room Temperature ($\mathbf{K_{cold}}$)

Three measurements were taken of the experimental ampoule permeating at room temperature; corresponding to the blue rows. The determined values can be found after the final columns of the "Raw Data and Corrections" tab. These

measurements solve for the permeation coefficient K at 26°C using Eq. (1.6) solved for K, given by Eq. (2.3).

$$K = -\frac{\ln(n_t) - \ln(n_0)}{GRTt} \tag{2.3}$$

Where n_t is the value of n_0 in the "3He n_0 (pccSTP)" column of the next row. Uncertainties for n_0 and n_t are propagated in quadrature to K and are relatively large, however the values of K for the three measurements are tightly clustered. The mean value of K determined from the three measurements is taken as K_{cold} and the standard deviation of the three measurements is taken as the 1σ uncertainty. These steps all added/subtracted the measured values m to determine the value of n_0 so that there was no dependence K_{cold} on their n_0 values. Their n_0 values are thus color-coded tan or brown.

The first of these three measurements (H30413) was slightly dependent on the value of K_{cold}, as all the permeation steps in that series had cooled for longer than 26 minutes and required Eq. (2.2) to determine n_0. This series was treated subtly different than the rest of the permeation steps, since it both required and contributed a value of K_{cold}. The informed bracketing step He30418 is given bold yellow text in addition to being colored dark magenta to signify that the n_0 was determined using Eq. (2.2). For this step, K_{cold} and its uncertainty was taken to be the average and standard deviations of the other two determinations of K_{cold} ($2.68\text{x}10^{10} \pm 8.50\text{x}10^{9}$). The rest of the magenta "informed bracketing" steps used the average and standard deviation of all three measured values of K_{cold} ($2.56\text{x}10^{10} \pm 7.63\text{x}10^{9}$). All measured values of K_{cold} are within uncertainty of one another.

The permeation coefficient K_{cold} determined here is *not* an effective permeation coefficient, because there was no change in temperature at the end of the measurement. K_{cold} is plotted along with the effective permeation coefficients determined with the naïve bracketing method in a smaller plot, but it was not incorporated into the regression and error of the estimation of the effective permeabilities used in Eq. (1.5) for the informed bracketing method, only for the darker magenta informed bracketing steps using Eq. (2.2).

Permeation Lag Corrections for 4He

Because of blank-level 4He signals for the permeation steps, 4He abundances within the ampoule could not be independently estimated. Instead, they were

estimated using the 3He n_0 abundances in the "3He n_0 (pccSTP)" column immediately following high-temperature experimental steps, assuming a high-uncertainty but temperature independent ratio of permeabilities $\left(\frac{K_{3He}}{K_4He}\right)$. This approach utilizes the steady-state n_t value n_{ss} already established in Chapter 1: for any long high-temperature experimental step where the helium source is assumed constant and the permeation coefficient is known, Eq. (1.9) gives $n^{ss} \leq m/(\lambda t - 1)$, which for these long steps ($\lambda t \gg 1$) is well approximated by $n^{ss} \approx \omega/(\lambda t)$, the exact steady-state balance between the sample's helium production rate Ω and permeative loss ($n^{ss} = \Omega/\lambda$, with $\omega = \Omega t$ the total production over the step).

Taking the ratio of the steady state abundances for 3He and 4He within the ampoule:

$$\frac{n_t^{4He}}{n_t^{3He}}_{ss} = \frac{\omega^{4He}\lambda^{3He}}{\omega^{3He}\lambda^{4He}}$$

Consider that $\lambda = KGRT$ and G, R, and T are constants, the above expression reduces. Additionally, the value for ω is not known, but it is related to the measured value in an experimental step according to Eq. (1.1): $\omega = m + n_t - n_0$. For long experimental steps at high temperature, $\omega \gg n_t,\ n_0$, so $\omega \approx m$. Furthermore, even when n_t and/or n_0 are significant relative to the measured value m, their relative significances $\left(\frac{n_t}{m}, \frac{n_0}{m}\right)$ would be expected to be very similar for both isotopes, because their permeabilities through silica glass are similar and both isotopes experienced the same heating schedule that would affect their abundances similarly. Therefore, both isotopes would be expected to differ from the assumption that $\omega \approx m$ in a similar fashion. In other words, the assumption $\frac{\omega^{4He}}{\omega^{3He}} \approx \frac{m^{4He}}{m^{3}He}$ is a much stronger assumption than $\omega \approx m$. Making this assumption and reducing:

$$\frac{n_t^{4He}}{n_t^{3He}}_{ss} \approx \frac{m^{4He}K^{3He}}{m^{3He}K^{4He}}$$

Which can be solved for n_t^{4He}

$$n_t^{4He} \approx \frac{m^{4He}K^{3He}}{m^{3He}K^{4He}} \bullet n_t^{3He} \tag{2.4}$$

In this work, $\frac{K_{3He}}{K_4He} = 1.10 \pm 0.19$ (2σ) and is not treated as a function of temperature, as conservatively measured by (Thyer et al., 2026). When used

in any equation in this work, its uncertainty is always propagated through quadrature.

Eq. (2.4) is solved in the column entitled "4He n0 by ratio (nccSTP)". Since Eq. (2.4) only applies to the n_t value of a high temperature experimental step (which is the same as the n_0 value of the permeation step immediately following), it is only used to assign n_0^{4He} values for the permeation step immediately following each high-temperature experimental step. $\frac{m^{4He}}{m^{3He}}$ refers to the ratio of measured helium values of the high temperature step, and the n_t^{3He} term is n_0^{3He} of the permeation step immediately following a high temperature step in the "3He n_0 (pccSTP)" column.

Similarly to 3He, n_0^{4He} values decrease for every permeation step. Because the measurements of 4He are blank level, the evolution of n_0^{4He} is tracked in the "4He n0 (nccSTP)" column using Eq. (2.2) to account for long cooling times for some permeation steps. Because the permeation coefficient of 4He is not known for any temperature for Win-06-01b, the permeation coefficients for 3He at the relevant temperature is multiplied by 1.10 ± 0.19 (2σ) and the errors are all propagated in quadrature. Because of large blank corrections and uncertainty in the permeation coefficient, the errors for corrected 4He yields are very high. Regardless, diffusivities calculated from these 4He data form a sensible array on an Arrhenius plot. Many of these measurements are within error of the 3He array, but some less uncertain data are resolved to lower apparent diffusivity values, suggesting that there is some degree of helium loss recorded even up to 14% cumulate yield of 3He.

Naïve-Method Isothermal Permeability Dataset

The effective permeability coefficients used by the informed bracketing method below – and the isothermal-permeation-fit methodology, cooling-tube numerical model, and Arrhenius regression used to determine them – are described in full in the main text (§2.2). The complete fitted dataset underlying that regression is reported here: Table 2.3 gives the raw per-step data for all 10 isothermal sequences, and Table 2.4 gives the lambda/n_0 fit to each sequence (Fig. 2.3) that together define the Arrhenius regression of Fig. 2.4.

Table 2.3: Raw step-heat data (‘DistilledNaiveLambdas’ sheet) for the 10 isothermal 3He permeation sequences used to determine the ampoule’s apparent permeability (Fig. 2.3, Fig. 2.4). Series are separated by horizontal rules; within a series, steps are listed in the order measured.

Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
1	105.00	150	26	0.0190	1.1×10^{-4}
1	105.00	150	26	0.0111	7.4×10^{-5}
1	105.00	150	26	0.0080	5.3×10^{-5}
2	109.99	150	26	0.0229	1.4×10^{-4}
2	110.00	150	26	0.0115	7.3×10^{-5}
2	110.00	150	26	0.0075	5.2×10^{-5}
3	99.86	120	26	0.0106	6.1×10^{-5}
3	100.00	120	26	0.0074	4.0×10^{-5}
3	100.00	120	26	0.0049	2.5×10^{-5}
3	99.99	120	26	0.0043	2.1×10^{-5}
4	95.00	150	26	0.0109	2.8×10^{-4}
4	95.00	150	26	0.0081	2.0×10^{-4}
4	95.00	150	26	0.0062	1.4×10^{-4}
5	104.99	120	26	0.0215	8.9×10^{-5}
5	104.99	120	26	0.0144	6.0×10^{-5}
5	104.99	120	26	0.0104	4.3×10^{-5}
6	100.00	120	26	0.0126	5.6×10^{-5}
6	100.00	120	26	0.0094	3.5×10^{-5}
6	100.00	120	26	0.0074	2.0×10^{-5}
7	95.00	120	26	0.0072	3.0×10^{-5}
7	95.00	120	26	0.0063	2.6×10^{-5}
7	95.00	120	26	0.0045	1.9×10^{-5}
8	90.00	150	26	0.0059	2.7×10^{-5}
8	89.90	150	26	0.0048	2.4×10^{-5}
8	90.00	150	26	0.0036	1.9×10^{-5}
9	90.00	180	26	0.0100	6.2×10^{-5}
9	89.92	180	26	0.0067	5.1×10^{-5}
9	90.00	180	26	0.0054	4.8×10^{-5}
10	99.99	150	26	0.0104	4.3×10^{-5}
10	99.98	150	26	0.0068	3.5×10^{-5}
10	100.00	150	26	0.0054	3.4×10^{-5}

Table 2.4: Fitted λ (permeation rate constant) and n_0 (residual helium at the start of the series) for each of the 10 isothermal sequences in Table 2.3, from the weighted non-linear least-squares fit of Eq. (1.8) (Chapter 1) described in the main text (§2.2). n is the number of steps in the sequence. These are the same per-sequence fits plotted in Fig. 2.3 and used to build the Arrhenius regression in Fig. 2.4.

Series	Temp (°C)	Duration (min)	n	λ (s^{-1})	$\pm 2\sigma$	n_0 (pccSTP)	$\pm 2\sigma$
1	105.00	150	3	4.9×10^{-5}	1.4×10^{-5}	0.0521	7.8×10^{-3}
2	109.99	150	3	6.3×10^{-5}	1.8×10^{-5}	0.0508	7.3×10^{-3}
3	99.86	120	4	4.3×10^{-5}	1.3×10^{-5}	0.0377	5.2×10^{-3}
4	95.00	150	3	3.1×10^{-5}	2.6×10^{-6}	0.0444	2.2×10^{-3}
5	104.99	120	3	5.0×10^{-5}	6.2×10^{-6}	0.0696	4.7×10^{-3}
6	100.00	120	3	3.7×10^{-5}	4.1×10^{-6}	0.0534	3.4×10^{-3}
7	95.00	120	3	3.3×10^{-5}	1.5×10^{-5}	0.0348	1.1×10^{-2}
8	90.00	150	3	2.7×10^{-5}	4.8×10^{-6}	0.0279	3.5×10^{-3}
9	90.00	180	3	3.0×10^{-5}	9.0×10^{-6}	0.0357	7.0×10^{-3}
10	99.99	150	3	3.8×10^{-5}	1.2×10^{-5}	0.0351	7.0×10^{-3}

Naïve-Method Step-Heat Dataset

The complete corrected step-heat dataset underlying the naïve bracketing method's apparent permeabilities (Fig. 2.4) is reported in Tables 2.5 and 2.6, reporting the central value and 2σ uncertainty for each quantity, analogous to Tables 2.1 and 2.2 in the main text for the informed bracketing ("treatment as uncertainty") method. Table 2.7 separately reports the bulk-normalized ^{4}He/^{3}He ratios and cumulative fractional ^{3}He yield for each step. "–" denotes steps for which a quantity could not be determined (see §2.3 in the main text).

Table 2.5: ^{3}He step-heat dataset for HT-Win-06-01b under the naïve bracketing method: measured and cumulative ^{3}He, the initial and residual ampoule content (n_0, n_t), the true step production (ω), and the apparent diffusivity $\ln(D/a^2)$, each with its 2σ uncertainty.

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.9630	0.0000	0.00×10^{0}	0.00×10^{0}	0.1968	2.22×10^{-2}	1.1597	2.63×10^{-2}	-21.910	0.045
180	463	0.5442	0.9630	7.26×10^{-2}	1.29×10^{-2}	0.0782	4.93×10^{-3}	0.5497	1.60×10^{-2}	-22.184	0.089
190	550	0.5292	1.6241	1.95×10^{-2}	2.54×10^{-3}	0.0679	8.20×10^{-3}	0.5777	1.30×10^{-2}	-21.976	0.085
200	500	0.5190	2.2112	3.28×10^{-2}	5.42×10^{-3}	0.0532	5.77×10^{-3}	0.5394	1.15×10^{-2}	-21.702	0.088
210	475	0.5770	2.7646	1.41×10^{-2}	2.11×10^{-3}	0.0552	5.71×10^{-3}	0.6182	1.17×10^{-2}	-21.311	0.082
220	450	0.6333	3.3799	1.10×10^{-2}	1.80×10^{-3}	0.0469	4.58×10^{-3}	0.6692	1.24×10^{-2}	-20.991	0.080
230	425	0.7899	4.0551	5.87×10^{-3}	1.86×10^{-3}	0.0639	3.48×10^{-3}	0.8479	1.25×10^{-2}	-20.514	0.066
225	908	0.7513	4.8885	1.95×10^{-2}	2.45×10^{-3}	0.0420	4.94×10^{-3}	0.7738	1.42×10^{-2}	-21.200	0.121
215	450	0.1415	5.6842	1.31×10^{-2}	1.85×10^{-3}	0.0278	3.54×10^{-3}	0.1563	6.48×10^{-3}	-22.015	0.316
205	520	0.0763	5.8529	7.18×10^{-3}	1.23×10^{-3}	0.0125	1.63×10^{-3}	0.0816	8.10×10^{-3}	-22.788	0.829
195	1106	0.0670	5.9491	2.12×10^{-3}	4.02×10^{-4}	0.0307	2.86×10^{-3}	0.0956	3.23×10^{-3}	-23.370	1.094
185	740	0.0227	6.0246	1.25×10^{-2}	1.64×10^{-3}	0.0090	1.06×10^{-3}	0.0192	2.00×10^{-3}	-24.567	2.822
175	920	0.0180	6.0610	3.68×10^{-3}	5.02×10^{-4}	0.0045	4.42×10^{-4}	0.0189	7.42×10^{-4}	-24.796	3.797
200	260	0.0224	6.0814	6.53×10^{-4}	1.17×10^{-4}	0.0095	1.23×10^{-3}	0.0313	1.65×10^{-3}	-23.022	1.495
210	520	0.0812	6.1081	7.48×10^{-3}	9.98×10^{-4}	0.0145	1.32×10^{-3}	0.0882	3.85×10^{-3}	-22.669	1.135
220	1167	0.5766	6.1914	2.82×10^{-3}	1.44×10^{-3}	0.0276	3.24×10^{-3}	0.6014	2.45×10^{-2}	-21.504	0.184
230	370	0.3577	6.7796	8.68×10^{-3}	3.96×10^{-3}	0.0422	5.13×10^{-3}	0.3912	1.48×10^{-2}	-20.712	0.146
240	320	0.6767	7.1563	1.72×10^{-2}	2.42×10^{-3}	0.0826	9.94×10^{-3}	0.7420	1.57×10^{-2}	-19.807	0.099
245	250	0.5549	7.8582	2.13×10^{-2}	8.87×10^{-3}	0.0516	5.65×10^{-3}	0.5852	1.46×10^{-2}	-19.710	0.125
250	210	0.6934	8.4743	1.08×10^{-2}	4.52×10^{-3}	0.0718	8.15×10^{-3}	0.7545	1.41×10^{-2}	-19.200	0.102
245	270	0.4659	9.2086	2.48×10^{-2}	3.48×10^{-3}	0.0498	5.84×10^{-3}	0.4909	1.01×10^{-2}	-19.810	0.197
240	320	0.2857	9.7206	2.07×10^{-2}	2.81×10^{-3}	0.0339	4.08×10^{-3}	0.2989	6.58×10^{-3}	-20.433	0.312
235	370	0.2199	10.0357	1.66×10^{-2}	2.20×10^{-3}	0.0273	3.29×10^{-3}	0.2307	5.23×10^{-3}	-20.810	0.401

continued on next page

(continued)

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
230	420	0.1400	10.2736	1.32×10^{-2}	1.76×10^{-3}	0.0255	2.53×10^{-3}	0.1523	4.00×10^{-3}	-21.332	0.603
225	652	0.1494	10.4279	1.26×10^{-2}	1.62×10^{-3}	0.0131	1.47×10^{-3}	0.1499	3.42×10^{-3}	-21.772	0.747
230	500	0.1550	10.5908	7.34×10^{-4}	1.67×10^{-3}	0.0128	1.32×10^{-3}	0.1671	3.37×10^{-3}	-21.383	0.461
235	450	0.1951	10.7582	2.57×10^{-3}	4.18×10^{-4}	0.0334	1.85×10^{-3}	0.2259	3.29×10^{-3}	-20.956	0.364
240	400	0.2425	10.9638	6.90×10^{-3}	1.45×10^{-3}	0.0251	3.16×10^{-3}	0.2607	1.23×10^{-2}	-20.672	0.318
245	350	0.3150	11.2328	5.79×10^{-3}	1.05×10^{-3}	0.0395	4.64×10^{-3}	0.3487	9.39×10^{-3}	-20.219	0.240
250	300	0.3716	11.5689	1.65×10^{-2}	2.24×10^{-3}	0.0467	5.24×10^{-3}	0.4018	9.62×10^{-3}	-19.888	0.209
250	300	0.3640	11.9627	2.43×10^{-2}	4.37×10^{-3}	0.0339	3.82×10^{-3}	0.3735	7.99×10^{-3}	-19.927	0.242
250	300	0.3452	12.3490	1.13×10^{-2}	1.60×10^{-3}	0.0395	4.48×10^{-3}	0.3733	7.91×10^{-3}	-19.895	0.244
250	10250	8.4829	12.7167	1.91×10^{-2}	2.53×10^{-3}	0.1165	7.38×10^{-3}	8.5804	1.28×10^{-1}	-19.958	0.189
240	785	0.2261	21.2187	2.90×10^{-9}	6.85×10^{-9}	0.0263	2.92×10^{-3}	0.2525	2.21×10^{-2}	-20.660	0.580
Total			146.1483								

Table 2.6: ^{4}He step-heat dataset for HT-Win-06-01b under the naïve bracketing method, structured as in Table 2.5.

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	n_0 (nccSTP)	$\pm 2\sigma$	n_t (nccSTP)	$\pm 2\sigma$	ω (nccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.0192	0.0000	0.00×10^{0}	0.00×10^{0}	3.10×10^{-3}	0.0014	0.0169	0.0057	-30.366	0.674
180	463	0.0305	0.0192	1.14×10^{-3}	5.30×10^{-4}	3.99×10^{-3}	0.0011	0.0281	0.0055	-28.998	0.446
190	550	0.0558	0.0660	1.22×10^{-3}	3.80×10^{-4}	7.42×10^{-3}	0.0016	0.0586	0.0039	-27.556	0.233
200	500	0.1024	0.1375	3.77×10^{-3}	8.85×10^{-4}	1.12×10^{-2}	0.0023	0.1066	0.0044	-26.114	0.134
210	475	0.1460	0.2500	3.32×10^{-3}	8.39×10^{-4}	1.48×10^{-2}	0.0030	0.1517	0.0071	-25.110	0.112
220	450	0.2091	0.4055	3.40×10^{-3}	9.21×10^{-4}	1.64×10^{-2}	0.0033	0.2142	0.0095	-24.217	0.098
230	425	0.3243	0.6319	2.79×10^{-3}	8.21×10^{-4}	2.83×10^{-2}	0.0051	0.3430	0.0097	-23.222	0.073
225	908	0.3483	0.9799	1.39×10^{-2}	2.76×10^{-3}	1.46×10^{-2}	0.0034	0.3517	0.0076	-23.575	0.114
215	450	0.0716	1.3487	7.32×10^{-3}	1.73×10^{-3}	5.64×10^{-3}	0.0025	0.0747	0.0075	-24.245	0.318
205	520	0.0385	1.4359	3.01×10^{-3}	8.65×10^{-4}	8.98×10^{-3}	0.0083	0.0338	0.0120	-25.142	1.022
195	1106	0.0336	1.4912	9.47×10^{-4}	4.68×10^{-4}	8.68×10^{-4}	0.0026	0.0258	0.0180	-26.143	2.313
185	740	0.0079	1.5467	4.94×10^{-3}	4.56×10^{-3}	3.39×10^{-3}	0.0011	-0.0021	0.0080	–	–
175	920	0.0152	1.6020	3.82×10^{-4}	1.16×10^{-3}	4.93×10^{-3}	0.0011	0.0152	0.0034	-26.474	3.286
200	260	0.0107	1.6351	5.77×10^{-4}	2.20×10^{-4}	1.07×10^{-2}	0.0322	0.0149	0.0011	-25.224	2.214
210	520	0.2166	1.6547	3.95×10^{-3}	8.87×10^{-4}	1.20×10^{-2}	0.0055	0.0612	0.1666	-24.476	3.011
220	1167	0.3187	1.8714	2.95×10^{-3}	8.90×10^{-3}	2.52×10^{-2}	0.0059	0.2365	0.0940	-23.836	1.736
230	370	0.2113	2.6766	4.31×10^{-3}	2.06×10^{-3}	4.90×10^{-2}	0.0136	0.2146	0.0199	-22.653	1.000
240	320	0.4286	3.1616	1.11×10^{-2}	2.82×10^{-3}	3.93×10^{-2}	0.0082	0.4018	0.0683	-21.726	0.680
245	250	0.3988	3.6430	1.43×10^{-2}	4.47×10^{-3}	4.89×10^{-2}	0.0102	0.4084	0.0195	-21.290	0.660
250	210	0.4440	4.2359	1.13×10^{-2}	2.88×10^{-3}	3.58×10^{-2}	0.0075	0.4663	0.0204	-20.824	0.594
245	270	0.3123	4.7263	1.85×10^{-2}	4.43×10^{-3}	2.45×10^{-2}	0.0052	0.3210	0.0132	-21.326	1.050
240	320	0.1927	5.0846	1.61×10^{-2}	3.71×10^{-3}	2.02×10^{-2}	0.0043	0.1955	0.0090	-21.918	1.654
235	370	0.1517	5.3031	1.27×10^{-2}	2.87×10^{-3}	1.90×10^{-2}	0.0039	0.1548	0.0071	-22.249	2.064
230	420	0.0991	5.4714	1.04×10^{-2}	2.35×10^{-3}	8.87×10^{-3}	0.0020	0.1031	0.0067	-22.749	3.071
225	652	0.0995	5.5835	9.79×10^{-3}	2.15×10^{-3}	9.15×10^{-3}	0.0019	0.0909	0.0087	-23.291	4.243
230	500	0.1051	5.6969	3.22×10^{-3}	8.17×10^{-4}	2.42×10^{-2}	0.0047	0.1063	0.0056	-22.845	2.498

continued on next page

(continued)

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	n_0 (nccSTP)	$\pm 2\sigma$	n_t (nccSTP)	$\pm 2\sigma$	ω (nccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
235	450	0.1369	5.8251	2.11×10^{-3}	5.73×10^{-4}	1.93×10^{-2}	0.0044	0.1502	0.0106	-22.363	1.886
240	400	0.1781	5.9762	4.79×10^{-3}	1.27×10^{-3}	3.09×10^{-2}	0.0065	0.1835	0.0115	-22.007	1.539
245	350	0.2305	6.1807	5.05×10^{-3}	1.47×10^{-3}	3.77×10^{-2}	0.0078	0.2493	0.0110	-21.519	1.128
250	300	0.2777	6.4385	1.39×10^{-2}	3.23×10^{-3}	2.66×10^{-2}	0.0055	0.2965	0.0110	-21.135	0.942
250	300	0.2654	6.7374	2.41×10^{-2}	5.21×10^{-3}	3.00×10^{-2}	0.0064	0.2619	0.0106	-21.204	1.142
250	300	0.2489	7.0231	9.79×10^{-3}	2.34×10^{-3}	1.02×10^{-1}	0.0186	0.2585	0.0150	-21.168	1.159
250	10250	6.8042	7.2901	1.54×10^{-2}	3.62×10^{-3}	2.15×10^{-2}	0.0054	6.8476	0.0767	-20.869	0.644
240	785	0.1853	14.1156	1.08×10^{-8}	3.61×10^{-8}	1.85×10^{-1}	0.0215	0.0054	0.0192	-21.540	1.209
Total			145.8145								

Table 2.7: Cumulative ^{3}He fractional yield and bulk-normalized ^{4}He/^{3}He ratios for HT-Win-06-01b under the naïve bracketing method. $(^4\text{He}/^3\text{He})_\omega$ is the ratio of the step's permeation-corrected ^{4}He and ^{3}He production (ω); F ^{3}He is the cumulative fractional ^{3}He yield; $(^4\text{He}/^3\text{He})_\omega$ is this step's ratio normalized to the bulk (whole-experiment) ^{4}He/^{3}He ratio. These data have much smaller uncertainties than those produced when permeation lag is treated as uncertainty.

Temp (°C)	Time (min)	$(^4\text{He}/^3\text{He})_\omega$	$\pm 2\sigma$	F ^{3}He	$\frac{(^4/^3\text{He})}{(^4/^3\text{He})_{\text{bulk}}}$	$\pm 2\sigma$
170	300	14.5470	4.91×10^{0}	0.0079	0.0146	0.0049
180	463	51.0439	1.02×10^{1}	0.0117	0.0512	0.0102
190	550	101.4852	7.20×10^{0}	0.0156	0.1017	0.0073
200	500	197.7263	9.21×10^{0}	0.0193	0.1982	0.0095
210	475	245.3349	1.24×10^{1}	0.0236	0.2459	0.0127
220	450	320.0824	1.54×10^{1}	0.0281	0.3208	0.0158
230	425	404.5353	1.29×10^{1}	0.0340	0.4055	0.0136
225	908	454.4434	1.29×10^{1}	0.0392	0.4555	0.0138
215	450	477.9049	5.18×10^{1}	0.0403	0.4790	0.0522
205	520	413.8402	1.53×10^{2}	0.0409	0.4148	0.1532
195	1106	269.9648	1.88×10^{2}	0.0415	0.2706	0.1887
185	740	–	–	0.0417	–	–
175	920	807.5143	1.84×10^{2}	0.0418	0.8094	0.1844
200	260	474.9353	4.38×10^{1}	0.0420	0.4760	0.0442
210	520	693.4629	1.89×10^{3}	0.0426	0.6951	1.8938
220	1167	393.2223	1.57×10^{2}	0.0467	0.3941	0.1575
230	370	548.6002	5.50×10^{1}	0.0494	0.5499	0.0554
240	320	541.4759	9.28×10^{1}	0.0545	0.5427	0.0932
245	250	697.8180	3.76×10^{1}	0.0585	0.6994	0.0385
250	210	618.0428	2.94×10^{1}	0.0636	0.6195	0.0302
245	270	653.8778	3.01×10^{1}	0.0670	0.6554	0.0310
240	320	654.1454	3.33×10^{1}	0.0690	0.6556	0.0341
235	370	671.0610	3.44×10^{1}	0.0706	0.6726	0.0352
230	420	677.0110	4.77×10^{1}	0.0717	0.6786	0.0484
225	652	606.4642	5.99×10^{1}	0.0727	0.6079	0.0604
230	500	636.2191	3.58×10^{1}	0.0738	0.6377	0.0365
235	450	664.7204	4.79×10^{1}	0.0754	0.6662	0.0486
240	400	704.0598	5.54×10^{1}	0.0772	0.7057	0.0560
245	350	714.8865	3.70×10^{1}	0.0795	0.7165	0.0379
250	300	737.8848	3.27×10^{1}	0.0823	0.7396	0.0337
250	300	701.0048	3.21×10^{1}	0.0849	0.7026	0.0330
250	300	692.4920	4.29×10^{1}	0.0874	0.6941	0.0436
250	10250	798.0564	1.49×10^{1}	0.1461	0.7999	0.0172
240	785	749.8748	1.01×10^{2}	0.1478	0.7516	0.1011
Total		997.716				

The finite-difference numerical model used to evaluate the contribution of the 26-minute cooling period to these apparent permeabilities – and to confirm that the ampoule's markedly elevated permeability is not an artifact of that cooling history – is described in full in the main text (§2.3).

Diffusivities

Apparent values of D/a^2 for 3He and 4He were calculated from times and corrected cumulative yields using the equations of Fechtig and Kalbitzer (1966). Uncertainties in the corrected values are propagated into the cumulative yield, and all errors are treated as uncorrelated for simplicity. Error bars plotted in the Arrhenius plots on this tab are likely exaggerated. Derivatives necessary to calculate uncertainty in the diffusivities using quadrature are tabulated to the right of the sheet. The Stage 1 reference-line linear regression (Fig. 2.7), its statistics, and the treatment of total-yield uncertainty in the regression are described in the main text (§2.3); the same regression underlies both the main text's treatment-as-uncertainty results and the naive-bracketing dataset reported here.

Totals

The main text (§2.2) already summarizes this final furnace-fusion step and its blank/re-extract corrections; the additional spreadsheet-level detail below (volume-split calibration, standard-averaging convention) is retained here for completeness. For the final step, the glass ampoule was broken and the remainder of the helium was completely degassed in a resistance furnace at 1100°C. The gas from this final step was split within the vacuum line on calibrated volumes so that purified helium from less than 1% of the evolved gas was inlet into a Thermo Fisher Helix SFT mass spectrometer. A gas standard with a known amount of 3He and 4He was measured twice preceding the final fusion measurement of the hydrothermal Win-06-01b, as was a split furnace hot blank. The gas standards were used to calculate sensitivities, which were taken to be the mean of the two measurements, with 1σ uncertainty approximated by their standard deviation. After the split measurement of the Win-06-01b ampoule, the furnace was reheated to 1100°C to ensure complete extraction of the helium in the sample. The final step was then taken to be the measured value minus the hot blank, plus the re-extract, all divided by their respective volume split fractions and multiplied by the calculated sensitivity. Uncertainties

were propagated in quadrature.

Chapter 3

SULFUR-BUFFERED HELIUM STEP HEAT EXPERIMENTS OF SULFIDES REVEAL REMARKABLE HELIUM RETENTIVITY AND COMPLEX BEHAVIOR: IMPLICATIONS FOR SULFIDE GEOCHRONOLOGY

Sulfide mineralization is a near-ubiquitous process in the formation of ore deposits of valuable chalcophile and siderophile elements. Helium isotope studies of sulfides have been applied to infer the origins of mineralizing hydrothermal fluids, and some pioneering studies have applied the U-Th/He method to date sulfides, especially pyrite (O. Yakubovich, Podolskaya, et al., 2020; O. Yakubovich, Vikentyev, et al., 2021). Yet, the interpretation of helium isotope data or U-Th/He ages requires an understanding of the conditions under which helium in sulfides behaves as an open or closed system, as well as the mechanisms of helium loss, a process which is poorly understood in sulfides. Furthermore, sulfides systematically reject U-Th and only produce appreciable radiogenic helium when they contain U-Th bearing inclusions, further complicating the interpretation of U-Th/He data in sulfides. We explore the systematics of U-Th/He in sulfides through two lenses: 1) an example application of the sulfide U-Th/He geochronometer to two test cases, and 2) helium step heat experiments of sulfide phases. We report pyrite/arsenopyrite U-Th/He ages for Au-ores in the Permian-Triassic Gympie formation of Queensland, Australia, and the Archean Kidd Creek VMS deposit of Ontario, Canada. The gympie sulfides yield a mean age of 253 ± 21 Ma and span the range between reported Ar/Ar and K/Ar ages of the the country rock and the mineralized quartz veins (Holcombe et al., 1997; Little et al., 1995). Pyrite and arsenopyrite from the 2.7 Ga Kidd Creek VMS deposit of Ontario contain large apatite inclusions ($r \leq 30$ μm) and span a range of ages from 465-860 Ma, which is significantly older than the region's ≈400 Ma apatite cooling age (Sturrock et al., 2024), but too young to suggest that the sulfides were closed to helium loss since crystallization at 2.7 Ga,

even when accounting for the limited helium retention of the large apatite inclusions.

Helium step heat experiments in sulfides are challenging because sulfides readily react/decompose in the high vacuum, high temperature conditions of the experiments, producing results that reflect phase decomposition/reaction kinetics that are not geologically relevant. We address this challenge by performing helium step heat experiments at temperatures up to 600°C on natural single-crystal pyrite, arsenopyrite, and galena implanted with uniform 3He through proton bombardment sealed in helium-permeable silica glass tubes stabilized under high sulfur (± arsenic) fugacity to prevent phase decomposition during analysis. We were unsuccessful in extending this method to stabilize chalcopyrite, chalcocite, or marcasite during step heat experiments: the helium retentivity of these phases in nature remains unclear. Retrograde-prograde thermal cycles on sulfur-stabilized galena support a Multiple Diffusion Domain (MDD) helium loss mechanism with an activation energy around 174.8 kJ/mol, but with an undefined, low helium retentivity that makes it unlikely to be useful as a target for helium isotope studies orU-Th/He geochronology. Comparisons of helium step heat data of sulfur-stabilized pristine and deformed pyrite point towards a multipath diffusion model for deformed pyrites, in which dislocations and fine fractures serve as high activation energy (300.6 ± 7.3 kJ/mol) fast pathways, with bulk volume diffusion following a lower activation energy that is constrained by step heats on pristine pyrite samples: 199.1 ± 3.5 kJ/mol. Helium step heat data of arsenopyrite are complex and may be complicated by exchange of sulfur and arsenic with the vapor phase, but qualitatively demonstrate very high helium retentivity. These experiments demonstrate that pyrite and is similarly helium-retentive to monzonite-structure phosphates and is more retentive of helium than natural zircon, apatite, hematite, or goethite. Arsenopyrite is less retentive, more similar to the most helium-retentive zircon.

3.1 Introduction

Sulfide minerals are nearly ubiquitous in the formation histories of economic ore deposits, and indeed are themselves the primary repositories for many base and precious metals in the crust. The question of when and how they formed is central to understanding ore-forming processes, crustal fluid circulation, and the tectonic contexts in which economic deposits develop. Despite decades of study, the timing of sulfide mineralization remains contested or poorly resolved across a wide range of deposit types, from shallow epithermal systems to deep, high temperature magmatic deposits in mafic intrusions. This uncertainty reflects both the genuine complexity of ore-forming systems — which are frequently multistage, hydrothermal, and overprinted — and the limitations of the geochronological tools available for dating sulfide phases directly.

An Alternative and Compliment to Re-Os Dates in Sulfide Geochronology

The rhenium-osmium isotope system is the most widely applied method for direct radiometric dating of sulfide minerals, exploiting the relatively high Re concentrations that pyrite, molybdenite, arsenopyrite, and pyrrhotite can accommodate. Re-Os geochronology has produced geologically meaningful ages across a wide range of sulfide-bearing deposit types, and remains indispensable for directly dating ore-forming events (M. Chen et al., 2015; Hnatyshin et al., 2026; Schoenberg et al., 1999; Ostendorf et al., 2019; Selby et al., 2002). Nevertheless, the Re-Os system is vulnerable to a set of well-documented failure modes, and a growing number of published ages exhibit scatter, implausible geologic interpretations, or internal inconsistencies that motivate the development of independent dating methods (Stein et al., 2003; Suzuki et al., 2001; Li et al., 2025; Lyu et al., 2020).

More practically, the highly specialized and costly n-TIMS method for Re-Os dating restricts its application to a select few labs, limiting the scale of its application (Ackerman, 2025). Furthermore, Re-Os dating of sulfides (other than the rare mineral molybdenite) requires gram-quantities (Qi et al., 2010) and is therefore poorly suited to resolve distinct mineralization episodes in multistage systems.

For multistage systems in particular, the co-application of independent radiochronometers is often useful to distinguish the timings for hydrothermal

alteration from uplift, cooling, exhumation, and potentially earlier emplacement of host rocks. For example, Selby et al. (2002) demonstrate that Re-Os molybdenite is more likely to accurately record the age of sulfide mineralization in the Tintina Gold Belt, Alaska, relative to the comparatively thermally-sensitive Ar-Ar mica ages. However, molybdenite is far from a ubiquitous phase in sulfide-bearing ores, unlike pyrite. A pyrite U-Th/He geochronometer would expand such opportunities to co-apply radiochronometers that interrogate processes operating at different times and temperatures to a much more diverse and expansive array of locales that experienced complex histories. Similar geochronometers developed for other sulfide phases such as arsenopyrite, galena, and sphalerite may offer the opportunity to disentangle multiple episodes of sulfide mineralization and alteration within an ore body, allowing geologists to better understand the sequence of events that lead to economic ore formation.

One such example of an opportunity for applying U-Th/He sulfide geochronology is in the world's largest known platinum-group element deposit, the Bushveld Complex of South Africa, which hosts the sulfide-bearing Merensky Reef and Platreef PGE deposits in the sulfide-bearing upper critical zone, which for which geochronometers yield a surprising diversity of ages: zircon U/Ph ages vary from ≈2057-2060 Ma, while rutile ages are significantly younger at ≈2053 Ma (Scoates et al., 2021), yet Re-Os systematics show evidence for multiple sulfide mineralization events within the same horizon (Schoenberg et al., 1999). The Re/Os and Rb/Sr isochron ages they report are unable to resolve these distinct mineralization events from the zircon U/Pb emplacement age. Unlike Re/Os dating, the U-Th/He system produces distinct ages for every dated aliquot, potentially offering the opportunity to resolve the timings of prolonged volatile activity or late hydrothermal Re-Os disturbance.

Similarly, at the Norilsk-Talnakh deposits of the Siberian Traps, the ore-bearing intrusions can be correlated with a specific transition in flood basalt source and eruptive style (Yao and Mungall, 2021), yet only three of many differentiated intrusions in the region are economic, and the precise timing and mechanism of sulfide saturation relative to trap emplacement is not fully understood. In the 1.1 Ga Sonju Lake Intrusion of Minnesota, USA, stable isotopic evidence suggests that meso/epithermal fluids reacted with primary magmatic sulfides raising questions about when redistribution occurred and how much metal was moved (Park et al., 2004). These are deposits where

independent thermochronometric constraints capable of distinguishing primary crystallization from later hydrothermal reworking are needed.

The Single-Jump Migration Model and its Application to Pyrite

The Russian school of noble gas geochronology spearheaded by Shukolyukov and Yakubovich, approaches the problem of helium migration in minerals through a theoretical framework that differs fundamentally from the volume diffusion model used for to describe noble gas migration dominant in the Western thermochronological literature (Dodson, 1973; Harrison et al., 1985; Fortier and Giletti, 1989; Lovera et al., 1991; Wolf, Farley, and Silver, 1996; Farley, 2000; Lee et al., 1997). In the Shukolyukov "single-jump model," helium atoms occupy discrete energy minima within the crystal structure, and migration proceeds through thermally activated jumps between these sites (Y. A. Shukolyukov et al., 2009; O. V. Yakubovich, Y. A. Shukolyukov, et al., 2010).

Within this framework, Y. Shukolyukov et al. (2012) developed the ^{190}Pt-^{4}He system as a new geochronometer for native platinum minerals, demonstrating through thermal desorption analysis that radiogenic ^{4}He in isoferroplatinum is stable to temperatures approaching the melting point of platinum, and obtaining ^{190}Pt-^{4}He ages from the Galmoenan and Kondyor massifs consistent with independent Sm-Nd and Rb-Sr estimates. He storage in native platinum was attributed to accumulation in vesicles and clusters at dislocation cores and grain boundaries. Western scientists similarly model helium migration in metals with single-jump mechanistic single-jumps, although these single jumps are not typically modeled to instantaneously liberate helium, instead the modeled single-jump allows the atom to spontaneously travel a distance of some angstroms with some frequency, introducing a spatial scale from which a diffusivity can be evaluated (Rätzke et al., 1992; Cost and Hickman, 1975; Wilson and Bisson, 1973).

O. V. Yakubovich, Mochalov, et al. (2015) extends this method to sperrylite ($PtAs_2$) a rare mineral found in magmatic PGE deposits that is isostructural with pyrite. ^{190}Pt-^{4}He ages from sperrylite at the Kondyor and Norilsk-Oktyabrsky deposits were geologically consistent, and the high He retentivity demonstrated in this arsenide prophesies similar behavior in pyrite.

Without providing any sulfur buffer, O. V. Yakubovich, Gedz, et al. (2019) interpreted ramped-heating experiments for a variety of silica glass encapsu-

lated sulfides under the single-jump step-heating method. Pyrite, in particular, was found to be anomalously high retentivity of helium. The first application of U-Th/He pyrite geochronometry to a natural ore deposit was reported by O. V. Yakubovich, Gedz, et al. (2019) and O. Yakubovich, Podolskaya, et al. (2020) for the Uzelga Cu-Zn VMS deposit of the South Urals, yielding a consistent weighted mean age of 382 ± 12 Ma that is older than regional prehnite-pumpellyite metamorphism, demonstrating that pyrite can survive a metamorphic overprint without complete isotopic resetting. While these results establish the viability of pyrite as a U-Th/He geochronometer, their analytical approach employs a simple, low-resolution monotonic heating schedule without grain-size scaling experiments, sulfur buffers, and proton-implanted ^{3}He domain mapping, leaves the diffusion/migration physics of sulfides undercharacterized. Unsurprisingly, given the rapid heating rates used in their experiments, their results for pyrite generally show abrupt helium loss at temperatures corresponding to the upper stability field of pyrite around 700°C Toulmin and Barton (1964), and therefore are unlikely to contain substantial information about helium loss in natural pyrite over geologic timescales. Extending these initial observations into a robust, predictive thermochronometric framework requires the more rigorous experimental and theoretical treatment that is one of the aims of this work.

Geologic Setting

Pyrite and Arsenopyrite for the geochronologic study was collected from two deposits that strongly differ in the timing and conditions of their formation: the Archean Kidd Creek volcanogenic massive sulfide (VMS) deposit, Ontario, Canada, and the Permo-Triassic orogenic gold system at Gympie, Queensland, Australia. Pairing a ∼2.7 Ga Cu-Zn-Ag VMS sulfide with a ∼235 Ma Au-bearing pyrite provides two independently well-constrained age brackets against which to test He retentivity and closure behavior in pyrite across very different formation temperatures, host lithologies, and post-depositional thermal histories.

Kidd Creek, Ontario, Canada

Kidd Creek is among the largest and most profitable VMS deposits in the world, with past production and reserves exceeding 138.5 Mt at average grades of 2.4 percent Cu, 6.5 percent Zn, 0.23 percent Pb, and 90 g/t Ag (Hannington et al.,

1999). The deposit occurs within the Late Archean Kidd-Munro assemblage of the Abitibi subprovince, Superior Province, hosted at the top of a thickened pile of massive rhyolite flows, felsic volcaniclastics, and coarse epiclastic rocks that attain a maximum thickness of ∼300 m beneath the mine (Hannington et al., 1999). The three main stratiform orebodies (North, Central, and South) formed largely by subseafloor infilling and replacement of permeable volcaniclastic rocks within a subsiding, rift-related graben, rather than by high-temperature venting at the seafloor (Hannington et al., 1999). U-Pb zircon geochronology brackets the ore-forming interval tightly: the oldest footwall rhyolite is dated at 2716.1 ± 0.6 Ma, the onset of bimodal komatiite-rhyolite volcanism at 2717 ± 1.1 Ma, and the youngest, uppermost quartz porphyry unit at 2711.5 ± 1.2 Ma, indicating that mineralization took place during at least one and possibly several million years of intermittent volcanic activity (Hannington et al., 1999). Oxygen isotope studies of the host rhyolites show that, while primary igneous zircon retains magmatic δ^{18}O values (5.4‰ ± 0.3‰), coexisting quartz phenocrysts are pervasively shifted to much higher and more heterogeneous values (14.2‰ to 16.1‰), recording extensive subsolidus recrystallization during the same long-lived hydrothermal convection that deposited the sulfides (King et al., 1997).

The interior of the Canadian shield has been a stable platform for hundreds of millions of years, as documented by U-Th/He and fission track dating of apatite (Sturrock et al., 2024). Kidd Creek in particular has been of particularly interest for its famously ancient groundwaters that sustain a low-biomass but metabolically active community of sulfate-reducing and alkane-oxidizing microorganisms (Lollar et al., 2019). The Kidd Creek deposit is thus an ideal natural laboratory for testing the long-term He retentivity of pyrite and arsenopyrite.

Gympie, Queensland, Australia

The Gympie pyrite comes from a very different setting: an accreted, Early Permian bimodal (basalt-dacite) submarine volcanic sequence, the Gympie Group, interpreted as a fragment of an intra-oceanic island arc that rifted from the Gondwana margin before docking against the New England Orogen of eastern Australia (Sivell and Waterhouse, 1988). Drillcore near the Gympie Eldorado Gold Mine resolves at least nine sequentially emplaced magmatic

suites and a >2 km thick volcano-sedimentary succession. This arc terrane is inferred to have accreted to the Gondwana margin by about 220 Ma (Early to Middle Triassic), based on cross-cutting granitoid relationships with adjacent New England Orogen terranes (Sivell and McCulloch, 2001).

Gold-bearing pyrite-quartz mineralization at Gympie occurs in fibrous extensional veins developed within sheared, carbonaceous black slates. Alteration associated with this vein set has yielded a K/Ar age of about 235 Ma (Holcombe et al., 1997). Refractory dolerite dykes and later microdiorite-shoshonite dykes cross-cut the mineralized structures and bracket the timing of the auriferous quartz veining from below and above, respectively, giving an independent field constraint on the age of mineralization that is broadly consistent with the K/Ar alteration ages (Sivell and McCulloch, 2001).

The nominal "Early Permian" age assigned to the host rocks to the Gympie deposit, the Highbury Volcanics and Rammutt Formation, however, rests mainly on sparse invertebrate and plant fossil biostratigraphy higher in the sequence and on lithologic correlation, rather than on direct radiometric dates on the volcanic pile itself. Holcombe et al. (Holcombe et al., 1997) note that the lower Gympie Block volcanics, like the correlative Berserker beds, "may contain older volcanic units," and fold the entire Gympie succession into a long-lived (∼320-280 Ma, peaking at ∼305 Ma) regional magmatic episode. These volcanics may very well be related to related to nearby ∼300-310 Ma granodiorite magmatism (306.5 ± 0.6 and 306.9 ± 1.2 Ma, $^{40}Ar/^{39}Ar$ hornblende) and exhumation of previously deep (>18 km), epidote-blueschist-facies rocks in the adjoining North D'Aguilar and Marlborough blocks (Holcombe et al., 1997).

3.2 Experimental Methods

U-Th/He Geochronology

Sample Preparation

Intact crystals of pyrite and arsenopyrite were extracted from samples from the Kidd Creek and Gympie deposits with a hammer and chisel at Caltech and SELFRAG high-voltage pulse fragmentation at the University of Queensland, respectively. A subsample of the gympie pyrites were leached in 1M HCl for 4 hours to remove any surface contamination and then rinsed in MilliQ water and dried. Individual crystals were washed in a sequence of acetone, ethanol, and MilliQ water baths, then promptly dried using clean, lint-free wipes. These

samples, along with the hematite MI-43 standard from Gogebic, Michigan (Farley and McKeon, 2015) and the Apatite standard from Durango, Mexico (Farley, 2000) were weighed, then encapsulated in silica glass ampoules similarly to the sulfides in section 3.2, but without any additionaly buffer materials. An empty tube was also encapsulated to serve as a procedural blank. Photos and masses of the prepared ampoules were recorded so that the helium and trace metal analyses could be correctly matched.

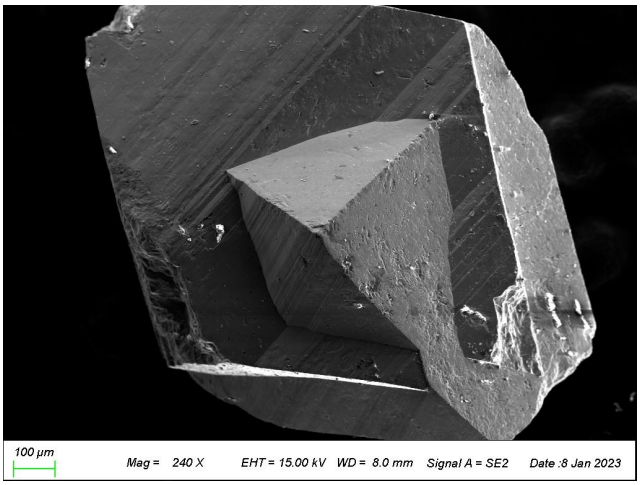


Figure 3.1: Secondary-electron SEM image of crystallographic arsenopyrite twins exceeding 1 cm across, recovered intact from Kidd Creek ore (sample ST05) with a hammer and chisel. Fine-scaled growth striations are preserved on the crystal faces. Euhedral crystals of this size were extracted routinely, showing that simple mechanical excavation preserved crystal faces well.

Helium Analysis

Encapsulated samples were wrapped in Ni-foil and loaded into a custom-built autosampler with the procedural blank loaded first. The autosampler was wrapped in heating tape, which was made to warm the samples to 80-100°C for at least 12 hours in ultra-high vacuum before the first step heat to allow for atmospheric helium to escape. Initial efforts in method development showed that atmospheric helium contamination was significant (Fig. 3.2). This atmospheric helium payload of an ampoule equilibrated with air is on the order of 10s of nccSTP helium per ampoule (Thyer et al., 2026). It is difficult to mathmatically correct for even when 3He can be accurately measured, because helium diffusion and permeation through silica glass fractionates 3He/4He ratios (Shelby, 1971; Jones, 1953), and the ampoules were each geometrically

unique and therefore had different permeation rates. Still, we found the best results when we applied a simple correction to the helium data after substantially mitigating the contamination with this warm-degassing step. To correct for the much smaller remaining atmospheric contamination, we subtract 4He proportionally to measured 3He by the atmospheric 3He/4He isotope ratio: 1.39×10^{-6}. The procedural blanks showed that atmospheric helium contamination was reduced by a factor of 5-10, which was sufficient for these helium-rich samples with 100s of nccSTP helium (Fig. 3.2). Future work may develop a more rigorous approach to experimentally remove atmospheric helium contamination or to mathematically correct for it using the first order framework of Thyer et al. (2026).

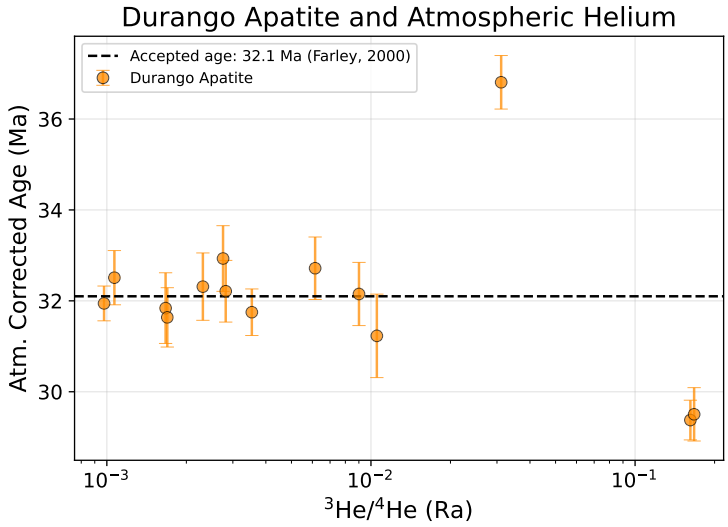


Figure 3.2: The atmosphere-corrrected age for encapsulated Durango apatite is plotted against the measured helium isotope ratio ($^3He/^4He$), normalized to atmosphere (Ra), illustrating the atmospheric-contamination problem motivating the experimental correction protocol described in 3.2. Aliquots with low measured Ra (i.e. a small atmospheric 4He contribution relative to radiogenic 4He) require insignificant mathmatical correction, preserving the accuracy of the ages and agreeing with the 32.1Ma accepted age (Farley, 2000). Aliquots measured before implementing the experimental atmosphere-degassing step contain more 3He and require stronger corrections. Because these corrections are unaware of the true isotopic composition of fractionated atmosphere in the ampoule, they yield inaccurate ages when they are large with respect to the sample's total measured helium. The weighted average age of the aliquots with mass >0.5 mg, 32.1 ± 0.2 Ma, agrees well with the accepted age, affirming the fundamental accuracy of the measurements.

After 12 hours, unheated furnace blank measurements were periodically taken

until the measured value was similar to such blanks taken for batches of samples not encapsulated in silica glass, then samples were individually dropped into a resistance furnace and baked at 1000-1200°C for 15 minutes to degas their helium contents. Although pyrite and arsenopyrite are unusually retentive of helium at lower pressures, these temperatures were far above these phases' stability field, and so they were readily degassed. Helium abundances and isotope ratios were measured on a Thermo Fisher Helix SFT mass spectrometer. Gasses evolved during the bake were passed through a charcoal filter held at 77°K, then exposed to an SAES-10 getter and a heated titanium getter before being cryofocused at 14°K, then released at 34°K. Sensitivities were quantified using a calibrated 3He-enriched air standard several times throughout the analysis.

Trace Metal Analysis

The encapsulated, now degassed samples were removed from the furnace and were matched to their pre-bake photos and masses. Around 20% of samples would be typically lost during the helium analysis process, with ampoules either broken by the autosampler or by thermal stresses during the bakes. Samples bearing residual nickel-foil that had fused to the surface of the glass were scratched clean with a dental pick, and the remaining glass ampoule was sonicated in a sequence of ethanol then MilliQ water baths. Samples were then placed in an acid-washed Teflon beaker and were broken underwater by an agate pestle, to prevent loss of sample material. The pestle was rinsed with MilliQ water between samples to avoid cross-contamination. Excess water was pipetted away from each sample and disposed of, then the remaining water was evaporated on a hot plate. The vast majority of the samples were then spiked with a U-232 and Th-230, and vanadium enriched spike and digested in concentrated HNO_3 in closed Teflon beakers, leaving the glass intact. The uranium and thorium concentrations of the glass ampoules were measured to be $\approx$ 50 and 30 ppb by mass respectively, but this contaminant payload was only delievered when the glass was dissolved in HF; the uranium and thorium contents of concentrated nitric acid in extended contact with broken glass were below detection limits.

After this point, the Kidd Creek pyrite and Durango Apatite solutions were pipetted and diluted to an ICPMS-ready solution. The gympie samples, how-

ever, required further processing to dissolve fine U-Th bearing rutile inclusions. The gympie and MS-43 hematite samples, now mostly digested in concentrated HNO_3, were dried down and put back up in small volumes of concentrated HCl, then were loaded into Parr vessels and baked at 200°C for 24 hours. The Gympie samples were then dried down again and put back up in concentrated HNO_3 before the concentrated acid was transferred and diluted to an ICPMS-ready solution. U and Th contents were quantified with isotope dilution using an Agilent 8800 ICP-MS. Other trace elements were measured as ratios with spike vanadium. The U-Th contents of the spikes and the sensitivities of the other quantified trace metals relative to vanadium were continuously montiored by analyzing a standard solution of known composition that was spiked with the same U-Th-V spike as the samples. The same empty tube procedural blank used for the helium analysis was also processed through the same acid digestion and ICPMS analysis, and the measured U, Th, and other trace metal contents were subtracted from the measured values of the samples. The U-Th contents of the procedural blank were below detection limits, and the other trace metals were typically 1-5% of the measured values of the samples.

The Gogebic Iron Range hematite standard (Farley and McKeon, 2015) was carried through this same Parr-vessel digestion step alongside the Gympie pyrite, serving as an in-run check on how completely this digestion protocol recovers a sample's true U-Th-Fe budget. Fig. 3.3 plots each Gogebic aliquot's measured ^{4}He against its eU, normalized by the sample mass calculated from measured Fe content where available and by the pre-analysis balance mass otherwise. Except for one clearly erroneous outlier (with an age in excess of 1.5Ga), these samples agree with two independently published ages for Gogebic, Michigain hematite: 453 Ma (U-Th/^{21}Ne, aliquot MI-45; (Farley and McKeon, 2015)) and 579 Ma, (U-Th/He, (Hofmann, Treffkorn, et al., 2020)). Here the outlier is interpreted to to have experienced U-loss, as its helium content is similar to the other aliquots, but it has low eU. Although incomplete digestion would also produce erroneous older ages, the apparent helium concentration would be higher and the eU would match the other samples if that was the case.

Encapsulated Sulfide Step Heat Experiments

Samples

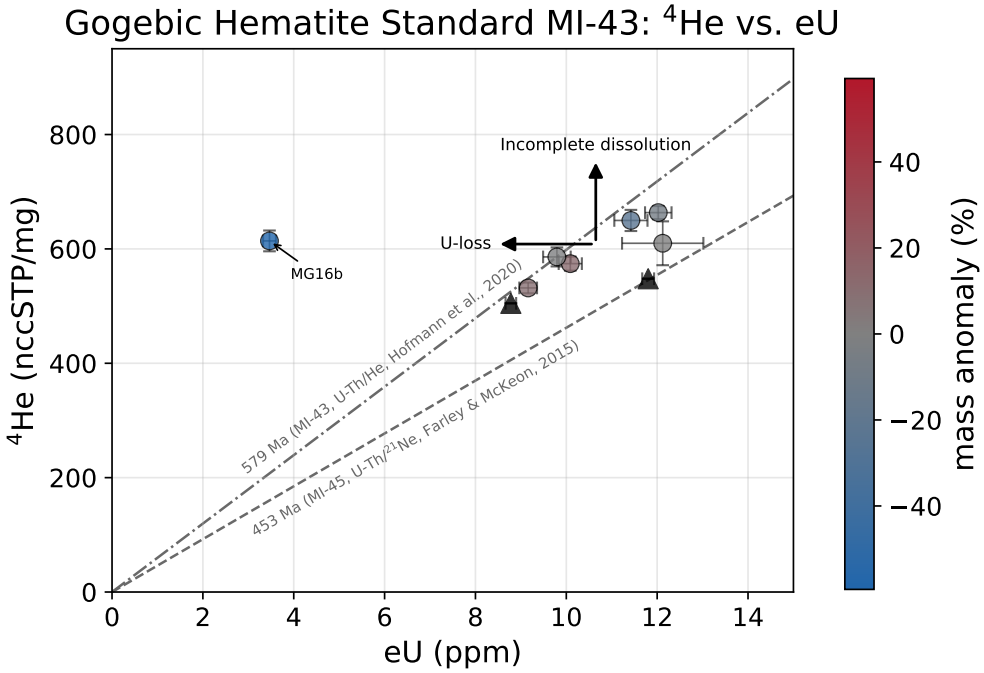


Figure 3.3: Atmosphere corrected ^{4}He (nccSTP/mg) vs. eU (ppm), with 2σ error bars, for the MI-43 Gogebic hematite standard aliquots analyzed alongside Gympie pyrite (Table 3.6), normalized by mass calculated from Fe molar abundaces measured on ICPMS. Fe was not measured for two samples: they are plotted as triangles using their balance-measured mass. These two measures of mass are compared to assess internal consistency within samples. Points are colored by "mass anomaly" in percent: $\frac{M_{Fe}-M_B}{M_{Fe}} \times 100$. Anomalies are generally small: with absolute values 11% or smaller for all but 2 samples. The old outlier sample 041323MG16b with low eU, however, has a mass anomaly of 59%, meaning that the balanced-measured mass was significantly higher than the mass inferred from the iron abundance. Two isochrons of independently published ages for Gogebic hematite are plotted: 453 Ma (MI-45, U-Th/^{21}Ne; (Farley and McKeon, 2015)) and 579 Ma (Hofmann, Treffkorn, et al., 2020). Only the low eU outlier 041323MG16b falls outside of this age range. With anomalously low eU at an otherwise unremarkable ^{4}He content, is interpreted here as an example of U-loss. Its large mass anomaly may point to the presence of an insoluble phase such as quartz, however, it is difficult to explain why an insoluble phase would affect only the eU concentration and not the helium concentration of the hematite.

Nine helium step heat experiments of sulfides (pyrite, arsenopyrite, chalcopyrite, galena, chalcocite, and sphalerite) sealed in silica glass ampoules performed at the Noble Gas Laboratory at Caltech are discussed here. Table **??** shows the dimensions, geometric factors, and encapsulated materials of this work. All sulfides of this study were sourced from the Caltech Mineral Collection from visually pure, well-formed euhedral crystals with half prism widths greater than 200 microns, often larger than a centimeter. Samples were broken into smaller pieces under ethanol to avoid irreversible trapping of atmosphere (Cox et al., 2022). Except for the Rustenburg pyrite, all step heats measured 3He evolution from an initially homogeneous distribution of spallogenic 3He. To produce a homogeneous distribution of 3He, the samples were irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster and Farley, 2005).

Table 3.1: Silica Glass encapsulated sulfides that were the subject of helium step heat experiments in this work. The Rustenburg Pyrite experiment had evolved radiogenic 4He quantified by isotope dilution using a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer, whereas evolved spallogenic 3He was quantified using a Helix-MC-Plus^{10K} multi-collector mass spectrometer for the remainder of the samples, which were proton-irradiated. Experiments Spal230707 and CCT230707 contained no steps for which the uncertainty due to helium permeation through the silica glass capsule could be estimated, and therefore are discussed in a limited, qualitative capacity only.

Experiment Name and Phase	Sample Locality	Tube Inner x Outer Diameters	Geometric factor G *(eqn 2.2, Ch.1)*	VR *(eqn 5.4, Ch.1)*	Encapsulated "Buffer" Materials	Mineral Half-prism Width (a)
LVCOPy230630 Pyrite	Leadville, Colorado	*Not Measured*	*3000 cm^{-2}*	-	~0.2 mg Fused Sulfur	~200 µm
BigLVCOPy230707 Pyrite	Leadville, Colorado	*Not Measured*	*3000 cm^{-2}*	-	~0.2 mg Fused Sulfur	~400 µm
112123LVCOPyIrr Pyrite	Leadville, Colorado	830 x 1550 µm	1859 cm^{-2}	2.5	~ 0.2 mg Fused Sulfur *(and >5 mg silica glass shards to separate sulfur melt from pyrite)*	~300 µm

Experiment Name and Phase	Sample Locality	Tube Inner x Outer Diameters	Geometric factor G (*eqn 2.2, Ch.1)*	VR (*eqn 5.4, Ch.1)*	Encapsulated "Buffer" Materials	Mineral Half-prism Width (a)
Rustenburg Pyrite Pyrite (4He on Quad)	Rustenburg, South Africa	520 x 810 µm	6675 cm^{-2}	1.4	~0.2 mg Fused Sulfur	106 µm
112123SalAspIrr Arsenopyrite	Mitterberg, Austria	530 x 970 µm	4711 cm^{-2}	2.3	~0.5 mg Crystalline Realgar (AsS)	~175 µm
GalTS230630 Galena	Tri-State District, Missouri	620 x 920 µm	5273 cm^{-2}	1.2	~0.2 mg Fused Sulfur	~100 µm
CCpTS230630 Chalcopyrite	Baxter Springs, Kansas	540 x 1010 µm	4382 cm^{-2}	2.5	~0.2 mg Fused Sulfur	~200 µm
Spal230707* Sphalerite	Chitwood, Missouri	1110 x 1490 µm	2205 cm^{-2}	0.8	~0.2 mg Fused Sulfur	~200 µm
CCT230707* Chalcocite	Messina, South Africa	1100 x 1520 µm	2044 cm^{-2}	0.9	~0.2 mg Fused Sulfur	~200 µm

Encapsulation

Before encapsulating the sulfide samples, glass tubes were cleaned and pulled into pipettes. 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen welding torch and the thin ends were fused closed.

Aliquots of sulfides were weighed and loaded into room temperature glass tubes along with fused sulfur in the presence of an ionizing fan to mitigate static. The arsenopyrite ampoule was loaded with crystalline realgar (arsenic sulfide) instead of fused sulfur, and one pyrite ampoule was loaded with silica glass pieces above and below the pyrite aliquot to separate the pyrite from the sulfur liquid during the experiment. To seal, the top of each pipette was interfaced with a vacuum line and the tube was pumped to a pressure of <60 millitorr, which greatly aided the ease and quality of the final seal. After attaining rough vacuum, a fine-flamed welding torch was used to seal the top of each ampoule. To determine the geometric factors for permeation of the

ampoules, the inner and outer diameters of most ampoules were measured using the measurement function of the Amscope stereo microscope software, which was calibrated and tested during early development of this method. Two experiments (LVCOpy230630 and BigLVCOPy230707) were completed without these geometric measurements. For these experiments, a geometric factor of 3000 cm^{-2} was arbitrarily assigned to estimate uncertainties of the measurements due to the convolution of the process of helium permeation through silica glass. This assumption is unlikely to affect the interpretation of the results.

The sealed ampoules were placed in copper foil packets and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in detail by Farley, Reiners, et al. (1999). Before the experiment began, each ampoule was washed with ethanol, air-dried, loaded in the lamp cell, pumped to high vacuum, and was held at 100°C for at least 30 minutes under dynamic vacuum to desorb ethanol, water, and other atmospheric species on the ampoule, copper packet, and lamp cell.

Helium Mass Spectrometry

Heating steps and 3He measurements for all samples were automated using custom LabView routines on a Helix-MC-Plus^{10K} multi-collector mass spectrometer, or a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer for the Rustenburg Pyrite ampoule. For all ampoules, a Watlow temperature controller held the ampoule at the temperature set point for a pre-programmed duration, after which the ampoule was allowed to cool while the evolved gas was prepared for measurement for about 25 minutes.

For the irradiated samples, Helium was purified by exposing evolved gasses to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. the Helix-MC-Plus^{10K} mass spectrometer was operating in peak hopping mode and the majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech has shown this approach to be reliable,

reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray. 4He signals were near-blank for all experiments of irradiated samples, and no attempt is made here to interpret those data. 4He overspray likewise did not affect 3He signals. Measurements of high 3He MM gas standard were analyzed every 8 hours to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility. These standards show 3He was generally internally reproducible to within 1% (2σ) for rolling sets of 5 consecutive standards, except for the latest measurements on the 112123SalAspIrr Arsenopyrite, during which the ion source began to fail. Those measurements are excluded from further analysis, and any helium evolved during those steps is likely to be a very small proportion of the total helium contents of the sample (<0.2%) and thus would not significantly affect the calculation of the total helium content of the sample.

Radiogenic 4He evolved during step heats of the encapsulated Rustenburg Pyrite was quantified using isotope dilution with a high purity 3He spike on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer. A standard gas with a known amount of 4He was measured before and after the experiment, to monitor drift in the relative sensitivities of 3He and 4He and to estimate reproducibility, which was about 6% (2σ). After each step, evolved gas from the ampoule in the lamp heating cell was allowed to mix with the 3He spike and the mixture was frozen on charcoal held at 14°K within a cryostat, then helium was selectively released to the mass spectrometer by raising the temperature to 36°K.

Scanning Electron Microscope Study

The methodology for the SEM study of the sulfides is described in detail in 1.6s.

Analytical Methods For Step Heat Experiments

Similarly to O. V. Yakubovich, Gedz, et al. (2019), these helium step heat experiments were designed according to the incorrect assumption that the silica glass ampoule was effectively “helium transparent”, or in other words that helium permeation through the silica glass ampoule was nearly instantaneous with respect to the rate of helium evolution from the encapsulated sulfides. Thus, permeation steps (as discussed in chapter 1 and 2), were rarely used. The step yields for steps at high temperatures were approximated as the the

measured value, with permeation lag constrained by upper bounds following the "Treatment as Uncertainty" approach of Thyer et al. (2026). This approach requires no permeation steps or knowledge of the helium content within ampoule from previous steps, instead calculating the maximum helium content remaining within the ampoule for a given measurement, provided the step was long and hot enough for the tube to have reached a steady state concentration and assuming that the rate of helium evolution from the encapsulated mineral is constant. The steady state assumption is taken to be effectively valid once the ampoule loses memory of the helium content remaining from the previous step. According to eqn. (1.56) in chapter 1, only 1.8% of the difference between the previous step's remaining helium content and the current step's steady state helium content remains when $\lambda t = 4$. In this work, the steady state assumption is only applied for steps long and hot enough for $\lambda t \geq 4$, where $\lambda = KGRT$, K is the temperature-dependent permeation coefficient for 4He or 3He through silica glass, G is the ampoule geometric factor, R is the gas constant, T is the temperature in Kelvin, and t is the time in seconds. The 4He permeation coefficient K is taken from Swets et al. (1961); and these coefficients are multiplied by the permeation coefficient ratio from Jones (1953) to yield the 3He permeation coefficient.

Many heating steps, especially near the beginning of the experiments where lower temperature steps were used, are not interpreted or plotted in this work because $\lambda t < 4$; however, some steps with $\lambda t < 4$ are used when eqn. (1.9) is used to place upper bound constraints on preceding steps, so that eqn. (1.56) can in turn be used to quantify the uncertainty on the yields step that has not reached steady state.

This method of calculating uncertainties produces an asymmetric upper and lower uncertainty for each step. These assymetric uncertainties are propagated through the diffusivity inversion of Fechtig and Kalbitzer (1966) with eqn. (2.1) in Chapter 2.

In general, the 2σ analytical uncertainties as calculated using the reproducibility of standards is greatly exceeded by the uncertainties due to the permeation lag of the ampoule as calculated above. For the Rustenburg pyrite experiment and LVCOpy230630, the uncertainty due to the permeation lag is less than a factor of 10 greater than the analytical uncertainty, and so the 2σ analytical uncertainty for a given measurement is added to the uncertainty by adding

it to the upper bound helium content of the tube, n_t^{UB}, for that step. For the remainder of the experiments, the analytical uncertainty is less than the uncertainty from the permeation lag by a factor of 10-100, and so the analytical uncertainty is neglected. No uncertainty in the total yields in incorporated.

Two of the experiments contained no steps for which $\lambda t \geq 4$: Spal230707 and CCT230707 for sphalerite (ZnS) and chalcocite (Cu_2S) respectively. No step heat results are reported here for those experiments, but they are discussed qualitatively.

3.3 Results

[TODO: Move SEM results to SI: Show marcasite pictures: the others have already been shared in chapter 1]

Effectiveness of the Ampoule Method in Stabilizing Sulfides

The methodology for the heating treatments and SEM imagery of pyrite and arsenopyrite surfaces following thermal treatments in different conditions is discussed in 1.6s. The SEM imagery reveals deformed, variably raised, ~10 µm cracked surfaces on pyrite and marcasite that are heated in open air, and for arsenopyrite heated in a closed capsule without a realgar buffer (Figures 3.2. . .). Pyrite and marcasite samples that are heated in ultra-high vacuum developed recessive ~1 µm rounded craters. Contrarily, images of thermally treated, encapsulated pyrite and arsenopyrite with added fused sulfur or realgar show no obvious distinction from the control group: both of which showcase conchoidal fracture (Figures 3.2. . .). The Rustenburg pyrite sample represents a potential exception; its pervasive fractures appear to have partially annealed after thermal treatment at high sulfur fugacity (Figures 3.2. . .). The buffered, encapsulated marcasite sample shows a raised layer of ≤2 µm angular (presumably pyrite) crystals growing atop its surface; evidence of phase transformation is present in all three heat-treated marcasite populations, demonstrating that the encapsulation method is not effective in preventing phase breakdown at elevated temperatures for marcasite. While these experiments are not conclusive evidence that thermal phase transformation is prevented for pyrite or arsenopyrite, they consistent with that interpretation, encouraging its application for helium step heat experiments.

Sulfide Step Heat Experiments

The apparent diffusivities calculated from the helium step heat data are reported in Tables 3.2–3.5 (one per pyrite/arsenopyrite experiment; the analogous galena and chalcopyrite tables are given in the Supplementary Information, §3.1s). These experiments qualitatively show remarkably high helium retentivity at high temperatures for pyrite and arsenopyrite. For no pyrite or arsenopyrite helium step heat experiment did the final cumulative fractional helium yield exceed 7%.

Table 3.2: ^{3}He step-heat dataset for Pyrite, Leadville, CO, 200 µm half-prism width (sample LVCOPy230630), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.5 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
31	475.0	100.0	0.0075	0.4382	0.0022	-24.682	-24.061	-23.805
32	465.0	120.0	0.0048	0.4430	0.0012	-25.283	-24.681	-24.457
33	455.0	140.0	0.0047	0.4476	0.0010	-25.149	-24.854	-24.650
34	510.0	60.0	0.0105	0.4581	0.0046	-23.282	-23.179	-22.810
35	520.0	60.0	0.0126	0.4708	0.0053	-23.412	-22.965	-22.612
36	530.0	60.0	0.0177	0.4885	0.0068	-22.943	-22.597	-22.263
37	539.9	60.0	0.0204	0.5089	0.0075	-22.815	-22.414	-22.096
38	555.0	60.0	0.0315	0.5404	0.0105	-22.193	-21.930	-21.632
39	545.0	60.0	0.0181	0.5585	0.0065	-23.302	-22.440	-22.127
40	535.0	80.0	0.0175	0.5760	0.0049	-23.190	-22.730	-22.480
41	525.0	100.0	0.0136	0.5895	0.0032	-23.622	-23.180	-22.962
42	515.0	120.0	0.0114	0.6009	0.0024	-23.848	-23.515	-23.319
43	505.0	140.0	0.0078	0.6087	0.0015	-24.400	-24.029	-23.849
44	560.0	60.0	0.0252	0.6339	0.0087	-22.046	-21.984	-21.682
45	570.0	60.0	0.0371	0.6710	0.0121	-21.810	-21.550	-21.258
46	580.0	60.0	0.0465	0.7176	0.0145	-21.553	-21.260	-20.979
47	590.0	60.0	0.0604	0.7780	0.0179	-21.188	-20.925	-20.654
48	600.0	60.0	0.0690	0.8470	0.0195	-20.999	-20.710	-20.450
49	604.7	60.0	0.0691	0.9161	0.0191	-20.945	-20.626	-20.372

continued on next page

(continued)

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
50	595.0	60.0	0.0381	0.9542	0.0109	-21.846	-21.163	-20.904
51	585.0	60.0	0.0244	0.9786	0.0073	-22.167	-21.576	-21.310

Table 3.3: ^{3}He step-heat dataset for Pyrite, Leadville, CO, 400 µm half-prism width (sample BigLVCOPy230707), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.5 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
16	400.0	160.0	0.0201	0.7190	0.0056	-26.075	-26.029	-25.779
17	420.0	140.0	0.0386	0.7576	0.0107	-25.357	-25.204	-24.951
18	440.0	120.0	0.0461	0.8037	0.0132	-25.072	-24.815	-24.554
19	460.0	100.0	0.0521	0.8558	0.0161	-24.734	-24.449	-24.170
20	480.0	80.0	0.0490	0.9048	0.0175	-24.619	-24.229	-23.914
21	500.0	60.0	0.0538	0.9586	0.0245	-24.174	-23.791	-23.402
22	510.0	60.0	0.0559	1.0145	0.0236	-24.262	-23.696	-23.332
23	490.0	70.0	0.0237	1.0382	0.0094	–	-24.668	-24.330
24	470.0	90.0	0.0109	1.0491	0.0036	-27.670	-25.682	-25.396
25	450.0	110.0	0.0053	1.0544	0.0016	-27.693	-26.586	-26.327
26	430.0	130.0	0.0033	1.0577	0.0009	-27.872	-27.226	-26.978
27	410.0	150.0	0.0007	1.0584	0.0002	–	-28.887	-28.642
28	450.0	160.0	0.0055	1.0640	0.0010	-26.951	-26.914	-26.743
29	470.0	140.0	0.0100	1.0739	0.0019	-26.295	-26.187	-26.012
30	490.0	120.0	0.0219	1.0958	0.0043	-25.321	-25.232	-25.049
31	510.0	100.0	0.0378	1.1336	0.0082	-24.596	-24.476	-24.276

Table 3.4: ^{3}He step-heat dataset for Pyrite, Leadville, CO, sulfur-isolated (glass-shard-separated) (sample 112123LVCOPyIrr), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.5 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
46	500.0	120.0	0.0447	0.1998	0.0138	-22.605	-22.576	-22.268
48	510.1	120.0	0.0311	0.2327	0.0090	-23.201	-22.736	-22.461
49	520.1	120.0	0.0259	0.2586	0.0070	-23.204	-22.796	-22.543
51	530.0	120.0	0.0325	0.2921	0.0082	-22.643	-22.453	-22.213
53	540.0	120.0	0.0277	0.3203	0.0066	-22.823	-22.507	-22.283
55	550.0	120.0	0.0326	0.3531	0.0073	-22.460	-22.249	-22.036
57	560.1	384.0	0.1064	0.4600	0.0061	-22.099	-22.041	-21.978
58	570.1	60.0	0.0156	0.4755	0.0080	-22.460	-21.968	-21.543
59	580.0	60.0	0.0263	0.5018	0.0125	-21.756	-21.400	-20.997
60	590.0	60.0	0.0596	0.5614	0.0263	-20.721	-20.497	-20.107
61	600.0	60.0	0.0630	0.6244	0.0265	-20.852	-20.332	-19.959
62	595.0	60.0	0.0468	0.6712	0.0206	-21.358	-20.542	-20.162
63	590.1	60.0	0.0318	0.7030	0.0146	-21.891	-20.867	-20.480
64	585.1	60.0	0.0259	0.7290	0.0122	-21.846	-21.031	-20.637
65	580.1	60.0	0.0165	0.7455	0.0082	-22.786	-21.452	-21.045
66	575.1	60.0	0.0114	0.7569	0.0058	-23.073	-21.808	-21.390
67	570.1	60.0	0.0101	0.7669	0.0053	-22.775	-21.915	-21.489
68	565.2	60.0	0.0075	0.7744	0.0041	-23.431	-22.203	-21.761
69	560.1	60.0	0.0058	0.7802	0.0033	-23.671	-22.444	-21.988

continued on next page

(continued)

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
70	555.2	60.0	0.0051	0.7853	0.0030	-23.638	-22.572	-22.104
71	600.1	60.0	0.0337	0.8190	0.0138	-20.747	-20.655	-20.303
72	551.7	79.4	0.0115	0.8305	0.0047	–	-21.986	-21.641
73	550.2	60.0	0.0044	0.8349	0.0028	–	-22.660	-22.161
74	550.2	60.0	0.0040	0.8388	0.0025	-23.985	-22.751	-22.266
75	500.0	420.0	0.0065	0.8453	0.0005	-24.680	-24.202	-24.131
76	500.1	180.0	0.0023	0.8476	0.0004	-24.621	-24.392	-24.220

Table 3.5: ^{3}He step-heat dataset for Arsenopyrite, Mitterberg, Austria (sample SalAspIrr), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.17 (delta, row 5). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
76	360.0	120.0	0.0052	0.3258	0.0017	–	-24.590	-24.302
78	370.0	120.0	0.0073	0.3342	0.0022	–	-24.224	-23.960
80	380.0	120.0	0.0059	0.3417	0.0016	–	-24.408	-24.166
82	390.0	120.0	0.0074	0.3504	0.0018	–	-24.159	-23.935
84	400.0	120.0	0.0111	0.3625	0.0025	–	-23.728	-23.520
86	395.0	120.0	0.0075	0.3704	0.0018	–	-24.086	-23.871
88	390.0	120.0	0.0049	0.3763	0.0012	–	-24.499	-24.277
90	385.0	130.0	0.0065	0.4182	0.0015	–	-24.194	-23.982
92	380.0	140.0	0.0044	0.4226	0.0010	–	-24.646	-24.443
93	410.0	120.0	0.0154	0.4379	0.0032	-23.277	-23.213	-23.020
95	420.0	120.0	0.0255	0.4693	0.0049	–	-22.647	-22.466
97	430.0	120.0	0.0547	0.5262	0.0097	–	-21.794	-21.621
99	440.0	120.0	0.1123	0.6419	0.0185	–	-20.915	-20.747
101	450.0	120.0	0.1449	0.7912	0.0222	–	-20.455	-20.297
102	445.0	120.0	0.0825	0.8737	0.0131	-21.171	-20.871	-20.716
103	440.0	120.0	0.0510	0.9247	0.0084	-21.566	-21.276	-21.119
104	435.0	130.0	0.0305	0.9552	0.0048	-22.142	-21.825	-21.678
105	430.0	140.0	0.0187	0.9739	0.0028	-22.651	-22.361	-22.221
106	425.0	150.0	0.0149	0.9888	0.0021	-22.849	-22.644	-22.510

continued on next page

(continued)

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
107	420.0	160.0	0.0091	0.9979	0.0012	-23.453	-23.189	-23.060
108	415.0	170.0	0.0062	1.0040	0.0008	-23.848	-23.624	-23.499
109	410.0	180.0	0.0048	1.0089	0.0006	-24.113	-23.926	-23.804
110	405.0	190.0	0.0035	1.0124	0.0004	-24.501	-24.304	-24.184
111	400.0	206.0	0.0054	1.0178	0.0007	-24.021	-23.937	-23.823
112	460.0	120.0	0.0588	1.0765	0.0084	-20.992	-20.981	-20.844
113	470.0	100.0	0.0750	1.1515	0.0123	-20.608	-20.493	-20.336
114	480.0	80.0	0.0732	1.2247	0.0147	-20.409	-20.230	-20.041
115	500.0	60.0	0.1425	1.3672	0.0353	-19.292	-19.189	-18.954
116	495.0	60.0	0.0693	1.4366	0.0178	-20.532	-19.831	-19.596
117	485.0	60.0	0.0300	1.4666	0.0083	-21.527	-20.634	-20.387
118	475.0	60.0	0.0100	1.4766	0.0030	-23.470	-21.718	-21.458
119	465.0	60.0	0.0046	1.4812	0.0015	-23.503	-22.482	-22.204
120	455.0	60.0	0.0054	1.4866	0.0019	-22.649	-22.328	-22.025
121	445.0	60.0	0.0022	1.4888	0.0009	-25.234	-23.223	-22.887
122	435.0	60.0	0.0015	1.4903	0.0006	-24.523	-23.620	-23.261
123	425.0	80.0	0.0004	1.4907	0.0001	–	-25.195	-24.929
125	405.0	120.0	0.0015	1.4922	0.0003	-24.385	-24.298	-24.102
126	395.0	150.0	0.0018	1.4940	0.0003	-24.512	-24.317	-24.151
127	395.0	150.0	0.0009	1.4950	0.0002	-25.402	-24.974	-24.808
128	395.0	150.0	0.0022	1.4972	0.0004	-24.203	-24.122	-23.956
129	395.0	150.0	0.0017	1.4989	0.0003	-24.635	-24.372	-24.206
130	400.0	150.0	0.0018	1.5007	0.0003	-24.527	-24.336	-24.176
131	410.0	415.0	0.0041	1.5048	0.0002	-24.597	-24.519	-24.468
132	420.0	130.0	0.0010	1.5058	0.0002	-25.025	-24.780	-24.619
133	430.0	120.0	0.0022	1.5080	0.0004	-24.001	-23.918	-23.754
134	440.0	593.0	0.0103	1.5183	0.0003	-23.988	-23.950	-23.921

The irradiated Leadville pyrite was the most retentive sample of the sulfide suite. Three separate helium step heat experiments were conducted on this sample, two with varying grain size and a third that attempted to isolate the pyrite from the sulfur liquid by packing glass shards above and below the pyrite grains within the ampoule (3.1). The Arrhenius plots for all three experiments show similar diffusivities (Fig. 3.4). The three experiments shared a similar slope, and a regression was fit to the most linear region of 9 measurements of the LVCOpy230630 experiment with an R^2 of 0.998, yielding an apparent activation energy of 47.59 ± 0.85 Kcal/mol (199.1 ± 3.5 kJ/mol; Fig. 3.4), implying a closure temperature of 294.3 ± 6.7°C assuming a cooling rate of 10°C/Ma. The delta plot constructed with this activation energy showed a general agreement between the smaller grain-size experiments, seemingly unaffected by the direct contact with sulfur liquid or lack thereof. The more retentive, larger ampoule yielded lower delta values at the same cumulative fractional yield, revealing that a smaller proportion of its helium resided in these least retentive domains that produce the common diffusivity trend on the arrhenius plot for the three experiments (Fig. 3.5).

The octahedral Rustenburg pyrite is less retentive of helium than the pyritohedral Leadville pyrite. On the down-temperature retrogrades of heating cycles, the Rustenburg pyrite shows clearly developed reproducible lines with the unusually high implied activation energy of 71.8 ± 1.7 Kcal/mol (300.6 ± 7.3 kJ/mol; Fig. 3.6: R^2 of 0.998 on 5 measurements). This feature is completely absent from the three experiments of proton-irradiated Leadville pyrite (Fig. 3.4). The delta plot constructed using this activation energy shows a monotonic decrease in delta values throughout the duration of the experiment (Fig. 3.7).

The encapsulated arsenopyrite helium step heat experiment produced an apparent diffusivity Arrhenius plot that showed generally high helium retentivity, but less so than the Leadville pyrites (Fig. 3.4). There were no representative linear regions produced by this experiment that were suitable for linear regression, although the general slope was similar to the Leadville pyrites' and so it was also used as a reference for this experiment. The delta plot using this reference showed complex behavior, where high delta values (far beyond the uncertainty related to the permeation lag of the ampoule itself) were observed for low temperature steps following high temperature steps (Fig. 3.17, row 5,

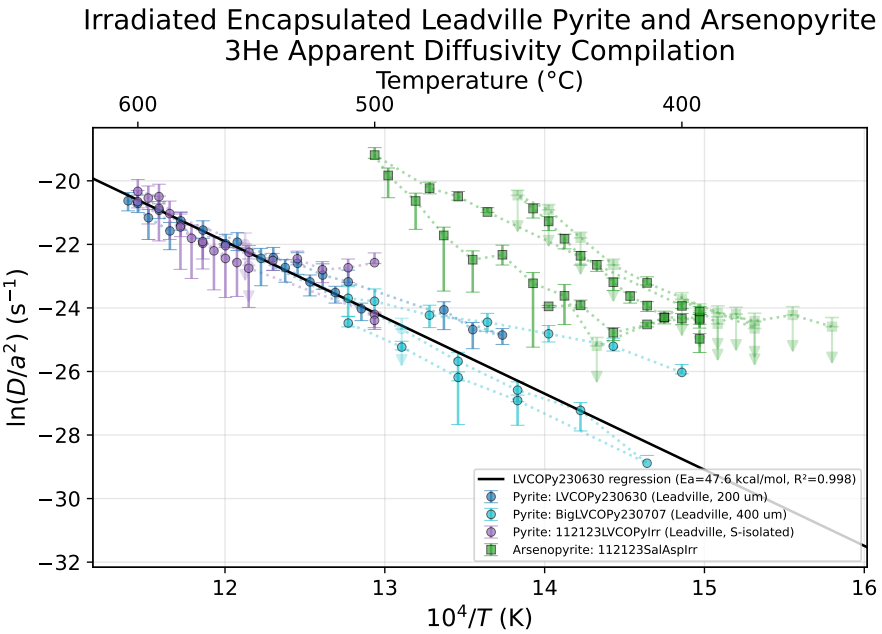


Figure 3.4: Compiled Arrhenius plot of apparent ^{3}He diffusivity for the three irradiated Leadville pyrite step-heat experiments (LVCOPy230630, BigLVCOPy230707, 112123LVCOPyIrr) and the proton-irradiated arsenopyrite experiment (112123SalAspIrr), each with permeation-lag uncertainty bars; downward-pointing arrows mark steps with an unconstrained lower bound (see §3.2). The solid black line is the regression fit to the most linear region of the LVCOPy230630 experiment (9 points, $R^2 = 0.998$, Ea = 47.59 ± 0.85 kcal/mol), used throughout this section as the shared reference line for all three Leadville pyrite experiments and for arsenopyrite.

Supplementary Information).

Step heat results for the two remaining, non-pyrite/arsenopyrite sulfides of this study – galena (GalTS230630) and chalcopyrite/chalcocite (CCpTS230630, CCT230707) – are reported in the Supplementary Information (§3.5). Neither mineral seems to be a promising U-Th/He geochronometer or helium isotope target.

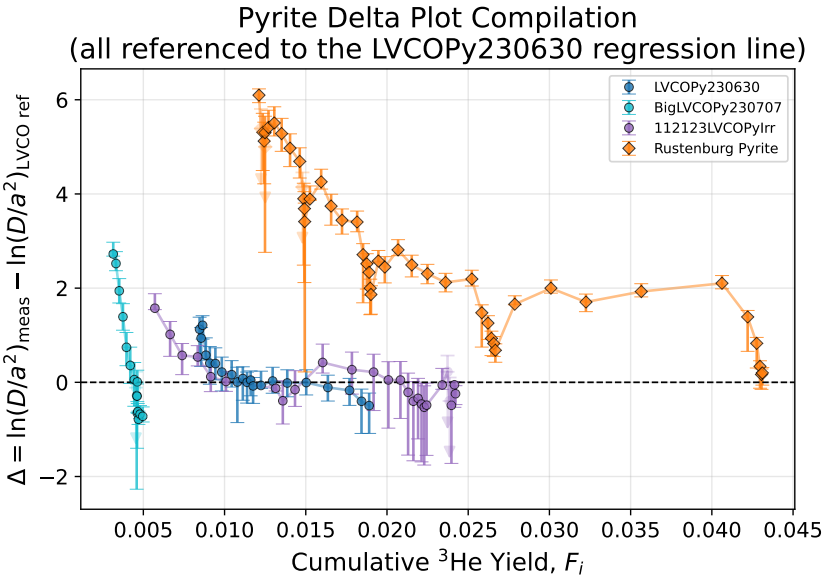


Figure 3.5: Delta plot ($\Delta = \ln(D/a^2)_{\text{meas}} - \ln(D/a^2)_{\text{LVCO ref}}$) vs. cumulative helium yield for all four pyrite step-heat experiments of this study, each referenced to the shared LVCOPy230630 regression line of Fig. 3.4. The three irradiated Leadville pyrite experiments (LVCOPy230630, BigLVCOPy230707, 112123LVCOPyIrr) cluster near $\Delta = 0$ throughout, with the sulfur-isolated 112123LVCOPyIrr aliquot running slightly below the other two; the Rustenburg pyrite (§3.4) departs sharply from all three, reaching Δ values up to +6 at low cumulative yield before decreasing toward $\Delta = 0$ at higher yield.

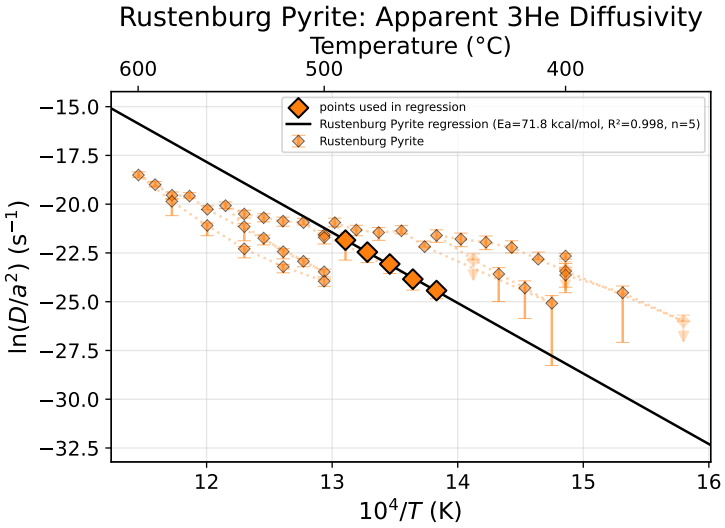


Figure 3.6: Apparent ^{4}He diffusivity Arrhenius plot for the Rustenburg pyrite step-heat experiment (isotope-dilution quantified on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer; §3.2). The solid black line is the regression fit to the five down-temperature retrograde steps used to define the Rustenburg activation energy (highlighted points; Ea = 71.8 ± 1.7 kcal/mol, R^2 = 0.998, n = 5), markedly steeper than the shared Leadville/arsenopyrite reference slope of Fig. 3.4.

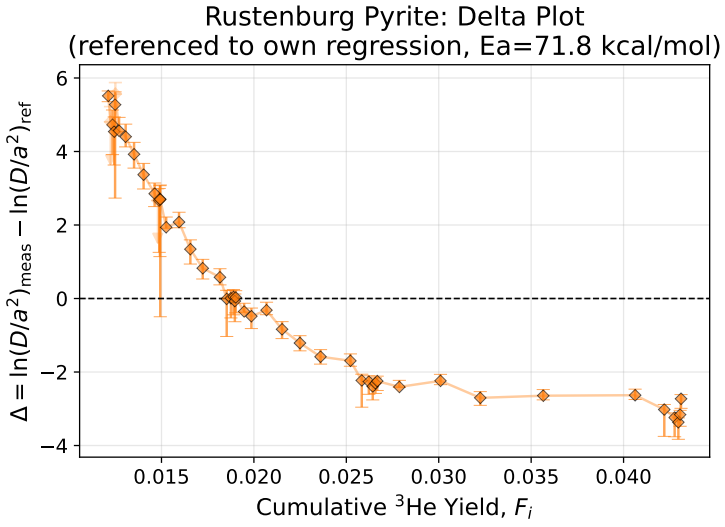


Figure 3.7: Delta plot for the Rustenburg pyrite step-heat experiment, referenced to its own regression (Fig. 3.6; Ea = 71.8 kcal/mol). Delta values decrease monotonically and near-linearly from about +5.5 at low cumulative ^{4}He yield to about −3.5 at high yield, in contrast to the near-flat delta trends of the three Leadville pyrite experiments (Fig. 3.5).

Table 3.6: U-Th/He geochronology results for the four sample suites discussed in the main text: Kidd Creek p
pyrite, and the Durango apatite and Gogebic (MI-43) hematite standards run alongside them, including all
quantified by ICPMS (§3.2): U, Th, and Sm (the radiogenic-budget elements) plus Cu, As, Zr, Ag, and Sb, a
ratios to spike vanadium; '–' denotes below detection or not measured for that aliquot. Ages are the atmospheri
'Age (Ma Atm=1Ra Corr.)' value; † marks the handful of early aliquots analyzed before a tube blank was avail
the uncorrected measured age is reported instead. Excluded aliquots (§3.3) are noted in the last column, as ar
Durango apatite aliquots (2211DApA, 2211DApC, 2211DApD, 2022StDApA), analyzed under the early "old me
that predated the warm atmosphere-degassing step (§3.2, Fig. 3.2). The complete compilation, including localitie
in the main text, is given in the Supplementary Information (Table 3.9).

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	Sb (ppm)	(pp
Kidd Creek Pyrite (Archean VMS, ~2716 Ma host)										
2211KC_Py_ST05_C	4.77	0.918 ± 0.054	1.641 ± 0.040	0.099 ± 0.006	–	–	–	–	–	1.3
2211KC_Py_ST05_A	3.23	0.686 ± 0.013	2.640 ± 0.050	0.134 ± 0.009	–	–	–	–	–	1.3
2211KC_Py_ST05_B	1.74	0.879 ± 0.052	1.593 ± 0.033	0.203 ± 0.012	–	–	–	–	–	1.2
2211KC_Py_ST05_D	1.98	0.659 ± 0.021	0.911 ± 0.029	0.037 ± 0.005	–	–	–	–	–	0.8
011123KCPyST05C	1.28	0.510 ± 0.019	0.732 ± 0.051	0.384 ± 0.070	248.36 ± 27.86	49.38 ± 12.69	9.13 ± 2.64	3.88 ± 0.98	15.91 ± 0.94	0.6
011423KCPyST05A	2.41	1.706 ± 0.024	4.047 ± 0.115	1.176 ± 0.071	555.10 ± 38.86	5922.20 ± 397.89	28.91 ± 2.25	7.82 ± 1.12	35.69 ± 0.95	2.6
011423KCPyST05B	4.04	1.999 ± 0.052	3.383 ± 0.086	2.205 ± 0.130	1365.56 ± 58.41	1604.14 ± 109.70	41.16 ± 3.86	12.62 ± 1.04	22.44 ± 0.41	2.8
011423KCPyST05C	3.93	0.834 ± 0.038	1.969 ± 0.062	0.492 ± 0.047	780.58 ± 52.49	8026.88 ± 487.81	15.46 ± 1.10	11.17 ± 1.64	21.97 ± 0.46	1.3
011423KCPyST05D	1.46	0.441 ± 0.024	0.850 ± 0.088	0.073 ± 0.035	146.71 ± 19.63	11440.33 ± 796.21	7.43 ± 2.53	2.13 ± 1.35	24.09 ± 0.73	0.6
013023KCPyST05As1	1.66	0.905 ± 0.019	7.933 ± 0.215	11.476 ± 0.420	17.41 ± 7.76	47121.03 ± 2800.88	6.85 ± 1.82	–	38.45 ± 0.81	2.8
013023KCPyST05Cu1	1.05	0.324 ± 0.074	1.001 ± 0.122	0.249 ± 0.085	314.00 ± 30.30	11923.72 ± 782.77	4.91 ± 2.77	3.52 ± 2.05	23.93 ± 1.33	0.5
013023KCPyST05Cu2	1.56	0.867 ± 0.018	9.902 ± 0.264	16.538 ± 0.625	102.88 ± 20.20	410116.35 ± 23437.59	8.81 ± 2.16	–	1235.42 ± 29.55	3.2
Gympie Pyrite (Permo-Triassic orogenic Au)										

(continued)

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	Sb (ppm)	(ppm)
013023PVGyPy-1	3.81	0.065 ± 0.004	0.064 ± 0.011	0.128 ± 0.022	42.50 ± 7.50	202426.65 ± 12489.88	1.22 ± 0.75	0.73 ± 0.41	138.00 ± 3.18	0.0
013023PVGyPy-2	1.15	0.272 ± 0.016	0.254 ± 0.040	0.320 ± 0.065	196.52 ± 14.56	1045127.96 ± 63721.03	3.58 ± 2.43	3.15 ± 1.26	733.74 ± 21.57	0.3
013023PVGyPy-3	1.81	0.128 ± 0.006	0.159 ± 0.024	0.173 ± 0.047	15.93 ± 8.91	433911.06 ± 27230.00	–	2.81 ± 0.86	345.22 ± 9.92	0.1
013023PVGyPy-4	1.82	0.280 ± 0.041	0.358 ± 0.031	0.647 ± 0.049	38.76 ± 16.89	616561.59 ± 36135.06	7.15 ± 3.71	1.67 ± 0.79	333.74 ± 7.83	0.3
0413PyGy-1	3.54	0.462 ± 0.006	0.587 ± 0.027	0.589 ± 0.069	43.08 ± 8.28	552.38 ± 44.66	56.95 ± 3.43	9.26 ± 1.92	70.17 ± 4.40	0.6
0413PyGy-2	1.67	0.232 ± 0.014	0.362 ± 0.032	0.375 ± 0.058	33.14 ± 8.87	164.64 ± 37.86	40.37 ± 3.17	8.78 ± 2.05	12.54 ± 3.70	0.3
0413PyGy-3	1.32	0.747 ± 0.018	0.721 ± 0.047	0.775 ± 0.124	95.81 ± 27.83	3445.18 ± 214.54	72.56 ± 3.28	7.91 ± 4.05	61.53 ± 5.65	0.9
0413PyGy-4	1.85	0.862 ± 0.016	0.791 ± 0.037	0.489 ± 0.071	30.30 ± 10.79	3814.85 ± 170.90	77.41 ± 5.18	7.41 ± 2.90	82.93 ± 4.53	1.0
062123GyPyHClb	2.15	0.636 ± 0.022	0.583 ± 0.050	0.498 ± 0.028	29.50 ± 5.71	3594.31 ± 91.81	89.48 ± 7.17	4.76 ± 1.64	186.80 ± 5.22	0.7
062123GyPyHClc	9.43	0.331 ± 0.007	0.394 ± 0.011	0.375 ± 0.022	21.71 ± 3.17	499.31 ± 18.48	65.54 ± 2.35	2.30 ± 1.11	6.36 ± 0.19	0.4
230625GyPyHCl	2.38	0.477 ± 0.007	0.463 ± 0.041	0.971 ± 0.053	30.60 ± 5.78	13394.39 ± 344.59	64.39 ± 2.99	4.16 ± 1.53	144.68 ± 5.06	0.5
230625GyPyUnAbr	2.32	0.615 ± 0.011	0.518 ± 0.044	0.782 ± 0.053	11.48 ± 2.49	101443.33 ± 2285.09	61.55 ± 2.84	2.85 ± 1.18	166.23 ± 5.89	0.7
230628GyPyHClab	6.17	0.627 ± 0.007	0.716 ± 0.034	0.785 ± 0.051	60.99 ± 6.34	297125.64 ± 8329.25	96.43 ± 3.82	72.30 ± 26.22	222.09 ± 12.53	0.7
230628GyPyHClc	4.03	0.585 ± 0.023	0.478 ± 0.025	0.650 ± 0.045	23.00 ± 4.19	33613.36 ± 1092.63	64.98 ± 6.53	7.65 ± 3.09	183.41 ± 11.95	0.6
230628GyPyUnAbrA	2.53	0.308 ± 0.013	0.371 ± 0.042	0.396 ± 0.032	30.71 ± 4.04	1615.76 ± 61.65	63.98 ± 3.08	4.60 ± 1.97	96.60 ± 6.99	0.3
230628GyPyUnAbrB	1.57	0.468 ± 0.012	0.410 ± 0.068	0.632 ± 0.049	12.00 ± 5.29	1498.79 ± 70.73	64.38 ± 6.01	7.27 ± 3.08	45.68 ± 3.40	0.5
Durango Apatite Standard (accepted age 32.1 Ma; Farley 2000)										
2211DApA	0.81	10.375 ± 0.104	193.585 ± 3.785	25.000 ± 0.909	–	–	–	–	–	56.8
2211DApC	0.82	10.830 ± 0.127	206.180 ± 4.095	24.512 ± 0.743	–	–	–	–	–	60.3
2211DApD	0.31	12.593 ± 0.188	272.219 ± 4.710	31.500 ± 1.123	–	–	–	–	–	77.9
2022StDApA	0.30	12.615 ± 0.216	230.680 ± 5.529	71.500 ± 1.955	–	–	–	–	–	67.9
011123DAp1	0.53	10.233 ± 0.205	206.604 ± 7.328	135.566 ± 5.575	–	924.97 ± 72.80	10.47 ± 7.17	118.08 ± 10.69	89.07 ± 4.68	59.8
011123DAp2	0.58	10.892 ± 0.116	200.372 ± 5.387	126.983 ± 5.017	–	910.07 ± 100.20	–	–	26.32 ± 2.87	58.9
011123DAp3	0.79	11.203 ± 0.092	226.208 ± 5.951	120.190 ± 4.386	–	803.92 ± 55.21	–	–	49.44 ± 1.01	65.4
011423DAp1	0.93	9.901 ± 0.107	196.551 ± 5.118	133.065 ± 4.774	–	787.88 ± 50.76	–	1.79 ± 1.42	28.71 ± 0.83	57.0
011423DAp2	1.51	11.131 ± 0.122	204.980 ± 4.486	129.139 ± 4.798	–	846.83 ± 57.62	–	–	17.66 ± 0.58	60.3

(continued)

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)		Sb (ppm)	(pp
011423DAp3	0.98	11.494 ± 0.150	223.171 ± 5.900	123.673 ± 4.393	–	820.23 ± 71.82	–	–	23.65	0.54	65.0
011423DAp4	0.81	11.773 ± 0.142	235.763 ± 6.994	125.741 ± 5.204	–	810.88 ± 61.16	8.92 ± 4.03	–	42.87	2.29	68.3
062123DAp	0.30	13.397 ± 0.173	266.573 ± 7.381	142.947 ± 10.021	–	665.14 ± 204.98	5.04 ± 2.79	–		–	77.3
062523DAp	0.85	11.074 ± 0.097	211.843 ± 5.315	123.774 ± 8.205	24.94 ± 8.99	799.92 ± 156.54	2.71 ± 1.71	–		–	61.9
062823DAp	–	–	–	–	–	–	–	–		–	
Average for > 0.5 mg	–	–	–	–	–	–	–	–		–	
Gogebic Iron Range BIF Hematite Standard (MI-43; 571 Ma)											
2211HemMSA	1.37	11.793 ± 0.131	0.039 ± 0.012	0.344 ± 0.029	–	–	–	–		–	11.8
2211HemMSB	2.12	8.765 ± 0.117	0.059 ± 0.007	0.343 ± 0.017	–	–	–	–		–	8.7
041323MG16a	0.46	9.199 ± 0.134	0.000 ± 0.000	1.559 ± 0.192	–	338.77 ± 109.12	–	–		–	9.1
041323MG16b	0.49	2.153 ± 0.057	0.089 ± 0.083	0.866 ± 0.154	77.04 ± 34.63	1310.34 ± 209.47	19.36 ± 5.03	10.81 ± 9.03		–	2.1
041323MG16d	0.28	11.281 ± 0.193	0.000 ± 0.000	1.548 ± 0.344	–	3960.06 ± 226.90	–	20.03 ± 15.53		–	11.2
041323MG16e	0.42	12.062 ± 0.467	0.441 ± 0.268	1.864 ± 0.457	–	8490.93 ± 1373.53	31.06 ± 24.33	–		–	12.1
062123MS	0.32	10.072 ± 0.102	0.000 ± 0.260	1.808 ± 0.101	–	2717.17 ± 112.31	–	3.63 ± 1.42	18.83 ± 0.96		10.0
230625MSHm	0.23	11.371 ± 0.121	0.000 ± 0.398	1.811 ± 0.183	–	1849.86 ± 128.15	–	78.00 ± 29.79	89.12 ± 4.54		11.3
230628MSHm	0.42	9.747 ± 0.111	0.000 ± 0.204	1.633 ± 0.167	–	443.71 ± 19.65	–	3.06 ± 1.63	93.15 ± 7.07		9.7

Pyrite U-Th/He Ages from Kidd Creek

Twelve aliquots of Kidd Creek pyrite were dated by U-Th/He step-degassing (Table 3.6) and give atmospherically corrected ages ranging from 465 to 860 Ma with the exception of one aliquot (011123KCPyST05C) that gives an anomalously old measured age of 1596 Ma Fig. 3.15. The mean age of the remaining eleven aliquots is 639 ± 36 Ma, 1σ. These dates are decisively older than the independent regional apatite cooling-age of 392 ± 32 Ma (Sturrock et al., 2024), yet far younger than the ∼2716 Ma age of the host rhyolite (Hannington et al., 1999). (Sec. 3.4).

Pyrite U-Th/He Ages from Gympie

Sixteen aliquots of Gympie pyrite were dated by the same method (Table 3.6; Fig. 3.8). The first four aliquots analyzed (013023PVGyPy-1 through -4) were prepared under an early iteration of the protocol that predated both the tube-blank atmospheric helium correction and the Parr-vessel HCl digestion step (Sec. 3.2) later adopted to fully dissolve fine U-Th-Sm-bearing rutile inclusions. The measured ages of these aliquots are unreasonable: ranging from 638 to 1008 Ma (mean 834 Ma, n=4), though it is unclear to what extent the bias is due to atmospheric contamination, incomplete digestion of rutile, or both. The ages of these four early aliquots are therefore considered unreliable and are excluded from further interpretation.

The remaining twelve Gympie aliquots, prepared after both protocol refinements were adopted, define a substantially younger and tighter age population. Eleven of these twelve give atmospherically corrected ages between 191 and 306 Ma (mean 258 ± 35 Ma, 1σ, n=11; these eleven are plotted in Fig. 3.8). The twelfth, 0413PyGy-4, is a clear outlier at 623 Ma and is also anomalous in its measured 4He concentration (68.4 ncc/mg), roughly three times that of the next most helium-rich aliquot in the accepted Gympie suite (230628GyPyHClab, 21.65 ncc/mg: supplementary Fig. 3.18). We interpret this combination as evidence for helium implantation from a U-Th-Sm-rich accessory phase concentrated at the sulfide grain margin and not fully digested during trace-element sample preparation. Aliquot 0413PyGy-4 is therefore also excluded from further interpretation.

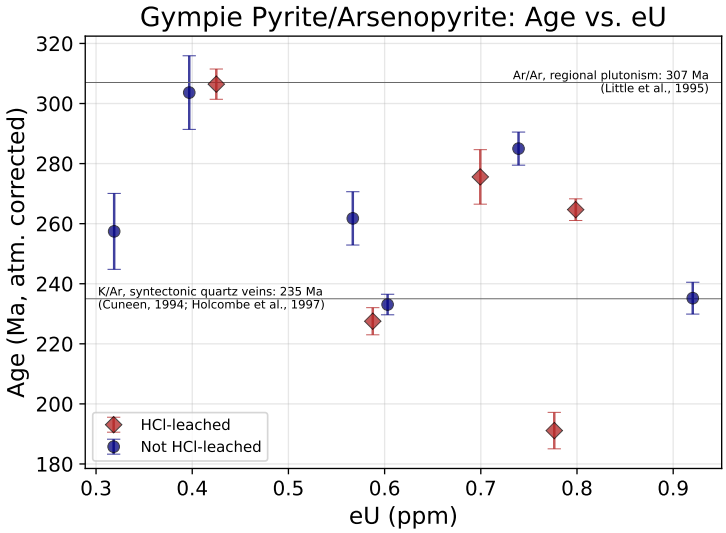


Figure 3.8: Atmospherically corrected age vs. eU for the eleven reliable, newer-protocol Gympie pyrite aliquots (Table 3.6), split by HCl-leaching. Reference lines mark the 235 Ma K/Ar syntectonic quartz-vein age (CUNEEN, 1994; Holcombe et al., 1997) and the 307 Ma Ar/Ar regional plutonism age (Little et al., 1995). The four early-protocol aliquots and the 0413PyGy-4 outlier (§3.3) are excluded here; see Fig. 3.18 (SI) for the outlier.

3.4 Discussion

Remarkable Helium Retentivity of Pyrite and Arsenopyrite

The central finding of this chapter's step heat experiments (§3.3) is that pyrite (and to a lesser extent, arsenopyrite) are extraordinarily retentive of helium: cumulative fractional yields never exceeded 7% for any pyrite or arsenopyrite experiment, and never exceeded 2% for the undeformed, inclusion-poor irradiated Leadville pyrite specifically, even after repeated heating cycles that reached temperatures well in excess of any plausible natural burial or hydrothermal setting. These cumulative fractional yields are meaningless without a reference diffusive timescale, so these step heat results are contextualized against the helium retentivity of other phases in Fig. 3.9, placing the sulfides of this study on the same diffusive-dose axis as several minerals with independently well-characterized helium retentivity: single-crystal and polycrystalline hematite, vacuum- and hydrothermally-degassed goethite, zircon, and monazite. To do this, the activation energy of hematite from Farley (2018) is used to convert the time and temperature of each step heat into a cumulative diffusive time, expressed as a Fourier number ($Fo = \frac{D}{a^2}_{\text{ref}} t$). The Fourier number can also be thought of as a dimensionless diffusive time, real time divided by the diffu-

sive timescale of a reference domain, $\tau = \frac{a^2}{D}$. τ here is referenced to a 1 μm single-crystal hematite domain. The aim of such a plot is to flatten complex heating histories into a single value. Incontrovertibly, this is not strictly correct because the acivation energy of different phases (and indeed diffusive domains within a single phase) are not identical. Such a plot is useful for understanding qualitatively the relative helium retentivity of different phases. Reference activation energies between 149-171 kJ/mol were tested, and they all yield similar results, preserving the relative positions of the different samples plotted.

On this comparison, pyrite and arsenopyrite sit at or beyond the retentive extreme of the entire compilation. Both plot at diffusive doses orders of magnitude beyond where the reference 1 μm single-crystal hematite domain (Farley, 2018) has gone fully to completion, and beyond essentially the full range spanned by zircon (Guenthner et al., 2013) and monazite (Farley, 2007) – two phases routinely treated as benchmark cases for high helium retentivity in the (U-Th)/He thermochronology literature. Only the most retentive of the Leadville pyrite aliquots (the experiment in which pyrite grains were isolated from the sulfur liquid by packing glass shards above and below them, §3.4) exceeds even this: a diffusive dose at which the reference hematite domain, zircon, and monazite have all long since gone to completion still corresponds to a cumulative pyrite yield of only a few percent. The polycrystalline hematite and vacuum-degassed goethite aggregates, by contrast, sit orders of magnitude further to the left (i.e. reach a given cumulative yield at a much lower diffusive dose) than either the single-crystal hematite reference or the sulfides – consistent with their retentivity being limited by short diffusion pathlengths to abundant grain boundaries rather than by the intrinsic volume-diffusion kinetics of hematite itself, exactly as this chapter's own SEM and step heat observations (§3.3, §3.4) argue is not the case for sulfur-buffered pyrite and arsenopyrite. The hydrothermally degassed goethite (Chapter 2) plots closer to, but still less retentive than, the sulfides here, and galena – the one sulfide of this study not discussed further as a geochronometer (§3.1s) – is markedly the least retentive phase on the whole plot, evolving the majority of its yield at diffusive doses where pyrite and arsenopyrite have released almost nothing.

Taken together, this comparison indicates that sulfur-buffered, sulfide-encapsulated pyrite and arsenopyrite are not merely retentive relative to other sulfides (galena, chalcopyrite, chalcocite; §3.1s), but are among the most helium-retentive phases

yet characterized by step heating, on par with or exceeding zircon and monazite and substantially exceeding both single-crystal and polycrystalline hematite and vacuum-degassed goethite. This is the empirical basis for treating pyrite (and, by extension, arsenopyrite) as a credible U-Th/He geochronometer in this chapter's Kidd Creek and Gympie case studies (§3.3, §3.3), notwithstanding the inclusion-hosted origin of the radiogenic budget actually being dated (§3.4).

Inferring Bulk Helium Migration Behavior from Low Fractional Yields

Many of the step heats on encapsulated sulfides presented here show very high helium retentivity, and for those experiments only a very small fraction of the total yield evolved (<5% for the arsenopyrite and Rustenburg pyrite; <2% for the Leadville pyrite), similarly to previous work on highly helium retentive phases such as baddeleyite (Baughman and Flowers, 2018). For these sulfides, a mechanism responsible for bulk helium loss besides desulfurization cannot be identified. For some well-studied phases like zircon and titanite, Arrhenius plots of the earliest fractions of evolved helium often reflect processes or conditions that are accessory to the bulk mechanism for helium migration through the mineral, such as radiation damage annealing or surface roughness (Reiners and Farley, 1999; Guenthner et al., 2013), where the primary mechanism for helium migration through the pristine mineral is volume diffusion. These apparent departures from the bulk helium migration behavior can be caused by properties of natural samples, like surface roughness or fine-scale cracks and fractures (Reiners and Farley, 1999) that produce complex grain size distributions even in carefully sieved, (apparently) single grains. Contrastingly, some other complexities in the helium loss behavior at early loss fractions can be ephemeral and altered during experimental heating, such as the annealing of radiation damage (Shuster and Farley, 2009). For radiation damaged samples, understanding the migration behavior of helium before and during annealing is essential to understand the sample's retentively in nature (citations) and therefore the earliest fraction of helium evolved in step heats is the focus of many studies (citations).

The fraction of total helium evolved before the predominant migration mechanism's behavior can be characterized in helium step heats varies: Guenthner et al. (2013) show that significantly non-zero delta or ln(r/r0) values persist in zircon until about 2% of the total helium is degassed for typical natural

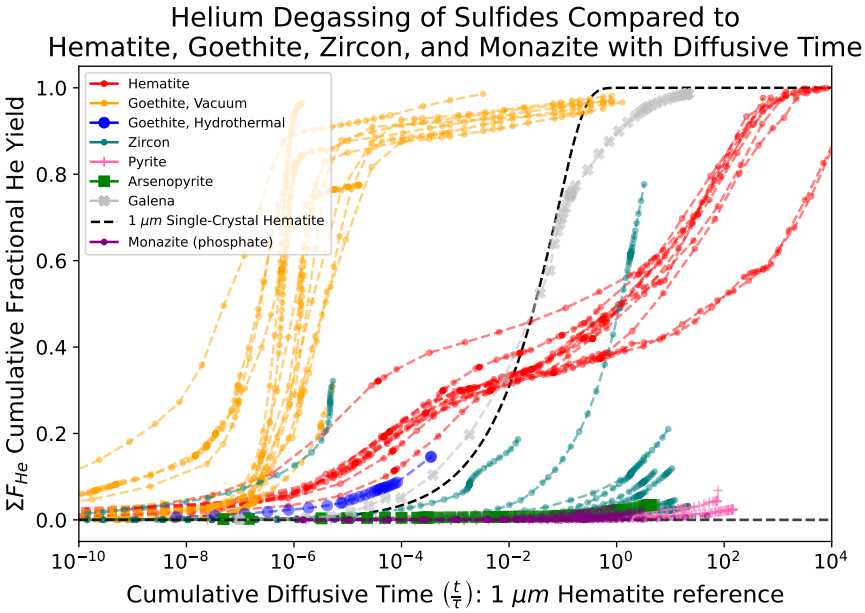


Figure 3.9: Cumulative fractional He yield vs. cumulative diffusive time (Fourier number, $Fo = D_{\mathrm{ref}}t/a^2_{\mathrm{ref}}$) for each heating step of every experiment plotted, referenced to a common 1 μm single-crystal hematite domain diffusivity (Farley, 2018) ($\ln(D_0/a^2) = 0.66$, $E_a = 171$ kJ/mol). Expressing step power this way, rather than as raw time or temperature, puts experiments run under very different heating schedules, temperatures, and durations onto one common, directly comparable dose axis: a phase reaching a given cumulative yield only at a much larger Fo is more helium-retentive than one reaching the same yield at a smaller Fo, regardless of how either was actually heated. Because cumulative fractional yield is dimensionless, isotope-specific normalization is not required to combine step heats of different helium isotopes on one axis: ^{3}He step heats of proton-irradiated pyrite, arsenopyrite, galena, zircon, monazite, and vacuum-degassed hematite/goethite are plotted alongside ^{4}He step heats of the (non-irradiated, isotope-dilution-quantified) Rustenburg pyrite and the hydrothermally degassed goethite. Plotted data: this chapter's own sulfur-buffered pyrite, arsenopyrite, and galena step heats (§3.3); 11 vacuum step heat aliquots of polycrystalline hematite from the Grand Canyon, Arizona (Farley and Flowers, 2012) and the Gogebic Range, Michigan (Farley and McKeon, 2015); 9 vacuum step heat aliquots of polycrystalline goethite (Win-06-01b and Capão, Minas Gerais, Brazil) (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024); the hydrothermal step heat of Win-06-01b goethite from Chapter 2 of this dissertation; and proton-irradiated zircon (Guenthner et al., 2013) and synthetic monazite (Farley, 2007) step heats. The dashed black reference curve is the forward-diffusion cumulative-yield curve for the reference 1 μm hematite domain itself (i.e. Fo vs. its own resulting cumulative yield, with no data plotted against it), included so every other curve's retentivity can be read directly relative to this single benchmark.

samples; Reiners and Farley (1999) observe apparent diffusivities of helium in titanite converging to linearity in titanite after 5% of the total helium is evolved. Similar experiments on proton-irradiated synthetic samples seem not to be marred by the same complexities: helium step heat data for synthetic monazite converge nearly immediately to a linear array at very loss cumulative helium loss fractions (Farley, 2007).

The step heat experiments on pyrite and arsenopyrite in this study would have benefitted from longer steps at high temperature to degas more helium to better understand the bulk behavior of helium, but even at small loss fractions some key observations about helium loss in sulfides can be made.

Comparison of Helium Step Heat Experiments of Natural Pyrites

In this work, four encapsulated, sulfur-buffered aliquots of pyrite taken from two localities are subjected to helium step heat experiments. Step heats for the three irradiated pyrites from Leavldville, CO are broadly similar, but contrast markedly from step heats of non-irradiated pyrite from Rustenburg, South Africa, therefore differences between pyrites of different localities is discussed before the variation within the aliquots from Leadville, CO.

Radiogenic helium evolution in step heats of octahedral pyrite from Rustenburg, South Africa show that this sample is less retentive than the pyritohedral pyrite from Leadville, Colorado (fig citation). Apparent diffusivities of these samples also have fundamentally distinct geometries when plotted in an arrhenius plot. The Rustenburg pyrite shows a striking feature of very steep slopes (corresponding to an activation energy of 300.6 ± 7.3 kJ/mol) on retrograde-prograde heating cycles at temperatures lower than the previous maximum temperature step heat, whereas the maximum apparent diffusivity for a given temperature seemed to follow a much shallower slope (lower activation energy of 199.1 ± 3.5 kJ/mol), which was broadly similar to the slope and activation energy of the Leadville, CO pyrite. The geometry of the step heats in an arrhenius plot for the irradiated Leadville, CO pyrite is generally simpler: generally slowly decreasing in a delta plot, or for isothermal steps (fig citation) as would be expected for helium release governed by MDD systematics (citation).

Natural pyrite is known to develop fractures and crystalline dislocations in metamorphic and experimental conditions (Gill, 1969; McClay and Ellis, 1983;

Barrie et al., 2011). Abundant fractures of micron-scale widths were clearly visible in the Rustenburg pyrite of this study (Fig. 3.10). Those pyrites were surprisingly easy to crush, presumably breaking easily along fine reexisting fractures. Although pyrites from Leadville are known to be associated with faulted dolomites, especially along deformation planes which likely served as pathways for mineralizing hydrothermal fluids (Nash, 1979), the samples used here were exceptionally sturdy and visually homogeneous under SEM. Rare veins and coatings of dolomite within and around the pyrites were present but were avoided when preparing the sample for irradiation.

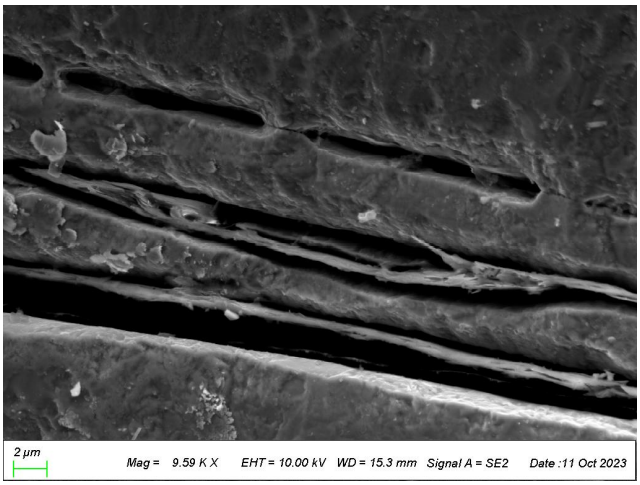


Figure 3.10: Secondary-electron SEM image of Rustenburg (Bushveld Complex) pyrite prior to heat treatment (control), showing the pervasive, micron-scale fractures characteristic of this sample (§3.4). These pre-existing fractures are interpreted as the pathways for the short-circuit helium diffusion mechanism invoked below to explain the Rustenburg pyrite's distinct retrograde Arrhenius behavior (Fig. 3.6).

Within this work, a pipe/short circuit diffusion mechanism is favored to explain the helium loss behavior of the Rustenburg pyrite, where helium populates the sample's fine fractures at increasing temperatures due to the primary helium migration mechanism for a system of encapsulated pyrite and sulfur fluid: volume diffusion or fine-scaled dissolution/reprecipitation of pyrite within sulfur fluid. This primary mechanism is interpreted to be the same for all pyrite samples regardless of locality. On retrograde heating steps, short circuit diffusion of helium out of fine fractures in the Rustenburg pyrite is hypothesized to become the rate-limiting process.

Leadville, CO Pyrite: Evidence for a Sulfur Liquid Control for Helium Loss in Step Heat Experiments

Three helium step heat experiments for the irradiated Leadville, CO pyrite are discussed here. Grain size was varied by a factor of 2 between a half-prism width of 200 microns for LVCOPy230630 and 400 microns for LVCOPy230707 to investigate the dependence of grain size on helium migration in pyrite. The aim of the third experiment, 112123LVCOPy, was to test the effect of isolating the pyrite from the liquid sulfur during the experiment to investigate the role of dissolution and (re)crystallization of pyrite by the ampoule's sulfur fluid. Silica glass shards were added to this ampoule below and above the pyrite aliquot, forcing the pyrite grains to remain in the middle of the ampoule, and not at the edges/bottom, where sulfur liquid is presumed to accumulate. Despite the similarities of the size fraction of the glass-filled 112123LVCOPy (300 micron half-prism width) to the other two aliquots of pyrite, it yielded apparent diffusities that were markedly more retentive. This suggests that reaction between the sulfur liquid and pyrite was the primary control for helium release from the Leadville samples. This effect is likely to be important for helium release in the Rustenburg pyrite as well.

Although iron solubility in sulfur liquids has only been sparsely studied, ferrous iron is known to be only modestly soluble in sulfur liquids. Clark (1960) reported that sulfur liquid can be up to 0.12% iron. Later, Barton (1969) reproduced this measurement and reported iron contents in pyrite-arsenopyrite saturated S-As liquids "less than 0.5, and probably less than 0.2, wt.%". In the step heat experiments of this work, the mass of the added sulfur was not recorded but was always small: less than or equal to the mass of the pyrite aliquot. By simple mass balance, only a maximum of 1.5% of the pyrite would be dissolved in an equal mass of sulfur fluid, however, if iron solubility in sulfur liquid is highly temperature dependent within the range of temperatures used in these experiments (24-500°C), then this effect could be multiplicative with each thermal cycle: the sulfur liquid could cyclically dissolve and recrystallize pyrite during thermal cycling. How this dissolution-recrystallization cycle might affect step heat results is unclear, but intuitively it would likely libertate helium near the grain surface early in the experiment, producing artificially high apparent diffusivities for 3He cumulative fractional yields less than 1.5%. This effect could potentially have been a primary control on the degassing behavior of all the irradiated pyrite within this work, but it is unlikely to be

significant for the majority of the 4He step heat data of the Rustenberg pyrite, which proceeded to beyond 4% total cumulative fractional yield.

The apparent diffusivities calculated for the two helium step heat experiments of Leadville pyrite with varying half-prism width, LVCOPy230630 and LVCOPy230707, both exhibit the same slope with monotonically decreasing delta values (fig citation). Intuitively, the smaller aliquot contains higher delta values at early helium fractional yields. These observations can be explained with MDD systematics (e.g. Farley, 2018) but they are also consistent with a fluid moderated dissolution-recrystallization mechanism discussed above, which would also proceed faster for smaller grains which expose more surface area for the same mass of material. Similarly to volume diffusion, for which irregular surfaces are degassed first (Reiners and Farley, 1999), dissolution-recrystallization would affect irregular surfaces first, leading to the intuition that smaller grains will contain more helium in less-retentive domains (at the grains' surfaces) than larger grains with less surface area for the same mass.

Relevance of Helium Step Heat Data of Encapsulated, Sulfur-Buffered Sulfides in Nature

Given the high helium retention of pyrite and arsenopyrite shown in this work, the predominant helium loss mechanism for these common sulfides in nature may indeed be recrystallization. Pyrite in metamorphic conditions commonly recrystallizes and may either shrink or grow (Craig et al., 1998; McClay and Ellis, 1983). Until more work has characterized helium loss (or gain) for recrystallizing pyrite or arsenopyrite in metamorphic conditions, helium should cautiously be considered to have behaved as an open or partially open system in sulfides during metamorphic events, although O. Yakubovich, Podolskaya, et al. (2020) report that helium in the Uzelga pyrite was retained through metamorphism.

Thermal Helium Migration Systematics for Other Sulfides

Discussion of the thermal helium migration systematics of the two remaining, non-pyrite/arsenopyrite sulfides of this study – chalcopyrite, chalcocite, and galena – is given in the Supplementary Information (§3.5); neither copper sulfide nor galena is considered a promising U-Th/He geochronometer or helium isotope target.

Rustenburg: Pipe/short circuit diffusion mode

(JKW Lee: K2/K1 correlated with temperature, retrogrades see grain size increase by annealing. (

Arsenopyrite

No helium step heat data for sulfides plotted in figure 3.3 provide evidence for a simple single domain volume-diffusion model, except perhaps for the lone translucent sulfide of this study: sphalerite. The sphalerite experiment (Spal230707) may allow a simple single domain volume diffusion model. Unfortunately, the heating schedule of this study was insufficient to deconvolve the process of helium permeation through fused silica from helium evolution of sphalerite

Helium step heat experiments for galena (GalTS230630) and pyrite (LVCOPy230630, BigLVCOPy230707, 112123LVCOPyIrr, Rustenburg Pyrite) are all consistent with a multi-diffusion domain

Arsenopyrite: Arsenic degrades symmetry, moves from monoclinic to triclinic with more arsenic, increases unit cell volume. (Morimoto and Clark, 1961)

Arsenopyrite becomes more arsenic rich at higher temperatures (Barton, 1969; Morimoto and Clark, 1961)

Origin of Radiogenic Helium in Sulfides

The pyrite and arsenopyrite of this study are not, themselves, appreciably radiogenic minerals: the pyrite structure has no crystal- chemical affinity for U, Th, or Sm, which are large, highly charged cations poorly suited to substitution for Fe^{2+} in octahedral or pyritohedral sites. Consistent with this, only the aliquots (of any locality or mineral in the compilation, main text or SI) that host measurable U-Th-Sm-bearing accessory phases returned appreciable eU and, correspondingly, appreciable radiogenic 4He; the purest, most visually inclusion-free grains selected for the sulfur- buffered step heat experiments of §3.2 are essentially non-radiogenic. The U-Th/He ages reported here for Kidd Creek and Gympie pyrite therefore do not date growth of the pyrite lattice directly; they date whatever population of accessory, U-Th-Sm-rich inclusions is hosted within (or immediately adjacent to) each dated aliquot, with the pyrite itself acting as the retentive helium reservoir.

Backscattered-electron imagery of the Kidd Creek pyrite shows this directly. Superficially, hand-picked crystals appear as beautifully formed, optically clean,

blocky euhedral grains set within a finer sulfide-silicate groundmass (Fig. 3.11), but in cross-section the same grains are pervasively spongy, riddled with abundant tiny inclusions rather than being the pure, homogeneous pyrite their exteriors suggest. These inclusions are dominantly apatite (Figs. 3.12, 3.13), ranging in size from sub-6 micron grains at the limit of SEM resolution up to ∼100 micron half-width, large enough to individually resolve and, in the largest cases, to have been visible (and avoidable) during hand picking had their presence been anticipated. Because apatite preferentially incorporates U, Th, and especially Sm relative to coexisting sulfide and silicate phases, even a small volume fraction of included apatite can dominate a pyrite aliquot's bulk eU budget and, correspondingly, its radiogenic 4He production.

The Gympie pyrite hosts a texturally distinct, but analogously inclusion-hosted, radiogenic budget. SEM imagery (Fig. 3.14) shows abundant fine, needle-like rutile inclusions – the same U-Th-Sm-bearing oxide phase implicated in §3.3 for the anomalously old ages of the first four, early-protocol aliquots and of the 0413PyGy-4 outlier – alongside much larger, blocky inclusions of alkali feldspar and carbonate. The needle-like habit of the rutile inclusions plausibly explains why they proved harder to fully digest than the coarser, more equant apatite inclusions of the Kidd Creek pyrite: acid attack of an acicular grain proceeds from a much larger surface-area-to-volume ratio, but complete dissolution still requires acid to penetrate the full length of an often deeply embedded, sub-micron-diameter needle, whereas the larger alkali feldspar and carbonate inclusions, though volumetrically substantial, contribute comparatively little U-Th-Sm and so are of secondary concern for dissolution completeness.

This inclusion-hosted origin of radiogenic helium has a direct methodological consequence: sulfide U-Th/He geochronology cannot be treated as a single, generic protocol applied uniformly across localities. The identity, size distribution, and digestion behavior of the accessory phase or phases actually carrying the U-Th-Sm budget is a property of each deposit's own mineral paragenesis, and differs sharply even between the two localities studied here – coarse, readily digested apatite at Kidd Creek versus fine, refractory rutile at Gympie. Reliable sulfide U-Th/He geochronology of a new locality therefore requires SEM characterization of the dated material to identify its radiogenic inclusion population before analysis, and a dissolution protocol tailored to that population's specific mineralogy and grain size, rather than an off-the-shelf

digestion procedure assumed to work generically across sulfide-hosted deposits.

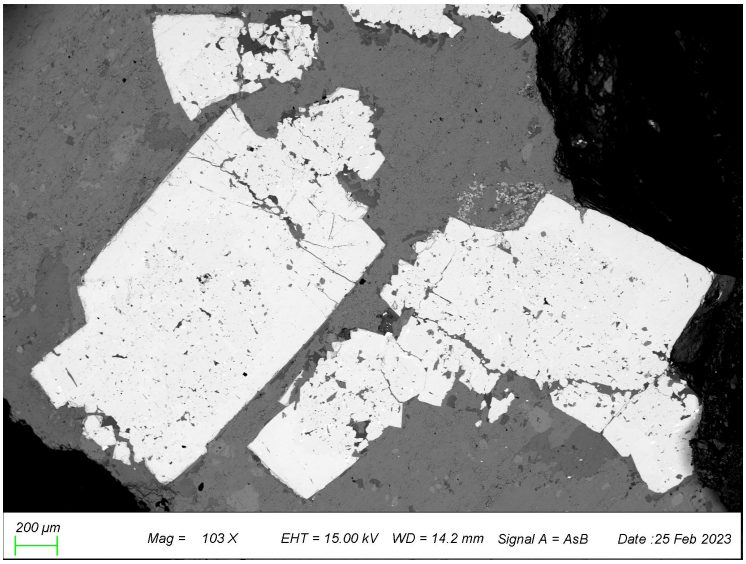


Figure 3.11: Backscattered-electron image of Kidd Creek pyrite (ST05) in cross-section: optically clean, blocky euhedral grains set within a finer-grained groundmass are, internally, spongy and riddled with abundant tiny (predominantly apatite) inclusions rather than homogeneous pyrite.

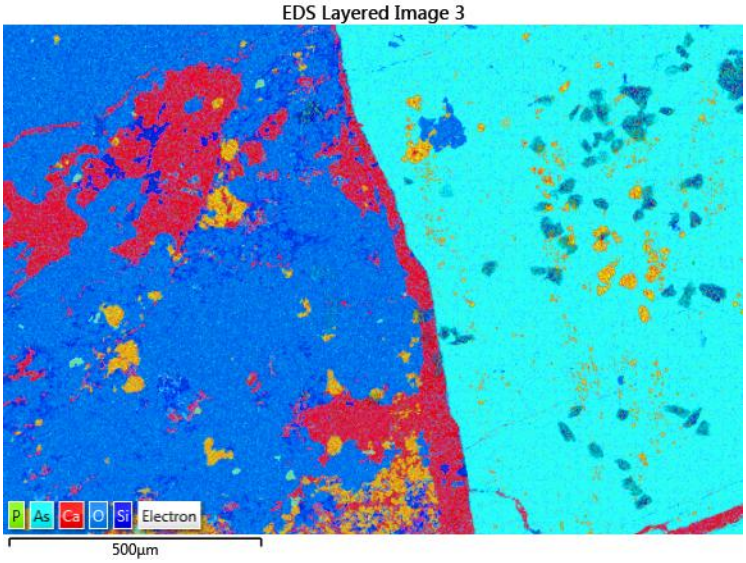


Figure 3.12: Backscattered-electron image of Kidd Creek pyrite showing apatite inclusions, the dominant host of the aliquots' U-Th-Sm budget.

Interpretation of Kidd Creek Pyrite Ages: Mixing Between a Retentive, He-Th-Poor Host and Apatite Inclusions

Excluding 011123KCPyST05C – treated above (§3.3) as an analytical outlier rather than a real age, on the grounds that its Th content is the lowest of the

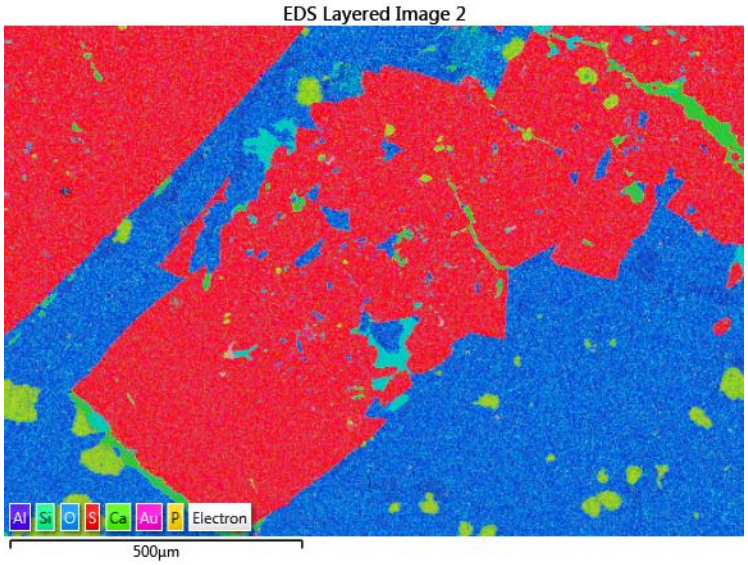


Figure 3.13: Backscattered-electron image of apatite inclusions in Kidd Creek pyrite, ranging from sub-6 micron grains up to ∼100 micron half-width.

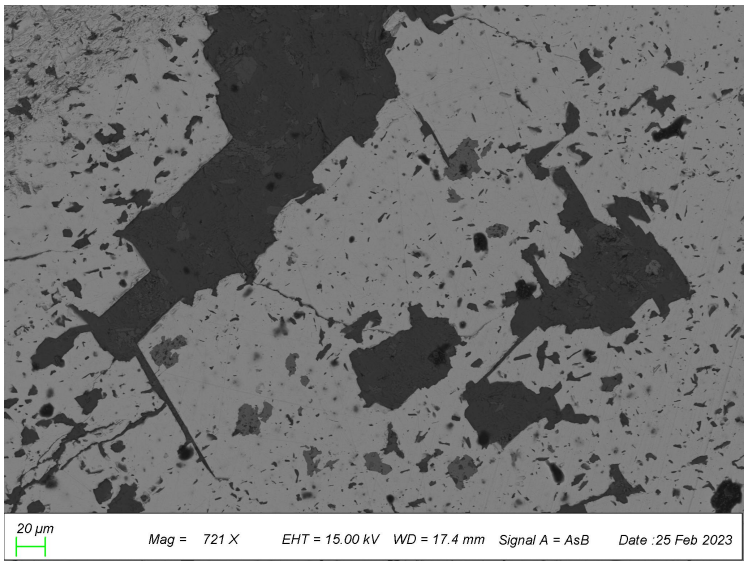


Figure 3.14: Backscattered-electron image of Gympie pyrite showing abundant fine, needle-like rutile inclusions and larger, equant feldspar inclusions. Ineffective rutile digestion is implicated in the anomalously old ages discussed in §3.3.

twelve-aliquot suite while its Sm and 4He contents are unremarkable, consistent with a spike-equilibration problem specific to Th (and not a bulk-dissolution problem that would have depressed Sm along with it) – the remaining eleven Kidd Creek pyrite ages are not scattered randomly around some single "true" age. They instead define a coherent array in which the Th- and He-richest aliquots give the youngest ages and the Th- and He-poorest aliquots give the oldest (Fig. 3.15). The youngest aliquots approach, without quite reaching, the 392 ± 32 Ma regional apatite (U-Th)/He cooling age reported by Sturrock et al. (2024) for sample 01-OE-69, sampled only ∼90 km from Kidd Creek; the oldest approach, without reaching, the ∼2716 Ma crystallization age of the host rhyolite (Hannington et al., 1999).

Given the inclusion-hosted origin of radiogenic helium established above (§3.4), the natural reading of this array is a two-component mixture: an old end-member that is essentially He-Th-poor but perfectly retentive, and a population of included apatite resembling Sturrock et al.'s regional standard, which itself only began retaining self-implanted helium at its own ∼392 Ma cooling age. A Kidd Creek pyrite aliquot's bulk age then falls between these two end-members depending on what fraction of its total measured Th and He budget was hosted in (and retained by) apatite inclusions since 392 Ma, versus alpha-ejected into the surrounding, always-retentive sulfide host and retained there since the ∼2716 Ma source age. It is not obvious, however, what the old end-member is compositionally: it could be pyrite that is simply intrinsically low in U-Th (variable trace-element zoning of the sulfide itself), or it could be inclusion-free (or inclusion-poor) domains of the same host rock in which essentially all the radiogenic budget of a given aliquot happens to have been alpha-ejected into inert, U-Th-free pyrite.

A simple model shows that this composition need not be invoked at all: the same qualitative array – lower age paired with higher Th and He – falls directly out of a single apatite population of variable inclusion size, via what is effectively a nugget effect. Fig. 3.16 extends the alpha-ejection retention framework of §3.4 to compute the apparent bulk He age of a pyrite aliquot hosting a single fluorapatite inclusion of half-prism radius R, assuming the inclusion crystallized (and the sulfide began retaining ejected alpha particles) 2700 Ma ago, but the apatite itself only began retaining its own self-implanted helium at a 380 Ma cooling age (essentially the Sturrock et al. value above). At small R, nearly

all the produced helium recoils out of the inclusion into the always-retentive sulfide, and the aliquot's bulk age approaches the old, 2700 Ma source age; at large R, an increasing fraction of the inclusion's own production is retained within the inclusion itself, only since its much younger cooling age, so the bulk age falls toward 380 Ma. Because a larger inclusion also simply contains more total U-Th-Sm mass, the same increase in R that drives the age down simultaneously drives up an aliquot's bulk Th and He content – exactly the anti-correlation seen in Fig. 3.15. Under this framing, the full spread of Kidd Creek ages could be produced by aliquot-to-aliquot variation in how large an apatite inclusion (or inclusions) happened to be captured during sample picking and crushing, drawn from a single apatite population with a single cooling age, rather than by mixing two distinct, compositionally separate end-members.

Bulk digestion data alone cannot distinguish between these two pictures – a genuine two-population mixture versus a single population subject to a size-dependent nugget effect – since both predict the same qualitative Th-He-age covariation. Resolving it would require in-situ characterization (e.g. SEM/EPMA or laser-ablation) of the specific inclusions hosted within each dated aliquot, rather than relying on the bulk trace-element and helium budgets used here.

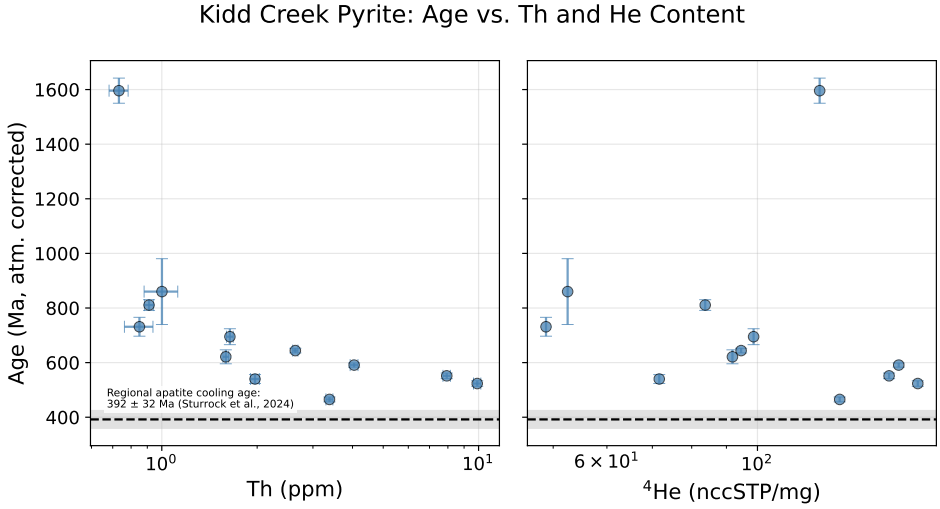


Figure 3.15: Atmospherically corrected Kidd Creek pyrite age vs. Th content (left) and vs. measured 4He content (right; nccSTP/mg), sharing a common age axis (§3.3). The shaded band is the independent regional apatite cooling-age estimate of 392 ± 32 Ma for sample 01-OE-69 (Sturrock et al., 2024). Both panels show the same sense of anti-correlation: Th- and He-rich aliquots give younger ages, Th- and He-poor aliquots give older ages.

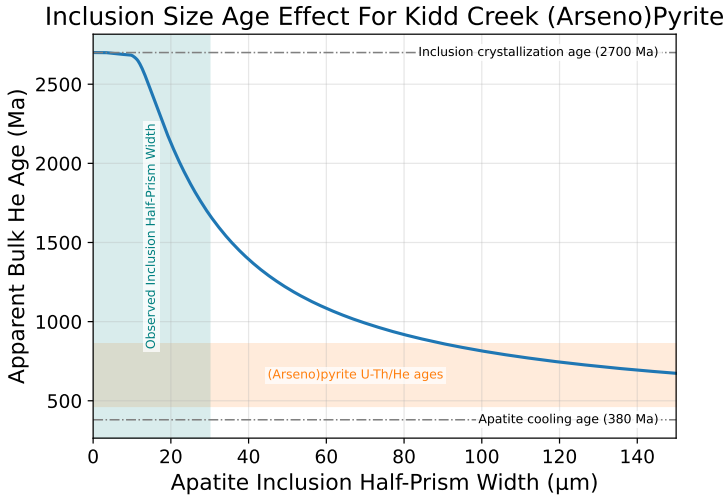


Figure 3.16: Modeled apparent bulk He age of a Kidd Creek pyrite aliquot hosting a single fluorapatite inclusion, as a function of the inclusion's half-prism radius R. The alpha-ejection retention fraction $F_t(R)$ (fraction of produced helium self-implanted and retained within the inclusion, rather than ejected into the surrounding sulfide) is computed via a Monte Carlo simulation of 10^6 decays uniformly distributed by volume within the inclusion with isotropic alpha-ejection directions, using per-isotope (^{238}U, ^{235}U, ^{232}Th, ^{147}Sm) stopping distances for fluorapatite and a fixed U:Th:Sm ratio of 1:2:2; this reproduces Farley, Wolf, et al. (1996)'s closed-form retention factor for $R \geq S$ (the stopping distance) and remains well behaved for $R < S$, where the closed form is not valid. Helium is then accumulated separately for the inclusion-retained fraction $F_t(R)$, assuming retention only since a 380 Ma apatite cooling age, and the sulfide-retained fraction $1 - F_t(R)$, assuming retention since a 2700 Ma inclusion/host source age, using the standard U-Th-Sm/He ingrowth equation; the two helium contributions (not the two ages) are summed and inverted back to a single apparent bulk age. Reference lines mark the 2700 Ma source age and 380 Ma cooling age end-members (the latter close to, though not identical to, the independently estimated 392 ± 32 Ma regional value of Fig. 3.15); the modeled age falls smoothly between them as R increases, in the same sense as the Th/He-age covariation of the real Kidd Creek data.

Pyrite/Arsenopyrite U-Th/He Ages for Gympie Gold Belt

Lithostratigraphic age constraints that rely on foraminifera microfossils suggest an Early Permian age for the Gympie Group rocks that host the gold-vein systems. However, as noted in §3.1, these biostratigraphic constraints are sparse and rest mainly on correlation higher in the sequence rather than direct radiometric dates on the volcanic pile itself, and Holcombe et al. (1997) note that the lower Gympie Block volcanics "may contain older volcanic units." If the Gympie Group is instead a structurally displaced, along-strike fragment of the same Bowen Basin/Connors-Auburn crustal domain, rather than a far-travelled exotic terrane – host-rock emplacement as old as ∼300-310 Ma is plausible for at least the base of the sequence, some 65-75 Myr prior to the ∼235 Ma gold-vein event, rather than the ∼45 Myr gap implied by a more conventional ∼280 Ma Early Permian age.

This older, ∼300-310 Ma end of the emplacement range provides a natural origin for the needle-like rutile inclusions identified in the Gympie pyrite (§3.4), and therefore for the U-Th-Sm-He budget actually being dated here (§3.4). Whether this ∼300 Ma age reflects the crystallization/emplacement of the regional granodiorite plutons themselves (306.5 ± 0.6 and 306.9 ± 1.2 Ma, $^{40}Ar/^{39}Ar$ hornblende; §3.1) or a basement cooling age recording exhumation of the adjoining North D'Aguilar/Marlborough blueschist-facies rocks is not resolvable from the U-Th/He data alone – the two are essentially contemporaneous (Holcombe et al., 1997) – but either way, the interpretation offered here is that the rutile inclusions' own radiogenic helium clock started running at approximately this time, well before the ∼235 Ma gold-vein event.

Under this interpretation, the eleven reliable Gympie ages (§3.3) record variable, partial resetting of that ∼300 Ma inclusion population during the ∼235 Ma hydrothermal mineralizing event, rather than two discrete pyrite generations. The oldest aliquots (up to 306 Ma) preserve close to the full inclusion age, essentially undisturbed by the vein-forming event; progressively younger aliquots record progressively more complete helium loss during that event, down to ages approaching, but not below, the ∼235 Ma event itself. Since ∼235 Ma, helium appears to have been essentially perfectly retained: no aliquot's age has continued to drift downward toward the present, which is itself consistent with this chapter's broader finding of remarkable helium retentivity in sulfides (§3.4) extending to their hosted inclusions as well.

The one clear exception is 062123GyPyHClb, at 191 Ma – the youngest age in the suite, and the only one below the ∼235 Ma mineralization age itself. A partial-resetting model bounded between a ∼300 Ma inclusion age and a ∼235 Ma resetting event cannot, on its own, produce an age younger than 235 Ma, so this aliquot requires some additional, more recent helium loss not shared by the rest of the suite. We have no strong constraint on what that additional process is. One speculative possibility is that this particular aliquot's radiogenic budget happens to be disproportionately hosted in one of the larger, coarser alkali feldspar inclusions identified alongside the rutile (§3.4, Fig. 3.14) rather than in rutile – feldspar is not normally considered a particularly helium-retentive phase, and if the aliquot's post-235 Ma budget were disproportionately feldspar-hosted rather than rutile-hosted, continued helium loss after the mineralizing event could plausibly depress its age below the rest of the suite. This remains speculative and is not independently tested here.

We note, however, that an alternative interpretation treats the Gympie Group as a far-travelled, exotic intra-oceanic arc terrane accreted only later to the Gondwana margin (Sivell and Waterhouse, 1988); under that view the North D'Aguilar/Marlborough ages would not directly constrain the age of the Gympie host rocks or its inclusion population, and the correspondence between the oldest pyrite ages and the ∼300-310 Ma regional magmatic/exhumation episode invoked above would need to be reassessed.

3.5 Conclusions

[TODO: Summarize He retentivity findings and geochronologic implications.]

Supplementary Information

for Chapter 3: *Sulfur-Buffered Helium Step Heat Experiments of Sulfides Reveal Remarkable Helium Retentivity and Complex Behavior: Implications for Sulfide Geochronology*

3.1s Chalcopyrite, Chalcocite, and Galena Step Heat Results and Discussion

This section reports the step heat results and discussion for the two non-pyrite/arsenopyrite sulfides of this study – the copper sulfides chalcopyrite (CCpTS230630) and chalcocite (CCT230707), and galena (GalTS230630) – moved here from the main text because neither mineral is a promising U-Th/He geochronometer or helium isotope target, and neither is otherwise part of this chapter's main narrative (Table 1 in the main text still lists all nine step heat experiments, including these three, for completeness).

Results

The majority of the helium total yield was evolved during the step heat experiment of the encapsulated galena, in stark contrast to the comparatively retentive pyrites and arsenopyrite. The reference line from the chalcopyrite experiment showed a similar activation energy to the activation energy implied by curved retrogrades of the galena experiment (Fig. 3.17, row 6), so it was used as the reference for the delta plot of galena. This delta plot shows a monotonic decrease in delta values as the experiment proceeds (Fig. 3.17, row 6).

Step heat data step heat experiments for chalcopyrite (CCpTS230630) and chalcocite (CCT230707) produced macroscopic evidence of phase transformation, including the development of an iridescent blue sulfide or oxide layer on both specimens and dramatic crystal growth on the ampoule walls for the chalcocite experiment (figs. 3.0xx). Although permeation lag was not sufficiently constrained to calculate apparent diffusivities for the chalcocite experiment, the data for CCT230707 are consistent with a primary phase-breakdown control of helium migration at temperatures $\leq$300°C.

A linear regression is fit to 9 measurements defining a linear region of the Arrhenius plot of chalcopyrite sample CCpTS230630 for use as a reference for the delta plot (Fig. 3.17, row 7). This regression has an R^2 value of 0.9811, an implied activation energy of 41.8 $\pm$ 2.2 kcal (174.8 $\pm$ 9.2 kJ/mol), and an implied closure temperature of 128 $\pm$ 13°C assuming a cooling rate of 10 °C/Ma. The delta plot generated using this reference line for chalcopyrite clearly shows a dramatic mid-experiment increase in diffusivity following steps above 400°C, followed by a continuous decrease in delta values (Fig. 3.17, row

7), similarly to that observed for the dehydrating goethite sample Win-06-01b (Farley, H. B. Monteiro, et al., 2024).

The full ^{3}He step-heat datasets for both experiments, in the same format as the main-text pyrite/arsenopyrite tables (§3.3), are given in Tables 3.7 and 3.8.

Table 3.7: ^{3}He step-heat dataset for Galena, Tri-State District, Missouri (sample GalTS230630), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.17 (row 6). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
15	385.0	100.0	0.2968	11.3209	0.0835	-14.758	-13.974	-13.716
16	375.0	120.0	0.2004	11.5213	0.0491	-15.018	-14.489	-14.264
17	380.0	110.0	0.2022	11.7235	0.0528	-14.616	-14.344	-14.105
18	390.0	90.0	0.1906	11.9141	0.0586	-14.471	-14.153	-13.877
19	400.0	70.0	0.3558	12.2699	0.1413	-13.379	-13.207	-12.853
20	410.0	60.0	0.0947	12.3646	0.0473	–	-14.316	-13.904
21	300.0	60.0	0.0374	12.4020	0.1171	–	-15.228	-13.793
22	350.0	60.0	0.0241	12.4262	0.0366	–	-15.656	-14.729
23	375.0	60.0	0.0459	12.4720	0.0327	-16.600	-15.005	-14.462
24	400.0	60.0	0.1039	12.5760	0.0525	-14.539	-14.166	-13.750
25	415.0	60.0	0.1722	12.7481	0.0744	-13.977	-13.621	-13.251
26	460.0	60.0	0.6155	13.3637	0.1765	-12.343	-12.226	-11.943
27	470.0	60.0	0.6037	13.9673	0.1605	-12.349	-12.034	-11.765
28	480.0	60.0	0.4856	14.4529	0.1200	-12.389	-12.024	-11.771
29	489.9	60.0	0.3938	14.8467	0.0908	-12.325	-11.997	-11.787
30	499.7	60.0	0.3285	15.1752	0.0709	-12.280	-11.985	-11.764
31	504.6	60.0	0.2357	15.4109	0.0493	-12.437	-12.108	-11.896
32	495.0	60.0	0.1264	15.5374	0.0282	-13.040	-12.570	-12.355
33	485.0	80.0	0.0935	15.6309	0.0158	-13.390	-13.047	-12.882

continued on next page

(continued)

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
34	475.0	100.0	0.0667	15.6976	0.0093	-13.777	-13.516	-13.380
35	465.0	120.0	0.0489	15.7465	0.0059	-14.144	-13.938	-13.820
36	455.0	140.0	0.0351	15.7816	0.0038	-14.549	-14.368	-14.262
37	510.0	60.0	0.0648	15.8464	0.0131	-12.894	-12.836	-12.643
38	519.9	60.0	0.0765	15.9229	0.0145	-12.740	-12.562	-12.376
39	529.9	60.0	0.0758	15.9987	0.0135	-12.638	-12.438	-12.261
40	539.1	60.0	0.0709	16.0696	0.0120	-12.557	-12.359	-12.190
41	535.8	60.0	0.0570	16.1266	0.0098	-12.650	-12.429	-12.258
42	538.6	60.0	0.0385	16.1651	0.0065	-12.977	-12.697	-12.531
43	532.9	80.0	0.0330	16.1980	0.0041	-13.243	-13.033	-12.908
44	525.0	100.0	0.0265	16.2245	0.0027	-13.542	-13.378	-13.276
45	515.0	120.0	0.0207	16.2452	0.0018	-13.857	-13.722	-13.633
46	503.7	140.0	0.0139	16.2590	0.0011	-14.348	-14.209	-14.129
49	480.0	60.0	0.0028	16.2618	0.0007	-14.929	-14.929	-14.706
50	489.9	60.0	0.0032	16.2651	0.0007	-15.010	-14.771	-14.562
51	500.0	60.0	0.0050	16.2701	0.0011	-14.468	-14.310	-14.113
52	505.0	60.0	0.0052	16.2753	0.0011	-14.495	-14.262	-14.070

Table 3.8: ^{3}He step-heat dataset for Chalcopyrite, Baxter Springs, Kansas (sample CCpTS230630), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.17 (row 7). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
16	374.9	120.0	0.0782	13.1758	0.0247	-18.321	-17.572	-17.296
17	380.0	110.0	0.0639	13.2396	0.0220	-18.167	-17.680	-17.383
18	390.0	90.0	0.0721	13.3118	0.0296	-17.713	-17.350	-17.005
19	400.1	70.0	0.0943	13.4061	0.0509	-17.198	-16.822	-16.388
20	410.0	60.0	0.0963	13.5024	0.0610	-17.385	-16.637	-16.143
22	291.8	60.0	0.0928	13.5961	3.0586	-17.704	-16.664	-12.980
23	345.2	60.0	0.2696	13.8657	1.0009	–	-15.578	-13.976
24	375.0	60.0	0.9931	14.8588	1.0408	–	-14.209	-13.439
25	400.0	60.0	2.0663	16.9251	1.4653	-13.975	-13.324	-12.714
26	415.0	60.0	2.2228	19.1480	1.3491	-14.056	-13.046	-12.507
27	429.9	60.0	2.8501	21.9981	1.4512	-13.143	-12.562	-12.080
28	420.0	60.0	2.4223	24.4204	1.3837	-13.328	-12.480	-11.959
29	410.0	60.0	1.4757	25.8962	0.9756	-15.497	-12.790	-12.232
30	400.1	60.0	0.7845	26.6806	0.6038	–	-13.309	-12.706
31	390.0	80.0	0.5091	27.1898	0.2693	–	-13.962	-13.523
32	380.0	100.0	0.3706	27.5604	0.1495	-15.738	-14.456	-14.108
33	370.0	120.0	0.2689	27.8293	0.0892	-15.728	-14.924	-14.632
34	360.0	140.0	0.1510	27.9803	0.0452	-16.521	-15.632	-15.367
35	355.0	150.0	0.1416	28.1219	0.0408	-16.131	-15.749	-15.493

continued on next page

(continued)

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
36	365.0	130.0	0.1775	28.2994	0.0557	-15.621	-15.362	-15.085
37	375.0	110.0	0.2245	28.5239	0.0808	-15.219	-14.937	-14.625
38	385.0	90.0	0.3152	28.8391	0.1360	-14.657	-14.365	-13.999
39	395.0	80.0	0.4454	29.2845	0.2065	-14.213	-13.857	-13.463
40	405.0	70.0	0.4926	29.7771	0.2543	-14.095	-13.565	-13.132
41	415.0	60.0	0.4935	30.2707	0.2950	-14.053	-13.346	-12.857
42	460.0	60.0	0.4521	30.7228	0.1759	-14.407	-13.370	-13.029
43	469.7	60.0	0.1397	30.8624	0.0517	–	-14.504	-14.186
44	465.4	60.0	0.1654	31.0278	0.0601	-14.684	-14.313	-13.999
45	473.5	60.0	0.1554	31.1832	0.0531	-14.837	-14.352	-14.054
46	480.0	60.0	0.1691	31.3523	0.0538	-14.617	-14.244	-13.963
47	491.9	60.0	0.1549	31.5072	0.0452	-14.729	-14.307	-14.047
48	491.0	60.0	0.1219	31.6291	0.0357	-14.985	-14.525	-14.265

Discussion: Thermal Helium Migration Systematics

Chalcopyrite and Chalcocite

Neither copper-bearing sulfide mineral studied here was successfully stabilized by encapsulation with sulfur, so these results provide little insight into helium migration within copper sulfides in nature, though they suggest high potential for helium retention at surface temperatures, similar to apatite. Still, they are unlikely to be promising targets for U-Th/He geochronology or helium isotope studies as there exists many copper-bearing sulfide phases stable at temperatures between 200-500°C, for which phase stability and Cu/Fe ratios are sensitive functions of temperature, composition, and sulfur fugacity (Zhao et al., 2014; J. H. Chen and Harvey, 1975; Yund and Kullerud, 1966). These reactions are likely to result in helium loss during reheating, burial, diagenesis, or metasomatism after crystallization. Copper sulfides are unlikely to be promising targets for U-Th/He geochronology, but may be suitable for helium isotope studies in young or unaltered samples.

Galena

Results from encapsulated Galena subjected to helium step heat experiments were consistent with a multiple diffusion domain helium loss mechanism. As expected for an MDD system, the delta plot for galena monotonically decreases as the fractional yield increases (Fig. 3.17, row 6; Farley (2018)). The majority of the 3He cumulative fractional yield was evolved at low temperatures where the uncertainties due to permeation lag could not be constrained, yet these early degassing domains were certainly less retentive than 150 µm half-prism width Durango apatite (Farley, 2000). Additionally, the total 3He concentration for the irradiated galena was less than half that of co-irradiated pyrite, implying poor helium retention even at ambient temperatures. Galena is unlikely to be a promising target for U-Th/He geochronology or helium isotope studies.

3.2s Complete U-Th/He Geochronology Compilation

This section is a data appendix, not a narrative: it reports the complete U-Th/He geochronology compilation (Chapter3/Tables/geochronology_ compilation_BJT.xlsx) for the sample suites and replicate aliquots not discussed in the main text – Kidd Creek replicate/mass-only aliquots, Franklin (NJ) pyrite and fluorite, Rustenberg/Pilanesberg pyrite, Magnet Cove carbonatite pyrite, French Creek magnetite/pyrite, and Salzburg arsenopyrite – so that the

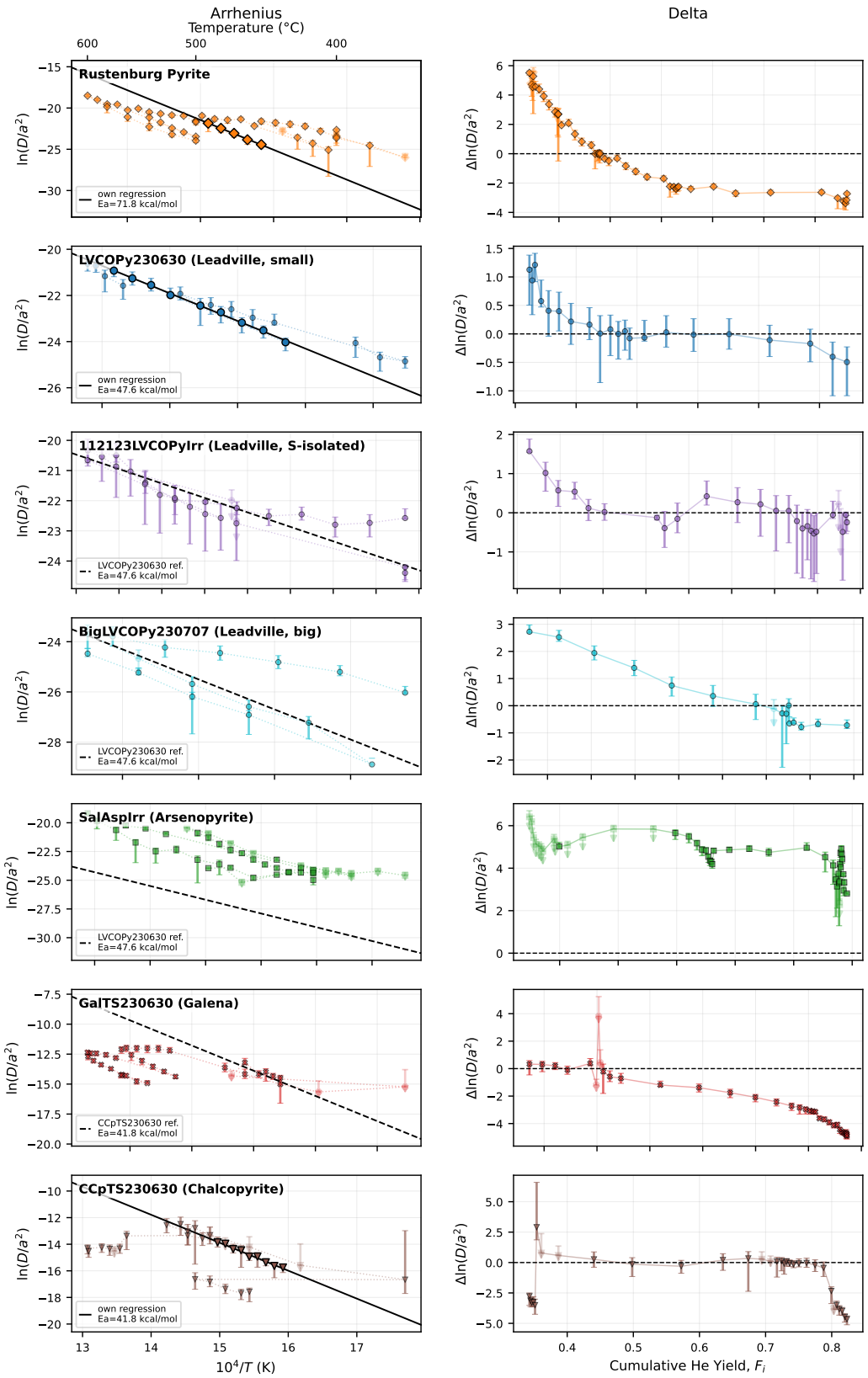


Figure 3.17: Compiled Arrhenius (left column) and delta (right column) plots for every plotable Chapter 3 sulfide step-heat experiment, one row per sample: (1) Rustenburg pyrite, (2) LVCOPy230630 (Leadville pyrite, small), (3) 112123LVCOPyIrr (Leadville pyrite, sulfur-isolated), (4) BigLVCOPy230707 (Leadville pyrite, big), (5) SalAspIrr (arsenopyrite), (6) GalTS230630 (galena),

full underlying dataset is available even where it is not part of this chapter's argument. Later work extending this chapter's geochronology (e.g. to these additional localities) can build directly on this section.

Table 3.9: U-Th/He geochronology results for all remaining sample suites in the compilation, not discussed in the main text: K Creek replicate/mass-only aliquots, Franklin (NJ) pyrite and fluorite, Rustenberg/Pilanesberg pyrite, Magnet Cove carbona pyrite, French Creek magnetite/pyrite, and Salzburg arsenopyrite. Reported here for completeness; columns and convention in Table 3.6.

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	S (ppm)	eU (ppm)	
Kidd Creek Pyrite (replicate / mass-only aliquots)											
2211KC_Py_ST05_DU1	2.09	0.781 ± 0.034	1.760 ± 0.045	0.081 ± 0.008	–	–	–	–		1.204	
2211KC_Py_ST05_DU2	2.98	0.801 ± 0.014	2.185 ± 0.062	0.021 ± 0.004	–	–	–	–		1.325	
2211KC_Py_ST05_DU3	1.77	0.766 ± 0.026	1.930 ± 0.056	0.103 ± 0.012	–	–	–	–		1.229	
12122022KCPyST05A	14.35	–	–	–	–	–	–	–		–	14
12122022KCPyST05B	10.52	–	–	–	–	–	–	–		–	9
12122022KCPyST05C	12.19	–	–	–	–	–	–	–		–	12
12122022KCPyST05D	9.60	–	–	–	–	–	–	–		–	8
Franklin, NJ Pyrite (marble/volcanics, ∼1.3–1.0 Ga)											
2211GFr_PyA	2.49	0.293 ± 0.013	0.049 ± 0.011	0.016 ± 0.001	–	–	–	–		0.305	
2211GFr_PyB	3.00	1.778 ± 0.021	0.008 ± 0.005	0.008 ± 0.002	–	–	–	–		1.780	
2211StFr_PyA	1.45	0.641 ± 0.031	0.022 ± 0.012	0.023 ± 0.004	–	–	–	–		0.646	
2211StFr_PyB	2.53	0.120 ± 0.014	0.000 ± 0.000	0.907 ± 0.035	–	–	–	–		0.120	
2211AtmFr_PyC	1.93	0.996 ± 0.020	0.012 ± 0.008	0.022 ± 0.005	–	–	–	–		0.999	
Franklin, NJ Fluorite											
011423FrFl2	1.40	3.193 ± 0.034	0.242 ± 0.070	0.211 ± 0.038	32.17 ± 10.03	–	1.82 ± 1.13	10.23 ± 1.72	35.81 ± 0.74	3.251	
011423FrFl3	1.03	5.352 ± 0.084	0.202 ± 0.060	0.342 ± 0.086	–	–	5.80 ± 1.56	–	30.63 ± 0.7	5.401	
Rustenberg/Pilanesberg(?) Pyrite											
013023BVPy H-3026-1	1.03	2.498 ± 0.309	1.040 ± 0.085	0.293 ± 0.066	173.63 ± 26.23	3328.64 ± 233.02	10.86 ± 5.67	–	152.60 ± 3.84	2.748	2
041323BVPy1	0.95	0.565 ± 0.015	0.167 ± 0.045	0.053 ± 0.050	105.69 ± 27.36	391.95 ± 59.38	–	–	–	0.605	

(continued)

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	Sb (ppm)	eU (ppm)	
041323BVPy2	0.53	2.069 ± 0.277	2.336 ± 0.410	0.770 ± 0.450	484.42 ± 455.64	22900.08 ± 4862.73	130.29 ± 59.21	–	680.02 ± 100.62	2.630	
062123BVPyHCla	1.22	3.132 ± 0.117	0.808 ± 0.075	0.255 ± 0.013	95.87 ± 10.25	582.38 ± 13.15	12.17 ± 2.15	2.43 ± 0.97	10.07 ± 0.42	3.326	2
062123BVPyHClb	0.53	1.460 ± 0.049	0.255 ± 0.178	0.163 ± 0.014	102.76 ± 11.05	1499.31 ± 44.51	7.89 ± 4.05	3.30 ± 1.37	10.48 ± 0.72	1.521	10
230628BVPyUnAbrB	0.75	2.146 ± 0.075	0.420 ± 0.129	0.145 ± 0.017	64.34 ± 12.08	996.13 ± 55.99	–	3.09 ± 1.39	3.54 ± 0.32	2.247	2
230628BVPyUnAbrC	1.11	0.500 ± 0.126	0.124 ± 0.090	0.093 ± 0.017	979.31 ± 62.60	1028.86 ± 34.43	–	7.63 ± 3.34	4.60 ± 0.29	0.530	7
230628BVPyUnAbrC	0.60	29.012 ± 0.404	5.444 ± 0.335	2.005 ± 0.170	138.48 ± 15.53	728.93 ± 53.84	202.97 ± 10.19	11.80 ± 4.77	10.59 ± 0.76	30.318	8
Magnet Cove Carbonatite Pyrite, Arkansas (~100 Ma)											
011123MCARPy1	3.38	0.061 ± 0.003	0.406 ± 0.025	0.311 ± 0.035	15.72 ± 4.66	3.86 ± 1.22	1.63 ± 0.65	1.10 ± 0.48	18.58 ± 0.78	0.159	
011123MCARPy2	3.90	0.011 ± 0.002	0.202 ± 0.014	0.023 ± 0.011	56.98 ± 6.51	62.10 ± 8.60	1.58 ± 0.65	–	11.49 ± 0.88	0.059	
French Creek, PA Magnetite and Pyrite (~130 Ma)											
062123FCHCl	4.36	0.010 ± 0.001	0.000 ± 0.019	0.007 ± 0.002	8.97 ± 2.49	29.58 ± 2.10	–	21.25 ± 6.89	7.65 ± 1.15	0.010	
062123FCMagHCl	3.84	0.059 ± 0.002	0.000 ± 0.022	0.050 ± 0.005	–	81.88 ± 3.18	–	0.71 ± 0.29	15.55 ± 0.41	0.059	
230625FCHFa	3.41	0.634 ± 0.046	0.709 ± 0.032	0.345 ± 0.022	209.62 ± 15.43	1118.04 ± 38.65	7.76 ± 0.88	5.71 ± 2.12	19.75 ± 0.71	0.804	2
230625FCMagHCl	4.07	0.055 ± 0.002	0.000 ± 0.020	0.023 ± 0.003	–	85.58 ± 8.67	–	0.74 ± 0.33	4.22 ± 0.19	0.055	
Salzburg, Austria Arsenopyrite											
230625SBAsPy	2.52	0.960 ± 0.012	1.295 ± 0.046	0.114 ± 0.009	78.87 ± 11.98	187365.67 ± 4344.31	24.74 ± 1.67	3.67 ± 1.41	94.85 ± 3.82	1.271	

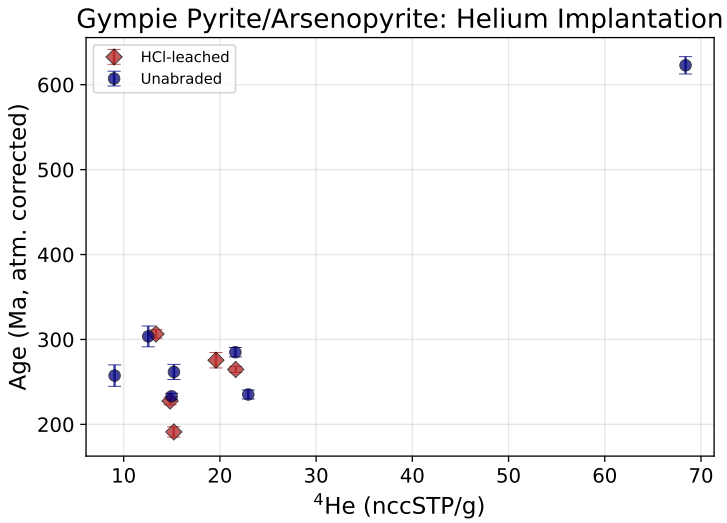


Figure 3.18: Companion to the main-text Gympie age-vs-eU figure (Fig. 3.8), plotting atmospherically corrected age against measured 4He content (nccSTP/g) for the newer-protocol Gympie pyrite suite with the 0413PyGy-4 outlier put back in (§3.3, main text). Its 4He content stands well off the trend defined by the other eleven aliquots, consistent with the helium-implantation interpretation offered there.

Chapter 4

CHEMICALLY MAPPING SATURN'S OUTER RINGS WITH DUSTMAP: AN AUTOMATED MASS CALIBRATION AND CATEGORIZATION PIPELINE FOR TOF MASS SPECTRA

[TODO: One-paragraph chapter summary.]

4.1 Introduction

[TODO: Introduce Cassini CDA and INMS context. State the challenge of mass-calibrating and categorizing large TOF spectral datasets. Motivate the need for an automated pipeline.]

4.2 Background

TOF Mass Spectrometry in Space Science

[TODO:]

Prior Approaches to Spectral Calibration and Classification

[TODO:]

4.3 The DustMAP Pipeline

Automated Mass Calibration

[TODO: Describe the calibration algorithm: reference peak identification, polynomial fitting, uncertainty estimation.]

Spectral Categorization

[TODO: Describe the classification approach: clustering algorithm, feature space, category definitions.]

Implementation and Computational Environment

[TODO:]

4.4 Validation

[TODO: Demonstrate calibration accuracy on known standards. Compare categorization results with manual classifications.]

4.5 Application: Chemical Map of Saturn's Outer Rings

[TODO: Present ring chemical maps. Identify compositional gradients or distinct populations.]

4.6 Discussion

[TODO: Interpret chemical structure of the rings. Discuss origin and evolution implications. Comment on pipeline generalizability to other TOF instruments.]

4.7 Conclusions

[TODO: Summarize pipeline contributions and key ring chemistry findings.]

Chapter 5

NEW FRONTIERS IN THERMODYNAMICALLY INFORMED GEODYNAMIC MODELING: A GENERAL DEEP LEARNING ACCELERATED FREE ENERGY MINIMIZER FOR MULTICOMPONENT SILICATE SYSTEMS

[TODO: One-paragraph chapter summary.]

5.1 Introduction

[TODO: Motivate the need for fast thermodynamic forward models in geodynamics. Review current tools (Perple_X, THERMOCALC, MAGEMin) and their computational bottlenecks. State the deep learning acceleration approach.]

5.2 Background

Gibbs Free Energy Minimization in Petrology

[TODO:]

Machine Learning Surrogates for Physical Models

[TODO:]

5.3 Model Architecture

Training Data Generation

[TODO: Describe the strategy for generating phase equilibrium training data from classical minimizers across relevant P–T–X space.]

Neural Network Design

[TODO: Describe the architecture: input/output spaces, layers, activation functions, loss function, training procedure.]

Uncertainty Quantification

[TODO:]

5.4 Validation

[TODO: Benchmark speed and accuracy against classical minimizers on held-out P–T–X points and on known phase diagrams.]

5.5 Application to Geodynamic Modeling

[TODO: Demonstrate integration into a geodynamic simulation. Show a representative case study.]

5.6 Discussion

[TODO: Discuss accuracy/speed tradeoffs, generalizability to other bulk compositions, and extension to systems relevant to the Bushveld Complex (foreshadow Appendix 5.7 and the Conclusion).]

5.7 Conclusions

[TODO: Summarize the ML minimizer performance and implications for future thermodynamically coupled geodynamic models.]

CONCLUSION

[TODO: Write short dissertation conclusion. No data, few or no figures. Synthesize contributions. Allude to the Bushveld Complex fieldwork (Appendix 5.7) as motivating nGibbs work and sulfide/goethite geochronology. Mention the laterite geomorphology at Draaikraal as an unexpected and compelling target for goethite geochronology.]

REFERENCES

Ackerman, L. (2025). "Re-Os Cookbook: From Sample Decomposition to Elemental and Isotopic Analyses". In: *Analytical Isotope Geochemistry: Techniques and Data Interpretation*. Ed. by G. R. Tripathy. Cham: Springer Nature Switzerland, pp. 267–289. DOI: 10.1007/978-3-031-88388-0_12.

Allard, T., C. Gautheron, S. Bressan Riffel, E. Balan, B. F. Soares, R. Pinna-Jamme, A. Derycke, G. Morin, G. T. Bueno, and N. do Nascimento (2018). "Combined dating of goethites and kaolinites from ferruginous duricrusts: Deciphering the Late Neogene erosion history of Central Amazonia". In: *Chemical Geology* 479, pp. 136–150. DOI: 10.1016/j.chemgeo.2018.01.004.

Altemose, V. O. (1961). "Helium Diffusion through Glass". In: *Journal of Applied Physics* 32, pp. 1309–1316.

Barrie, C. D., M. A. Pearce, and A. P. Boyle (2011). "Reconstructing the Pyrite Deformation Mechanism Map". In: *Ore Geology Reviews* 39.4, pp. 265–276. DOI: 10.1016/j.oregeorev.2011.03.006.

Barton, P. B. (1969). "Thermochemical Study of the System Fe-As-S". In: *Geochimica et Cosmochimica Acta* 33.7, pp. 841–857. DOI: 10.1016/0016-7037(69)90031-3.

Bassal, F., B. Heller, J. Roques, H. Balout, L. Tassan-Got, T. Allard, and C. Gautheron (2022). "Revealing the radiation damage and Al-content impacts on He diffusion in goethite". In: *Chemical Geology* 611, p. 121118. DOI: 10.1016/j.chemgeo.2022.121118.

Baughman, J. S. and R. M. Flowers (2018). "Deciphering a 2 Gyr-Long Thermal History From a Multichronometer (U-Th)/He Study of the Phalaborwa Carbonatite, Kaapvaal Craton, South Africa". In: *Geochemistry, Geophysics, Geosystems* 19.5, pp. 1581–1594. DOI: 10.1029/2017GC007198.

Carman, P. C. (1956). *Flow of Gases through Porous Media*. New York: Academic Press.

Carslaw, H. S. and J. C. Jaeger (1959). *Conduction of Heat in Solids*. 2nd. Oxford: Oxford University Press.

Chen, J. H. and W. W. Harvey (1975). "Cation Self-Diffusion in Chalcopyrite and Pyrite". In: *Metallurgical Transactions B* 6.2, pp. 331–339. DOI: 10.1007/BF02913577.

Chen, M., J. Mao, C. Li, Z. Zhang, and Y. Dang (2015). "Re–Os Isochron Ages for Arsenopyrite from Carlin-like Gold Deposits in the Yunnan–Guizhou–Guangxi "Golden Triangle", Southwestern China". In: *Ore Geology Reviews* 64, pp. 316–327. DOI: 10.1016/j.oregeorev.2014.07.019.

Cherniak, D. J. (2019). "Diffusion of helium in radiation-damaged zircon". In: *Chemical Geology* 529, p. 119308.

Clark, L. A. (1960). "The Fe-As-S System–Phase Relations and Applications". In: *Economic Geology* 55.7, pp. 1345–1381. DOI: 10.2113/gsecongeo.55.7.1345.

Cost, J. R. and R. G. Hickman (1975). "Helium Release from Various Metals". In: *Journal of Vacuum Science and Technology* 12.1, pp. 516–519. DOI: 10.1116/1.568578.

Cox, S. E., H. B. D. Miller, F. Hofmann, and K. A. Farley (2022). "Short Communication: Mechanism and Prevention of Irreversible Trapping of Atmospheric He during Mineral Crushing". In: *Geochronology* 4.2, pp. 551–560. DOI: 10.5194/gchron-4-551-2022.

Craig, J. R., F. M. Vokes, and T. N. Solberg (1998). "Pyrite: Physical and Chemical Textures". In: *Mineralium Deposita* 34.1, pp. 82–101. DOI: 10.1007/s001260050187.

Crank, J. (1979). *The Mathematics of Diffusion*. Clarendon Press.

CUNEEN, R. (1994). "New Insights into the Gympie Goldfield". In: *Queensland Exploration Potential 1994, Symposium Abstracts*, p. 26.

Danišík, M., N. J. Evans, E. R. Ramanaidou, B. J. McDonald, C. Mayers, and B. I. A. McInnes (2013). "(U-Th)/He chronology of the Robe River channel iron deposits, Hamersley Province, Western Australia". In: *Chemical Geology* 354, pp. 150–162. DOI: 10.1016/j.chemgeo.2013.06.012.

Davis, K. M. and M. Tomozawa (1995). "Water diffusion into silica glass: Structural changes in silica glass and their effect on water solubility and diffusivity". In: *Journal of Non-Crystalline Solids* 185, pp. 203–220.

Deng, X.-D., J.-W. Li, and D. L. Shuster (2017). "Late Mio-Pliocene chemical weathering of the Yulong porphyry Cu deposit in the eastern Tibetan Plateau constrained by goethite (U-Th)/He dating: Implication for Asian summer monsoon". In: *Earth and Planetary Science Letters* 472, pp. 289–298.

Dodson, M. H. (1973). "Closure temperature in cooling geochronological and petrological systems". In: *Contributions to Mineralogy and Petrology* 40, pp. 259–274.

Doremus, R. H. (1966). "Physical Solubility of Gases in Fused Silica". In: *Journal of the American Ceramic Society* 49, pp. 461–462.

Farley, K. A. (2000). "Helium diffusion from apatite: General behavior as illustrated by Durango fluorapatite". In: *Journal of Geophysical Research: Solid Earth* 105, pp. 2903–2914.

Farley, K. A. (2007). "He diffusion systematics in minerals: Evidence from synthetic monazite and zircon structure phosphates". In: *Geochimica et Cosmochimica Acta* 71, pp. 4015–4024.

– (2018). "Helium diffusion parameters of hematite from a single-diffusion-domain crystal". In: *Geochimica et Cosmochimica Acta* 231, pp. 117–129.

Farley, K. A. and R. M. Flowers (2012). "(U–Th)/Ne and multidomain (U–Th)/He systematics of a hydrothermal hematite from eastern Grand Canyon". In: *Earth and Planetary Science Letters* 359-360, pp. 131–140.

Farley, K. A. and R. McKeon (2015). "Radiometric dating and temperature history of banded iron formation-associated hematite, Gogebic iron range, Michigan, USA". In: *Geology* 43, pp. 1083–1086.

Farley, K. A., H. B. Monteiro, P. M. Vasconcelos, and K. Waltenberg (2024). "Dehydration of goethite during vacuum step-heating and implications for he retentivity characterization". In: *Chemical Geology* 663, p. 122254.

Farley, K. A., P. W. Reiners, and V. Nenow (1999). "An Apparatus for High-Precision Helium Diffusion Measurements from Minerals". In: *Analytical Chemistry* 71, pp. 2059–2061.

Farley, K. A., R. A. Wolf, and L. T. Silver (1996). "The effects of long alpha-stopping distances on (U-Th)/He ages". In: *Geochimica et Cosmochimica Acta* 60, pp. 4223–4229.

Fechtig, H. and S. Kalbitzer (1966). "The Diffusion of Argon in Potassium-Bearing Solids". In: *Potassium Argon Dating*. Ed. by O. A. Schaeffer and J. Zähringer. Berlin, Heidelberg: Springer, pp. 68–107.

Flowers, R. M., R. A. Ketcham, D. L. Shuster, and K. A. Farley (2009). "Apatite (U-Th)/He thermochronometry using a radiation damage accumulation and annealing model". In: *Geochimica et Cosmochimica Acta* 73, pp. 2347–2365.

Flowers, R. M., D. L. Shuster, B. P. Wernicke, and K. A. Farley (2007). "Radiation damage control on apatite (U-Th)/He dates from the Grand Canyon region, Colorado Plateau". In: *Geology* 35, pp. 447–450.

Fortier, S. M. and B. J. Giletti (1989). "An Empirical Model for Predicting Diffusion Coefficients in Silicate Minerals". In: *Science* 245.4925, pp. 1481–1484. DOI: 10.1126/science.245.4925.1481.

Gill, J. E. (1969). "Experimental Deformation and Annealing of Sulfides and Interpretation of Ore Textures". In: *Economic Geology* 64.5, pp. 500–508. DOI: 10.2113/gsecongeo.64.5.500.

Goldsmith, A. S., R. A. Ketcham, and D. F. Stockli (2020). "Simulating effects of heterogeneous 4He concentration profiles and radiation damage annealing on whole-grain zircon diffusivity analyses". In: *Geochimica et Cosmochimica Acta* 284, pp. 239–253.

Goss, C. J. (1987). "The kinetics and reaction mechanism of the goethite to hematite transformation". In: *Mineralogical Magazine* 51, pp. 437–451.

Guenthner, W. R., P. W. Reiners, R. A. Ketcham, L. Nasdala, and G. Giester (2013). "Helium diffusion in natural zircon: Radiation damage, anisotropy, and the interpretation of zircon (U-Th)/He thermochronology". In: *American Journal of Science* 313, pp. 145–198.

Guo, H. and A. Barnard (2011). "Thermodynamic modelling of nanomorphologies of hematite and goethite". In: *Journal of Materials Chemistry* 21, pp. 11566–11577.

Haar, L., J. S. Gallagher, and G. S. Kell (1984). *NBS/NRC Steam Tables: Thermodynamic and Transport Properties and Computer Programs for Vapor and Liquid States of Water in SI Units*.

Hannington, M. D., W. Bleeker, and I. Kjarsgaard (1999). "Sulfide Mineralogy, Geochemistry, and Ore Genesis of the Kidd Creek Deposit: Part I. North, Central, and South Orebodies". In: *Economic Geology Monograph* 10, pp. 163–224.

Hanson, G. N., K. R. Simmons, and A. E. Bence (1975). "40Ar/39Ar spectrum ages for biotite, hornblende and muscovite in a contact metamorphic zone". In: *Geochimica et Cosmochimica Acta* 39, pp. 1269–1278.

Harrison, M. T. (1982). "Diffusion of 40Ar in hornblende". In: *Contributions to Mineralogy and Petrology* 78, pp. 324–331.

Harrison, M. T., I. Duncan, and I. McDougall (1985). "Diffusion of 40Ar in biotite: Temperature, pressure and compositional effects". In: *Geochimica et Cosmochimica Acta* 49, pp. 2461–2468.

Heim, J. A., P. M. Vasconcelos, D. L. Shuster, K. A. Farley, and G. Broadbent (2006). "Dating paleochannel iron ore by (U-Th)/He analysis of supergene goethite, Hamersley province, Australia". In: *Geology* 34, pp. 173–176.

Hnatyshin, D., D. van Acken, S. Paradis, R. A. Creaser, and M. Brodbeck (2026). "Pyrite Re-Os Geochronology of Carbonate-Hosted Zn-Pb Deposits of the Canadian Cordillera". In: *Economic Geology* 121.1, pp. 75–92. DOI: 10.5382/econgeo.5215.

Hofmann, F., B. Reichenbacher, and K. A. Farley (2017). "Evidence for >5 Ma paleo-exposure of an Eocene–Miocene paleosol of the Bohnerz Formation, Switzerland". In: *Earth and Planetary Science Letters* 465, pp. 168–175.

Hofmann, F., J. Treffkorn, and K. A. Farley (2020). "U-loss associated with laser-heating of hematite and goethite in vacuum during (U–Th)/He dating and prevention using high O2 partial pressure". In: *Chemical Geology* 532, p. 119350.

Holcombe, R. J., C. J. Stephens, C. R. Fielding, D. Gust, T. A. Little, R. Sliwa, J. Kassan, J. McPhie, and A. Ewart (1997). "Tectonic evolution of the northern New England Fold Belt: the Permian-Triassic Hunter-Bowen event". In: *Tectonics and Metallogenesis of the New England Orogen: Alan H. Voisey Memorial Volume*. Ed. by P. M. Ashley and P. G. Flood. Special Publication 19. Geological Society of Australia, pp. 52–65.

Jones, W. M. (1953). "Permeability and Solubility of He^3 and He^4 in Vitreous Silica". In: *Journal of the American Chemical Society* 75, pp. 3093–3096.

Ketcham, R. A., C. Gautheron, and L. Tassan-Got (2011). "Accounting for long alpha-particle stopping distances in (U-Th-Sm)/He geochronology: Refinement of the baseline case". In: *Geochimica et Cosmochimica Acta* 75, pp. 7779–7791.

King, E. M., C. T. Barrie, and J. W. Valley (1997). "Hydrothermal alteration of oxygen isotope ratios in quartz phenocrysts, Kidd Creek mine, Ontario: Magmatic values are preserved in zircon". In: *Geology* 25.12, pp. 1079–1082.

Lee, J. K. W. (1993). "The argon release mechanisms of hornblende in vacuo". In: *Chemical Geology* 106, pp. 133–170.

Lee, J. K. W., I. S. Williams, and D. J. Ellis (1997). "Pb, U and Th Diffusion in Natural Zircon". In: *Nature* 390.6656, pp. 159–162. DOI: 10.1038/36554.

Li, Y., S. Glorie, and D. Selby (2025). "Re–Os Geochronology for Sulfides and Organic-Rich Sediments". In: *National Science Review* 12.9, nwaf300. DOI: 10.1093/nsr/nwaf300.

Little, T. A., M. O. McWilliams, and R. J. Holcombe (1995). "40Ar/39Ar thermochronology of epidote blueschists from the North D'Aguilar block, Queensland, Australia: Timing and kinematics of subduction complex unroofing". In: *GSA Bulletin* 107.5, pp. 520–535. DOI: 10.1130/0016-7606(1995)107<0520:AATOEB>2.3.CO;2.

Lo, C.-H., J. K. W. Lee, and T. C. Onstott (2000). "Argon release mechanisms of biotite in vacuo and the role of short-circuit diffusion and recoil". In: *Chemical Geology* 165, pp. 135–166.

Lollar, G. S., O. Warr, J. Telling, M. R. Osburn, and B. Sherwood Lollar (2019). "'Follow the Water': Hydrogeochemical Constraints on Microbial Investigations 2.4 km Below Surface at the Kidd Creek Deep Fluid and Deep Life Observatory". In: *Geomicrobiology Journal* 36.10, pp. 859–872. DOI: 10.1080/01490451.2019.1641770.

Lovera, O. M., F. M. Richter, and T. M. Harrison (1991). "Diffusion domains determined by 39Ar released during step heating". In: *Journal of Geophysical Research: Solid Earth* 96, pp. 2057–2069.

Lyu, C., J.-F. Gao, L. Qi, and X.-W. Huang (2020). "Re-Os Isotope System of Sulfide from the Fule Carbonate-Hosted Pb-Zn Deposit, SW China: Implications for Re-Os Dating of Pb-Zn Mineralization". In: *Ore Geology Reviews* 121, p. 103558. DOI: 10.1016/j.oregeorev.2020.103558.

Majzlan, J., K.-D. Grevel, and A. Navrotsky (2003). "Thermodynamics of Fe oxides: Part II. Enthalpies of formation and relative stability of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3)". In: *American Mineralogist* 88, pp. 855–859.

Masaryk, J. S. (1968). *Permeation in Fused Silica*. As cited in the fused-silica helium permeability literature (e.g., Shelby, 1972); original venue not confirmed.

McClay, K. R. and P. G. Ellis (1983). "Deformation and Recrystallization of Pyrite". In: *Mineralogical Magazine* 47.345, pp. 527–538. DOI: 10.1180/minmag.1983.047.345.14.

McInnes, B. I. A., N. J. Evans, F. Q. Fu, and S. Garwin (2005). "Application of Thermochronology to Hydrothermal Ore Deposits". In: *Reviews in Mineralogy and Geochemistry* 58, pp. 467–498.

McMurtry, G. M., J. R. DeLuze, D. R. Hilton, and J. E. Blessing (2019). "Differential Diffusion of Helium Isotopes in Glass, Quantum-tunneling 3He Enrichment, and Portable 3He/4He Monitoring of Mantle Processes". In: *Scientific Reports* 9, p. 5213.

Monteiro, H. (2017). "Paleoenvironmental Evolution of Continental Landscapes through Combined High-Resolution Geochronology and δ^{18}O Ion Microprobe Analysis of Goethite". PhD thesis. The University of Queensland. DOI: 10.14264/uql.2018.6.

Monteiro, H. S., P. M. Vasconcelos, K. A. Farley, C. A. Spier, and C. L. Mello (2014). "(U-Th)/He geochronology of goethite and the origin and evolution of cangas". In: *Geochimica et Cosmochimica Acta* 131, pp. 267–289.

Monteiro, H. S., P. M. P. Vasconcelos, K. A. Farley, and C. A. M. Lopes (2018). "Age and evolution of diachronous erosion surfaces in the Amazon: Combining (U-Th)/He and cosmogenic 3He records". In: *Geochimica et Cosmochimica Acta* 229, pp. 162–183.

Morimoto, N. and L. A. Clark (1961). "Arsenopyrite Crystal-Chemical Relations". In: *American Mineralogist* 46.11–12, pp. 1448–1469.

Naono, H., K. Nakai, T. Sueyoshi, and H. Yagi (1987). "Porous texture in hematite derived from goethite: Mechanism of thermal decomposition of goethite". In: *Journal of Colloid and Interface Science* 120, pp. 439–450.

Nash, J. T. (1979). *Geology, Petrology, and Chemistry of the Leadville Dolomite: Host for Uranium at the Pitch Mine, Saguache County, Colorado*. Tech. rep. 79-1566. US Geological Survey.

Navrotsky, A., L. Mazeina, and J. Majzlan (2008). "Size-Driven Structural and Thermodynamic Complexity in Iron Oxides". In: *Science* 319, pp. 1635–1638.

Nomade, S., P. R. Renne, and R. K. W. Merkle (2004). "40Ar/39Ar age constraints on ore deposition and cooling of the Bushveld Complex, South Africa". In: *Journal of the Geological Society* 161, pp. 411–420.

Norton, F. J. (1953). "Helium Diffusion Through Glass". In: *Journal of the American Ceramic Society* 36, pp. 90–96.

Oehler, A. and M. Tomozawa (2004). "Water diffusion into silica glass at a low temperature under high water vapor pressure". In: *Journal of Non-Crystalline Solids* 347, pp. 211–219.

Ostendorf, J., F. Henjes-Kunst, T. Seifert, and J. Gutzmer (2019). "Age and Genesis of Polymetallic Veins in the Freiberg District, Erzgebirge, Germany: Constraints from Radiogenic Isotopes". In: *Mineralium Deposita* 54.2, pp. 217–236. DOI: 10.1007/s00126-018-0841-1.

Ostrogorsky, A. G. and M. E. Glicksman (2015). "Segregation and Component Distribution". In: *Handbook of Crystal Growth*. 2nd. Vol. IIA. Amsterdam: Elsevier. Chap. 25.

Park, Y.-R., E. M. Ripley, J. D. Miller, C. Li, J. Mariga, and P. Shafer (2004). "Stable Isotopic Constraints on Fluid-Rock Interaction and Cu-PGE-S Redistribution in the Sonju Lake Intrusion, Minnesota". In: *Economic Geology* 99.2, pp. 325–338. DOI: 10.2113/gsecongeo.99.2.325.

Protin, M., P.-H. Blard, Y. Marrocchi, and F. Mathon (2016). "Irreversible adsorption of atmospheric helium on olivine: A lobster pot analogy". In: *Geochimica et Cosmochimica Acta* 179, pp. 76–88.

Qi, L., M.-F. Zhou, J. Gao, and Z. Zhao (2010). "An Improved Carius Tube Technique for Determination of Low Concentrations of Re and Os in Pyrites". In: *Journal of Analytical Atomic Spectrometry* 25.4, pp. 585–589. DOI: 10.1039/b919016c.

Rätzke, K., P. W. Hüppe, and F. Faupel (1992). "Transition from Single-Jump Type to Highly Cooperative Diffusion during Structural Relaxation of a Metallic Glass". In: *Physical Review Letters* 68.15, pp. 2347–2349. DOI: 10.1103/PhysRevLett.68.2347.

Reiners, P. W. and K. A. Farley (1999). "Helium diffusion and (U-Th)/He thermochronometry of titanite". In: *Geochimica et Cosmochimica Acta* 63, pp. 3845–3859.

Rogers, W. A., R. S. Buritz, and D. Alpert (1954). "Diffusion Coefficient, Solubility, and Permeability for Helium in Glass". In: *Journal of Applied Physics* 25, pp. 868–875.

Sander, R. (2015). "Compilation of Henry's law constants (version 4.0) for water as solvent". In: *Atmospheric Chemistry and Physics* 15, pp. 4399–4981.

Schoenberg, R., F. J. Kruger, T. F. Nagler, T. Meisel, and J. D. Kramers (1999). "PGE Enrichment in Chromitite Layers and the Merensky Reef of the Western Bushveld Complex; a Re–Os and Rb–Sr Isotope Study". In: *Earth and Planetary Science Letters* 172.1, pp. 49–64. DOI: 10.1016/S0012-821X(99)00198-3.

Schwertmann, U., P. Cambier, and E. Murad (1985). "Properties of Goethites of Varying Crystallinity". In: *Clays and Clay Minerals* 33, pp. 369–378.

Scoates, J. S., C. J. Wall, R. M. Friedman, D. Weis, E. A. Mathez, and J. A. VanTongeren (2021). "Dating the Bushveld Complex: Timing of Crystallization, Duration of Magmatism, and Cooling of the World's Largest Layered Intrusion and Related Rocks". In: *Journal of Petrology* 62.2, egaa107. DOI: 10.1093/petrology/egaa107.

Selby, D., R. A. Creaser, C. J. R. Hart, C. S. Rombach, J. F. H. Thompson, M. T. Smith, A. A. Bakke, and R. J. Goldfarb (2002). "Absolute Timing of Sulfide and Gold Mineralization: A Comparison of Re-Os Molybdenite and Ar-Ar Mica Methods from the Tintina Gold Belt, Alaska". In: *Geology* 30.9, pp. 791–794. DOI: 10.1130/0091-7613(2002)030<0791:ATOSAG>2.0.CO;2.

Shackelford, J. F. (1999). "Gas solubility in glasses – principles and structural implications". In: *Journal of Non-Crystalline Solids* 253, pp. 231–241.

Shackelford, J. F., P. L. Studt, and R. M. Fulrath (1972). "Solubility of Gases in Glass. II. He, Ne, and H2 in Fused Silica". In: *Journal of Applied Physics* 43, pp. 1619–1626.

Shelby, J. E. (1971). "Diffusion of Helium Isotopes in Vitreous Silica". In: *Physical Review B* 4, pp. 2681–2686.

– (1972). "Helium Migration in Natural and Synthetic Vitreous Silica". In: *Journal of the American Ceramic Society* 55, pp. 61–64.

– (1976). "Pressure dependence of helium and neon solubility in vitreous silica". In: *Journal of Applied Physics* 47, pp. 135–139.

Shukolyukov, Y., O. Yakubovich, A. Mochalov, A. Kotov, E. Sal'nikova, S. Yakovleva, S. Korneev, and B. Gorokhovskii (2012). "New Geochronometer for the Direct Isotopic Dating of Native Platinum Minerals (Pt-He Method)". In: *Petrology* 20.6, pp. 491–505. DOI: 10.1134/S0869591112060033.

Shukolyukov, Y. A., M. M. Fugzan, I. P. Paderin, S. A. Sergeev, and D. P. Krylov (2009). "Geothermochronology Based on Noble Gases: I. Stability of the U-Xe Isotopic System in Nonmetamict Zircons". In: *Petrology* 17.1, pp. 1–24. DOI: 10.1134/S0869591109010019.

Shuster, D. L. and K. A. Farley (2005). "4He/3He Thermochronometry: Theory, Practice, and Potential Complications". In: *Reviews in Mineralogy and Geochemistry* 58, pp. 181–203.

Shuster, D. L. and K. A. Farley (2009). "The Influence of Artificial Radiation Damage and Thermal Annealing on Helium Diffusion Kinetics in Apatite". In: *Geochimica et Cosmochimica Acta* 73.1, pp. 183–196. DOI: 10.1016/j.gca.2008.10.013.

Shuster, D. L., K. A. Farley, J. M. Sisterson, and D. S. Burnett (2004). "Quantifying the diffusion kinetics and spatial distributions of radiogenic 4He in minerals containing proton-induced 3He". In: *Earth and Planetary Science Letters* 217, pp. 19–32.

Shuster, D. L., R. M. Flowers, and K. A. Farley (2006). "The influence of natural radiation damage on helium diffusion kinetics in apatite". In: *Earth and Planetary Science Letters* 249, pp. 148–161.

Shuster, D. L., P. M. Vasconcelos, J. A. Heim, and K. A. Farley (2005). "Weathering geochronology by (U-Th)/He dating of goethite". In: *Geochimica et Cosmochimica Acta* 69, pp. 659–673.

Sivell, W. J. and M. T. McCulloch (2001). "Geochemical and Nd-isotopic systematics of the Permo-Triassic Gympie Group, southeast Queensland". In: *Australian Journal of Earth Sciences* 48, pp. 377–393.

Sivell, W. J. and J. B. Waterhouse (1988). "Petrogenesis of Gympie Group volcanics: evidence for remnants of an early Permian volcanic arc in eastern Australia". In: *Lithos* 21, pp. 81–95.

Stein, H., A. Schersten, J. Hannah, and R. Markey (2003). "Subgrain-Scale Decoupling of Re and ^{187}Os and Assessment of Laser Ablation ICP-MS Spot Dating in Molybdenite". In: *Geochimica et Cosmochimica Acta* 67.19, pp. 3673–3686. DOI: 10.1016/S0016-7037(03)00269-2.

Studt, P. L. (1962). "Mechanism of Gaseous Permeation through Glass, Single-Crystal Silicon, and Germanium; and Stress-Enhanced Gaseous Permeation through Alumina Bodies". PhD thesis. University of California. URL: https://escholarship.org/uc/item/9pj1h2x1.

Sturrock, C. P., R. M. Flowers, B. P. Kohn, and J. R. Metcalf (2024). "Phanerozoic Burial and Erosion History of the Southern Canadian Shield from Apatite (U-Th)/He Thermochronology". In: *Minerals* 14.1, p. 57. DOI: 10.3390/min14010057.

Suzuki, K., M. Feely, and C. O'Reilly (2001). "Disturbance of the Re-Os Chronometer of Molybdenites from the Late-Caledonian Galway Granite, Ireland, by Hydrothermal Fluid Circulation". In: *Geochemical Journal* 35.1, pp. 29–35. DOI: 10.2343/geochemj.35.29.

Swets, D. E., R. W. Lee, and R. C. Frank (1961). "Diffusion Coefficients of Helium in Fused Quartz". In: *The Journal of Chemical Physics* 34, pp. 17–22.

Thyer, B. J., J. Treffkorn, and K. A. Farley (2026). "A Simple Method for Helium Step Heat Experiments at High Water or Sulfur Fugacity". In preparation.

Toulmin, P. and P. B. Barton (1964). "A thermodynamic study of pyrite and pyrrhotite". In: *Geochimica et Cosmochimica Acta* 28, pp. 641–671. ISSN: 0016-7037. DOI: 10.1016/0016-7037(64)90083-3.

Tsuru, T., R. Igi, M. Kanezashi, T. Yoshioka, S. Fujisaki, and Y. Iwamoto (2011). "Permeation properties of hydrogen and water vapor through porous silica membranes at high temperatures". In: *AIChE Journal* 57.3, pp. 618–629. DOI: 10.1002/aic.12298.

Vasconcelos, P. M., J. A. Heim, K. A. Farley, H. Monteiro, and K. Waltenberg (2013). "40Ar/39Ar and (U-Th)/He–4He/3He geochronology of landscape evolution and channel iron deposit genesis at Lynn Peak, Western Australia". In: *Geochimica et Cosmochimica Acta* 117, pp. 283–312.

Verhallen, P. T. H. M., L. J. P. Oomen, A. J. J. M. v. d. Elsen, J. Kruger, and J. M. H. Fortuin (1984). "The diffusion coefficients of helium, hydrogen, oxygen and nitrogen in water determined from the permeability of a stagnant liquid layer in the quasi-steady state". In: *Chemical Engineering Science* 39, pp. 1535–1541.

Virtanen, P. et al. (2020). "SciPy 1.0: Fundamental algorithms for scientific computing in Python". In: *Nature Methods* 17, pp. 261–272.

Waltenberg, K. M. (2012). *Mineral Physics and Crystal Chemistry of Minerals Suitable for Weathering Geochronology: Implications to 40Ar/39Ar and (U-Th)/He Geochronology*. PhD thesis; institution not confirmed in the source draft.

Wilson, W. D. and C. L. Bisson (1973). "Atomistics of helium diffusion in copper and tungsten". In: *Radiation Effects* 19, pp. 53–58.

Wolf, R. A., K. A. Farley, and D. M. Kass (1998). "Modeling of the temperature sensitivity of the apatite (U-Th)/He thermochronometer". In: *Chemical Geology* 148, pp. 105–114.

Wolf, R. A., K. A. Farley, and L. T. Silver (1996). "Helium diffusion and low-temperature thermochronometry of apatite". In: *Geochimica et Cosmochimica Acta* 60, pp. 4231–4240.

Yakubovich, O., M. Podolskaya, I. Vikentyev, E. Fokina, and A. Kotov (2020). "U-Th-He Geochronology of Pyrite from the Uzelga VMS Deposit (South Urals) – New Perspectives for Direct Dating of the Ore-Forming Processes". In: *Minerals* 10, p. 629.

Yakubovich, O., I. Vikentyev, E. Ivanova, M. Podolskaya, I. Sobolev, E. Tyukova, and A. Kotov (2021). "U-Th-He Geochronology of Pyrite from Alteration of the Au-Fe-Skarn Novogodnee-Monto Deposit (Polar Urals, Russia) – The Next Step in the Development of a New Approach for Direct Dating of Ore-Forming Processes". In: *Geosciences* 11.10, p. 408. DOI: 10.3390/geosciences11100408.

Yakubovich, O. V., A. M. Gedz, I. V. Vikentyev, A. B. Kotov, and B. M. Gorokhovskii (2019). "Migration of Radiogenic Helium in the Crystal Structure of Sulfides and Prospects of Their Isotopic Dating". In: *Petrology* 27, pp. 59–78.

Yakubovich, O. V., A. G. Mochalov, and S. F. Sluzhenikin (2015). "Sperrylite ($PtAs_2$) as a ^{190}Pt-^{4}He Geochronometer". In: *Doklady Earth Sciences* 462.1, pp. 472–474. DOI: 10.1134/S1028334X15050098.

Yakubovich, O. V., Y. A. Shukolyukov, A. B. Kotov, S. Z. Yakovleva, and E. B. Sal'nikova (2010). "Geothermochronology based on noble gases: II. Stability of the (U-Th)/He isotope system in zircon". In: *Petrology* 18, pp. 555–570.

Yao, Z. and J. E. Mungall (2021). "Linking the Siberian Flood Basalts and Giant Ni-Cu-PGE Sulfide Deposits at Norilsk". In: *Journal of Geophysical Research: Solid Earth* 126.3, e2020JB020823. DOI: 10.1029/2020JB020823.

Yapp, C. J. and H. Poths (1995). "Stable hydrogen isotopes in iron oxides: III. Nonstoichiometric hydrogen in goethite". In: *Geochimica et Cosmochimica Acta* 59, pp. 3405–3412.

Yapp, C. J. and D. L. Shuster (2011). "Environmental memory and a possible seasonal bias in the stable isotope composition of (U-Th)/He-dated goethite from the Canadian Arctic". In: *Geochimica et Cosmochimica Acta* 75, pp. 4194–4215.

Yund, R. A. and G. Kullerud (1966). "Thermal Stability of Assemblages in the Cu–Fe–S System". In: *Journal of Petrology* 7.3, pp. 454–488.

Zeitler, P. K., A. L. Herczeg, I. McDougall, and M. Honda (1987). "U-Th-He dating of apatite: A potential thermochronometer". In: *Geochimica et Cosmochimica Acta* 51, pp. 2865–2868.

Zhao, J., J. Brugger, Y. Ngothai, and A. Pring (2014). "The Replacement of Chalcopyrite by Bornite under Hydrothermal Conditions". In: *American Mineralogist* 99.11–12, pp. 2389–2397. DOI: 10.2138/am-2014-4825.

FIELD STUDIES REVEAL SYNMAGMATIC DEFORMATION AND EXTENSIVE CRUSTAL CONTAMINATION OF THE UPPER CRITICAL ZONE, RECESSIVE LOWER MAIN ZONE, AND FLOOR ROCKS AT THE MARGIN OF THE BUSHVELD COMPLEX AT DRAAIKRAAL, MPUMALANGA, SOUTH AFRICA

[TODO: One-paragraph summary of the field campaign and key findings.]

Introduction

[TODO: Introduce the Bushveld Complex as one of the world's largest and best-preserved layered intrusions. State the scientific motivation for fieldwork at Draaikraal: tectonic setting, unexplained recessive topography, and the potential for goethite laterite geochronology.]

Literature Review

[TODO: Review Bushveld stratigraphy, magmatic layering, and prior work on the Critical Zone, Main Zone, and floor rock relationships. Summarize existing interpretations of crustal contamination and syn-magmatic deformation. Review goethite laterite formation in the context of Bushveld-derived lithologies.]

Geology of the Bushveld Complex

[TODO:]

The Upper Critical Zone and Merensky Reef

[TODO:]

The Lower Main Zone and Its Recessive Expression

[TODO:]

Floor Rocks and Crustal Contamination

[TODO:]

Laterite Formation and Goethite Geochronology Potential

[TODO:]

Field Methods

[TODO: Describe traverses, mapping strategy, sampling, and photographic documentation at Draaikraal.]

Field Observations

[TODO: Describe structural and lithological observations: deformation fabrics, contamination textures, lateral facies changes.]

Sample and Station Data

[TODO: Insert data table of field stations, GPS coordinates, lithologic descriptions, and samples collected.]

Field Map

[TODO: Insert field map figure.]

Annotated Field Photographs

[TODO: Insert annotated field photographs with captions. Use `subcaption` for multi-panel figures.]

Discussion

[TODO: Interpret field evidence for synmagmatic deformation and crustal contamination. Discuss the recessive topography at Draaikraal as a geomorphological signature of the Lower Main Zone. Connect to the motivating synthesis described in the Conclusion.]

Conclusions

[TODO: Summarize key field findings. Point to the application of ML-accelerated thermodynamic models and goethite/sulfide geochronology (Chapters 5, 2, 3) as the next research steps.]